

The 100-year life

How longer lives reshape work, money, health, and society through 2050 and beyond

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The demographic transition has two motors: fewer births and longer lives. Fertility decline is what shrinks the base of the population pyramid, while rising life expectancy stretches its top; together they produce the structural age-composition shift Section 1 described. Net migration can cushion the shift, but only modestly and temporarily, because migrants are disproportionately young.

The dependency ratio is the mechanism through which demography becomes economics: it measures the number of people aged 65 and above relative to the number of people of working age, defined as 20-64. When the numerator grows and the denominator shrinks, fewer contributors support each pensioner — and population ageing has been the main driving force behind changes in pension policies. The OECD's standard monitoring of this pressure pairs the demographic indicator with two labour-market ones: the employment rates of older workers, and the effective age of labour market exit combined with the life years expected from that age.

The oldest-old are the age group where the 100-year life is no longer an abstraction: people aged 80 and above are the fastest-growing segment of the population in advanced economies, and the previous section established the scale of their expansion. What the evidence set supplies for this section is not more headcounts but the modelling and policy reality of very old age: how long people might live, how uncertain that horizon is, and how societies are — or are not — adapting work and retirement to it. Longevity risk: the horizon to 110 Retirement modelling now treats extreme old age as a planning parameter, not a curiosity.

Demographic foundations of the 100-year life

Global population ageing and the demographic transition

Population ageing is a structural shift in the age composition of societies: the share of older people rises while the share of younger people falls. In the European Union, declining fertility and increasing longevity are the two key drivers of this process. The outcome is already visible in Europe's population counts: in 2018 there were 101 million people aged 65 years or more in the EU-28, of whom 42% lived in predominantly urban regions, 38% in intermediate regions, and 20% in predominantly rural regions.¹

The concentration of ageing is uneven across regions. In 2018, 14 EU regions had older populations exceeding one quarter of their total residents, principally in eastern Germany and northern and central Italy; the highest shares were recorded in Chemnitz in eastern Germany at 28.5% and Liguria in north-western Italy at 28.4%, with Ipeiros in north-western Greece at 26.6%. Older people were also disproportionately present in rural areas: they were generally more inclined than their fellow compatriots to live in predominantly rural regions, with Slovakia, Belgium, and Poland the only exceptions in 2018. In some cases, high rural shares reflect younger generations leaving sparsely populated regions for jobs and better opportunities while older people remain.¹

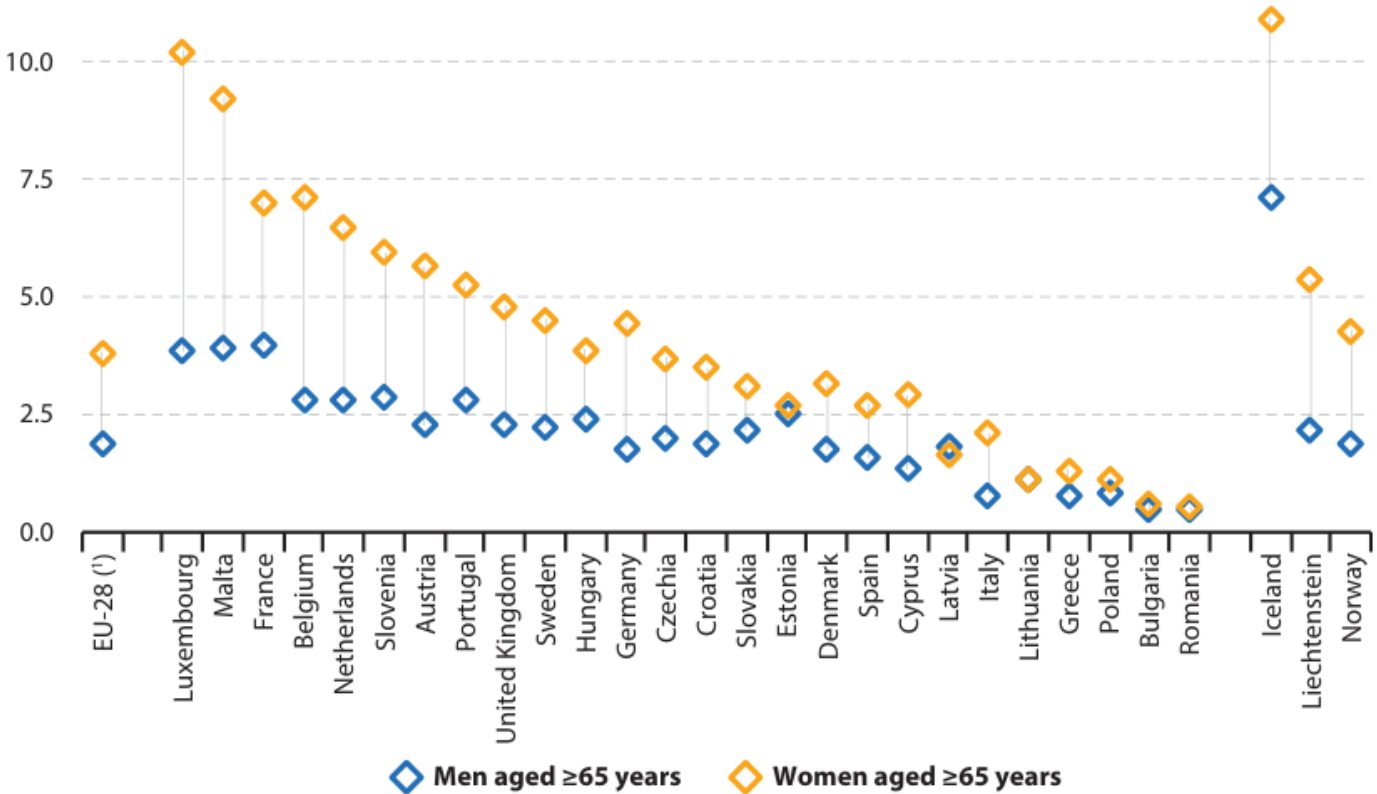
A third driver, international migration, can slow the ageing process because most immigrants are relatively young, often seeking work and a better quality of life. The EU data confirm this pattern: in 2018, foreign citizens of another EU Member State accounted for 3.4% of the EU-28 population but only 1.7% of people aged 65 years or more, while citizens of non-EU countries accounted for 4.4% of the total population but just 1.5% of the older population. France and Croatia were the only Member States where older people included higher shares of other-EU citizens, suggesting retirement migration, and Estonia and Latvia showed higher shares of older non-EU citizens linked to former So-

viet Union residents classified as recognised non-citizens. In Luxembourg, foreign citizens made up almost half (47.8%) of the total population but a considerably lower 28.4% of older people.¹

Ageing has a pronounced gender dimension because women live longer. In 2011, 3.8% of older women (aged 65 or more) in the EU-28 lived in institutional households such as retirement or nursing homes, twice the 1.9% share of older men, a gap repeated in every Member State except Latvia. The pattern reflects both longevity and the responsibility for financing institutional care often residing with the older person or their family, since long-term social care is rarely covered to the same extent as healthcare. At the extremes, the share of older women in institutional households reached around 7% in France and Belgium, 9.2% in Malta, and peaked at 10.2% in Luxembourg.¹

Looking ahead, the demographic transition reshapes labour supply as well as age structure. EU employment is projected to rise from around 75% of the working-age population in 2022 to around 79% in 2070, a change mostly reflecting higher employment rates among older people and women, alongside the assumption of declining unemployment. The employment rate for the population aged 20-64 is expected to rise from 74.5% to 78.5%, including a nearly 6-percentage-point increase for women and about 2 points for men, and the share of older workers in total employment at EU level would climb from around 20% in 2022 to about 23% in 2070. These gains are tied directly to recent pension and labour-market reforms in many Member States that incentivise later retirement.²

The transition also exposes unequal footing. Employment rates among 55-to-64-year-olds rise sharply with educational attainment: in 2015 the OECD average was 51% for men and 36% for women with no upper-secondary schooling, compared with 75% for men and 65% for women with tertiary degrees. Among those who exit early, 24% of poorly educated men had not worked before retirement for health reasons, versus 7% of highly educated men. Pension coverage mirrors this gradient: with the exception of Chile, Korea, and Mexico, which have relatively large informal economies, mandatory earnings-related pension coverage is high for employees, but far fewer self-employed enjoy coverage — in Latin America and the Caribbean, about two-thirds of salaried workers contribute to pension schemes compared with only one-sixth of the self-employed. Only eight OECD countries have voluntary private pension coverage exceeding 40% of the working-age population, and in the United States only 13% of the lower half of the income distribution receive private pensions compared with over 50% of the upper half. How governments respond to these pressures has measurable fiscal consequences: in Spain, the removal of the Index for Pension Revaluation and the Sustainability Factor is expected to substantially increase pension spending in 2050, while legislation raising retirement ages in the Slovak Republic and Sweden is expected to limit spending increases significantly.^{3,4}



Fertility decline, rising life expectancy, and net migration

The demographic transition has two motors: fewer births and longer lives. Fertility decline is what shrinks the base of the population pyramid, while rising life expectancy stretches its top; together they produce the structural age-composition shift Section 1 described. Net migration can cushion the shift, but only modestly and temporarily, because migrants are disproportionately young. This section examines the engine where the evidence is strongest — the scale, composition, and inequality of longevity gains — and how differential longevity is already built into retirement policy.

The scale of longevity gains

Between 2000 and 2015, life expectancy at birth increased by an average of 3.7 years across the OECD. The gains were overwhelmingly good years: WHO estimates indicate that 85% of the increase — 3.2 of the 3.7 years — was healthy life, with only the remaining 15% (about half a year) marked by bad health from disease or injury. Alternative WHO estimates put it even more positively, with over 90% of life expectancy gains since 2000 translating into disability-free life years. The gains were not evenly shared. They were strongest in countries that had relatively low levels in 2000, such as Estonia and Turkey, while Mexico grew by less than two years — held back by a greater prevalence of risk factors and persistent barriers to high-quality care — and the United States grew by only 2.5 years, more than one year below the OECD average.³

The gendered split of healthy years

Aggregate longevity conceals who gets the healthy years. In the 25 European OECD countries, an average of almost 40% of 50-year-old men's remaining life expectancy in 2014 was affected by limitations on activities; the share among women was even higher, at 47%. Between 2005 and 2014, life expectancy at age 50 rose by about two years, but only around 40% of those gains were healthy years for men and only about 30% for women — meaning the ratio of healthy life years to remaining life expectancy at 50 fell slightly for both sexes over the decade. This matters for the 100-year life because the years after 50 are exactly the years when extended working lives and active ageing must happen.³

Longevity is occupationally unequal

Life expectancy gains also divide along the fault line of occupation. Manual workers face higher mortality than other workers: differences in mortality rates after age 30 leave workers in professional occupations living about five years longer than manual workers in Italy. At age 65, the gap between high- and low-earnings occupations is estimated at two to five years in Germany, France, and the United States; across occupational groups at 65 the differences run about two to three years in Germany, Norway, and Sweden and almost five years in France. Among French hospital workers, life expectancy at 65 for blue-collar workers is shorter by 3.7 years for men and 2.2 years for women compared with executives; in Norway, cleaners at age 67 have about one year less than all professions on average and three years less than academic professions, while differences at age 62 between occupational groups were estimated at 3.5 years.⁴

These gaps survive controls for education, income, and habits. Finnish research finds occupation has an independent effect on disability and mortality; occupational differences in life expectancy at 60 remain substantial in Spain once education and habits are controlled for; and in the Netherlands, workers in academic professions have a life expectancy at 65 that is 3.5 years longer than that of low-skilled workers after accounting for gender, age, and education. The causes concentrate in lung cancer, breast cancer, and cardiovascular disease, and transitions from work into disability are also more common among the low-skilled. Critically, differences in healthy life expectancy are larger than differences in life expectancy itself. One caveat: such evidence is based on past data, since mortality records take decades to accumulate, and working conditions are changing rapidly through regulation and technology.⁴

What is killing people now

The disease burden behind longevity tells the same unequal story. Across OECD countries in 2021 there were 12,818,369 deaths: cancers accounted for 28% and diseases of the circulatory system for 21%. Avoidable-mortality figures show how much of this is policy-malleable: in 2021, age-standardised mortality from preventable causes averaged 158 per 100,000 population across OECD38 countries (79 for treatable causes), ranging from 83 per 100,000 in Israel to 435 in Mexico; lung cancer is classified as preventable, while breast and colorectal cancers are classified as treatable.⁵

The surge of the oldest-old

The most direct expression of rising life expectancy is the growth of the oldest-old. The share of people aged 80 and over in OECD populations is projected to more than double by 2050, when it will account for an average of 9.5% of the total population — and up to 16% in Japan and Spain; in 2015, by contrast, the over-80s accounted for less than 1% of the population in Indonesia, India, and South Africa. Life expectancy gains beyond age 80 in the 1970s were chiefly attributable to progress in preventing cardiovascular diseases, and 80-year-olds' average life expectancy continued to climb steadily in every OECD country between 2000 and 2015.³

Longevity, work capacity, and the design of retirement ages

Longer lives are the premise of extended working lives — and the evidence supports the premise up to a point. Labour force participation among workers in their 50s and 60s tends to be higher when expected remaining lifetimes are longer, which is why older and less well-educated people, who have lower remaining life expectancies, participate less. The upper frontier of participation is concave: when remaining life expectancy is short, improving health has a stronger effect on maximum work capacity. But health explains low participation at older ages only to a limited extent: participation rates differ substantially between population groups with similar remaining life expectancy, implicating financial incentives in the pension system, wealth, and education as well.³

Pension systems already encode differential longevity through special provisions for hazardous and arduous work. Ballet dancers in France, Norway, and Portugal are allowed to retire very early, in their forties; workers covered by arduous-work provisions can retire at 55 or earlier in Belgium, Chile, Colombia, Estonia, Greece, Latvia, Poland, Portugal, the Slovak Republic, Slovenia, Spain, and Türkiye. The effective gaps are large: covered workers in Latvia claim benefits almost 13 years earlier than other workers, at age 50; in Slovenia they retire at 54, 8.5 years earlier; in Colombia and Czechia around 6 years earlier — while the difference is under 3 years in Austria, Chile, Finland, France, Greece, Hungary, and Türkiye.⁴

The qualifying conditions reveal how finely calibrated these systems are. Austria requires 10 of the last 20 years in hazardous or arduous jobs plus 45 years of contributions to retire at 60 (versus 62 for other long-career workers), while nightshift workers can retire at 57 for men and 52 for women. Italy offers three early-retirement options for arduous occupations — at 61 years and 7 months with 35 years of contributions, a 41-year career started before age 19, or 63 with 36 years via the so-called Social APE — against a statutory age of 67. Chile lets hazardous workers anticipate retirement by up to 10 years with at least 20 years of contributions; Spain reduces retirement age by between 5% and 50% of years worked in hazardous occupations — 10% for local police officers, 15% for train drivers, 40% for pilots, 50% for some miners — with a general floor at 52 that miners and marine workers are exempted from. Norway applies an “85-year rule” allowing retirement three years before the special age limit when age plus tenure exceeds 85. Tenure bars vary: Poland requires 15 years in hazardous jobs, Greece 12, Portugal 10 to 22, the Slovak Republic 10 to 20, Estonia and Korea 50% and 60% of the total contribution period respectively, while Türkiye credits up to 50% more contribution period for up to 5 years. In Korea, hazardous workers cannot retire below the minimum age but their pensions are not subject to early-retirement penalties.⁴

The three drivers together

Fertility decline sets the long-run trajectory — it is the irreversible component of the transition — while rising life expectancy is what is currently pushing the oldest-old share toward 9.5% of OECD populations by 2050. Net migration is the only component policymakers can steer in the short run, but as established in the previous section, the youth bias of migrants means it can only slow, not reverse, the ageing process. The policy conclusion from the longevity

evidence is sharper than any single retirement age: because longevity gains divide by gender, education, and occupation, any retirement architecture for the 100-year life must be differentiated by work capacity — not merely pushed upward uniformly — or it will convert differential mortality into differential welfare.³

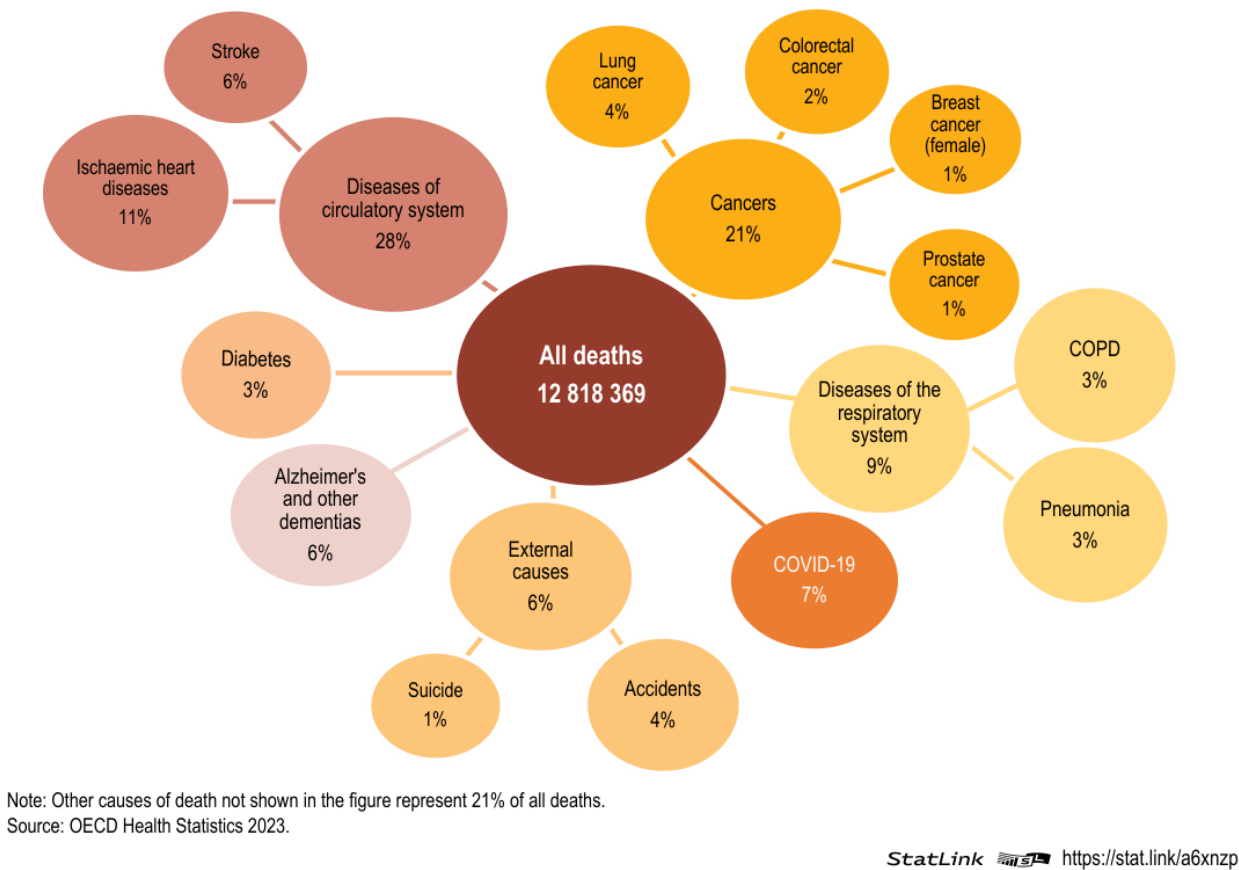


Figure 3.6. Main causes of mortality by country, 2021 (or nearest year)

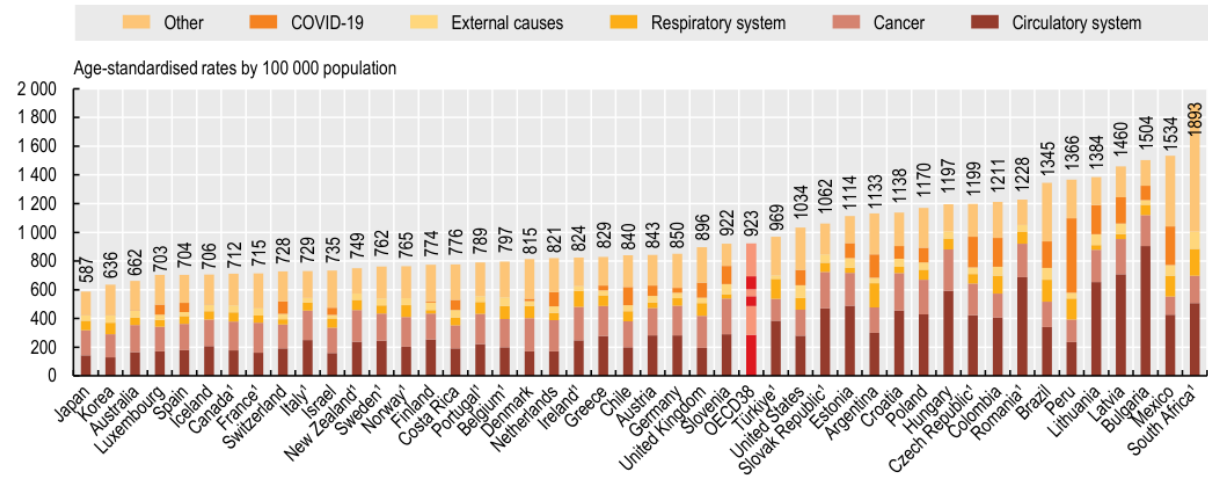


Exhibit. Figure 3.5. Main causes of mortality across OECD countries, 2021 (or nearest year) Stroke Colorectal Lung cancer 6% cancer 2% Breast 4% cancer (female) Ischaemic heart 1% diseases Diseases of 11% circulatory system Cancers 28% 21% P. Source: S0026.

Dependency ratios and a shrinking working-age population

The dependency ratio is the mechanism through which demography becomes economics: it measures the number of people aged 65 and above relative to the number of people of working age, defined as 20–64. When the numerator grows and the denominator shrinks, fewer contributors support each pensioner — and population ageing has been the main driving force behind changes in pension policies. The OECD’s standard monitoring of this pressure pairs the demographic indicator with two labour-market ones: the employment rates of older workers, and the ef-

fective age of labour market exit combined with the life years expected from that age. That triad defines the policy toolkit: if the ratio cannot be fixed by demography, it must be eased by work — more people working, and people working longer.⁴

The participation response is already large

The most striking lever is that older-worker employment has already moved dramatically. In the EU, participation rates for the 55-64 age bracket increased from 38% in 2000 to 64% in 2021. Enacted pension reforms are projected to push this much further: adopted reforms should lift the projected participation rate among both men and women aged 55-64 by about 10 percentage points on average by 2070. The effect extends beyond 64: for the 65-74 age group, legislated reforms push up projected participation rates by an average of 8 percentage points for men and 7 for women by 2070. Countries that introduced an automatic link between retirement ages and gains in life expectancy show the highest increases. In Austria, Czechia, Croatia, and Bulgaria the projected increase for women exceeds that for men because of a progressive convergence of women's legal retirement ages to men's.²

Summarised through average exit ages — derived from the participation rates of the oldest brackets — the projections show an average increase of around two years in the age at which people exit the labour market by 2070, a reasonable approximation for the effective retirement age. This is the denominator strategy in action: the working-age population may shrink, but the working population can expand if exit ages track life expectancy.²

The coverage response: widening who contributes

A second denominator strategy is to widen who counts as a contributor. Expanding mandatory earnings-related pensions is the most effective way to widen coverage, but many mandatory schemes do not apply to all contract types, demand minimum contribution years, and have limited portability — and the trend toward shorter contracts and part-time work lowers the share of the labour force actually covered by traditional pensions. Where mandatory coverage is wide, voluntary private schemes can top up adequacy, and these have grown recently — but voluntary private coverage exceeds 40% of the working-age population in only a handful of OECD countries, reaching as much as 70% in Germany and New Zealand, while Mexico and Poland sit at the other extreme, combining low replacement rates for low-wage workers from mandatory schemes with low voluntary coverage. Replacement rates are also projected to be low for private-sector workers in Canada, Sweden, Slovenia, the United Kingdom, and the United States.³

The coverage gap has a steep income gradient. In the United States in 2012, under 20% of 25-to-59-year-old full-time workers in the lowest earning decile participated in a defined contribution plan, against over 80% in the highest earnings decile. The self-employed illustrate the structural problem: they contribute 18.5% compared with a combined 22.8% for employee and employer, with the state financing the remaining 4.3% in some designs — Chile is completing a phase-in reform that moves the self-employed to employee-like coverage. Untargeted tax incentives for private pensions may even be regressive: such incentives exist in Chile, Denmark, Israel, the Netherlands, the Slovak Republic, and Switzerland, and can reduce the relative attractiveness of voluntary private pensions for low-income households.³⁴

What the ratio can and cannot absorb

The arithmetic of the dependency ratio is unforgiving, but it is not destiny. Two levers are demonstrably moving: older-worker participation has already risen 26 percentage points in two decades in the EU, and reforms are projected to add another 10 points by 2070 with exit ages rising around two years. A third lever — widening coverage to non-standard contracts, the self-employed, and low earners — remains largely unpulled: voluntary private coverage remains thin at the bottom of the income distribution, and non-standard work keeps growing.²³

The limits are those established earlier in this paper: health does not sustain work capacity equally across education and occupation groups, pension incentives and wealth shape who actually works longer, and informal and self-employed workers sit outside contribution systems. The dependency ratio tells policymakers how many contributors each pensioner needs; the evidence says the contributors exist — the policy challenge is making their employment, coverage, and exit ages line up with it.

The oldest-old and the rise of centenarians

The oldest-old are the age group where the 100-year life is no longer an abstraction: people aged 80 and above are the fastest-growing segment of the population in advanced economies, and the previous section established the scale of their expansion. What the evidence set supplies for this section is not more headcounts but the modelling and policy reality of very old age: how long people might live, how uncertain that horizon is, and how societies are — or are not — adapting work and retirement to it.

Longevity risk: the horizon to 110

Retirement modelling now treats extreme old age as a planning parameter, not a curiosity. The OECD's stochastic model for defined-contribution pension outcomes assumes the individual survives with 100% certainty until age 66 — retirement lasts at least one year — and thereafter derives the probability of death at each age from the average mortality rates across 33 OECD countries, with the mortality rate set equal to 1 at age 110. The ceiling matters: centenarians and the near-centenarian years are inside the planning horizon, and the risk being modelled is idiosyncratic longevity risk — the uncertainty around how many years any one individual will live in retirement. The policy implication is direct: as the oldest-old grow as a share of the population, retirement systems must finance not an average lifespan but the right tail of the lifespan distribution, where the costs are concentrated and the uncertainty is greatest.⁶

Capability, not calendar age, ends working lives

A second piece of evidence reframes what old age means for work. Arduousness and hazardousness are not only about health damage: in a broad sense they include the depreciation of the capabilities needed to work safely and efficiently in some occupations with age — ballet dancers and air traffic controllers, for instance, who are not likely to be in worse general health than many other workers of similar age. The OECD's position is explicit: incapacity to continue in one very specific job is not enough to justify special old-age pension provisions, and not being able to continue work in a very specific occupation should not be dealt with by early retirement. Over recent decades, special pension schemes for hazardous or arduous jobs have been reduced in scope, and the argument for allowing very early retirement based solely on not being able to continue a specific career has lost ground. Finland and France have instead improved scheme design by linking eligibility to actual hazardous or arduous job characteristics — such as working at night — rather than to occupational groups, limiting mistargeting and making entitlements transferable across occupations; these measures reduce retirement age by two years at most.⁴

For the oldest workers, the policy answer is adaptation rather than exit. Continuous training throughout working lives is critical to staying in employment as people age, particularly where changing jobs is required to limit health effects or because a job cannot be done at older ages. Mid-life career reviews are a simple and cost-effective way to identify reskilling needs; in 2021 France introduced a medical examination at age 45 intended to raise employees' awareness of ageing at work and prevent occupational wear and tear, during which a reskilling agreement can be proposed. Age-management practices in firms do the rest: in France, older bus drivers are sometimes moved to easier tasks, and night shifts for older drivers can be drastically limited. The oldest-old workforce — where it exists — is built at mid-life, not at 75.⁴

What the evidence does not yet answer

The retrieval for this section exposes a genuine gap: there is abundant modelling of longevity risk and detailed design of working-age interventions, but the corpus supplied little on the oldest-old themselves — their care needs, their economic activity, or the fiscal arithmetic of a centenarian population. What can be said with confidence is structural. The mortality ceiling used in OECD planning is 110; the fastest-growing age group is 80-plus; and the systems built for the twentieth century — designed around retirement at 60-65 and life expectancies in the 70s — are being stretched at both ends, with the tail end the least provisioned. The retirement architecture for a 100-year life must therefore price two things previous generations never faced: the uncertainty of how long any individual will live, and the systematic depreciation of job-specific capabilities long before general health fails.⁶

East Asia and Pacific: the fastest-ageing region on record

No region is ageing as fast as East Asia and Pacific. The region already has more people aged 65 and above than any other region in the world, and its economies are compressing a demographic transition that took Europe a century into a matter of decades. Declining fertility combined with steady increases in both life expectancy and healthy life expectancy — in contrast to Eastern Europe and Central Asia — is driving the shift. But the region is not ageing as one: three distinct groupings face radically different challenges.⁷

Three regions in one

The World Bank's framing divides the region into three groups. The wealthy "red" economies — Hong Kong SAR, Japan, the Republic of Korea, and Singapore — already had 14% of their population aged 65 or older in 2010. The middle- and lower-middle-income "orange" countries — China, Indonesia, Malaysia, Mongolia, Thailand, and Vietnam — are ageing very quickly from lower income levels. The low-income "green" countries — Cambodia, Lao PDR, Myanmar, Papua New Guinea, the Philippines, Timor-Leste, and the Pacific Islands — were still young, with just 4% of the population aged 65 or older in 2010, but will begin to age quickly in coming decades.⁷

The speed is what makes the region unique. Measured in years to move from 7% to 14% of the population aged 65 and older, Vietnam needs just 15 years, Korea, Thailand, Indonesia, Malaysia, and Lao PDR need 20, and China, Japan, and Singapore need 25 — against 45 years for the United Kingdom, 69 for the United States, and 115 for France. The record of demographic projections in the region is mixed, and fertility decline has tended to consistently outpace projections; assumptions about future fertility reversals in richer economies risk extrapolating from unstable short-run dynamics.⁷

Getting old before getting rich

The orange countries are getting old before getting rich: they have been ageing rapidly at low levels of GDP per capita, which intensifies the pressure to raise productivity growth. Thailand's pace of ageing is outstripping per-capita income growth — it is following a path closer to Poland, which is now struggling with fiscal pressures from ageing — and Mongolia combines stagnant GDP per capita with rapid ageing. The stakes are high because the demographic dividend powered the region's growth miracle: around one-third of the high per-capita income growth in East Asia from 1960 to the 1990s can be attributed to the demographic dividend, with some estimates reaching 44%. For the green countries still in their youth bulges, the lesson is to make productivity-enhancing investments now — to maximise the dividend and to avoid adopting pension and health policies that are affordable today but would rapidly become unsustainable when ageing accelerates.⁷

The shrinking working-age population

The labour-force consequences are already quantified. Between 2010 and 2040, the share of the working-age population is projected to fall by more than 15% in Korea and by at least 10% in China, Japan, and Thailand; in green and some orange countries, the working-age share is not expected to shrink until after 2040, and in absolute terms the Philippines and Indonesia will account for the largest share of the regional increase. The traditional elderly dependency ratio itself is coming under scrutiny: the static definition of the working-age population as 15 to 64 is increasingly questioned, and alternative measures — years of remaining life, or ratios based on actual age profiles of labour-force participation — may be more relevant for policy as people live longer and the young spend more time in education.⁷

Older workers work — because many cannot afford not to

In much of the region, the policy problem is not how to get older people to work longer but how to stop them working until they drop. For those still working at 65, urban men average 40 or more hours per week and women 30–45 hours depending on the country; rural men continue working 30–40 hours per week even at age 75, and women 20–35 hours at the same age. Self-employment dominates older-age work in China, Indonesia, Thailand, and Timor-Leste. What determines whether an older person works is largely whether a pension exists: receiving a formal-sector pension is associated with 52% and 54% reductions in the probability that men and women work, after controlling for age, education, health, wealth, and family characteristics. In other words, where pensions are absent or

inadequate, the elderly work long hours out of necessity — and where adequate pensions exist, they retire. Urban China and Vietnam show a distinctive pattern: hours decline little and then drop off sharply at ages 70 and 65 respectively, in contrast to the gradual decline elsewhere.⁷

The fiscal squeeze

The biggest risk to aggregate savings in the region will run through government saving — rising pension, health, and long-term care spending — and it is most acute in developing economies where projected fiscal space is very limited. IMF projections point to a significant increase in pension spending between 2010 and 2030: over 3% of GDP in China, with more modest increases for Indonesia, the Philippines, and Thailand; Japan shows a slight fall, reflecting its already-aged population and cost-containment measures, while Korea faces an increase approaching 4.5% of GDP as its relatively young pension system matures alongside continued rapid ageing. The fiscal risks are compounded because governments are simultaneously pursuing rapid coverage expansion of health insurance and pensions and exploring more active public financing of long-term care.⁷

Old-age poverty and the care burden

The first tier of pension protection is thin in the region. Safety-net pension payments amount to less than 10% of average earnings in China, Korea, India, Turkey, Mexico, and South Africa (versus 50% in Brazil), though China and Turkey also provide minimum pensions above 40% of average earnings. The consequences are visible: old-age poverty rates reach as high as 50% of all over-66s in Korea, and low-income workers face old-age poverty risk — with projected replacement rates below 60% at half the average wage even for a full career — in Japan as well as Mexico, Chile, the UK, Germany, Poland, the US, Sweden, Slovenia, and Canada.³

Care costs compound the pressure. Working parents who take time out for care — particularly mothers — may retire with 22% less in retirement savings; in the US nearly 70% of family carers report difficulty balancing career and caring, and nearly 30% have shifted from full-time to part-time work or reduced their hours. Nursing home care in the US can surpass \$100,000 per year, and low- and middle-income countries often lack robust long-term-care infrastructure, leaving families dependent on informal networks. Japan, the world's fastest-ager, offers the most developed response: its Community-Based Integrated Care System enables ageing in place with localised healthcare, nursing care, and daily living assistance within a 30-minute radius, supplemented by AI-enabled fall monitoring and companion robots such as PARO.⁸

Divergent priorities

The World Bank's policy triage follows the groupings. Red economies must slow the structural decline of the labour force — through female labour-force participation, extended working lives, and, in some countries, immigration of younger workers from within the region. Orange countries must sustain high productivity growth while reforming pensions, health, long-term care, and labour markets, accepting that they will become aged societies at much lower income levels than the red economies — though more educated cohorts will be better prepared for longer working lives. Green countries must realise the demographic dividend now and avoid locking in commitments that will become unsustainable when ageing accelerates over the next two to three decades. The region's speed is its distinguishing feature — and speed is what makes preparation, not reaction, the only viable strategy.⁷

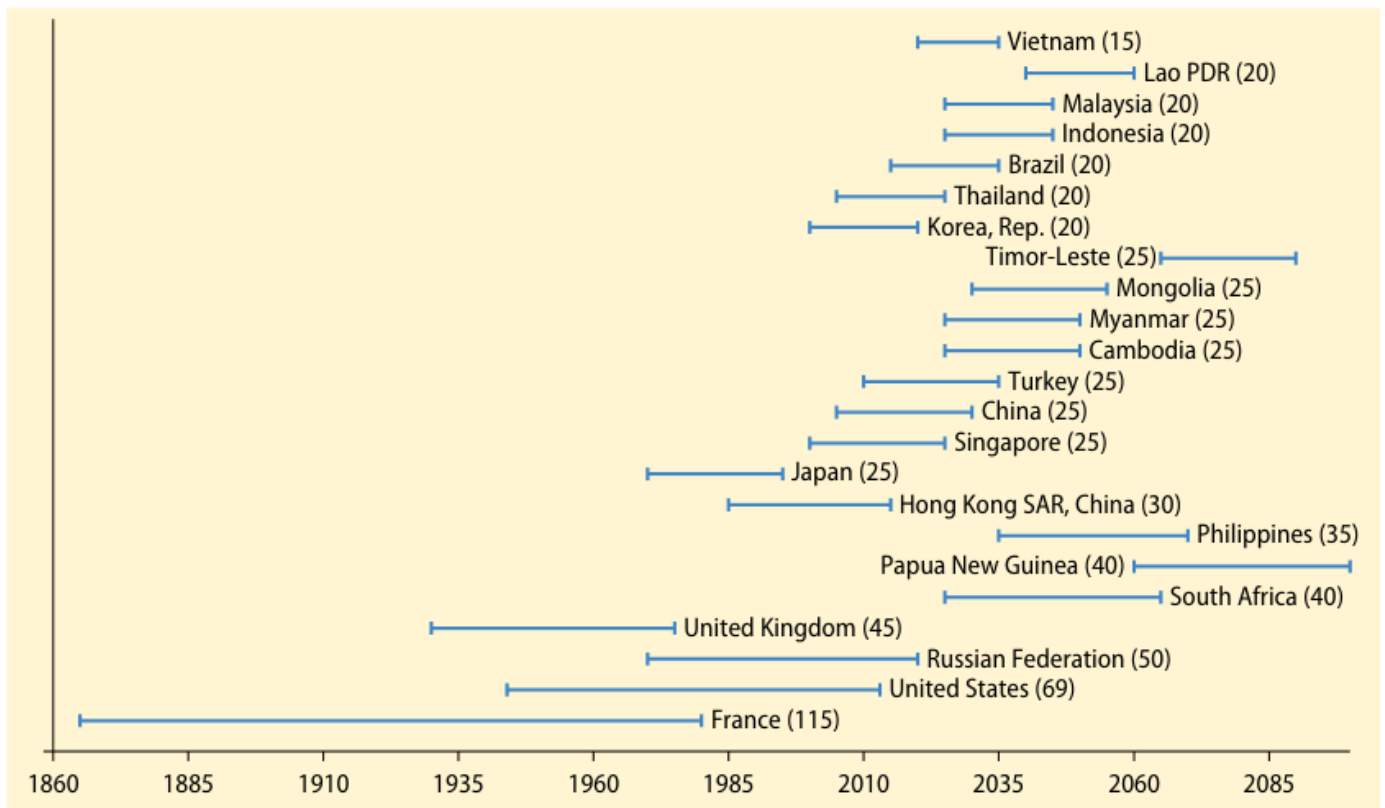


Exhibit. **Figure O.2 east Asian and Pacific economies are aging more rapidly than economies elsewhere** Years to move from 7 to 14 percent population share 65 years and older and the start and end years of transition Vietnam (15) Lao PDR (20). Source: S0029.

Getting old before getting rich: the developing-economy challenge

The previous section showed East Asia and Pacific ageing at income levels far below Europe's. The developing-economy challenge is broader than one region: it is what happens when the demographic transition arrives before the institutions that cushion it — comprehensive pensions, universal health coverage, formal labour markets — are built. The evidence shows the mechanisms through which old age becomes unequal, and they bite hardest where systems are thinnest.

What lies ahead: the dependency arithmetic

The EU's 2024 Ageing Report provides the cleanest statement of the trajectory every ageing economy faces. The EU population is projected to peak at 453 million in 2026 and fall to 432 million by 2070; the old-age dependency ratio rises from about 29% in 2010 to 36% in 2022 and 59% in 2070 — the EU goes from nearly thirty people aged 20-64 for every ten people over 65 in 2022 to fewer than twenty by 2045. Fertility stays below the 2.1 replacement rate everywhere (1.5 in 2022, projected 1.6 by 2070); male life expectancy at birth rises 7.7 years to 86.1 and female 6.4 years to 90.4, with remaining life expectancy at 65 up about five years for both. Even with participation rising — 3 percentage points for the working-age population, 10 points for 55-64-year-olds — the EU labour force is projected to shrink 12% (25 million people) between 2022 and 2070, and by more than a quarter in several Member States. Developing economies face the same arithmetic on compressed timelines and with far less fiscal room.²

Wage inequality becomes pension inequality

Whether working-life inequality turns into old-age poverty depends on how progressive the pension system is. The OECD's progressivity index — where a universal basic scheme scores 100% and a scheme mirroring wage inequality scores 0% — averages just 37% across the OECD; Ireland, New Zealand, and the United Kingdom score 100%, while Sweden, Turkey, Latvia, Hungary, and Finland score very low, tying pensions closely to wages. On average, about two-thirds of any increase in wage inequality is transmitted into pension inequality — but in Turkey, Latvia, Hungary, Finland, Italy, Poland, Spain, Portugal, Sweden, the Netherlands, Austria, France, and Germany, more than 85% passes through. Voluntary private pensions amplify the effect: in the seven countries with coverage above

40% (Belgium, Canada, Germany, Ireland, New Zealand, the UK, and the US), accounting for voluntary pensions raises the transmission of wage inequality from 0.38 to 0.46 — a 20% increase in pass-through. In developing economies, where wage inequality is typically wider and mandatory coverage thinner, the transmission runs close to its maximum.³

Education divides lifetimes, then deaths

Education is the deepest axis of inequality in the 100-year life. Across 23 OECD countries around 2011, the gap in life expectancy at age 25 between highly and low-educated men averaged 7.5 years (4.6 for women); at 65, the gaps were 3.5 and 2.4 years — and as a share of remaining life, the inequality grows with age (18.5% vs 13.4% for men). The gaps are widest in Chile, Hungary, Latvia, and Poland, and comparatively narrow in Canada and Italy. Smoking explains much of the mechanism: in 2014, 45% of low-educated men aged 18–64 smoked versus 23% of the highly educated (women 32% versus 17%), and smoking accounts for up to half of observed mortality inequalities in some countries. Education's effects compound through earnings too: the lifetime-earnings ratio of highly to low-educated women born in 1970–74 is 3.10, ranging from 1.7 in Denmark to 4.3 in Greece. The gender gap in lifetime earnings has shrunk by about a third across cohorts — from 45% to about 30% among the highly educated — but remains wide.³

Incomplete careers, incomplete pensions

In developing economies, the textbook full career from age 20 is the exception. The OECD modelled a late start (labour-market entry at 25) plus ten years of unemployment between 35 and 45: without any offsetting mechanisms, pension benefits would fall about 35%; under legislated rules the average projected fall is 22%, meaning redistributive devices offset about 37% of the shortfall. But the average hides the countries where the fall is steepest — 30% or more in Hungary, Greece, Chile, Turkey, Mexico, Poland, Israel, Iceland, and Italy. In six countries, workers with gaps must work longer to qualify for a full pension: two extra years in Germany and Spain, four in France, five in Greece, Luxembourg, and Slovenia. Australia's Age Pension makes up over two-thirds of the fall, leaving the late-comer with 92% of a full-career pension; in Israel, with its strong defined-contribution component and no means-tested element, the figure is 70%. Where careers are informal and intermittent — the norm in developing economies — the 35% un-offset fall is the more relevant number.³

The assets the poor don't have

Housing is the largest asset most households own, and home equity release can supplement retirement income — in Australia, individuals can access around 20–25% of their home equity at age 65 through reverse mortgages, and the government runs its own Home Equity Access Scheme; in Canada, reverse mortgages allow borrowing up to 55% of property value. But this lever presupposes homeownership with clear title and functioning mortgage markets — conditions absent for much of the developing world's elderly, who rent informally or hold untitled land. Labour-market risk compounds the problem: the OECD's stochastic modelling finds around 40% of individuals in any cohort suffer unemployment spells, and 55% experience real-wage profiles that fall in the second half of the career.⁶

The challenge stated plainly

Getting old before getting rich means facing the dependency arithmetic — 59% old-age dependency ratios, shrinking labour forces, five more years of life at 65 — with systems that transmit wage inequality into old-age poverty at rates above 85%, education gradients that carve years off the lives of the poor, and careers too broken to earn full pensions. The policy agenda for developing economies is therefore not a copy of Europe's: it is building progressivity into pension systems before the age wave arrives, extending coverage to informal and intermittent workers, and treating education and health inequality as pension policy — because in the 100-year life, the inequalities of age 25 decide the poverty of age 75.

Long-run population projections to 2050 and beyond, and their uncertainty

Population projections to 2050 and beyond shape every pension, health, and labour-market decision in this paper — but they are not forecasts. They are conditional scenarios, and their conditions are worth reading as carefully as

their conclusions. The EU's 2024 Ageing Report, built on Eurostat's EUROPOP2023 demographic projections, shows how the machinery works and exactly where the uncertainty lives.²

How the projections are built

The labour-force projections rest on the cohort simulation model (CSM), developed by the European Commission and used since the 2006 Ageing Report. Its central device is a "synthetic" cohort: because no individual longitudinal data on labour-market transitions exist, participation data by single age are used to estimate whether people entered or exited the labour market between consecutive ages, computed as a ten-year average (2013–2022 for the 2024 report) so that no single year's cyclical fluctuations or statistical errors unduly influence the rates. For Member States with legislated pension reforms, average exit rates for ages 51–74 are adjusted for the reform impact; otherwise entry and exit rates are held constant — a no-policy-change assumption. The base year is 2022 and the projection period runs 2023–2070.²

Two technical choices reveal the model's sensitivities. First, a corrective mechanism for young cohorts (15–24): participation rates at ages 15–19 are frozen at the ten-year average so that rising education enrolment does not feed mechanically into projected declines in prime-age participation. Second, the split between full-time and part-time work and weekly hours worked are frozen at 2022 averages per age group for the entire projection period — total hours worked change only through compositional effects, mostly the rising female participation given women's higher incidence of part-time work. The headline labour-force identity is simple: employment equals population times participation rate times one minus the unemployment rate ($E = P \times PR \times (1-UR)$). On these mechanics, the EU's 20–64 participation rate is projected to rise from around 79% in 2022 to 83% in 2070, driven by older workers and women — yet total hours worked still fall 9%.²

Where the uncertainty lives

The demographic side carries its own named assumptions. EUROPOP2023 models emigration as country-specific probabilities converging linearly between 2027 and 2100 toward the EU average of 2013–2022. The Ukrainian displacement exposed how fragile such assumptions are: the projections assume 33% of refugees arriving under temporary protection in 2022–2023 remain permanently and 67% return over ten years, with returns decreasing linearly — but surveys disagree sharply, with 81% of refugees in a 43-country UNHCR survey saying they wished to return eventually, while a German survey found 37% wanting to stay permanently, 34% staying at least until the war's end, 27% undecided, and 2% planning to leave within a year, and a Finnish survey showing 27% would never return, 33% would return after the war, and 39% uncertain.²

Policy itself is a source of uncertainty the model handles by rule, not prediction. Pension reforms enter only as legislated adjustments to exit rates, through a typology of measures: increases in statutory or early retirement ages, convergence of lower female ages to men's, linking the statutory age to life expectancy, and changes in financial incentives for early or delayed retirement. The countries with the most significant new reforms since the 2021 report are France, Slovakia, and Sweden. But politics cuts both ways: Czechia's 2017 reform capped the statutory retirement age at 65, reversing the 2011 reform that had linked increases to life expectancy; Slovakia is raising the age to 65 (men by 2029, women by 2037) with a post-2037 life-expectancy link that has no clear legislated rule and is therefore excluded from the projections. A reform that exists only as intention does not exist in the model.²

What the uncertainty means

Three lessons follow. First, the projections are most reliable where behaviour is most inertial: the cohort mechanism captures the genuine, slow-moving replacement of older women with low participation by younger women with much stronger labour-force attachment — while also registering the small decline in men's participation across recent generations in several countries. Second, they are least reliable where shocks concentrate: migration, where a single war moved a year's inflow from an assumed 0.2% of the EU population to 1%, and politics, where legislated links to life expectancy can be reversed. Third, the no-policy-change convention means the projections are a baseline against which to measure action, not a destiny to accept: they assume no further reforms beyond those already legislated, so every additional reform — or reversal — moves the outcome.²

The 100-year life makes this uncertainty structural rather than incidental. When remaining life expectancy at 65 is rising by about five years and the oldest-old are the fastest-growing group, small errors in mortality, migration, or

participation assumptions compound over a 45-year horizon into very different fiscal worlds. The honest use of long-run projections is therefore not to pick the central scenario and plan to it, but to identify which levers — retirement-age links to life expectancy, female participation, coverage of non-standard workers — move the outcome most, and to build systems that stay solvent across the range of plausible futures rather than betting on one.

The longevity economy: the economic power of the 50-plus

The 50-plus population's contribution to GDP, jobs, and labor income

The conventional framing of population ageing as a fiscal burden misses the scale of what the 50-plus cohort already contributes. Across every industry and sector, the activities of people aged 50 and older generate widespread economic impacts adding trillions of dollars worldwide, and their weight is still growing. This is the “longevity economy”: not a cost center to be managed, but an engine of market growth whose expansion is being driven by rapid ageing in middle-income economies as well as continued ageing in high-income ones.⁹

A share of workers already past one in four

The 50-plus population made up more than one in four workers (27%) in 2020 across 101 economies with available data — rising to one in three (34%) in high-income economies, versus 24% in middle-income and 17% in low-income economies. More than 60% of adults aged 55–64 participate in the labor force on average (compared with 78% of 25–54-year-olds), and almost one in four (23%) of those 65 and older remain active. Participation at 65-plus rises dramatically in lower-income nations, where household incomes are more stretched and safety nets less reliable: 41% in low-income economies versus 13% in high-income ones, with 55–64 participation running 65% in low-income economies against 63% in high-income ones.⁹

Older workers spur multigenerational workforces, which can lift productivity by reducing worker turnover and bolstering overall work and management experience — making age-inclusive strategies a direct competitive interest for employers, not merely a social-policy concern. The policy corollary is that avoiding discrimination and designing strategies that use older workers' expertise become more crucial as societies age and face rising dependency ratios.⁹

Labor income: the median economy's 50-plus share is heading past one-third

The 50-plus population's footprint in labor income is both large and rising. For the median economy, the 50-plus contribution to labor income stood at \$66 billion — 30.5% of the base — in 2020, and is projected to reach \$162 billion, or 35.1% of the base, by 2050, a gain of 4.2 percentage points. Country patterns vary with demographic position. In China, the 50-plus labor-income impact rises from \$2,439 billion (18.8%) in 2020 to \$6,545 billion (22.9%) by 2050, up 4.1 percentage points; in India, from \$1,501 billion (29.8%) to \$8,093 billion (35.9%), up 6.1 points. In already-old economies the share is higher but still climbing: Japan moves from \$1,025 billion (31.0%) to \$1,290 billion (33.9%); Germany from \$900 billion (30.3%) to \$1,313 billion (32.5%); France from \$571 billion (27.9%) to \$889 billion (30.2%); Italy — the most aged major economy — from \$557 billion (34.7%) to \$922 billion (43.9%), a 9.2-point gain.⁹

These gains mirror the demographic shift itself. In the median economy, the 50-plus share of the population rises from 34.3% in 2020 to 43.1% in 2050, up 9.1 percentage points. China jumps from 32.8% to 47.2% (up 14.4 points), India from 19.4% to 32.8% (up 13.4 points), Japan from 47.4% to 55.3%, and Italy from 45.7% to 54.2%. Notably, the labor-income share grows more slowly than the population share in the largest economies — a reminder that converting demographic weight into economic weight depends on participation rates, not just headcount.⁹

The consumption engine underneath

The labor-market contribution is reinforced by spending power: the 50-plus population accounts for the majority of spending in half of all product and service categories. Essentials — housing and utilities, food and beverages, and transport — were the three largest categories of 50-plus spending in 2020 and are expected to remain the top categories through 2050. But the composition is shifting: the average household's share of spending on recreation and culture is projected to rise 2.4 percentage points, on housing and utilities 1.3 points, and on furnishings and house-

hold maintenance 1.5 points, while the food share falls by 4 percentage points on average — and often much more in rapidly developing economies. Growth in non-essentials such as recreation, culture, restaurants, and hotels reflects rising incomes accompanying ageing; alcohol, by contrast, is projected to grow sluggishly as habits shift toward health and life expectancy.⁹

Sector-level impacts follow the money. In 2020, the industries with the largest 50-plus impact relative to their size were real estate (58% of the sector), financial and insurance activities (49%), and telecommunications (48%) — unsurprising given the importance of accumulating and managing assets as people move toward retirement. By 2050, real estate is projected to keep the largest relative contribution, with several more sectors seeing impacts equivalent to over 50% of their GDP: agriculture, manufacturing (pharmaceuticals and chemicals), electricity and gas, accommodation and food services, telecommunications, and financial and service activities — growth driven disproportionately by the rapid expansion and ageing of middle-income economies.⁹

A global, not a national, contribution

The longevity economy crosses borders. Globally, close to 30% of the 50-plus population's total GDP impact occurs outside their home economy — through direct overseas spending or indirectly via supply-chain components and services — and for the median economy, close to half (44%) of the total impact typically originates from abroad, with differences driven mainly by economic structure and trade reliance rather than by age structure.⁹

The aggregate picture is therefore one of expansion, not contraction: a workforce more than one-quarter over-50, a rising share of labor income, consumer spending that already dominates half of all categories, and impacts that ripple across borders. The question for the decades ahead is not whether older people can afford to age, but whether economies capture the full value of what they already produce.

50-plus consumer spending across product and service categories

The 50-plus population accounts for half of global consumer spending — a consumer bloc whose weight varies sharply by product, by income level, and by how each economy converts household spending into GDP.⁹

Where the money goes

In the five largest consumer product categories, people aged 50-plus are responsible for roughly half or more of global spending: housing and utilities (51%, or \$7.8 trillion of \$15.1 trillion), food and non-alcoholic beverages (49%, \$6.1 trillion of \$12.4 trillion), miscellaneous goods and services such as professional and financial services (52%, \$3.7 trillion of \$7.0 trillion), transport (49%, \$3.4 trillion of \$6.9 trillion), and health (60%, \$3.6 trillion of \$6.0 trillion). Beyond the top five, 50-plus consumers account for the majority of spending in recreation and culture (53%, \$2.3 trillion of \$4.3 trillion) and furnishings and household maintenance (50%, \$1.7 trillion of \$3.5 trillion). Smaller categories still show a heavy 50-plus hand: restaurants and hotels (\$2.4 trillion of \$5.4 trillion), clothing and footwear (\$1.1 trillion of \$2.6 trillion), communications (\$1.0 trillion of \$2.0 trillion), and alcoholic beverages and recreational drugs (\$1.0 trillion of \$1.9 trillion). Education is the outlier — the only category where the 50-plus share is lowest, at 37% (\$0.6 trillion of \$1.7 trillion). All values are PPP-adjusted international dollars, inflation-adjusted post-2020.⁹

Health deserves a closer look. The 50-plus population's 60% share of consumer health spending is the highest of any category — yet more than half of that 50-plus health spending (\$1.8 trillion) is attributed to the United States alone. This does not imply a lack of health care services elsewhere; it reflects that health care in other economies is more often provided through public mechanisms, so it shows up less in household spending. The same logic applies to sector-level impacts: there is no one-for-one relationship between consumer spending and GDP impact, especially in health, where much of the sector's GDP is driven by public and non-profit expenditure rather than household outlays.⁹

The income gradient: food vs. everything else

Spending patterns diverge dramatically across income levels, and the single biggest fault line is food. Food and beverages make up 51% of the typical 50-plus household budget in low-income economies — against 16% in high-income economies. The difference is not that older people in wealthy economies eat less; it is that as incomes rise,

they spend much more on other essentials such as health and transport, and on non-essentials like recreation, restaurants, and hotels.⁹

Housing is the striking constant. In high-income economies it is by far the largest 50-plus expense, at a quarter of total expenditure; at lower income levels it is also the largest share aside from food, at just under 20%. The share of spending allocated to housing stays high across all economies studied, indicating that as incomes rise, 50-plus expenditure on housing rises with them. Health expenditure generally rises as incomes grow and people live longer, but levels off for the highest income group — households in less wealthy economies are more likely to pay for health care directly (out-of-pocket or through insurance), while wealthy-economy households more often receive it as a government-provided service. Spending on miscellaneous goods and services also increases substantially with income; on average almost half of that category is financial services and insurance purchases, the other half personal items and professional services.⁹

How the numbers are built — and their limits

These estimates rest on household-level data, using spending by households headed by someone aged 50 or older as the best available approximation for spending by the 50-plus population. The approximation is imperfect: households can straddle the age cutoff (a 50-plus household head with younger members, or vice versa), and no attempt is made to allocate expenditures at the individual level. For economies without consumer surveys or age-and-product consumption data, 50-plus spending patterns were estimated econometrically from income statistics, tax data, age distribution, urbanization, historical trends, and patterns in similar economies.⁹

The forecasts draw on Economist Intelligence Unit country and product forecasts, the previous U.S. Longevity Economy long-range forecasts, projected demographic shifts, and projected changes in 50-plus consumption propensities. Population projections use the U.N. World Population Prospects 2019 release (medium variant) and do not account for COVID-19; the newer 2022 projections — released after the analysis — imply slightly slower total population growth but slightly faster ageing, with a 50-plus population share about 0.7 percentage points higher in 2050 at the global level. A final caveat: the link between 50-plus spending and domestic GDP impact is not uniform — it ranges from the U.S. at the top to Luxembourg at the bottom, driven by each economy's structure (how much consumption drives GDP versus government spending, investment, and exports) and overall economic diversification.⁹

The picture that emerges is of a consumer bloc that dominates the essentials everywhere — housing above all — while its discretionary spending is a luxury good in the economic sense: it rises with income, and it is in the rising economies of the developing world that its shape is changing fastest.

Industry impacts: where older consumers create jobs and growth

Section 8 established the aggregate scale of the longevity economy; this section asks where that value lands. The answer varies sharply by industry — and the industries that benefit most are not always the ones older people spend on directly, because input-output linkages propagate consumer spending through supply chains.⁹

Jobs: the median economy's 50-plus share heads toward two in five

The employment footprint of 50-plus spending is large and growing. For the median economy, 50-plus activities supported 2.9 million jobs in 2020 — 32.6% of the base — and are projected to support 4.1 million jobs, or 37.1% of the base, by 2050, a gain of 4.8 percentage points (with 2.0 points of that gain arriving by 2030). Country trajectories reflect both demographic position and labor intensity. In India, 50-plus spending supported 161 million jobs (35.6%) in 2020, rising to 239 million (42.0%) by 2050, up 6.4 points; Indonesia moves from 41 million (32.0%) to 62 million (36.1%, +4.1pp); Brazil from 37 million (38.2%) to 50 million (46.9%, +8.7pp). In China, the jobs total peaks mid-period — 161 million (21.5%) in 2020, 179 million in 2030, then 164 million (26.7%) by 2050 as the population itself shrinks while the 50-plus share keeps rising (+5.2pp). In already-old economies the share is high and still climbing: Italy from 8.5 million (38.0%) to 8.8 million (47.6%, +9.6pp); Germany from 13.0 million (31.2%) to 12.9 million (33.4%, +2.2pp); France from 8.0 million (28.3%) to 8.4 million (30.6%, +2.3pp).⁹

Technology sectors punch above their weight

Technology-heavy sectors saw strong 50-plus impacts in 2020: telecommunications at \$845 billion (48% of the sector's GDP), computer and electronics manufacturing at \$461 billion, and IT and other information services at \$665 billion (25% each). The dollar values are comparable across these tech sectors even though their GDP shares differ, because total GDP in the latter two is substantially elevated by high levels of private business investment. Note the aggregation caveat: when summed globally, some sectors appear to hold steady between 2020 and 2050 — but every individual economy actually sees rising impacts; the flat global picture is an artifact of aggregation.⁹

Health and education: where spending and impact diverge

The 50-plus impact in the health sector was 34% of its GDP in 2020 — lower relative to GDP than the spending shares would suggest, because much of the sector's GDP is driven by public and non-profit expenditure rather than household spending. Education tells a different story: 50-plus impacts are projected to rise from 15% of the sector's GDP in 2020 to 21% in 2050 — not a systematic transformation, but a recovery from the stark downturn in 50-plus education spending during the worst of the COVID-19 pandemic.⁹

The income-level split in sector impacts

Industry impacts follow each economy's underlying structure. In low-income economies, where populations are relatively young, the 50-plus population paradoxically has much greater impacts than in high-income economies on professional services, education, and health — because those sectors are far more reliant on consumer spending to power production at lower income levels. Conversely, in high-income and upper-middle-income economies, 50-plus impacts are stronger for real estate and agriculture, driven by the cohort's comparatively large share of total spending on housing and food.⁹

The care economy: demand is certain, provision is not

One sector whose 50-plus impact is almost guaranteed to grow is long-term care. Across OECD countries, lower socio-economic groups are both more likely to need care and less able to afford it: social protection for home care at moderate needs covers as little as 6% of costs in some U.S. states, versus about 45% in the Czech Republic and Israel and nearly 100% in Sweden, Iceland, and the Netherlands. Informal care — the default everywhere — is not costless: carers have 20% more mental health problems than other people, are more likely to stop working or reduce hours, and between 55% and 70% of them are women. In emerging economies, the traditional family support system is eroding as birth rates fall and younger generations become more mobile — precisely as larger cohorts advance into older ages more exposed to non-communicable-disease disability, while formal long-term-care systems are still nascent. That combination — rising demand, eroding informal supply, thin formal provision — makes care one of the clearest growth frontiers of the longevity economy, and one where the policy response (public support for care homes, community health services) will shape whether the growth is captured formally or imposed informally on women.³

The industry picture, then, is not a single story of “health and housing.” It is a mosaic: real estate and finance riding asset accumulation, technology sectors capturing disproportionate value, care emerging as the growth frontier, and each economy's sector mix determined less by how old it is than by how its economy is built.

Cross-border effects: foreign-driven impacts of older populations

The longevity economy does not respect national borders. Spending by 50-plus consumers abroad routinely generates significant cross-border impacts — through direct overseas spending on products and services, and indirectly via supply-chain components and services — even in economies that are very young. But how those cross-border effects combine with domestic structure and demographics produces strikingly different outcomes across economies — and over time, economies that start in the same place can end up in very different ones.⁹

Same starting point, different destinations

Consider three economies — Costa Rica, the DRC, and Laos. In 2020, the 50-plus population generated economic impacts of about 35% of GDP in each, very close to the median value across the 76 economies studied. That similarity masks differences in demographic and economic composition that are projected to drive strong divergence by 2050.⁹

In the DRC, the 50-plus impact on the economy is projected to expand very little by 2050. The reason is demographic: its 50-plus population share will grow by only three percentage points over the next 30 years — with only Denmark projected to see a smaller demographic shift across all 76 economies. Contrast Costa Rica and Laos, where the 50-plus share expands by 18 and 14 percentage points respectively — the fifth and 17th highest rates of increase in the sample. Both will see a corresponding increase in the 50-plus proportion of consumer spending — yet Costa Rica's lack of increased impact on GDP stands out starkly.⁹

Why spending weight and GDP weight decouple

The major reason for the Costa Rica–Laos divergence is that consumer spending is becoming significantly more important to the overall economy in Laos while becoming much less important in Costa Rica. Although both economies shared a similar consumption-to-GDP ratio in 2020 (62% for Laos, 64% for Costa Rica), by 2050 Laos's ratio is projected to rise to 76% while Costa Rica's falls to 51%. The lesson: the 50-plus population's economic impact is a product of three things — how old the population is, how much of GDP flows through consumption, and how open and structured the economy is — and demographics alone cannot predict it.⁹

Shifts in an economy's industry mix and changes in its trade relationships can also contribute to such divergences. These are harder to disentangle and predict, but in 2020 Costa Rica and Laos at least shared similar levels of foreign-driven impacts.⁹

What determines the foreign share

The variation in how 50-plus consumer spending translates into domestic-driven GDP impact is wide — ranging from the United States at the top to Luxembourg at the bottom — and is driven largely by each economy's underlying structure: the degree to which consumption spending drives GDP (as opposed to government spending, investment, and exports), and overall economic diversification. Differences in the size of foreign-driven impact are predominantly driven by economic structure and reliance on international trade, and differ little between younger and older economies as such.⁹

The implication

The cross-border dimension reframes the longevity economy as a trade story as much as a demographic one. Young economies with open, consumption-driven structures can capture outsized foreign-driven impacts from older consumers abroad — while economies where consumption's share of GDP is shrinking will see their ageing produce less domestic economic weight than their demographics suggest. For policymakers, this means the fiscal arithmetic of ageing cannot be read off a population pyramid: what matters is the interaction of demography with consumption structure, industry mix, and trade integration — all of which evolve over the projection horizon.⁹

The 50-plus economy in 2050: 3.2 billion people, more than double the impact

Pulling together the projections traced through the preceding sections, the 2050 longevity economy is not simply a scaled-up version of today's. It is larger, differently composed, and distributed through economies whose structures will determine how much of the demographic weight converts into economic weight.⁹

The demographic engine

The core driver is the expanding 50-plus cohort. In the median economy, the 50-plus share of the population rises from 34.3% in 2020 to 43.1% in 2050 — a 9.1 percentage-point gain. The largest movements are in the developing world: China's 50-plus share jumps from 32.8% to 47.2% (up 14.4 points), India's from 19.4% to 32.8% (up 13.4 points), while Japan moves from 47.4% to 55.3% and Italy from 45.7% to 54.2%. Because the growth is fastest

where incomes are rising most rapidly, the economic expansion outpaces the demographic one: for the median economy, 50-plus labor income grows from \$66 billion (30.5% of the base) to \$162 billion (35.1%), a 4.2-point gain, and 50-plus-supported jobs from 2.9 million (32.6%) to 4.1 million (37.1%), up 4.8 points.⁹

Spending shifts toward discretionary

The composition of 50-plus spending changes as the cohort grows and incomes rise. Essentials — housing and utilities, food and beverages, transport — remain the top categories through 2050, but the average household's spending share shifts toward recreation and culture (+2.4 percentage points), housing and utilities (+1.3 points), and furnishings and household maintenance (+1.5 points), while the food share falls by 4 points on average — often much more in rapidly developing economies. Growth concentrates in non-essentials such as recreation, culture, restaurants, and hotels, reflecting the rising incomes that accompany ageing; alcohol consumption, by contrast, grows sluggishly as habits shift toward health.⁹

Industries follow the money

By 2050, real estate is projected to keep the largest relative 50-plus contribution, with several more sectors seeing impacts equivalent to over half of their GDP: agriculture, manufacturing (pharmaceuticals and chemicals), electricity and gas, accommodation and food services, telecommunications, and financial and service activities — growth driven disproportionately by the rapid expansion and ageing of middle-income economies. Technology sectors already punch above their weight (telecommunications at \$845 billion, 48% of its GDP in 2020), and while some sectors' global impacts look flat between 2020 and 2050, every individual economy sees rising impacts — the flatness is an aggregation artifact.⁹

The conversion mechanism: how much spending becomes GDP

Whether this demographic and spending weight turns into GDP impact depends on economic structure. Total consumer spending as a share of GDP ranges massively across the 76 economies — from 24% in Brunei to 86% in Zimbabwe. Older, wealthier economies tend to be more diversified and rely less on household consumption: high-income European economies with strong social support systems, like Denmark, Norway, Sweden, and the Netherlands, lean more on public spending. The United States stands out with consumer spending near 70% of GDP — driven by its privatized healthcare system and large real estate markets — while younger, lower-income economies in Sub-Saharan Africa cluster where private consumption dominates.⁹

There are important exceptions: Saudi Arabia and Brunei are oil-dominated; Ireland and Luxembourg attract multinationals whose income dwarfs domestic activity; Singapore draws heavy foreign investment alongside higher public spending; China's economy is driven by capital investment and public-sector involvement, with exports and imports roughly balancing so the net trade contribution is minimal.⁹

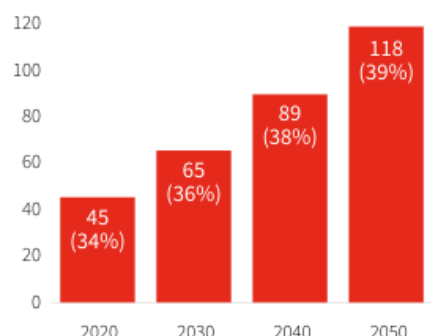
This is the same mechanism behind the Costa Rica–Laos divergence in Section 11: consumption's share of GDP can rise or fall over the projection horizon, amplifying or muting the demographic wave. The 2050 longevity economy will therefore be concentrated where three forces coincide — fast-ageing populations, rising incomes, and consumption-driven economic structures. Where they don't coincide, the same demographics produce far less economic weight.

The bottom line for 2050

The 50-plus economy of 2050 is defined by four facts: a population share approaching half in the median economy and exceeding it in the oldest ones; labor income and jobs shares climbing several percentage points everywhere; spending composition tilting toward discretionary categories; and industry impacts concentrating in housing, health-adjacent manufacturing, utilities, hospitality, telecom, and finance. But none of this converts automatically into GDP: the economic structure of each country — how diversified it is, how consumption-driven, how trade-integrated — is the final gate between demographics and economic destiny.⁹

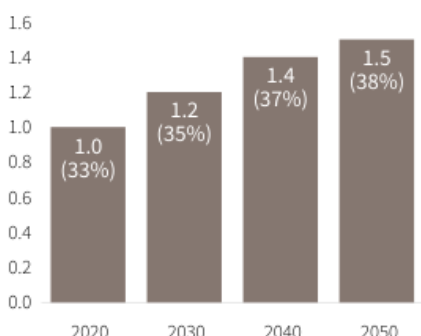
a) The impact of people age 50-plus on the **global economy** will more than double between 2020-2050

Impact on GDP (\$ trillion) and share of GDP (%)



b) **Jobs** supported²⁶ by people age 50-plus will increase by nearly 50% between 2020-2050

Impact on jobs (billion) and share of jobs (%)



c) The impact of people age 50-plus on **labor income** will more than double between 2020-2050

Impact on labor income (\$ trillion) and share of labor income (%)

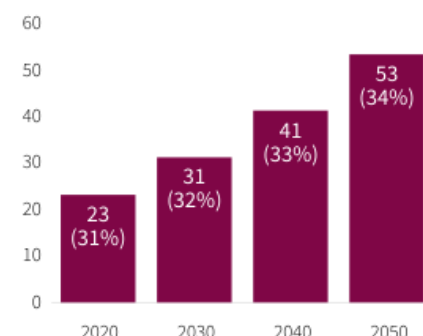


Exhibit. Figure 3. The 50-plus population has an outsized impact 25 on the economy, supporting about one third of global GDP, employment, and labor income in 2020. Source: S0030.

Six principles for a financially resilient longevity economy

The preceding sections documented the scale of the longevity economy; this section turns to what to do about it. The World Economic Forum's Longevity Economy Principles (2024), developed by a multistakeholder consortium across finance, health, academia, and government, offer six principles as a compass for societies preparing for the demographic transformation — a unifying language for a topic where countries' pension rules, retirement ages, and benefit designs differ widely but individuals everywhere want the same outcomes: meeting basic financial needs, enjoying a healthy life, accessing employment, and living with purpose.¹⁰

Principle 1: Ensure financial resilience across key life events

Nearly 40% of people globally face financial instability after unplanned career interruptions — career breaks, illness, or unexpected retirement — making public-private collaboration crucial to support individuals through these shocks. This connects directly to the paper's earlier findings: incomplete careers can cut pension entitlements by 35% or more without offsetting mechanisms, and ~40% of cohorts suffer unemployment spells over their working lives. Resilience means building systems that absorb shocks rather than transmitting them one-for-one into old-age poverty.¹⁰

Principle 2: Provide universal access to impartial financial education

Only 33% of the global population is deemed financially literate — a gap that contributes to wealth inequalities, which are in turn strongly correlated with life expectancy inequalities. Comprehensive, impartial financial education empowers individuals to make informed decisions across decades of saving and decumulation. (Sections 28–30 develop the education, literacy, and nudge architecture in detail.)¹⁰

Principle 3: Prioritize healthy ageing as foundational

Around one-fifth of life is expected to be lived with illness, and 80% of adults in developing countries are concerned about the cost of medical expenses. Equitable access to health services facilitates well-being for individuals and broader society alike — and, as Section 2 established, health is the binding constraint on extending working lives: the 24%-vs-7% health-exit gap between low- and high-educated workers shows that without healthy ageing, longer working lives are a privilege of the healthy. (Sections 47–50 expand the health agenda.)¹⁰

Principle 4: Evolve jobs and lifelong skill-building for a multigenerational workforce

Internationally, up to 25% of individuals aged 55 and older wish to work in old age but face barriers to finding opportunities. Demographic shifts and technological change require jobs and skill-building to adapt so people can ex-

tend their working years as desired. The evidence supports the urgency: average job tenure fell between 2012 and 2019 across most OECD countries — by 9.5% for young workers, 7.2% for prime-age, and 5.7% for mature workers — with the largest declines for low-educated workers in every age group. Policy levers include promoting phased retirement, better balancing work and care, permitting pension–work–income combinations, and restricting publicly funded early-retirement schemes that push healthy workers out of employment. (Sections 33–45 develop the employment agenda.)^{10 11}

Principle 5: Design systems and environments for social connection and purpose

Social connection is integral to healthy longevity: socially isolated older adults face higher risks of poor health and earlier death, and intentional design of systems and environments for connection can mitigate these impacts. This principle reframes ageing policy beyond money and medicine — purpose and belonging as infrastructure.¹⁰

Principle 6: Intentionally address longevity inequalities — gender, race, class

The benefits of longevity are not distributed equitably, and the disparities are widening as societies age. Across all OECD countries, women aged 65 and above receive 26% less retirement income than men — driven significantly by caregiving: 57% to 81% of informal, unpaid caregivers are women, eroding their lifetime earnings and career continuity. In the United States, approximately 1 in 6 Black Americans over 65 live in poverty versus 1 in 15 white Americans, and older Hispanic women living alone have the highest poverty rates at roughly 1 in 3. Life expectancy gaps are stark: overall U.S. life expectancy at birth is 76 years, but only 65 for Indigenous Americans — the same as the total U.S. population in 1950 — while the richest Americans live up to 30 years longer than the poorest, and in Denmark the richest men outlive the poorest by about nine years. These gaps have only increased as income inequality has grown.¹⁰

The economic stakes of inaction are large: improving gender parity could add as much as \$13 trillion to global GDP, and closing the living-wage gap could generate an additional \$4.56 trillion every year through increased productivity. Thirty percent of caregivers from low-income households provide daily care, perpetuating old-age poverty — workplaces can respond with caregiver-targeted benefits and leave, and as gig work grows, employment benefits need to reach workers regardless of job status. Without collaborative public-private action now, the authors warn, disadvantaged groups will be compounded for generations, safety nets will strain, and intergenerational inequity will fuel political polarization that blocks the very adjustments needed.¹⁰

A compass, not a blueprint

The principles are deliberately cross-industry: financial resilience, education, health, work, connection, and equity cut across pension systems, employers, and public policy alike. The remaining sections of this paper give them institutional flesh — pension architecture (14–17), coverage and decumulation (18–22), labor markets (33–40), skills (41–45), and health (47–50) — each translating one or more principles into concrete system design.¹⁰

Pensions and retirement income for longer lives

Architecture of national pension systems: public, asset-backed, and private

National pension systems are built from the same basic parts arranged in very different proportions. The standard architecture distinguishes public pensions — old-age and early pensions, whether basic/flat-rate, earnings-related, or non-contributory minimum pensions — from private pensions, which comprise occupational schemes, individual mandatory schemes, and individual voluntary schemes. Over the past two decades, asset-backed pensions — plans in which retirement savings are invested, earn a return, and accumulate to help finance retirement — have diversified the sources of retirement financing and strengthened multi-pillar systems, a key OECD recommendation.²⁶

The public pillar: pay-as-you-go and its variants

Public pensions remain the backbone in most countries, financed largely on a pay-as-you-go basis from current contributions and taxes. Their design varies in how redistributive they are: across East Asia and Pacific, target replacement rates are significantly higher for lower-income workers, with simulated net replacement rates for hypo-

theoretical low-income workers (half average wage) exceeding those for high-income workers (twice average wage) in most economies.⁷

Where contributory coverage is thin, social (non-contributory) pensions determine adequacy. EAP's social pensions fall into three groups: Japan, Malaysia, Mongolia, the Philippines, and Vietnam pay benefits of 6–24% of income per capita to small numbers of recipients; China, Hong Kong, Korea, and Thailand pay low benefits (5–9% of per capita) but cover 54–94% of the population aged 65+; only Kiribati and Timor-Leste combine relatively high benefits with high coverage — Timor-Leste's recipient ratio exceeds 100% because eligibility starts at age 60. In Japan, Malaysia, and Mongolia, mature contributory systems make social pensions merely supplementary; in Indonesia and the Philippines, large cash/in-kind transfer programs reach many poor elderly through high coresidence rates — yet a large coverage gap remains in these countries and Vietnam.⁷

Contribution rates in EAP split into two groups: Hong Kong, Indonesia, Korea, Lao PDR, the Philippines, and Thailand have modest total rates of 5.7–10.3%, while China, Japan, Malaysia, Mongolia, Singapore, and Vietnam charge 18–28%. The higher rates risk discouraging formalization — especially among smaller firms and the self-employed — a particular concern in rapidly ageing China and Vietnam, where extending coverage is most urgent. Retirement ages in EAP national schemes are relatively low relative to life expectancy at retirement: women face more than 25 years of life expectancy at the normal retirement age in most countries except Korea and the Philippines.⁷

The asset-backed pillar: scale, design, and participation

Both the assets earmarked for retirement and the share of the working-age population participating in asset-backed pensions have grown over the last two decades. The scale varies enormously: assets as a share of GDP range from near-zero in countries like Greece and Türkiye to around 200% in Denmark, Iceland, and Switzerland, with the Netherlands, Canada, the United States, Australia, and Sweden also near the top.⁶

Participation comes in three flavors. Mandatory or quasi-mandatory systems cover Latvia, Sweden, Bulgaria, the Netherlands, Croatia, Costa Rica, Chile, Mexico, Israel, Iceland, Switzerland, Australia, Norway, Denmark, Estonia, Romania, Colombia, and Peru. Auto-enrolment operates in New Zealand, Lithuania, the United Kingdom, the Slovak Republic, Türkiye, and Poland. Voluntary personal and occupational plans exist across dozens more countries, from Czechia and Japan to Brazil and Luxembourg.⁶

"Pension plans" in the OECD taxonomy are plans individuals access via an employer or financial institution in which they accumulate rights or assets — assets that belong to plan members and finance their own future retirement, held in pension funds, pension insurance contracts, or other savings vehicles. Alongside them sit public pension reserve funds (PPRFs): reserves established to support unfunded, pay-as-you-go public arrangements — a hybrid that backs the public pillar with funded assets.⁴

Good design, per OECD guidance, means plans that are coherent and inclusive, with contributions aligned to objectives and supported by appropriate incentives, good value for members, adequate investment options, protection from longevity risk, and effective communication and financial education.⁶

How entitlements are measured

Comparability rests on the OECD pension models, which calculate theoretical entitlements for workers entering the labor market at age 22 in 2022, incorporating the full impact of legislated measures. The indicator suite starts with the gross replacement rate in mandatory schemes (pensions relative to individual earnings), then mandatory-plus-voluntary where coverage is broad, then tax treatment, net replacement rates, and pension wealth — the lifetime discounted value of the benefit flow, accounting for retirement age, indexation rules, and life expectancy. Early, normal, and late retirement ages interact with penalties and bonuses: Denmark's bonus in the basic/targeted scheme is tied to life expectancy at first receipt; Finland allows partial early retirement of 25% or 50% of accrued rights from age 61; Greece and Latvia impose temporary early-retirement penalties of 10% and 50% respectively until normal retirement age; Luxembourg offers no bonus but a higher accrual rate once age plus contribution years exceed 100; and in defined-contribution schemes, benefits adjust to retirement age in an actuarially neutral way by construction.

The 2022 stress test

Asset-backed systems’ exposure to markets was laid bare in 2022, when bond and equity prices fell simultaneously: pension plans recorded negative nominal investment rates of return in most OECD countries, with the lowest in the Netherlands (–21.1%) and the United Kingdom (–18.5%), and negative real returns in every reporting country. Strong long-term gains cushioned the blow in most countries, and all reporting public pension reserve funds remained positive in real terms over the previous 15–20 years. Defined-benefit funding ratios are tracked across a dozen jurisdictions from 2012 to 2022, reflecting the ongoing shift of risk between sponsors and members.⁴

Returns are not guaranteed to track wages — a critical assumption in DC adequacy modeling. Simulations typically assume net DC returns exceed wage growth by 1.5 percentage points annually, but reality diverges: Hong Kong’s returns beat the assumption (with high volatility), while China’s DC returns — linked to short-term bank deposits — have barely exceeded inflation and lagged far behind wage growth since long before individual accounts were introduced, and Singapore’s nominal return has been essentially fixed since inception.⁷

The architecture, then, is a spectrum from pure PAYG public systems to heavily funded multi-pillar ones — with the dividing lines running through who bears market risk, how redistributive the public tier is, and whether the growing asset-backed pillar is designed well enough to deliver on its promise.

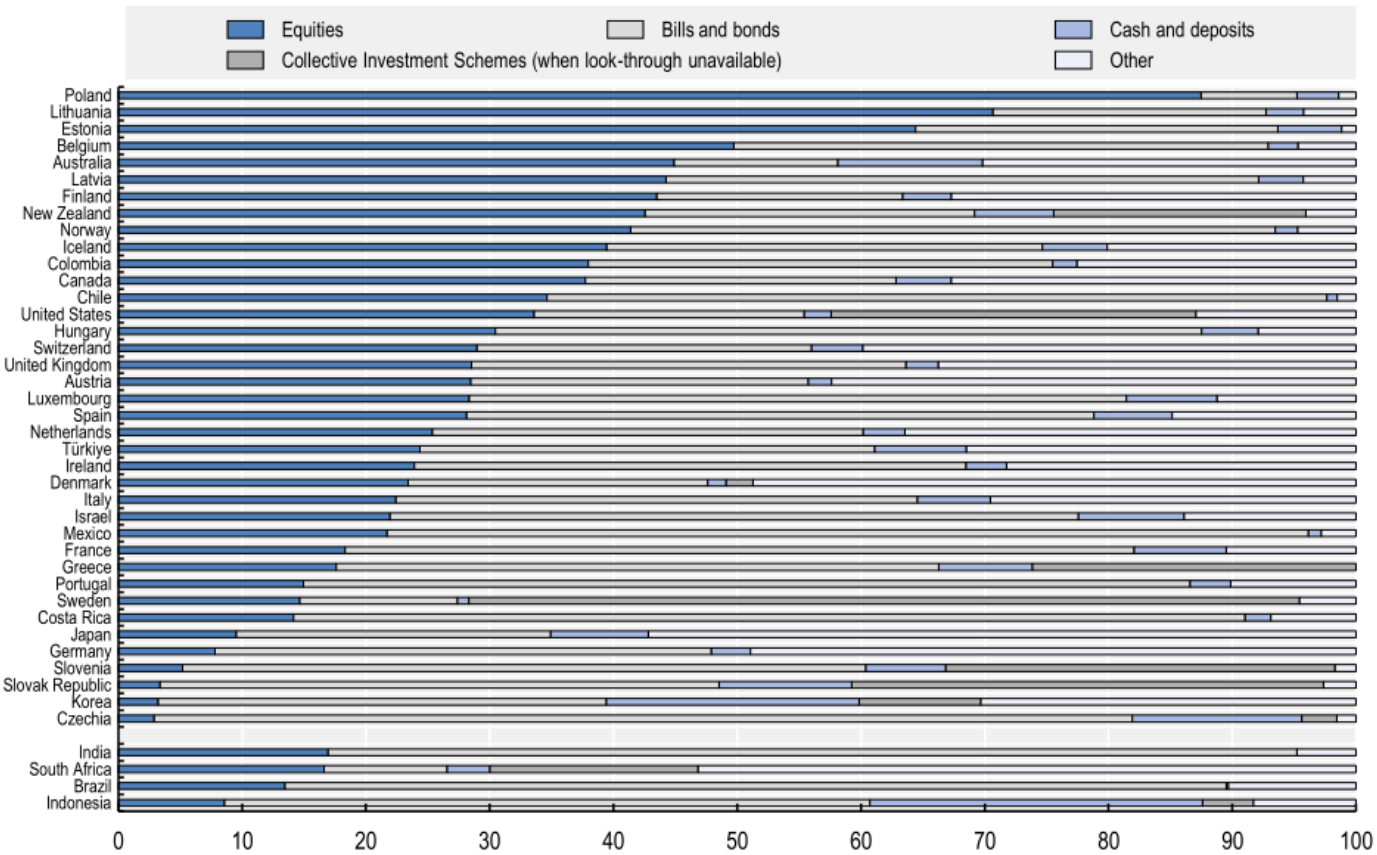


Exhibit. Figure 9.3. Allocation of assets in pension plans in selected asset classes and investment vehicles, at the end of 2022 or latest year available As a percentage of total investment Equities Bills and bonds Cash and deposits Collecti.
Source: S0002.

Public pension expenditure and long-run fiscal sustainability

Public pension expenditure as a share of GDP is the product of three forces: how many retirees there are relative to workers, how generous benefits are, and how many people are covered. Population ageing pushes the first term relentlessly upward — the EU’s old-age dependency ratio climbs from 29% (2010) to 59% (2070) while the labour force shrinks 12% — which is why the long-run sustainability question is not whether spending pressure rises, but which levers governments pull to contain it, and whether those levers hold politically.

The reform levers that move expenditure

In the EU's projection machinery, pension reforms enter expenditure arithmetic through their estimated impact on the labor-market exit rates of older workers aged 51–74 — a “best-reasoned judgment” built on the probabilistic cohort simulation model, country-specific evidence on how retirement behavior responds to pension parameters, and cross-country evidence. The typology of measures that moves the fiscal needle is now standard: increases in the statutory or early retirement age, convergence of lower female statutory ages to men's, linking the statutory age to life-expectancy gains, and changes in financial incentives for early or delayed retirement. These can be legislated as one-off measures or phased in progressively.²

The most significant new reforms since the 2021 Ageing Report come from France, Slovakia, and Sweden. Slovakia's statutory age rises to 65 — by 2029 for men, 2037 for women — though the post-2037 life-expectancy link was not legislated and is therefore excluded from projections. The framework also captures one-off national designs: early retirement within 12 months of the statutory age with a 0.4%-per-month benefit reduction, and early exit for hazardous work or special groups such as teachers, military, and police at ages 55–60.²

Politics cuts both ways

Reforms that cut expenditure can be reversed, and reversals flow straight back into the spending path. Czechia's 2017 reform capped the statutory retirement age at 65 — reversing the 2011 reform that had linked increases approximately to life-expectancy gains — so that by 2036 the retirement age is 65 for all. Spain's removal of the Index for Pension Revaluation and the Sustainability Factor is projected to substantially increase spending by 2050. The lesson for sustainability analysis: legislated savings are only as durable as the politics behind them, and the projection convention of “no further reforms” cuts both ways — it excludes future savings and future reversals alike.²

Where the pressure is most acute

The arithmetic is harshest where ageing is fastest and systems are still expanding. In East Asia and Pacific, China's pension spending is projected to rise by more than 3% of GDP between 2010 and 2030, with Korea approaching a 4.5% increase — in economies that are getting old before getting rich, so the fiscal room to absorb it is narrower. Japan's growing reliance on earmarked value-added taxes to finance pensions shows one adaptation: shifting the base from payroll contributions (which shrink with the workforce) to consumption taxes (which the 50-plus population keeps paying, as Sections 8–9 established).

The sustainability condition

Two structural facts bound the problem. First, weekly hours worked are frozen at 2022 levels per age group in the projections — total hours change only through composition effects (more women, more part-time). So expenditure ratios cannot be eased by assuming a future surge in hours; the adjustment must come from participation (later exit), generosity (indexation and valorisation rules, Section 22), or financing (broader tax bases). Second, the exit-rate adjustments that reforms buy are probabilistic judgments, not guarantees — they assume the relationship between pension parameters and retirement behavior holds as cohorts change.²

Long-run fiscal sustainability, then, is not a single number but a conditional statement: spending paths stay manageable if retirement ages keep rising with life expectancy, if female participation keeps converging, if indexation rules restrain generosity growth, and if the politics of reform survives contact with electorates that are themselves ageing. Every one of those conditions is contestable — which is why the expenditure projections are best read as baselines for measuring the action still required, not as forecasts of what will happen.

Recent pension reforms across OECD, EU, and East Asia

Between September 2021 and September 2023, OECD governments reformed pensions on nearly every margin at once — retirement ages, benefit formulas, indexation rules, first-tier generosity, and emergency inflation support. The pattern is not one of convergence on a single model, but of shared pressures met with different national toolkits.⁴

Pensioners' starting position

People aged 66–75 average 93% of the total population's disposable income across the OECD, dropping to 81% for those over 75. National positions diverge sharply: the 66+ income ratio falls below 75% in the Baltic states and Korea, while reaching 100% or more in Costa Rica, France, Israel, Italy, Luxembourg, and Mexico. Relative income poverty touches 12.5% of 66–75-year-olds and 16.6% of the 76+, against 11.4% for the whole population — above 25% for the 66+ in the Baltic states and Korea, below 6% in Czechia, Denmark, France, Iceland, Luxembourg, and Norway.⁴

The life-expectancy link spreads

A quarter of OECD countries now tie retirement ages to life expectancy — Denmark, Estonia, Finland, Greece, Italy, the Netherlands, and Portugal. The Slovak Republic reintroduced a one-to-one link; Sweden raised its retirement age and will connect it to two-thirds of life-expectancy gains, which also lifts payouts from its notional defined-contribution scheme. Denmark's link runs on five-year parliamentary approval cycles with a 15-year lead time; the statutory age has been approved to reach 69 in 2035. The Netherlands chose a different systemic path: converting its funded private pensions from defined benefit to defined contribution.⁴

Raising — and sometimes lowering — retirement ages

France lifted the minimum retirement age in its main mandatory scheme from 62 to 64 and began phasing out special pension schemes; Costa Rica's early-retirement tightening raises the normal age three years, to 65 for men and 63 for women; Czechia narrowed early-retirement eligibility from five to three years before the statutory age. Italy went the opposite direction, extending early-retirement options that were meant to expire — though take-up data show design matters: only 7,536 new pensions under the "women's option" were paid in the first half of 2023 versus 24,559 in all of 2022, and uptake before age 60 fell 63%. Türkiye is the extreme outlier: eliminating the statutory retirement age for workers who entered employment before it was legislated in 1999 brought the normal retirement age down to 47 for men and 46 for women in that group — though it will climb rapidly, reaching 65 for men who entered the labor market in 2022. Gender gaps are narrowing elsewhere: Switzerland will close its gap in normal retirement ages, Israel will shrink its from five years to two, and Slovakia now grants early retirement only with a pension worth at least 1.6 times the subsistence minimum (€375 in the first half of 2023, up from 1.2 times).⁴

Protecting the bottom, stretching the formula

Chile replaced its means-tested public pension with a quasi-universal scheme, significantly raising low pensions; Canada, Estonia, France, Italy, Lithuania, Sweden, and Türkiye also raised first-tier pensions substantially. Spain combined the removal of its automatic sustainability adjustments with higher contributions — particularly on high earners — and stronger protection for low-income pensioners and workers with irregular careers, including mothers. Earnings-related formulas are being computed over longer careers: Costa Rica moved from the last 20 years to the best 25, and Spain will move from 25 to 27 years starting in 2044. Only five countries still calculate on part of the career: Colombia (10 years), Slovenia (24), France (25), the United States (35), and Portugal (40).⁴

Inflation as a reform teacher

The 2021–23 inflation surge pitted pension finances against adequacy: protecting purchasing power means indexing to prices, but wages — and scheme revenues — lagged. Australia temporarily eased its targeted-pension means test (July 2022–June 2024); the Netherlands relaxed funding rules so funds could index ahead of the systemic reform; Hungary pulled its 13th-month pension forward to the full amount in 2022. Eight countries made ad hoc pensioner payments (Austria, Estonia, Germany, Poland, Portugal, Slovak Republic, Slovenia, United Kingdom), with Denmark and Greece targeting low-income pensioners; broader support — heating allowances, lower energy VAT — reached pensioners without being pension-specific.⁴

Three indexation lessons emerged. First, speed matters: Belgium's 2% threshold trigger adjusts quickly after price surges, though frequent adjustments complicate budgeting. Second, partial indexation above a threshold — Austria's and Italy's approach — can fully protect low-income pensioners while capping the fiscal cost, leaving open whether higher pensions later catch up. Third, pensioner-specific price indices are of doubtful value: a retiree bas-

ket would have changed French indexation by only -0.3% cumulatively over 1998–2015, and the U.S. elderly CPI outpaced the standard index by 0.4 points a year over 1983–2002 but showed essentially no gap over 2002–21. Because consumption baskets differ more across income levels than between age groups, a low-income basket would be the better-targeted reform.⁴

East Asia's different sequence

East Asia's reform frontier runs on a different track: national-scheme retirement ages remain low relative to life expectancy at retirement, contribution rates are already high (18–28% in China, Japan, Malaysia, Mongolia, Singapore, and Vietnam), and the urgent race is extending coverage to informal workers — the “win the race” agenda of Section 5. The OECD toolkit of life-expectancy linking, notional accounts, and auto-enrolment is increasingly the reference point, but the sequencing differs: build coverage first, calibrate generosity second.

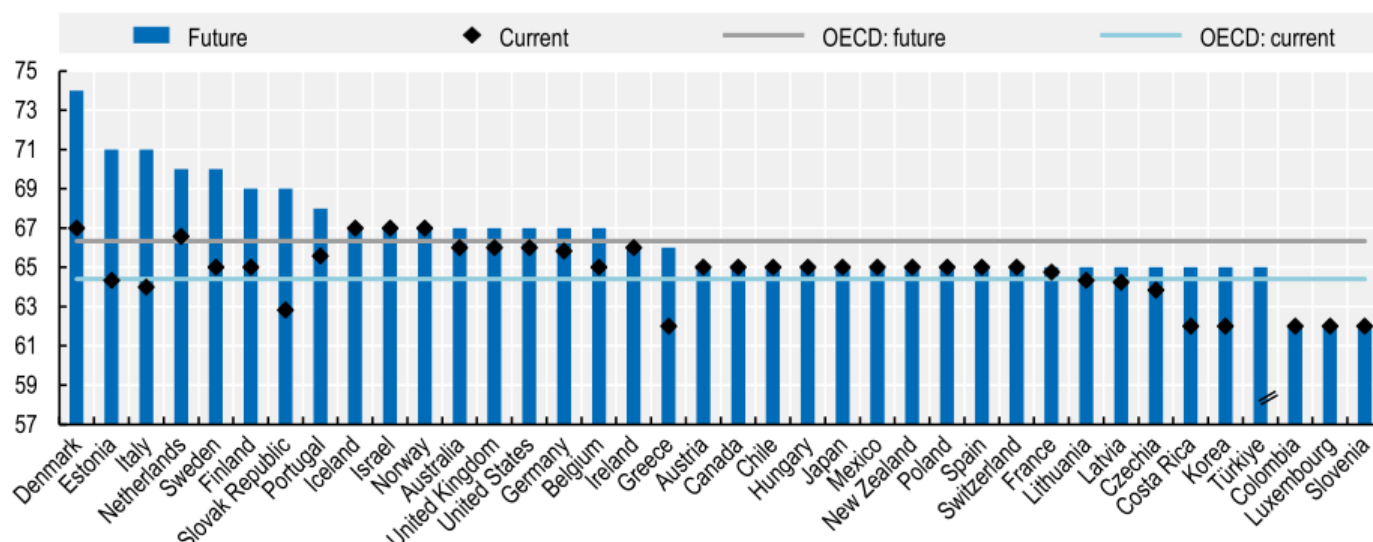


Exhibit. Normal retirement age for men entering the labour market at age 22 with a full career Future Current OECD: future OECD: current 75 73 71 69 67 65 63 61 59 57. Source: S0002.

Retirement ages: raising them, linking them to life expectancy, effective exit ages

Statutory retirement ages are policy; effective exit ages are behavior. The two have followed very different trajectories — and the gap between them, plus the unequal life expectancies behind it, is where the equity debate over raising retirement ages lives.

Exit ages fell, then rose — while life expectancy only rose

Combining long-term trends in life expectancy and labor-market exit age shows that health is currently not a key obstacle to higher participation at older ages. Life expectancy at age 65 rose steadily from 1975, gaining 5.3 years for both men and women by 2019 (before dropping half a year in 2020 from COVID-19). Yet people were leaving the labor market earlier than in 1975: across 25 OECD countries, men exited 3.4 years earlier and women 4.2 years earlier in 2000 than in 1975. The trend reversed in the first half of the 2000s — exit ages have since risen 2.3 years for men and 3.2 years for women — and since the mid-2000s the exit age has increased roughly at the same pace as remaining life expectancy at 65.⁴

The longevity gains feeding this dynamic are slowing. After faster growth between the mid-1990s and early 2010s (about 1.5 years per decade for men, 1.4 for women at age 65), the pace has fallen to 1.0 and 0.8 years per decade since around 2012, with COVID-19 magnifying the break. New UN projections of life expectancy at 65 for 2050–55 are stable on average versus pre-COVID projections — at least half a year higher for New Zealand and the Slovak Republic, at least half a year lower for Colombia, Costa Rica, Lithuania, Poland, and Türkiye — and cross-country differences have widened.⁴

Linking retirement ages to life expectancy

One in four OECD countries now links retirement ages to life expectancy, with Slovakia and Sweden the latest to strengthen their links (Section 16). The logic is fiscal sustainability on autopilot: as people live longer, the retirement age rises to hold the ratio of working years to retirement years roughly constant. But the autopilot has a distributional problem — because life expectancy itself is unequal.

Assuming a three-year difference in life expectancy at retirement, low earners are penalized by 12.8% in their total pension benefits relative to high earners across the OECD; high earners enjoy more than 10% greater relative pension wealth in every OECD country, from over 10% in Greece to nearly 17% in Latvia. The penalty bites hardest where expected retirement is shortest — the Czech Republic, Estonia, Latvia, and the Slovak Republic — and least where it is longest: France, Greece, Japan, and Korea. Income redistribution from those who die early to those who die late is the core insurance function of pension systems, but it is regressive and reduces the progressivity of the system.³⁴

What happens when the retirement age itself rises in line with life-expectancy gains? If the effective retirement age rose by three years between 2015 and 2060, the pension wealth of low earners relative to high earners would fall by a cross-country average of just 1.2% — relatively small. The OECD's key analytical point is sharper: for the distributive impact of linking retirement ages to life expectancy, what matters is not the existence of longevity gaps but their change over time. If gaps are broadly stable, life-expectancy improvements benefit socio-economic groups roughly equally and the link is distributionally neutral. If gains favor high-income groups — as they have in Denmark and the United States, where longevity inequality has risen — a higher retirement age raises genuine equity concerns.³⁴

The evidence on trends is mixed. There is no clear international trend in the educational gap in life expectancy: it is stable for most OECD countries, grew in Belgium during the 1990s, in Canada from the mid-1990s to around 2010, and in Lithuania from 2001 to 2014, while decreasing in Czechia between 2010 and 2016. Occupational gaps at 60–65 were stable for French women and rose at most slightly for French men (about one week per year), rose 1.5 months per year for German men, and one month per year for Swedish men (0.4 for women). Differentiating retirement ages by socio-economic group would be the theoretical fix, but the groups are impractical to define and differentiated rules nearly impossible to implement — so policymakers are advised instead to account for longevity inequality in benefit levels for low-income workers, where large gaps can justify high redistribution in benefit formulas.⁴

The real barrier is jobs, not health

Leaving health aside, the main reason a higher retirement age might harm disadvantaged groups is the shortage of employment opportunities as they near retirement. That is a labor-market failure, not a pension-design failure — and it reframes the policy response: raising the retirement age is not the wrong policy, but it must be paired with labor-market policies that foster inclusive employment at older ages. The exit-age data support the point: since the mid-2000s, effective exit ages have kept pace with life expectancy at 65, meaning workers have absorbed rising statutory ages without a health crisis — where jobs exist.³⁴

The retirement-age story, then, has three layers: statutory ages are rising and increasingly linked to longevity; effective exit ages have rebounded since 2000 and track life expectancy; and the equity cost of the linkage is small on average but concentrated where longevity gaps are widening — which is why the design of redistribution in benefit formulas, and the availability of jobs at older ages, matter more than the age number itself.

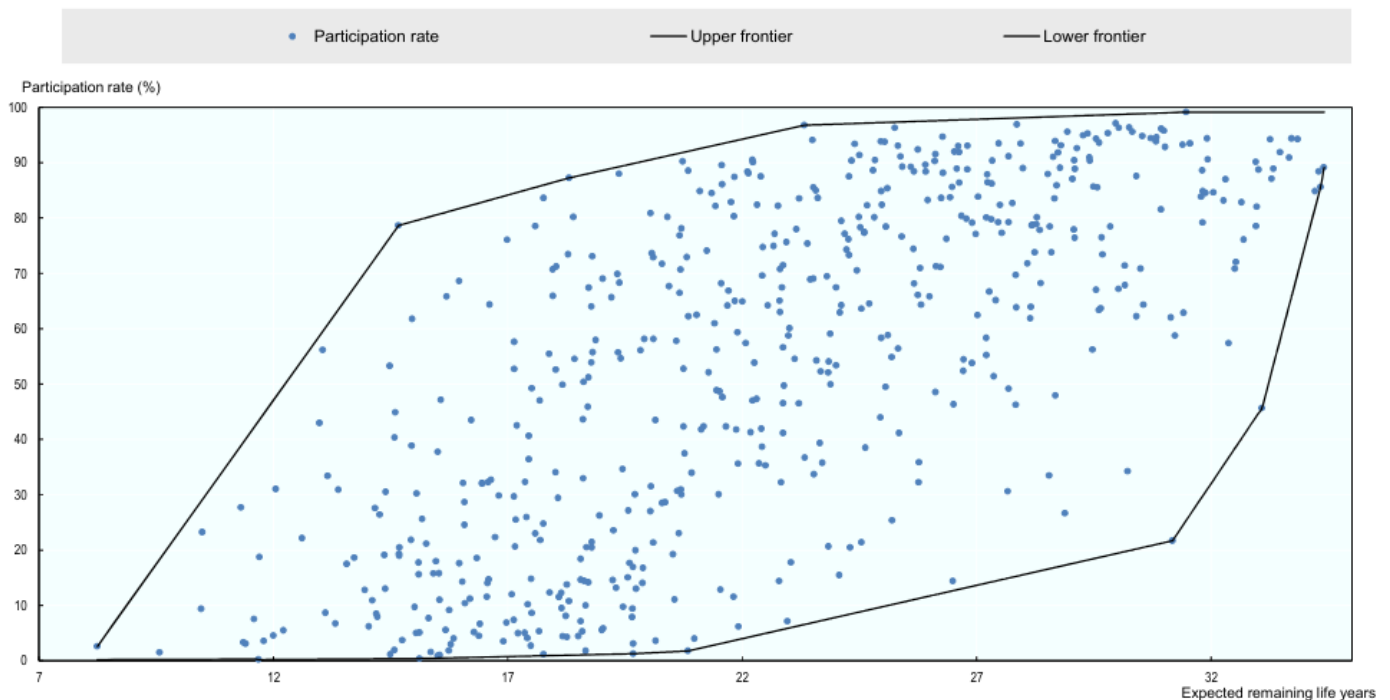


Exhibit. Figure 5.11. Labour market participation among older workers in their 50s and 60s and remaining life expectancy, 2015 or latest Participation rate Upper frontier Lower frontier Participation rate (%) 100 90 80 70 60 50 40 30 20 10 0. Source: S0016.

Asset-backed and defined-contribution systems: coverage, fees, and investment

As public pay-as-you-go systems strain under ageing, asset-backed pensions — where individuals or funds accumulate invested assets — carry more of the retirement-income burden. How much people save, what it is invested in, and what it costs to administer decides outcomes as much as benefit formulas do.

Coverage: wide for employees, thin for the self-employed and low earners

Participation in asset-backed plans is the first indicator of their reach, and it is highly unequal. Only eight OECD countries have voluntary private-pension coverage above 40% of the working-age population (Section 14). Within countries, coverage follows income: in the United States, only 13% of the lower half of the income distribution has private pensions versus over 50% of the upper half, and under 20% of full-time workers in the lowest earnings decile participated in a defined-contribution plan in 2012 against over 80% in the highest decile.³

The self-employed are the sharpest coverage gap. They are not required to join earnings-related schemes in Australia, Denmark, Germany, Japan, Mexico, and the Netherlands; where not required to contribute, their pensions are limited to first-tier benefits — about half of employees' pensions in Denmark, Germany, Greece, and the Netherlands, 17% in Mexico, and 34% in Japan. Even where mandatory, reduced contribution rates bite: self-employed pensions reach only 51% of employees' in Switzerland, 66–70% in Italy and Sweden, 74–87% in Chile, Costa Rica, Colombia, Czechia, Israel, Portugal, and Slovenia, and over 90% in Estonia, France, Iceland, Korea, Lithuania, and Norway. Australia is the outlier: its means-tested basic pension gives the self-employed 109% of what average-wage employees get from mandatory schemes. In Latin America, two-thirds of salaried workers contribute versus one-sixth of the self-employed (Section 16).⁴

Investment: equities win on average, lose on certainty

Equity exposure is the dominant design question in defined-contribution schemes, and it has been rising: at end-2022, total equity exposure (public and private) exceeded 40% of total DC investment in 13 of 38 jurisdictions analyzed, was below 20% in 7, and rose by more than 20 percentage points between 2001 and 2022 in five OECD countries.⁶

The historical verdict is decisively pro-equity. Across 19 OECD countries, at least 80% of the 83 cohorts analyzed would have accumulated more at retirement by investing part of their contributions in diversified equities rather than

only in domestic fixed income — 100% in Australia, Austria, Belgium, Finland, New Zealand, Portugal, and the United Kingdom. In Sweden, cohorts better off with diversified equities accumulated 6.1 times the sum of contributions more than with domestic fixed income on average; the few cohorts for whom fixed income won gained only 0.4 times contributions — the upside is large and the downside small. Stochastic modeling agrees: an 87–94% probability of a higher replacement rate with equity exposure, improving replacement rates by 12–26 percentage points where equities win; where fixed income wins (9–13% of cases), it improves them by at most 5 points.⁶

But the equity case carries three caveats. First, it needs time: compounding does the work, so it pays most when people start saving early or delay retirement — though even with only 20 years of contributions, at least 69% of cohorts were still better off in equities. Second, it is volatile: longer and higher equity exposure produces more volatile outcomes across cohorts, so two people born a year apart with identical earnings, contributions, and strategy can retire with very different pots. Third, benefits become sensitive to market falls just before retirement: the median lifelong-annuity replacement rate from an equity portfolio falls from 39% to around 30% if markets drop in the year before retirement.⁶

Life-cycle strategies — high equity exposure early, tapering before retirement — are the standard mitigation. The strategy that reduces equity exposure 10 years before retirement beats the 20-year version (median replacement rate 31% versus 28%, with a 76% probability of a higher outcome). When markets crash just before retirement, life-cycle portfolios hold up far better than pure equity: in Belgium, a diversified-equity portfolio would have fallen from 12.3 to 5.6 times contributions between the 1999 and 2002 retiring cohorts, while the 10-year life-cycle strategy stayed stable at around 7 times.⁶

Decumulation: drawdowns beat annuities when bequests count

What members do at retirement matters too. Regular drawdowns — dividing remaining assets each year by a factor based on remaining life expectancy — leave a bequest, unlike a traditional lifelong annuity, and the median bequest rises with equity exposure during retirement. Combining benefits received while alive with bequests, total pension payments are higher with drawdowns than annuities, and the gap grows with equity exposure: the probability of being better off with drawdowns is 57% with the fixed-income portfolio, 70% with life-cycle strategies, 84% with a 60/40 portfolio, and 85% with the equity portfolio.⁶

Fees: the known leak the evidence doesn't quantify

Fees are the remaining design lever the retrieved evidence does not quantify: the Pensions at a Glance 2023 asset-backed chapter tracks fees charged in selected defined-contribution plans as one of its eight indicators, but the retrieved chunks contain no fee figures to report. The mechanism, however, is direct — a percentage-of-assets charge compounds against the member exactly as returns compound for them — which is why fee transparency sits alongside coverage and investment strategy as a first-order outcome variable even where the numbers are not in hand.⁴

The asset-backed story, then: equities have delivered far more at retirement for the vast majority of historical cohorts, life-cycle tapering buys insurance against late-market crashes cheaply, and drawdown designs turn leftover assets into bequests — but all of this accrues to people who are in the system, while the self-employed, low earners, and workers outside mandatory schemes remain on the outside looking in.

Coverage gaps for informal, gig, and self-employed workers

Pension systems were designed for the standard employment relationship: one employer, a regular wage, contributions deducted at source. The workers falling outside that model — informal, gig, and self-employed — are the fastest-growing and least protected segment of the labor force.

The scale of the excluded

Globally, an estimated 60% of workers operate in the informal economy, and the gig workforce is growing rapidly. These workers frequently lack access to contributory pension schemes, which depend on stable employer–employee relationships; the irregular nature of their work makes consistent contributions difficult, leaving them vulnerable in old age and perpetuating inequalities. Pension systems also often lack portability, making it hard for workers who

change jobs frequently or move across borders to build adequate savings — and as gig and informal work reshapes labor markets, public systems must evolve to include these workers or risk leaving a growing segment of the population unprotected in retirement.⁸

The self-employment gap within rich countries is just as stark. Many mandatory earnings-related pensions do not apply to all contract types, demand minimum contribution years, and have limited portability — and the trend toward shorter contracts and part-time work lowers the share of the labor force actually covered by traditional pensions (Sections 14, 18). Where the self-employed are not required to contribute at all, their pensions can collapse to first-tier benefits only: about half of employees' pensions in Denmark, Germany, Greece, and the Netherlands, 17% in Mexico, 34% in Japan (Section 18).

The opt-out machine

Even where arrangements exist, voluntary participation stratifies by sector and worker type. In Italy, multi-employer occupational arrangements are mandatory for employers but voluntary for employees: participation reaches 70–90% in industrial sectors like energy, chemicals, and pharmaceuticals but sits around 10% in retail, tourism, and services. The sectors with low participation have smaller firms, weaker union representation, and more vulnerable workers — women, younger workers, and low-income earners. Smaller employers may actively encourage opt-outs to keep severance-pay contributions (the *Trattamento di Fine Rapporto*) as cheap company financing, while vulnerable workers prefer keeping it as a guaranteed-return severance allowance accessible on job change.⁶

Collective-agreement-based multi-employer arrangements also structurally exclude workers outside the agreements: the self-employed, employees in companies that avoid collective bargaining (Sweden's young IT firms are the classic example), and — in Denmark — around 10% of employees with no occupational pension at all. Small employers face another disadvantage under provider-driven models: some UK master trusts refuse small employers outright as uneconomical, charge them relatively more per member, and offer them fewer product choices — which is why the UK's Nest carries a legal requirement to accept all employers.⁶

Why untargeted fixes backfire

Expanding mandatory earnings-related pensions is the most effective coverage tool, but where governments prize choice, extending voluntary coverage — especially to low-income groups — matters more (Section 18). The catch is that untargeted incentives skew regressive: strong tax deductions for private pension saving favor higher earners, producing regressive systems in countries like Chile, Denmark, Israel, the Netherlands, the Slovak Republic, and Switzerland [S0016 reference in Section 18 evidence]. Without targeting, subsidies for voluntary pensions benefit high incomes more than low ones.³

Three design principles emerge from the evidence. First, defaults beat incentives: automatic enrolment overcomes the inertia that keeps vulnerable workers out — which is exactly why the Italian opt-out design collapses to 10% in vulnerable sectors while auto-enrolment systems (Section 14) sustain coverage. Second, portability and multi-employer pooling must be built for workers who cross employers constantly: pooling arrangements reduce costs through scale, but only if they accept small and irregular employers rather than screening them out. Third, coverage rules must follow the worker, not the contract: as non-standard work grows, tying pension rights to stable employment relationships guarantees that the fastest-growing segment of the workforce retires unprotected.^{6,8}

The coverage story is ultimately about compounding exclusion: informal and gig workers miss the contributory base, the self-employed face reduced or absent mandates, and voluntary systems replicate every labor-market disadvantage — low pay, small firms, weak bargaining power — as low pension coverage. By 2050, the question is not whether coverage gaps close on their own; it is whether pension systems are redesigned for the workforce as it exists rather than the workforce of the 1970s.

The decumulation challenge: payout options, annuities, and longevity risk protection

Accumulating a pension pot is only half the problem. The harder half is spending it down over a lifespan of unknown length: too fast and you outlive your money, too slow and you forgo consumption you could have enjoyed. This decumulation challenge is where defined-contribution systems are weakest — and where the design of payout options decides how much retirement income a given pot actually buys.

The core trade-off: insurance versus flexibility

The textbook answer to longevity risk is the lifelong annuity: hand the pot to an insurer, receive guaranteed income until death. But annuitization rates remain low almost everywhere, and regular drawdowns have a measurable appeal. As Section 18 showed, combining lifetime benefits with bequests, drawdowns beat lifelong annuities with a probability of 57% for fixed-income portfolios, 70% for life-cycle strategies, 84% for 60/40 portfolios, and 85% for equity portfolios.⁶

The OECD's payout-phase framework pushes policymakers to start from financial needs, not products: the regulatory framework should first determine what role DC assets play in financing retirement alongside other income sources, then allow the products that fit those needs — distinguishing essential spending (which needs guaranteed income), unexpected shocks (which need liquidity), and discretionary goals like bequests. No single product covers all three, which is why blended approaches — partial annuitization for essentials, drawdowns for the rest — increasingly look like the rational default rather than a compromise.⁶

The advice gap

Good payout decisions require personalized guidance, but advice is the most resource-intensive delivery channel — and the people who need it most get the least. In the UK, around three-quarters of people who received financial advice said it helped them understand complex matters and make better decisions, yet around 30% of non-seekers say they don't take advice because they don't earn enough, it's too expensive, or they lack sufficient savings — and UK advisors have been dropping less profitable clients they can no longer afford to serve.⁶

Digital advice could fill the gap cheaply, but uptake is tentative: in Australia only 37% of people who considered advice were open to a digital channel, 40% felt it would be more valuable if a person explained the digital tool's recommendations, 35% of UK respondents preferred in-person support, and 36% said they would never take advice from an AI tool even if the technology improves. Digital solutions for the payout phase have also been slower to emerge than those for saving. The result is a decumulation market where complex products exist but the guidance infrastructure to match people to them does not.⁶

Housing: the reserve army of retirement income

For many retirees, home equity is the largest untapped reserve — and home equity release products (reverse mortgages, home reversions, sell-and-rent-back) are the mechanism to draw on it. Common uses include paying off outstanding debt (a pattern in the UK and Japan), funding large one-off expenses like home repairs, bridging early or gradual retirement, and financing of last resort for long-term care or divorce.⁶

The distributional pattern is telling. In the United States, HECM reverse-mortgage loans have high penetration among low-income groups; nearly two-thirds of HECM borrowers are single, and around 70% of those are women — the equity release market serves people with thin pensions. In Korea, by contrast, JTYK loans concentrate among wealthier high-rise owners; in Italy, people view equity release as a way to meet basic needs where income falls short, not to live more comfortably. The products are most suitable for older homeowners who intend to stay in their homes for life and don't prioritize leaving the house as a bequest — they are poorly suited to younger retirees (debt compounds longer), to those who plan to downsize (downsizing earlier releases capital more efficiently), and to anyone who might later need the equity for aged care (Australia's ASIC found significant reverse-mortgage borrowers at risk of lacking funds for average upfront aged-care costs).⁶

Uptake barriers remain formidable: poor product reputation, debt aversion after a lifetime of paying off a mortgage, emotional attachment to the home, unwillingness to reduce heirs' bequests, inflexibility, and thin access to advice.⁶

The developing-world floor

The decumulation debate in rich countries assumes a pot to decumulate. In much of the developing world, there is no pot at all. In Rwanda, informal labor-market participation is around 75% among the half-million people aged 65 and over; with no social pension in place, evidence suggests old-age poverty runs higher than official statistics show — and many older people, disproportionately widowed women, live without family support. Where governments invest strategically in youth for the demographic dividend while fiscal capacity is limited, the balance of in-

vestment across generations becomes the binding policy question. For these populations, the “decumulation challenge” is not payout design but the absence of anything to pay out — making non-contributory social pensions the prerequisite for any payout-phase architecture.¹²

The decumulation challenge, then, stacks in layers: rich-country retirees need payout designs that balance insurance against flexibility, advice they can access, and honest treatment of housing wealth — while in much of the world the task is still to create a floor from which any decumulation could begin.

Replacement rates and retirement income adequacy

The replacement rate — retirement income as a share of pre-retirement earnings — is the headline measure of whether a pension system delivers on its promise. It is also the measure most sensitive to design choices: who contributes, how much, for how long, how assets are invested, and what gets charged along the way.

What drives replacement rates in funded systems

The OECD’s cohort analysis of 83 successive retiring generations (1939–2021) across 19 countries models a typical worker: entering at 25, wages growing with inflation plus 1.25% productivity, contributing a constant 5% for 40 years, paying a 1% annual asset-based fee, retiring at 65. The spread of outcomes is enormous. Portfolios invested only in domestic fixed income accumulated between 1.9 and 3.5 times total contributions depending on the country; diversified-equity portfolios accumulated between 4.5 and 11.2 times — roughly two to three times more. Since replacement rates scale with the accumulated pot, investment strategy alone can be the difference between an adequate pension and a thin one.⁶

Three parameters deserve attention. First, the contribution rate: the model’s 5% reflects voluntary systems that top up public pensions; where defined-contribution plans are mandatory and the main retirement-finance source, rates run 10–20%. Second, fees: the assumed 1% annual charge compounds against the member for 40 years — fee levels are a first-order adequacy variable even though (as noted in the asset-backed section) the retrieved evidence does not quantify them. Third, equity exposure: in Chile, every 10 percentage points of additional equity exposure raised average annual real returns by 22 basis points over 20 years. But equity is not the only driver — Australian Super’s Indexed Diversified option earned 4.3% over 2009–22 versus 4.6% and 4.7% for its Balanced and Socially Aware options despite higher equity exposure (70% vs 58%), because indexing strategies, fixed-income weight, and diversification into infrastructure, private equity, and property matter too. Geography matters as well: the US Thrift Savings Plan’s globally invested I Fund underperformed its US-equity C and S funds over 2013–22.⁶

Adequacy for workers in hazardous or arduous jobs

Adequacy looks different for workers whose jobs shorten their lives. Occupational life-expectancy gaps are wide — several years at age 65 between high and low occupational groups — and part of the difference survives controls for education, income, and habits. Yet special pension provisions for hazardous work have been broadly mistargeted and are being wound back: Estonia, Hungary, and Slovenia found covered jobs no more hazardous than others, no higher mortality, and many early pensioners continuing to work.⁴

The reform trend is unmistakable. Finland, Ireland, Lithuania, and Luxembourg eliminated most occupation-specific provisions in the 1990s; Chile and Poland adjusted theirs; Latvia left them largely untouched; Sweden’s system has been universal for over a century with no special entitlements. The better recent designs link eligibility to actual job characteristics rather than occupational titles — Finland’s 2017 years-of-service pension allows retirement two years early based on individually assessed arduousness; France’s 2010 scheme accrues points for periods in strictly defined arduous work, usable for training, reduced hours, or up to two years’ early retirement.⁴

The adequacy lesson generalizes: inability to continue one specific job does not justify permanent labor-market exit — many such workers can perform other jobs — and special provisions financed publicly weaken firms’ incentives to reduce hazards. The OECD’s prescription is to handle hazardous work at working age — health and safety regulation, reskilling, career transitions — rather than through pension privileges.⁴

The adequacy frontier

Pulling the threads together, retirement income adequacy in 2050 rests on four levers: contribution adequacy (rates and career completeness), investment adequacy (equity exposure with life-cycle risk management), fee discipline, and targeted rather than occupation-based exceptions. The groups at risk are familiar — incomplete-career workers, the low-paid, the self-employed, informal and gig workers — and the policy menu is the same one that runs through this report: keep people contributing (defaults over incentives), keep assets growing (equities for the long young, tapering for the near-retired), keep costs low, and protect the floor for those the contributory system never reached.

Inflation, indexation, and valorisation of pension benefits

A pension is a promise of purchasing power over decades. Two design choices determine whether that promise holds: valorisation — how past earnings are uprated when the initial benefit is calculated — and indexation — how benefits in payment keep pace with prices or wages afterward. The 2021–23 inflation wave was the first serious stress test of these mechanisms in a generation, and it exposed how much design detail matters.

Indexation in payment: the devil in the timing

Most systems index benefits in payment to prices, wages, or a mix — but the exact rule decides how much inflation retirees actually feel. Timing is the first fault line. Greece now uses an estimate of CPI growth for the previous year to index pensions already in January, with a correction once confirmed statistics arrive; Germany reformed its targeted-benefit indexation in 2023 so the benefit would not lag inflation too far before the next adjustment, adding a supplementary one-year adjustment approximating within-year inflation on top of the permanent basic adjustment (70% price increases, 30% net nominal wage increases). Belgium's rule indexes prices five months after a 2% threshold is exceeded — and can trigger indexations above the threshold if prices keep rising over those five months.⁴

Some countries tie indexation to the economy, not just prices: Portugal adjusts indexation to CPI both upward and downward depending on past real GDP growth and individual pension levels. Others index to living standards directly: Latvia sets its targeted benefit at 20% of median income three years prior (the minimum pension at 25%), so benefit adequacy is anchored to the income distribution rather than to last year's prices. Japan indexes its targeted benefits only every five years, based on local cost-of-living surveys — infrequent by design, and therefore vulnerable to real erosion between resets.⁴

When indexation rules are too rigid to deliver, governments improvise. The inflation wave of 2021–23 brought a flurry of ad hoc responses: Hungary brought forward its 13th-month pension, Austria, Estonia, Germany, Poland, Portugal, Slovakia, Slovenia, and the UK made ad hoc payments, and Denmark and Greece targeted low-income pensioners specifically — emergency measures that sit outside the indexation rulebook and reveal its gaps (established in Section 16).

What should the index measure?

Whether benefits should track prices, wages, or a pensioner-specific basket is a substantive choice, and the evidence cautions against fashionable answers. Pensioner-specific indices look intuitively right but perform questionably: France's pensioner-specific index fell 0.3% in cumulative terms relative to the headline index over 1998–2015, and the US experimental CPI-E ran only 0.4 percentage points above headline inflation per year from 1983–2002 and essentially matched it from 2002–21 — while a low-income consumption basket is better-targeted to the people who actually face inflation risk (established in Section 16). Partial indexation above a threshold (Austria, Italy) and GDP-linked rules (Portugal) trade adequacy against fiscal sustainability.⁴

The Netherlands offers the cautionary tale of indexation denied. Implied funding-ratio requirements for final-benefit schemes produced little or no indexation of pensions in payment for more than a decade despite high returns — a major source of discontent that helped drive the systemic shift from DB to DC. Ahead of the transition, indexation rules were relaxed in July 2022, lowering the funding-ratio threshold for full price indexation from around 130% to 105%.⁴

The erosion most people never see

Indexation protects purchasing power year to year — but what about erosion over a 20- or 30-year retirement? Pension dashboards almost universally show estimated retirement income in real terms so individuals can compare it with their current salary, yet the OECD identifies a major shortfall: individuals do not see how their purchasing power may erode over the course of retirement, and Sweden is considering how to present this information. The point generalizes: indexation rules are calibrated to annual inflation, but retirees' welfare depends on cumulative adequacy — the compound of the initial benefit level (set by valorisation of past earnings) and decades of subsequent adjustments.⁶

Valorisation: the quieter lever

Valorisation — revaluing the earnings of earlier career years when computing the initial pension — matters as much as indexation but gets far less political attention. It determines whether a worker's contributions from age 25 still count at full value at 65, and it interacts with every design choice: in NDC systems, the annuity factor plays this role implicitly; in earnings-related systems, explicit revaluation rules decide how much early-career earnings are worth at retirement. The retrieved evidence does not quantify valorisation rules across countries, but the mechanism is direct: weak valorisation silently taxes early contributors and long careers — the very workers coverage policy most wants to protect.

Inflation, then, tests pensions twice: first at award, through how past earnings are valued; then every year after, through how benefits are adjusted. The 2021–23 wave showed that rules built for a low-inflation world needed emergency patches — and that the patches, not the rules, did much of the protecting.

Special provisions for hazardous and arduous jobs

Workers whose jobs wear down their bodies have historically been granted earlier retirement and better terms. These special pension provisions — among the oldest carve-outs in social insurance — are now in broad retreat, replaced by a harder question: should pensions compensate hazardous work at all, or should the response come earlier, at working age?

What the provisions look like

OECD countries can be classified into four groups in their treatment of pensions for hazardous or arduous jobs, with eligibility defined either by occupational titles or by measurable job characteristics — lifting heavy weights, extreme temperatures, chemical risks, night shifts — sometimes both. The reductions are substantial: Latvia's special pensions start at age 50 (13 years early), Slovenia's at 54, Colombia's and Czechia's six years early, Italy's at 61 years 7 months with a 41-years-worked-before-19 option, Spain's reductions of 5–50% depending on sector (established in Section 2).⁴

The coverage is overwhelmingly male: men make up more than 75% of beneficiaries in Austria, Chile, Colombia, Czechia, Finland, Latvia, Poland, and Slovenia, because the schemes cover male-dominated occupations like mining. Norway is the exception — women are 70% of covered workers, likely because care workers are included. And many covered workers don't use the early-retirement option at all: in Chile and Slovenia, only part of those with additional entitlements retire early; the majority prefer to boost their income after normal retirement — hazardous workers are 3.0% and 5.1% of workers but only 1.3% and 0.9% of new retirees in the two countries.⁴

The case against — and the better designs

The OECD's assessment is blunt. Special provisions have been seriously mistargeted — covering jobs of questionable hardship (Estonia, Hungary, Slovenia), weakening firms' incentives to reduce hazards when publicly financed, discouraging workers who could keep working, and reducing job mobility when changing jobs forfeits entitlements. Incapacity to do one specific job is not incapacity to work: ballet dancers in Australia, Switzerland, and the UK have no special provisions and many successfully pursue alternative careers, though about half had to finance their own retraining. Career reorientation for soldiers (Czechia, Korea, US) shows evaluated programs improving employment outcomes.⁴

Reforms have followed this logic. Finland, Ireland, Lithuania, and Luxembourg eliminated most occupation-specific provisions in the 1990s; Chile and Poland adjusted theirs; Sweden's universal system has never had them (established, Section 21). The better surviving designs link eligibility to actual, measurable job characteristics rather than occupational titles: Finland's 2017 years-of-service pension allows retirement two years early based on individually assessed arduousness; France's 2010 scheme accrues points for strictly defined arduous periods, spendable on training, reduced hours, or up to two years' early retirement.⁴

The working-age alternative

The central policy message is that hazardous work is primarily a labor-market problem. Health and safety regulation should limit exposure; reskilling and age-management policies should enable career shifts — reskilling absorbed the phase-out of coal mining in Germany, Poland, and the UK and of fisheries in Canada. Where risks materialize, disability and work-injury insurance (many schemes pay up to 100% wage compensation for work-related injuries) should bridge to retirement age. There is a genuine tension: early-retirement options for hazardous jobs clash with activation policies aimed at keeping disability beneficiaries in work, and with labor shortages that make every worker's participation valuable.⁴

One caveat survives the critique. Delayed health effects — cancers and chronic conditions appearing years after exposure — are typically not covered by disability or sickness insurance, which require proof of a causal link that latency periods make hard to establish. For those cases, well-targeted, evidence-backed early-retirement options may complement disability schemes — but any such program must be narrow, measurable, and revisited as working conditions change. The debate has also widened beyond physical strain: psychological stress at work is now part of hardship discussions, and it is far harder to measure.⁴

The trajectory is clear: occupation-based pension privileges are being replaced by universal rules with narrow, evidence-tested exceptions — and by the harder, earlier work of making hazardous jobs safer and career change feasible.

Gender gaps in pensions and lifetime earnings

Pension systems mirror the labor market, and the labor market is not gender-neutral. Women's pensions are lower than men's in every OECD country — a gap built from decades of lower pay, more part-time work, career breaks for caregiving, and, historically, earlier statutory retirement ages. The result is one of the most persistent inequalities in retirement.

Measuring the gap

The OECD defines the gender gap in pensions as the relative difference between the mean retirement income of men and women aged 65 and over, as a percentage of men's pensions, among pension beneficiaries. The WEF's estimate puts women's retirement income at 26% below men's across the OECD (established in Section 13). The gap starts in working life: employment rates diverge between men and women across age groups, with the gap widest at older ages (Figure 6.8,) — and it compounds through every mechanism this report has documented. Women are overrepresented among low earners, part-time and non-standard workers, and informal workers; they face the steepest coverage gradients in voluntary pensions and the sharpest penalties from career breaks.⁴

Retirement-age history widened the gap directly. In most countries women could retire earlier than men — Austria, Colombia, Greece, Hungary, Luxembourg, and Slovenia still had lower female retirement ages paired with lower exit ages in the early 2020s — and earlier retirement mechanically means fewer contribution years and longer benefit durations. The OECD average effective exit age was 64.4 years for men and 63.1 for women in 2022, ranging from 60.5/58.4 in Luxembourg to 68.3 for Icelandic men and 67.4 for Korean women. Convergence of female retirement ages to men's — underway across the EU — removes one structural source of the gap but cannot undo the career histories behind it.⁴

How systems respond

Pension systems have developed a toolkit for the gender gap, centered on compensating caregiving. Four countries' designs illustrate the range. Chile pays women aged 65 or older a government subsidy for each child, equivalent to

18 months of mandatory contributions (10% over the minimum wage at the child's birth) earning fund-type-C returns — directly converting motherhood into pension capital. Estonia credits one parent with monthly contributions of 4% of the national average wage for up to three years per child, whether or not the parent returned to work, for children born after January 2013. Germany pays its Riester child subsidy into the account of the parent receiving child allowances — the mother by default — at up to €185 per year per child born before 2008 and €300 per year after. Lithuania has the government contribute 1.5% of the average gross salary into the account of parents with children up to three years old receiving maternity benefits.⁶

A second instrument targets low earners, among whom women are overrepresented. Government matching contributions are designed for this: Australia's super co-contribution gives low-income earners a 50% match up to AUD 500; Chile matches young low-wage workers' first 24 months of contributions; Colombia's BEPS program pays a 20% government match at retirement for voluntary low-income contributions. Fixed-amount subsidies work the same distributional logic — the fixed payment is worth more as a share of a small income.⁶

Derived rights remain a third, older instrument: survivors' pensions and family-pension rules. Belgium's rule illustrates the design tension — when one partner's pension rights are below 25% of the other's, the smaller pension is not paid and the other receives a family pension calculated at 75% of wages instead of 60%. Such rules protect women who never built independent entitlements, but they also reduce the incentive to build them — and they leave divorced and never-partnered women uncovered.⁴

The deeper arithmetic

None of these instruments changes the underlying arithmetic: pensions built on lifetime earnings will reproduce lifetime earnings gaps. Three structural facts decide how much they reproduce. First, how progressive the benefit formula is — the countries with universal basic pensions (Ireland, New Zealand, the UK, scoring 100% on the progressivity index) transmit none of the wage gap into pensions (established in Sections 13–14). Second, whether caregiving is credited as contributions rather than compensated afterward — credits protect the contributory record itself. Third, whether the system insures against the events that hit women's careers hardest: divorce (the Netherlands' pension dashboards show divorce's impact on pensions precisely because assets must be split), widowhood, and part-time work.⁶

Closing the gender pension gap, then, is less a pension problem than a labor-market one wearing pension clothes. Until earnings, hours, and caregiving are shared more equally during working life, pension design can only blunt the gap — through progressivity, caregiving credits, targeted matches, and derived rights — not erase it.

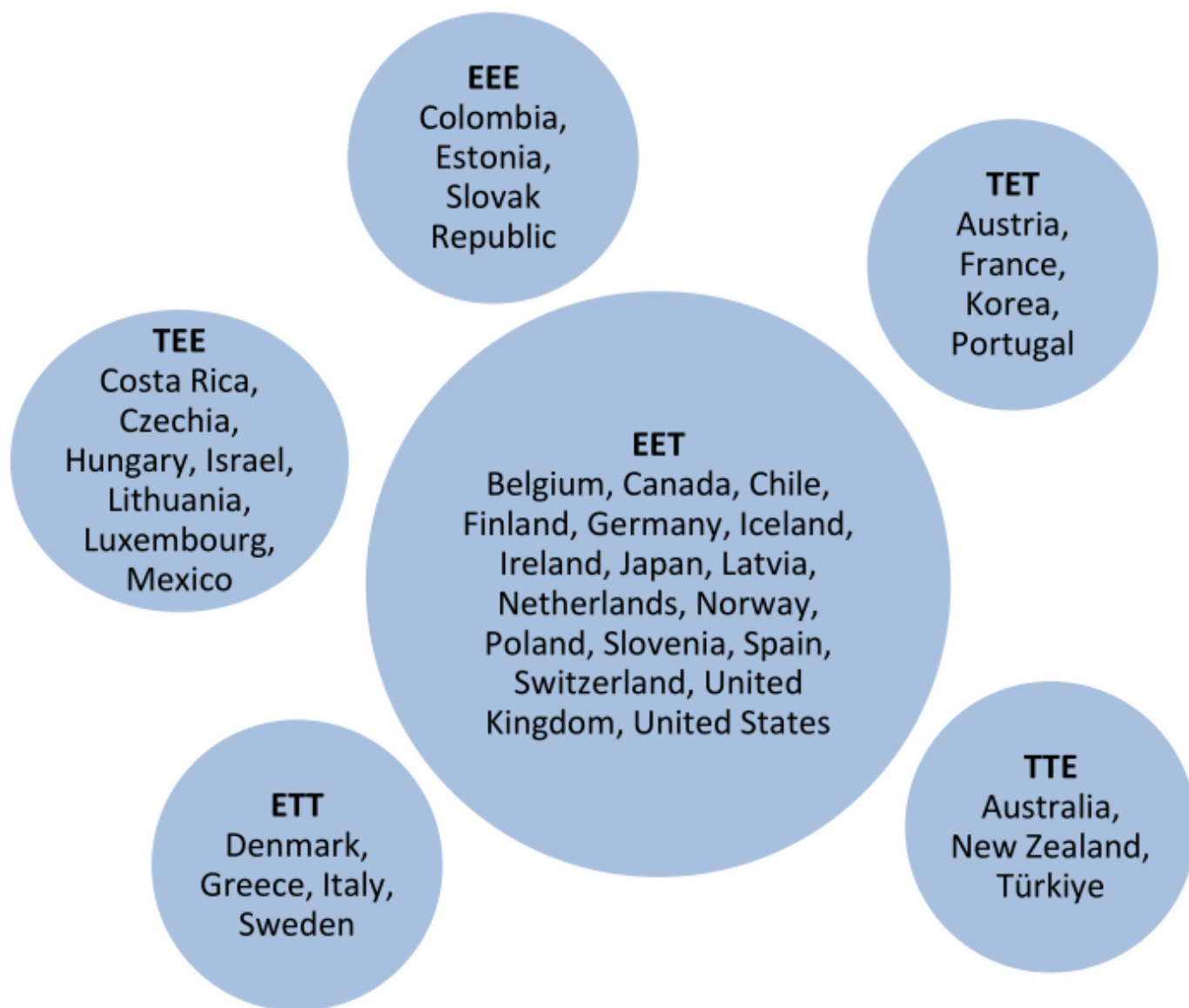


Exhibit. Figure 1.3. Overview of the tax treatment of retirement savings in OECD countries, 2023 EEE Colombia, Estonia, Slovak TET Austria, Republic France, Korea, TEE Portugal Costa Rica, Czechia, EET Hungary, Israel, Belgium, Canada, Chile. Source: S0013.

Individual pension dashboards and impartial financial guidance

Retirement decisions are among the most consequential financial choices people make — irreversible, high-stakes, and made once, with no chance to learn from experience. Yet most people arrive at them with little understanding of their options. Bridging that gap — through dashboards that show people what they have, and guidance that helps them decide what to do with it — is the newest frontier of pension policy.

Generic information doesn't work

The evidence is stark: a UK survey found that 27% of people felt general communication about retirement options did not help their decision-making at all, with another 54% saying it helped only a little. An experimental study found that only 14% of people made the best decision about how much to withdraw from their pension pot — balancing financial needs against tax — when given general information about withdrawal tax implications. Generic leaflets fail because retirement choices are personal: the right payout depends on health, housing, other income, family, and risk tolerance.⁶

Personalized financial advice is the gold standard, but as the decumulation section showed, it is resource-intensive, costly, and increasingly rationed — UK advisors are dropping less profitable clients, and around 30% of non-seekers say they can't afford it or lack the savings to justify it. Guidance occupies the middle ground: engaging individuals in their decision and helping them decide for themselves, through digital tools, choice architecture, and personalized suggestions.⁶

Dashboards: seeing the whole picture

Individual pension dashboards — consolidated views of all of a person's pension entitlements in one place — are the infrastructure that makes guidance possible. The OECD's review shows wide variation: most dashboards show accrued amounts plus an estimated pension income at the statutory retirement age; Chile, the Netherlands, and the Slovak Republic show good-and-bad investment scenarios; Belgium, Chile, the Netherlands, and Sweden let users adjust parameters like retirement age, hours, and salary; Norway offers an advanced calculator covering marital status, partial retirement, and work abroad; Australia's Moneysmart lets users compare up to four funds' performance and fees (established in Section 22).

Sweden's Minpension offers the clearest evidence that dashboards change behavior: its interactive retirement-planning tool for users over 54, showing the impact of different payout options, contributed to an increase in people choosing lifetime income. But dashboards have limits: centralized tools often show average outcomes rather than what a specific provider's product will deliver, and most display income in real terms — hiding the cumulative purchasing-power erosion over retirement that Sweden is now considering how to present (established in Section 22).⁶

Choice architecture and defaults

How options are presented shapes what people choose. The Dutch financial-markets authority (AFM) published guidelines requiring pension providers' communications to be aware of behavioral biases, use relevant variables to distinguish member groups, and give people tools to identify their own needs. The UK's FCA "investment pathways" require firms to present four objective-aligned options to people approaching retirement — with a 50% take-up rate in the first quarter of 2023. A Scottish Widows survey found 38% of respondents preferred being nudged toward a specific option after answering questions. Providers are building hybrid tools: Aviva's Guided Retirement splits assets across a guaranteed lifetime income account, a flexible drawdown account, and a liquid account, letting people visualize allocations.⁶

Defaults are the most powerful nudge — and the most dangerous if mis-set. The OECD's guidance: any decumulation default must not cause undue harm, should be reversible (life annuities are generally inappropriate as defaults because they are irreversible), should adapt to market context (don't default people into products that lock in recoverable losses), and shouldn't push small pots into lifetime income products. Managed drawdown products with cooling-off periods thread the needle: regular income with retained flexibility.⁶

Designing for the unexpected

Good payout guidance must also leave room for shocks. Over 40% of US retirees under 80 have experienced an unexpected financial shock, nearly a quarter from major home repairs; in Costa Rica, 61% of lump-sum withdrawers used the money for home improvements. The death of a spouse reduced the surviving parent's financial security in a third of US families surveyed, a quarter "much worse" — and divorce can hit harder than death. Long-term care is the shock people are least prepared for: around 40% of lower-income Americans and Canadians are financially unprepared for it, 35% for divorce, and an estimated quarter of elderly Europeans needing long-term care couldn't finance even 10% of expenses from all their resources. Home equity functions as the precautionary reserve — in the US it is the primary resource for health shocks after insurance — but using it is a last resort, often involving selling the home.⁶

The design implication: payout guidance must preserve liquidity for shocks alongside lifetime income for essentials — which is exactly the three-layer structure (guaranteed income, flexible drawdown, liquid reserve) that the best new tools implement.⁶

Impartiality as a regulatory requirement

The final word is supervision. Behavioral insights in the hands of providers can serve consumers or commercial gain; supervisors must ensure the former. Nudges reduce engagement even as they improve choices, so they must be paired with tools that keep people aware of what they decided and why. Impartial guidance — free from product sales incentives — is the institutional answer, and dashboards are its delivery vehicle. As defined-contribution systems mature and more retirees face genuine choices, the quality of guidance may matter as much as the quality of investment returns.⁶

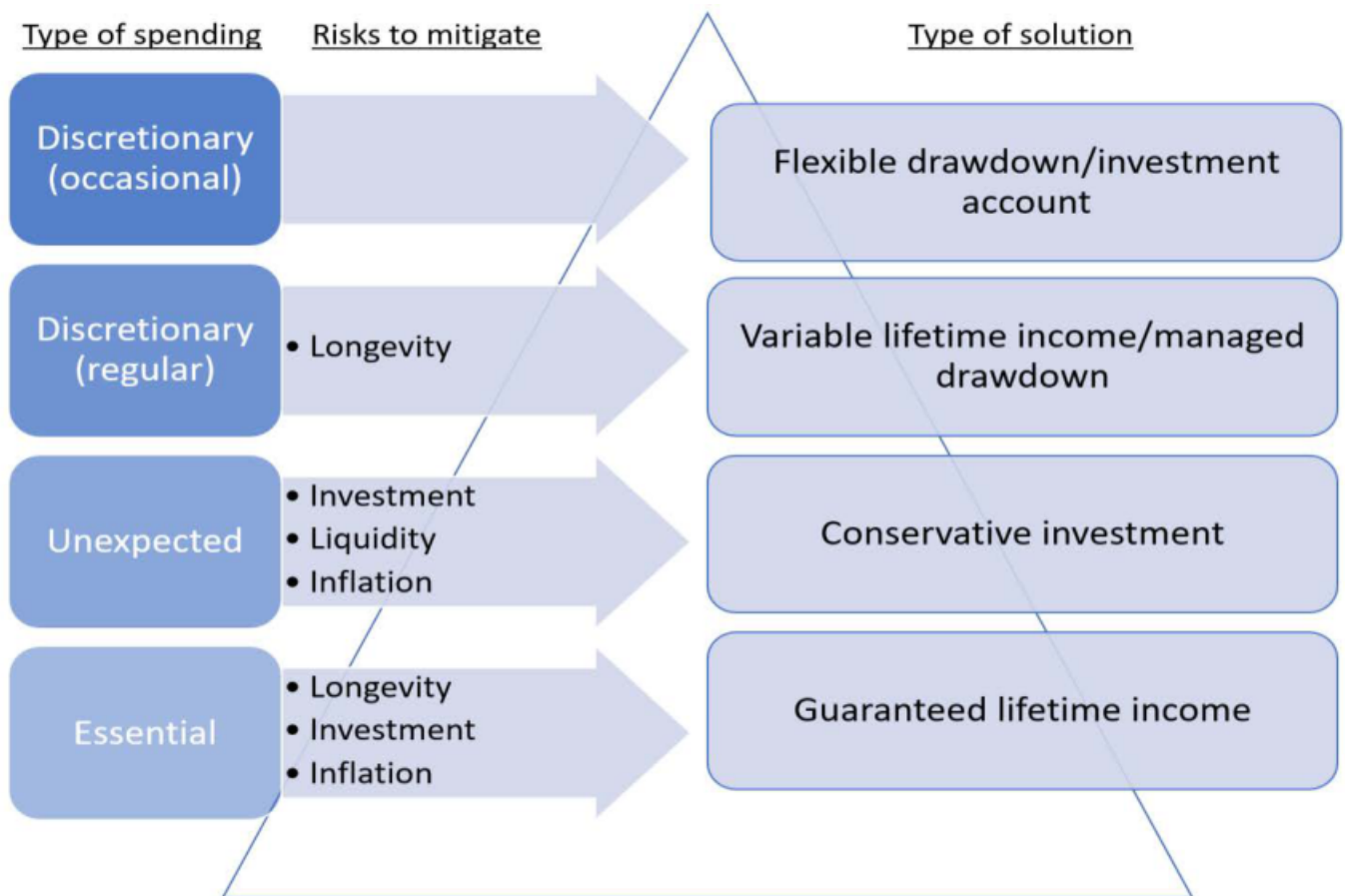


Exhibit. Figure 5.3. Illustration of the risks that should be mitigated for different spending needs in retirement and the corresponding type of solutions Essential spending to cover basic needs will require a regular, stable source of finan. Source: S0013.

Wealth, savings, and financial resilience

Household wealth, debt, and savings over the life course

Pensions are only part of how households finance old age. The full picture includes housing wealth, financial savings outside pensions, and the debts carried into retirement. The retrieved evidence for this section is thinner than for others — the corpus speaks most clearly about how tax systems shape the savings decision — so this section says what the evidence supports and names what it does not.

How tax systems shape the savings decision

Retirement savings everywhere enjoy preferential tax treatment relative to ordinary savings — which are typically taxed on contributions and investment income with withdrawals exempt (the “TTE” regime). When pension plans deviate from TTE, individuals pay less tax over their lifetime: tax incentives are indirect subsidies through the tax code, not money paid into the account.⁶

The dominant design is “EET” — exempt contributions, exempt investment income, taxed benefits — used by 17 of 38 OECD countries for their main asset-backed pension plan. But the range is wide: EEE (Colombia, Estonia, Slovak Republic), TET, ETT, TEE, and TTE (Australia, Denmark, Greece, Italy, New Zealand, Sweden, Türkiye), and only 16 of 38 countries apply a single consistent treatment — most vary it by plan type, contribution type, and source.⁶

Contribution limits put a ceiling on the subsidy, and they vary enormously: Australia’s voluntary-contribution limit of AUD 110,000 represents 119% of the average wage, while in most countries the fixed limit is under 40% of the average wage — down to 1% in the Slovak Republic (EUR 180). Several countries tilt limits toward older workers: Ireland lets the deductible share rise from 15% of earnings under 30 to 40% at 60; the Netherlands from 3.9% to 27.5%; Denmark’s annual limit jumps from DKK 8,800 to DKK 56,900 within seven years of retirement age; the US allows catch-up contributions of USD 1,000–7,500 for the over-50s.⁶

Employers get their own incentives: in Australia, Austria, Belgium, Ireland, Italy, Poland, Slovenia, and the UK, employer pension contributions are excluded from the social-contribution base or taxed at a reduced rate — in Italy, 10% instead of the standard 24% — making it cheaper for an employer to fund a pension than to pay the same amount as salary.⁶

Who actually saves

The incentive architecture is generous, but the savings follow income. As established in the coverage sections, voluntary pension coverage tracks the income distribution steeply — in the US, 13% of the lower half of the income distribution has private pensions versus over 50% of the upper half — and tax deductions deliver their value to higher earners first, making untargeted incentives regressive. The fixed-amount subsidies and matching contributions that target low earners (Australia's co-contribution, Colombia's BEPS, Chile's young-worker match) exist precisely because the main tax incentives don't reach them.

Housing is the other great store of household wealth — and the one most retirees actually hold. Home equity functions as the precautionary reserve of last resort: in the US it is the primary resource for health shocks after insurance, and equity-release products are commonly used to pay off outstanding debt at retirement (established in Section 20). The 50-plus generation's spending is dominated by housing (51% of the category total, Section 9), reflecting both the wealth embedded in homes and the cost of occupying them.

The evidence gaps

Two gaps matter for honesty. First, the retrieved evidence contains no direct figures on household debt levels or trajectories — mortgage debt, consumer debt, or how indebtedness at retirement varies across countries. The title's debt dimension cannot be quantified from these chunks. Second, non-pension financial savings (bank deposits, equities held directly) appear only through the tax-regime lens, not as measured stocks. What the evidence does establish is the policy architecture around savings — generous, complex, and skewed toward higher earners — and the central role of housing wealth as the de facto retirement reserve for households the pension system reaches least.

Unlocking housing wealth: home equity release products and reverse mortgages

A note on evidence first: this prompt's retrieved chunks contain no new material on home equity release — the corpus's treatment of it sits in the Pensions Outlook 2024 chapter on home equity release products, already drawn on in the decumulation section. What follows carries that evidence forward without pretending to new findings.

The products

Home equity release comes in three forms: the loan-type reverse mortgage (debt secured on the home, repayable on death or sale); home reversion (selling all or part of the home's equity while retaining the right to live there); and sell-and-rent-back schemes (selling the home and renting it back). Several jurisdictions run government-backed programs alongside private providers.⁶

Who uses them and why

The established pattern is distributional: in the US, HECM reverse mortgages have high penetration among low-income groups, nearly two-thirds of borrowers are single, and around 70% of those are women — the market serves people with thin pensions. In Korea, JTYK loans concentrate among wealthier high-rise owners; in Italy, people view equity release as a way to meet basic needs where income falls short. Common uses: paying off outstanding debt (a UK and Japan pattern), funding one-off costs like home repairs (France has a product targeting lower-means households for this), bridging early or gradual retirement, managing sequencing risk via a line of credit drawn during market downturns, and financing of last resort for long-term care or divorce. Australia has a product specifically targeting those needing long-term care, designed to buy time before a home sale.⁶

Suitability is narrow

The products are best suited to older homeowners who intend to stay in their homes for life and do not prioritize leaving the house as a bequest — equity release reduces what heirs receive, either by increasing the debt they must clear or by shrinking owned equity. They are poorly suited to younger retirees (debt compounds over a longer period, releasing less equity at higher cost), to those who plan to downsize (selling earlier and taking the capital directly is more efficient than paying for an equity-release product first), and to anyone who may later need the equity for aged care — Australia's ASIC found a significant share of reverse-mortgage borrowers at risk of lacking funds for average upfront aged-care costs. Using equity for other purposes can leave nothing for the moment of last resort.⁶

Why uptake stays low

Barriers are well documented: poor product reputation and distrust of providers; debt aversion after a lifetime of paying off a mortgage; emotional attachment to the home; unwillingness to reduce bequests; product inflexibility when circumstances change; and thin access to financial advice — with advisors sometimes reluctant to recommend these products for fear of regulatory liability. In several jurisdictions, financial advice is effectively required to obtain a reverse mortgage, which raises the bar further for the low-income borrowers the products most suit.⁶

The regulatory bottom line

The OECD's conclusion: equity release can be a valuable additional retirement resource, but regulation must ensure suitability, control homeowner risks, and let providers manage theirs. Products should be targeted at those most likely to benefit — older, stay-in-place homeowners with genuine income shortfalls — not sold as lifestyle enhancers. And where housing equity is the financing of last resort, spending it early on discretionary purposes risks leaving nothing when there are no other options. For the 100-year life, housing wealth is the largest untapped reserve in most household balance sheets; unlocking it safely remains an unsolved design problem, not a solved one.⁶

Financial resilience across key life events and transitions

A 100-year life is not a 40-year career followed by a 20-year retirement. It is a sequence of transitions — education, jobs, caregiving, illness, job loss, divorce, disability, long-term care — each carrying financial consequences that compound over decades. Financial resilience means being able to absorb these shocks without permanent damage to lifetime income. The evidence shows that most systems are built for the smooth path, not the shocks.

The shocks that matter most

Long-term care is the largest financial shock most households will ever face, and the one social protection covers most unevenly. Someone with median income receiving home care for moderate needs may have as little as 6% of the cost paid by the social protection system in some US states, compared with about 45% in the Czech Republic and Israel and almost 100% in Sweden, Iceland, and the Netherlands. The impact is felt most acutely by the poor: low-income older people in Slovenia and Korea are already below the poverty threshold before care costs, and would need to spend around a third of their disposable income on long-term care; even in France and the Czech Republic, out-of-pocket costs would push them well below the poverty line. In no OECD country is anyone denied institutional care for low income — but inadequate protection makes staying at home, the option most countries aim for, inaccessible to the poor.³

Informal care is not free either. Carers give up paid work and leisure: informal carers have 20% more mental health problems than others and are more likely to stop working or reduce hours — costs borne disproportionately by women, who make up 55–70% of informal carers in OECD countries. In countries spending less than 1% of GDP on publicly funded long-term care, women are 41% more likely than men to provide daily informal care, versus 23% where spending exceeds 2% of GDP. And the care burden does not fade with women's careers: women with full careers are no less likely to provide informal care than those who spent a decade or more out of work — meaning longer careers plus continued caregiving could deepen women's long-term impoverishment without carer support. In emerging economies, where formal long-term care systems are still nascent, the traditional family support system is eroding as birth rates fall and younger generations move away — demanding a faster policy response.³

Prevention and the working-age buffer

Many shocks are cheaper to prevent than to absorb. Only around 3% of health spending in OECD countries goes to prevention, yet well-designed prevention policies are effective — and fiscal measures like taxes on tobacco and alcohol are the one intervention producing consistently larger health gains among poorer groups, thanks to greater price responsiveness. Prevention also protects financial resilience directly: better health means fewer sick-leave and disability claims and longer working lives.³

Job quality is the other working-age buffer. Enhancing it pays what the OECD calls a triple dividend: healthier and longer working lives for individuals, more productive workers for firms, and a lower burden on social protection. Gradual withdrawal through phased or partial retirement — with pension systems that don't penalize combining pensions with reduced-hours work — extends working life and enhances entitlements. Flexible work arrangements for carers (Canada's right to request flexible working; England's Care Act duties on local authorities) fight the employment, earnings, and pension losses caregiving imposes, particularly on women.³

The multistage frame — and a new risk

The deeper reframing is the multistage life itself: everyone contributes and benefits at different times — entrepreneurial ventures, career shifts, informal care — and systems must support transitions rather than assume one continuous career. Financial resilience across life events means flexible, inclusive systems: lifelong learning, carer support, adequate savings, and health — the WEF's first longevity principle (established in Section 13).⁸

A new transition risk is emerging on top of the familiar ones. The WHO projects climate change will cause approximately 489,000 additional deaths per year between 2030 and 2050 from heat-related illness, occupational hazards, and extreme weather; the ILO warns heat stress could cost the equivalent of 80 million full-time jobs by 2030, with a 2.2% decline in global working hours and \$2.4 trillion in economic losses. Older adults, outdoor workers, and those with pre-existing conditions face the greatest health risks, and marginalized communities the greatest financial ones. As working lives lengthen, climate-driven career disruptions — in agriculture, construction, manufacturing, energy — become a retirement-income issue too.⁸

Financial resilience, then, is not a personal virtue but a system property: the capacity of pensions, health systems, labor markets, and social protection to carry people through the shocks a century of life guarantees — including ones, like extreme heat, that previous generations of pension policy never had to price.



Exhibit. Climate change is reshaping economies, communities and public health systems worldwide, presenting significant challenges and considerations for the longevity economy. As extreme weather events become more frequent and e. Source: S0010.

Impartial financial education and longevity literacy

Evidence note: this prompt's retrieval was sparse — most chunks were table-of-contents pages, reference lists, or endnotes. This section builds on the WEF's second longevity principle (universal access to impartial financial education, established in Section 13) plus the advice and shock-preparedness evidence in the retrieved chunks.

Financial education is the second of the WEF's six longevity-economy principles, and its absence is visible everywhere the paper has documented it. The dashboard and guidance section (Section 25) showed people failing the basic decisions: general communication helps little, and most retirees cannot pick the best withdrawal option on their own. What people lack is not only information but impartial information — education separated from sales.

What the evidence shows

The advice gap is partly a supply problem: around 30% of UK survey respondents don't seek advice because they don't earn enough, find it too expensive, or don't have enough savings — and advisors have been dropping less profitable clients they can no longer afford to serve. The main drivers of reluctance are lack of trust, lack of awareness, and cost.⁶

Digital channels could close the gap cheaply, but engagement is shallow. In Australia, only 37% of people who recently considered advice were open to a digital channel, and 40% felt it would be more valuable if a person explained the tool's results. In the UK, 35% prefer in-person support; asked about taking advice from an AI tool, 36% said they never would, even as the technology improves. Impartial guidance, then, must be human-backed to be trusted — purely automated education leaves the least engaged behind.⁶

The cost of low literacy shows up in shock preparedness. Around 40% of lower-income people in the US and Canada report being financially unprepared for long-term care; among UK retirees, 31% have no idea how to prepare for LTC costs and 27% have no plans to set money aside; in Europe, an estimated 25% of elderly people needing long-term care could not finance even 10% of their expenses with all their financial resources. Most over-85s have no realistic view of how they will pay for care. This is not just a savings problem — it is a literacy problem: people cannot prepare for risks they do not understand.⁶

Even the willing struggle with opacity. The actual fees members bear are hard to compare across countries: aggregated fee figures reflect fee structures and system maturity, not lifetime cost to members; fees pay for different service levels; and some indirect charges that reduce pension pots may still need to be uncovered and disclosed. Impartiality without comparability is an empty promise — consumers cannot be educated on numbers they cannot see.⁴

Longevity literacy

Longevity literacy is the specific form financial education must take in a 100-year life: understanding how long one might live, what income a given pot produces over decades, and how compounding shocks — caregiving, divorce, LTC — erode it. The dashboards section (Section 25) argued that dashboards fail if they show nominal balances; financial education fails if it teaches accumulation without decumulation, returns without fees, and life expectancy without inequality in life expectancy.

The design principles emerging from this evidence: education must be impartial (separated from product sales, with supervisors ensuring nudges serve consumers rather than commercial gain); it must be human-backed where trust requires it; it must teach decumulation, shock preparation, and fee literacy — not just saving; and it must reach the people the advice market drops. The least capable, as the OECD noted, need advice most and get it least. Closing the advice gap is a financial-education problem before it is a product problem.

Behavioral nudges: auto-enrolment and better payout defaults

Evidence note: this prompt's chunks contained no new behavioral evidence — demographics tables, pension-reform catalogues, and reference lists. This section carries forward the nudge evidence established in the coverage, decumulation, and dashboards sections, stated rather than repeated, and adds the default-architecture logic that ties them together.

The most effective retirement policy of the last two decades was a nudge. Auto-enrolment — defaulting workers into pension plans with an opt-out — raised coverage dramatically in the countries that adopted it (New Zealand, Lithuania, the United Kingdom, Slovakia, Türkiye, Poland; established in Sections 14 and 19), because defaults beat incentives and inertia beats intention. The UK's Nest goes further: it is legally required to accept any employer that wants to use it, removing the last barrier for small firms. The design lesson from the coverage-gap evidence: coverage follows the worker when the default follows the worker.⁶

Defaults matter even more at the payout stage, where the decision is irreversible and the cognitive load heaviest. When UK retirees are given full freedom, only 8.3% of those accessing a pension for the first time (30,395 of 367,592 in Q4 2023) chose an annuity — freedom defaults to drawdown, for better or worse. The OECD's payout framework (Section 20) implies a better default: essential spending financed by guaranteed income, shocks covered by liquid buffers, discretionary and bequest goals funded through drawdown. The rational default architecture is therefore blended — partial annuitization plus managed drawdown — exactly what products like Nest's Guided Retirement Fund, Aviva's Guided Retirement, and Smart Pension's Smart Retire already combine.⁶

Investment pathways show nudges can work within freedom. The UK FCA's investment pathways — four simple options presented to non-advised drawdown customers — achieved 50% take-up in Q1 2023, a rare success for guided choice at the point of withdrawal (established in Section 25). The Dutch AFM's choice-architecture guidelines add the governance layer: nudges must be designed around how people actually decide, not how rational agents would.⁶

The limits of nudging are equally instructive. Defaults must be reversible and harm-free — the Dutch guidelines' twin tests — because one person's good default is another's trap (lifetime income defaults suit large pots, not small ones), and the OECD warns that defaults rely on inertia that perpetuates disengagement and can be irreversible at payout. Nudging also presumes engagement that the advice-gap evidence shows is missing: the people who most need defaults are the least likely to understand them, and supervisors must ensure that nudges serve consumers rather than the providers' commercial gain (Section 25). Auto-enrolment's success is partly that it works without any engagement at all; payout nudges, by contrast, require comprehension to be genuinely consensual.⁶

Three design principles follow. First, make the good outcome the default at every transition — enrolment, contribution escalation, investment allocation, payout shape — since each defaulted decision loses a fraction of the population to inertia. Second, segment defaults: auto-enrolment works universally, but payout defaults must vary by pot size, health status, and homeownership, the dimensions where one-size-fits-all breaks — the OECD's own default design uses accumulated-savings thresholds such as Nest's GBP 10,000 floor. Third, pair defaults with opt-out friction, not opt-out difficulty: the Nest accept-all rule shows that the right nudge removes barriers for providers too, not just for savers.⁶

Nudges are not a substitute for adequate systems — a well-designed default in a thin plan still produces a thin pension. But in the coverage and decumulation evidence, defaults were the single most powerful lever identified: stronger than tax incentives, stronger than communication campaigns. In a 100-year life with dozens of financial transitions, designing the defaults well is the closest policy gets to choosing for people without choosing at all.

Financial needs in retirement: essential, unexpected, and discretionary spending

The payout problem in retirement is a matching problem: different kinds of spending need different kinds of income. The OECD's framework divides retirement needs into three buckets — essential spending, unexpected shocks, and discretionary spending — and matches each to the product features that serve it best.⁶

Essential spending: guaranteed income

Essential needs — housing, food, transportation, healthcare — call for guaranteed lifetime income: a constant stream of income for life, well suited to covering these costs and in line with retirees' preferences. In a survey

across 15 European countries, security was the top priority when investing for retirement — around 80% chose safety over performance (women more strongly) — and 62% preferred an annuity solution for at least part of their savings, though most still preferred a lump sum when shown projections; in the UK, 92% felt a guaranteed regular income was important; in the US, guaranteed lifetime income was the top improvement identified for workplace retirement plans.⁶

The guaranteed income the PAYG public pension provides often covers basic needs — but not for everyone. The self-employed, for example, receive a median public pension of only 78% of what regular employees get across 15 OECD countries, and in Germany only around half. Where public pensions fall short, annuities complement them — immediate lifetime annuities, deferred annuities starting payments later, or payments rising by a fixed amount (a cheaper inflation hedge for providers than full indexation, and often more cost-effective than indexed annuities).⁶

Delayed claiming is the cheapest way to buy longevity and inflation protection together. Public pensions generally provide inflation protection, and boosting them through later claiming is more cost-effective than buying the same protection privately: up to 36% more in Quebec's public pension through delayed claiming, and 75% more in the US by claiming at 70 instead of 62. Using retirement savings to bridge the first few years — via a fixed-term annuity or programmed withdrawals — until the public pension is claimed can be more efficient than buying an immediate annuity. For early retirees (often forced out by job loss, poor health, or caregiving, before public-pension eligibility), interim solutions like fixed-term annuities or drawdown fill the gap. And where savings cannot purchase a meaningful lifetime income at all, lump sums or drawdown are the better choice.⁶

Unexpected spending: liquid buffers

Retirees should hold safe, liquid resources for shocks that regular income cannot finance — health events, home repairs, loss of a spouse. Over 40% of US retirees below 80 have experienced an unexpected financial shock, nearly a quarter of them from major home repairs; precautionary saving for health reasons is the main savings motive for retirees in Australia and the Netherlands, stronger for women than men. Spouse loss and divorce are among the most damaging: a third of US adults with a recently deceased parent said it reduced the surviving parent's financial security, and divorce can harm retirement finances even more than spousal death. Insurance is the cost-effective answer where risks are insurable at affordable prices — particularly valuable for low-to-middle-income households. But the two risks retirees are least prepared for — long-term care and divorce — are also the least insurable.⁶

Discretionary spending: flexibility, but with structure

After essentials and security needs are met, remaining resources fund discretionary lifestyle spending — regular comforts plus occasional holidays and large purchases. People value flexibility almost as much as security: nearly 90% of the same UK respondents who called guaranteed income important also called flexible access important. Regular income still matters for wellbeing: annuitized retirees report higher and more durable satisfaction; 90% of US annuity holders said their budget became more predictable and paying for needs easier; those with defined-benefit income draw down wealth more slowly, suggesting people budget against annuitized income. But reference points mislead: over half of Australian retirees over 65 draw only the minimum required rate from superannuation, and 90% of US retirees treat Required Minimum Distribution rates as important guides to withdrawal — often underspending and living more frugally than they could afford. Variable lifetime income products (low or no guarantees) can raise expected income while preserving a reference income for budgeting, accepting volatility as the trade-off — reasonable for spending that is not strictly necessary. Managed drawdown, where the provider sets withdrawals dynamically against investment and longevity risk (possibly combined with a deferred annuity), can optimize income while limiting the risk of running out. And discretionary spending needs less inflation protection: consumption declines with age across all income levels, mainly because enjoyment of spending declines — travel and leisure peak early in retirement then fall.⁶

The framework's power is architectural: essentials guaranteed, shocks buffered in liquidity, discretionary spending given flexible products with reference incomes for budgeting. Section 20's decumulation analysis and Section 30's payout defaults implement exactly this logic. The failure mode is also clear: matching works only when retirees have enough resources in each bucket — thin public pensions, uninsurable shocks, and underspending against bad reference points are where the framework breaks.

Financial security for gig, contract, and informal workers

Globally, an estimated 60% of workers operate in the informal economy, and the gig workforce is growing rapidly. These workers sit outside the architecture pension systems were built around: contributory schemes depend on stable employer–employee relationships, and the irregular income of gig and informal work makes consistent contributions hard — leaving workers vulnerable in old age and perpetuating inequalities. This section addresses financial security; the coverage mechanics were covered in Section 19.⁸

The design gap has three layers. First, portability: pension systems often lack it, making it hard for workers who change jobs frequently or move across borders to build adequate savings. For platform and migrant workers, contributions fragment across jobs and jurisdictions — each small balance on its own insufficient, and lost where systems don't talk to each other. Second, contribution regularity: schemes assume payroll-deducted monthly contributions from a single employer; workers paid per gig, seasonally, or in cash need flexible contribution windows and low minimums. Third, behavioral architecture: auto-enrolment, the most powerful coverage lever, has no employer to attach to when there is no employer.⁸

The OECD's policy agenda directly addresses this group: increase pension coverage especially for the self-employed and those with nonstandard jobs — including through improved financial literacy — and move towards a unified pension framework for all workers, targeting adequate retirement income through a combination of old-age safety nets, mandatory pensions, annuities in private schemes, and pension credits. The unified-framework recommendation matters: separate schemes for the self-employed and informal workers, where they exist, tend to offer lower benefits — the self-employed receive a median public pension of only 78% of employees' across 15 OECD countries, and around half in Germany (established in Section 31).³

Financial incentives can be retargeted to irregular earners where tax relief alone fails them. The OECD's inventory of non-tax incentives shows the available machinery: government matching contributions — Australia's 50% co-contribution up to AUD 500 a year, Chile's young low-wage worker matching, Colombia's 20% BEPS matching — and fixed nominal subsidies such as Chile's 18 months at 10% of the minimum wage and Estonia's 4% of the national average wage. Matching and flat subsidies, unlike tax deductions, pay value to low earners with irregular incomes who have little tax liability to deduct against. The design question is eligibility plumbing: making these incentives accessible without an employer as intermediary — through individual accounts, platform withholding, or mobile-money rails.⁶

The scale of the challenge is generational. In OECD countries, nonstandard work — part-time, temporary, self-employment, gig — keeps expanding, while in developing economies most of tomorrow's retirees are in today's informal economy. A pension system that secures only the standard employment relationship will cover a shrinking share of workers as that relationship shrinks. The direction is clear: universal individual accounts that follow the worker, portable across jobs and borders; flexible, low-minimum contributions; matching incentives that reach irregular earners; and an adequate first-tier floor so that workers whose contributions are unavoidably thin still retire above poverty. As gig and informal work reshapes labor markets, public pension systems must evolve to include those workers or risk leaving a growing segment of the population unprotected in retirement.⁸



Exhibit. For gig workers or those taking time off as carers, this model can be especially beneficial by providing a dedicated emergency fund that supports them during periods of uncertainty. Addressing both the need for liquidity. Source: S0010.

Work and multi-stage careers

Labor-force participation of older workers

Economic activity rates for older people in the EU have gradually increased during the last three decades. This is the supply side of the longevity economy: longer lives only pay for themselves if more years are spent working. The rise reflects pension reforms that tightened early-retirement pathways, healthier ageing, and — increasingly — flexible working patterns that let people remain in work until a later age, increasing their income and reducing reliance on government support. The effective exit age, after falling for most of the 20th century, has rebounded since around 2000 and now tracks life expectancy at 65 (established in Section 17).¹

The barriers that remain

Care responsibilities are the largest structural barrier to longer participation, and they bite hardest at the ages where participation gains are needed. On average across OECD countries, 61% of those providing daily informal care are women — and as populations age, the care burden threatens to undo recent gains in female labor-force participation. The high cost of childcare and long-term care prevents re-entry or pushes women out of current positions toward work-from-home alternatives; more generous childcare support (public provision or subsidies) is key to sustaining participation. The price of care is a participation cost: the typical gross childcare fee for two children (ages two and three) in full-time centre-based care averages just under 26% of earnings across the OECD — but reaches 50% or more in Australia, Greece, Japan, Luxembourg, the Netherlands, New Zealand, and Switzerland, versus as little as 1% in Germany.¹¹

Care leave exists but is underused. Most OECD countries provide paid leave for sick children or family members; 19 of 35 provide paid leave to care for an older dependent (mostly in Europe — outside it only Japan, and only Canada, Japan, and Korea offer unpaid leave), with 22 of 35 providing paid or unpaid leave for older dependents. But stigma surrounds its use: firms and governments must make benefits visible and advertise them, foster internal caregiver communities, and model senior leaders' own use of care leave.¹¹

Longer leave is not automatically better: most European evidence finds that longer maternity leave delays mothers' return to work without lasting effects — positive or negative — on pay and participation. The lever is affordable care, not extended absence.¹¹

Extending careers: what governments can do

In the last decade, many OECD countries have raised the retirement age above 65 — to 67 and higher. But the OECD's message is that ages alone are insufficient; governments should encourage longer and more satisfying careers for those who are able through flexibility: promoting phased retirement, better balancing work and care, and permitting combinations of pensions with work income. Canada, Denmark, Iceland, Norway, Portugal, Switzerland, and the United States have maximum pension deferral ages of at least 70. Conversely, countries should restrict publicly funded early-retirement schemes that pay workers to leave employment while still healthy and able to work.¹¹

The deeper point concerns the intergenerational bargain. Older people in the EU are relatively well off — often at lower poverty risk than younger age groups — but this divide is likely to grow, with younger generations under pressure to save more and work longer. Participation of older workers is not just a fiscal measure; it is what legitimizes the rising retirement ages and contribution rates the younger generation is being asked to accept. And it pays the participants: retirees in good health who keep working, learn, volunteer, and stay socially active report higher life satisfaction than their inactive peers — in several western and northern EU Member States, older people reported higher life satisfaction than other age groups.¹

The frontier is no longer getting people to stay — participation is rising. It is getting them to stay on good terms: flexible patterns, care support that doesn't force exit, and retirement transitions designed around capability rather than the calendar.

Longer working lives and the lump-of-labor fallacy

Evidence note: this prompt's retrieval contained no direct econometric test of the lump-of-labor hypothesis. This section builds on the older-worker participation evidence (Sections 17, 33), the World Bank's East Asia and Pacific ageing analysis in the retrieved chunks, and the logic of the fallacy itself.

The lump-of-labor fallacy is the belief that work is a fixed quantity: if older people stay employed longer, younger people lose jobs. It underpins much resistance to raising retirement ages and to longer working lives — and it is wrong in its premise. Labor markets are not zero-sum: economies adjust, wages clear, and experienced workers complement rather than substitute for younger ones. The World Bank's East Asia and Pacific ageing report — the most relevant evidence in this prompt — frames the question the right way: the issue is not whether longer working lives crowd out the young, but whether countries can augment their labor forces enough to sustain growth as populations age.⁷

The stakes are clearest where ageing outruns income. Developing countries in East Asia and the Pacific are getting old before getting rich: Thailand's pace of ageing is outstripping per capita income growth, and orange (largely ageing middle-income) countries, even with sustained productivity growth, will not reach the income levels of richer ageing economies at similar demographic stages. The demographic dividend — estimated at around one-third of East Asia's high per-capita income growth from 1960 to the 1990s, with some estimates as high as 44% — is fading; younger countries need deeper reforms to capture similar effects. The relevant lever is not rationing jobs by age but productivity-enhancing reform plus labor-force augmentation across channels: longer working lives, higher female participation, immigration, and skills.⁷

The EAP evidence is reassuring on feasibility: people in the region already have long working lives; entitlements are still modest, leaving scope to build sustainable systems now; household savings are high and people save late into life; and healthy life years have been rising steadily. The key message is that rapid ageing can be managed while

sustaining economic dynamism — but it requires politically difficult choices and complementary reforms across labor markets, pensions, health, and long-term care.⁷

The OECD experience reinforces the point differently. The last two decades saw both rising employment of older workers and, through most of the period, strong prime-age employment — exit ages rebounding since 2000 while labor markets absorbed younger cohorts (established in Sections 17 and 33). Retirement ages are rising in more than half of OECD countries. The real constraints on longer working lives are on the supply side, not the demand side: care responsibilities, health, and age discrimination — not a shortage of jobs to go around. The binding problem identified in Section 17 was employment shortage near retirement: people who want to work longer can't find suitable work, while others are pushed out through publicly funded early-retirement schemes that pay healthy workers to leave (Section 33).⁸

None of this means every older worker staying on is costless for younger workers in every context. The fallacy is in the “fixed lump”; the reality is that labor markets adjust imperfectly and slowly. When older workers stay, they also spend, invest, and create demand — and the pension savings they generate reduce fiscal pressure that would otherwise fall on the young. The countries that have combined rising retirement ages with strong youth employment — the Nordics, Switzerland, Japan — did so with flexible labor markets and lifelong learning, not with job rationing.

The policy conclusion follows Section 33: encourage longer and more satisfying careers for those who are able, through flexible transitions and care support — while restricting early-retirement schemes that pay the healthy to leave. The lump-of-labor fear is a reason to reform labor markets, not to defend retirement ages the demography can no longer afford.

Barriers: health, job quality, ageism, and premature exit

Longer working lives are demanded by demography, encouraged by policy, and desired by many workers. They are blocked by four things: bodies that wear out, jobs that don't, employers that don't want older workers, and exit routes that pay people to leave early. The binding constraint on longer working lives, as Section 17 established, is not health in the abstract — it is employment: workers who could and would continue cannot find suitable work.

Health: capability ends unevenly

The capability gap is steepest where it matters most. About 30% of men aged 50–64 with low education report being limited in paid work because of a health problem, against only 10% of those with high education. Disability inequalities by gender and education are similar across most countries studied; women report disabilities more often in both the 50–64 and 65+ groups (non-fatal but disabling conditions like arthritis and depression), while men suffer more fatal diseases. Disabling chronic disease concentrates among the lower-educated through lower incomes, worse working and living conditions, behavioral risk factors, and poorer healthcare access.³

The backdrop is a stalled longevity engine. Life-expectancy gains had been slowing markedly before COVID-19 — most sharply in the US, France, the Netherlands, Germany, and the UK — driven by fading improvements in heart-disease and stroke mortality, rising obesity and diabetes, and population ageing. The pandemic then wiped out a decade of gains in seven OECD countries (Czech Republic, Greece, Hungary, Latvia, Poland, Slovak Republic, United States). The assumption that each generation will be healthier for longer — the premise of rising retirement ages — is no longer advancing on schedule.⁵

Job quality and the tenure erosion

Job quality is deteriorating fastest for the workers who most need to stay. Average job tenure declined between 2012 and 2019 for mature and prime-aged workers in almost all OECD countries studied — and the largest declines hit low-educated workers across all age groups and both sexes, with reductions exceeding 10% in Lithuania, Ireland, Luxembourg, Estonia, and Sweden. Tenure also varies sharply by sector and firm size: around 16 years in agriculture, forestry, and fishing versus about six years in accommodation and food services; elementary occupations average about eight years; and large firms retain staff better than small ones. The sectors with the worst tenure — hospitality, elementary occupations — are also the ones employing the workers with the worst health, lowest education, and weakest pension coverage: disadvantage concentrates.¹¹

Ageism and premature exit

Direct evidence on ageism is not in these chunks, but its footprints are. When exit ages rebounded after 2000, the gains were uneven by education and occupation: tertiary-educated employment at older ages surged while below-upper-secondary employment barely moved (established in earlier sections). Older workers who lose jobs face the longest unemployment durations and the steepest re-employment wage cuts in the labor market — a pattern consistent with hiring discrimination, not with productivity decline alone. The policy literature's recommendations read as a diagnosis: eliminate special employment protection and benefit rules for older workers that backfire, match labor costs to productivity, and fund return-to-work and rehabilitation — all barriers the OECD's retaining-talent agenda identifies.¹¹

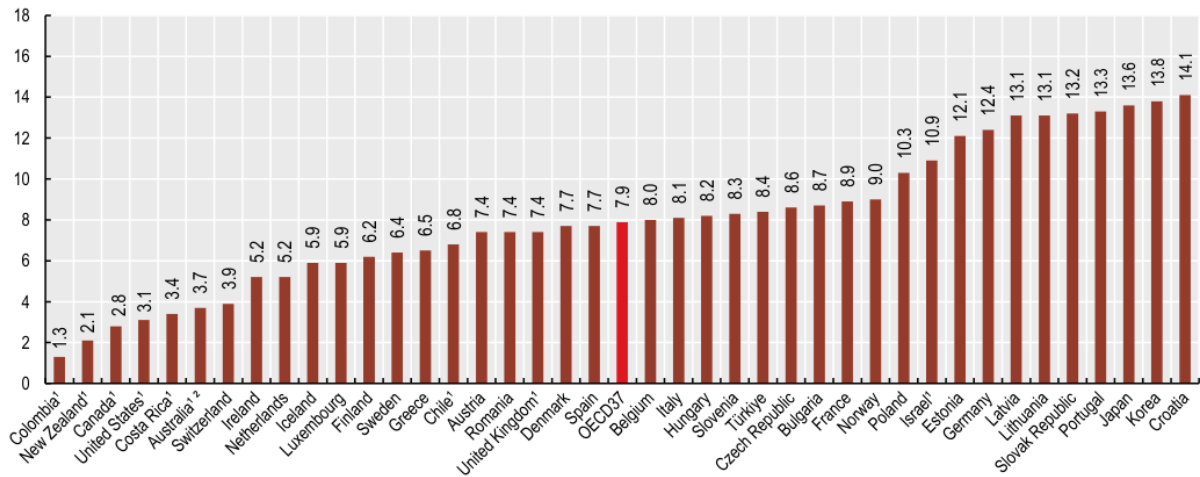
Premature exit also remains subsidized. Publicly funded early-retirement schemes that pay healthy workers to leave employment persist in several countries, and Section 33 documented the OECD's call to restrict them. The lump-of-labor logic of Section 34 would suggest early exit helps the young; the evidence shows it mainly helps no one — draining pension funds while the “freed” jobs mostly disappear.

The combined barrier

These four barriers compound. A low-educated worker in a physically demanding job faces higher disability risk, shorter job tenure, worse re-employment prospects if displaced, and a pension system whose rising retirement age assumes a capability she may not have. Raising the statutory age without removing the barriers punishes precisely this worker — while the healthy, tertiary-educated worker who could retire later often retires at the same age anyway. The OECD's policy agenda (established in Section 32's source material) prescribes the working-age response: job-quality regulation, rehabilitation and return-to-work, skills upgrading for low-skilled and older workers, and hiring-barrier removal — because by retirement age, the damage is done.

Longer working lives will be achieved through working-age policy or not at all.

% of population aged 15 years and over



Notes: 1. Results for these countries are not directly comparable with those for other countries owing to methodological differences in the survey questionnaire resulting in a bias towards a more positive self-assessment of health. 2. Most recent data point corresponds to 2017.
Source: OECD Health Statistics 2023 (EU-SILC for EU countries).

StatLink <https://stat.link/qf0ej7>

Figure 3.23. Adults rating their own health as good or very good, by income quintile, 2021 (or nearest year)

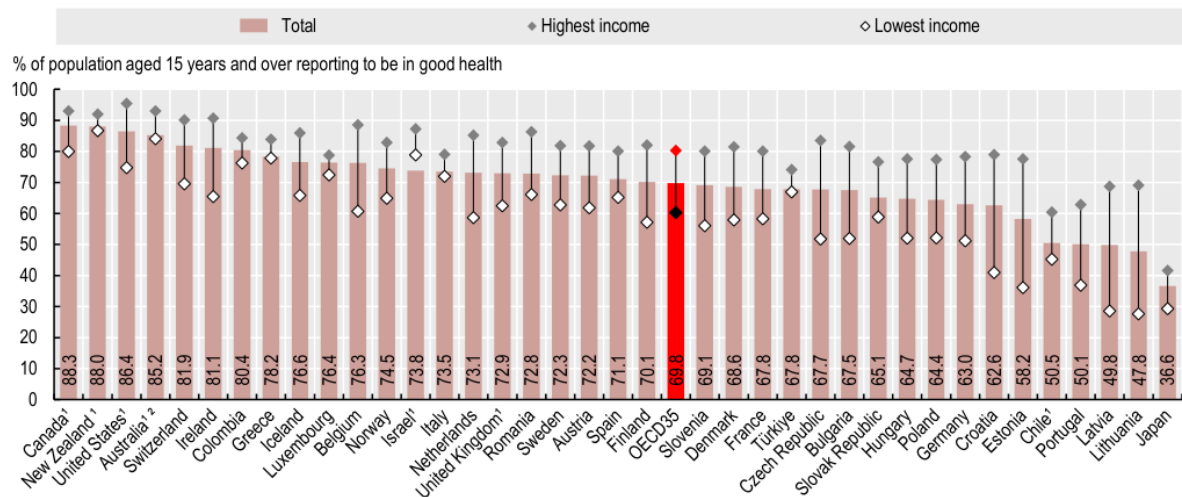


Exhibit. Figure 3.22. Adults rating their own health as bad or very bad, 2021 (or latest year) % of population aged 15 years and over 18 16 14 12 10 8 6 4 2 0 Notes: 1. Source: S0026.

Retaining talent at all ages: pay, flexibility, and job design

Keeping older workers employed is the most direct answer to ageing. It raises pension entitlements, shrinks the fiscal burden, and — as Section 33 documented — correlates with higher life satisfaction. The OECD's Retaining Talent at All Ages agenda shows what retention actually requires: it is not retirement ages but pay, working conditions, flexibility, health, and skills.

Pay: generosity retains, unfairness repels

Pay is a core component of job quality and a key determinant of quits. New OECD evidence shows workers are less likely to quit in firms with more generous wage-setting practices: on average across selected countries, the quit rate is about 50% higher in firms in the bottom quintile of firm-wage premia (the firm-specific element of wage setting, independent of worker composition). Firm wage premia reflect performance, wage-setting power, and personnel policy — all levers employers control.¹¹

Mature workers (aged 50–59) behave differently from younger ones: they are much less likely to leave their firm for another firm — but much more likely to exit the labor market altogether. This is the retention margin that matters: when older workers are dissatisfied, they don't job-hop; they retire. Average tenure for mature workers (55–64) is 18

years and 10 months — deep firm-specific capital that, once walked out the door, is expensive to replace. Retaining them is cheaper than recruiting them.¹¹

Flexibility: the care double-bind

Older workers face the sharpest care squeeze with the least flexibility. The 2022 AARP Global Employee Survey shows mature workers have much less flexibility to care for children and dependent adults than younger workers — with the least flexibility of all. Combined with the 61% female share of daily informal carers and the childcare-cost barrier documented in Section 33, this is the channel through which ageing most directly undoes participation gains. Parental leave and paid caregiving, alongside flexible working practices, are the retention tools — but only where stigma doesn't neutralize them (Section 33).¹¹

Health and skills: the two multipliers

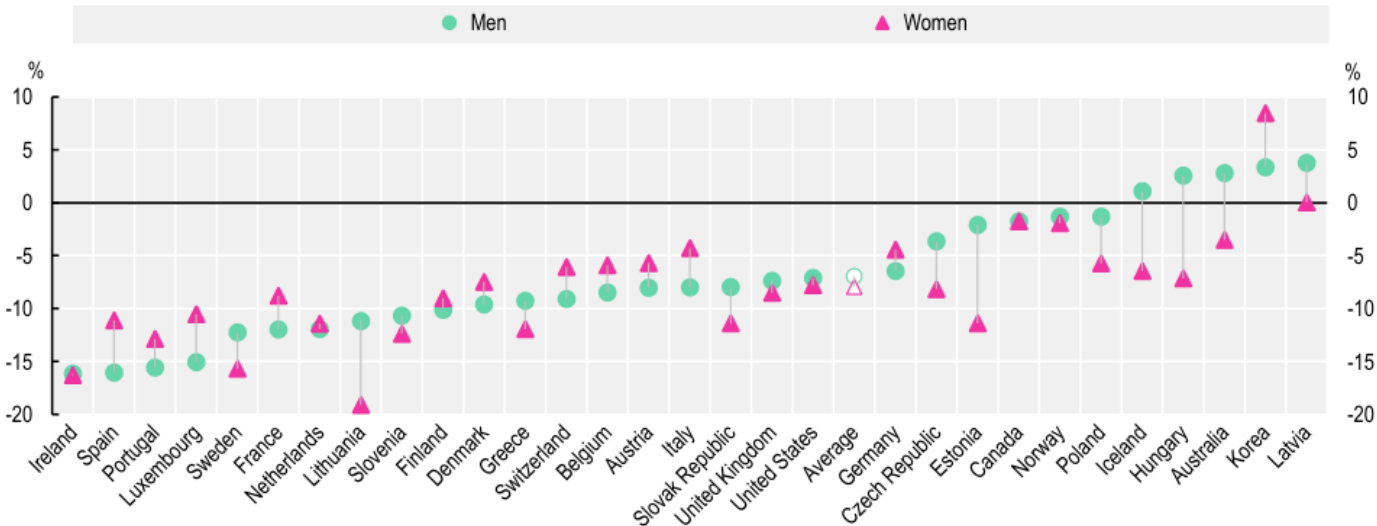
Health is the gatekeeper. Among surveyed adults in European countries, Australia, Korea, and the US, about one-third report less than good health; the share rises from 11% at age 20 to 58% at age 64 (39% among the employed). At all ages, people in bad health work less and earn less when they work — and the health-wage gap widens with age, compounding disadvantage, because healthy workers' wages rise more steeply. Health-employment gaps (in percent) run three times larger than health-wage gaps, stable at 25–35% until age 45, after which employment falls sharply for men and women in both good and bad health. Work damages health too: the causation runs both ways, which is why job-quality regulation and occupational health are retention policies, not just safety policies.³

Skills are the second multiplier, and digital skills the sharpest gap. COVID-19 drove millions into remote work and learning, making digital skills essential to access work and training at all; PIAAC's problem-solving-in-technology-rich-environments assessment shows large potential to improve older workers' digital skills. Training participation for mature workers remains low — 24% of 55–65-year-olds in job-related training, versus 46% of 35–44-year-olds (established in Section 33's source material) — with the low-skilled receiving least. Skills and retention reinforce each other: workers who train stay employable; workers who are expected to stay get trained.¹¹

Job design: the retention package

Pulling the levers together, retaining talent at all ages means: generous and fair wage-setting (cuts quits, especially the exit-to-retirement margin); flexible working patterns that accommodate care (Section 33) and phased retirement rather than cliff-edge exit; occupational health that treats prevention as a retention investment (Section 28's triple dividend); and lifelong learning targeted at the low-skilled and digitally excluded. Large firms already retain better (Section 35) — the retention deficit is concentrated in small firms, hospitality, and elementary occupations, exactly where pension coverage is thinnest.

The macro payoff is the inverse of the lump-of-labor fear (Section 34): retaining experienced workers does not take jobs from the young; it preserves firm knowledge, keeps pension systems solvent, and lets older workers finance their own longer lives. Retention is the cheapest pension reform available.



Employers' role in multigenerational workplaces

Governments set retirement ages; employers decide whether longer working lives are real or nominal. Four and five generations now share workplaces, and employers control the levers that matter most for retention: job quality, training, flexibility, phased transitions, and financial wellbeing programs that support employees at all life stages. The OECD's retaining-talent evidence (Section 36) shows what works; this section assigns it to employers.

Train across the working life — especially older workers

Promoting employability throughout working lives is a particularly important requirement for preventing unequal ageing: workers now switch jobs more often and adapt to frequently changing tasks, so upgrading skills is crucial for employment and earnings — yet groups at greater risk of labor-market disadvantage, including older workers, receive less training, compounding their disadvantage. Digitalization sharpens the gap: on average only 27% of older workers (55–64) use email or the internet at work daily, versus 49% of prime-age workers (35–44), and the age gap widens with technological complexity — digitalization accelerates the obsolescence of older workers' skills. Workers in technology-intensive occupations faced lower unemployment risk in 2008–12, but older workers were less likely to be in those occupations — and almost three times more likely than younger workers to lose their jobs in highly technology-intense occupations (vs 1.7× in low-tech ones). The case for employer-funded, high-quality professional training for older workers is empirical, not rhetorical.³

Equal-opportunity employment policies should not be age-specific; they should support at-risk groups regardless of age — a key message of the OECD Recommendation on Ageing and Employment Policies. Older workers are a diverse group: many do well, others struggle — very often because they lack skills. The policy task is twofold: mitigate the risks of obsolete skills and reduced work capacity, and harness older workers' assets — their experience — which are often not fully exploited.³

Design the transition, not just the job

Retention requires the full employer package: generous fair wage-setting (Section 36's 50% quit-rate differential); flexible patterns that accommodate care (Section 33) — critical since mature workers have the least flexibility for caregiving (Section 36); phased or partial retirement that lets pensions combine with reduced-hours work; and occupational health treated as a retention investment (Section 28's triple dividend). The Future-Proofing report assigns employers an explicit role in financial wellbeing: programmes supporting employees at all life stages — through career changes, caregiving, and the accumulation-to-decumulation handoff (its third focus area, noted in the introduction chunk).

Employers also have a de-biasing role. The tenure and exit evidence suggests hiring and retention practices quietly disadvantage older workers; organizational culture determines whether caregiving leave is actually used (Section 33); and affirmative action through information campaigns can counter negative employer attitudes toward older hires (the retaining-talent report's Box 2.5, listed among its key messages). Visible senior-leader use of care leave, internal caregiver communities, and age-neutral hiring practices are employer actions no government can mandate into existence.

The business case

The multigenerational workplace is not a compliance exercise. Mature workers average 18 years and 10 months of tenure — deep firm-specific capital (Section 36). Losing it to premature exit is a business loss as well as a fiscal one; replacing it is harder in the tight labor markets ageing itself creates. Employers who invest in training, flexibility, and age-management reap the productivity; those who don't subsidize their competitors' hiring — and the pension system absorbs the exits.

The 100-year life's labor market is multigenerational by definition. Employers who design for all ages capture the experience premium; those who design for the median 35-year-old will keep losing talent they can't afford to lose.

Flexible work, phased retirement, and partial pensions

Evidence note: this prompt's chunks contained no flexible-work or phased-retirement material — dashboards, pension taxonomy, and projection methodology. This section assembles the phased-retirement evidence established across earlier sections, stated rather than repeated.

The retirement transition need not be a cliff. Flexible work, phased retirement, and partial pensions are three versions of the same idea: letting people reduce work gradually while drawing income partially, rather than forcing a binary choice between full employment and full retirement. The OECD's retaining-talent evidence and its preventing-unequal-ageing agenda converge on this as the key policy design for longer working lives, with phased retirement and flexible work-retirement transitions among the recommended levers.¹¹

The mechanism is straightforward. Gradual withdrawal through phased or partial retirement — with pension systems that don't penalize combining pensions with reduced-hours work — extends working life and enhances entitlements (established in Section 28). The financial design: partial pensions that pay a fraction of the benefit while the worker continues contributing on part-time earnings; deferral bonuses for late claiming (up to +36% in Quebec's public pension, +75% in the US from 62 to 70 — Section 31); and bridge products — fixed-term annuities or programmed withdrawals — that finance the first years of retirement until the public pension is claimed (Section 31). Those who continue working part-time may not need additional income at the normal retirement age at all — the relevant policy question is whether the system permits the combination.⁶

The barriers are regulatory, not financial. Many systems still penalize the pension-plus-work combination: earnings tests, mandatory retirement ages, and accrual rules designed for single-career, full-time, cliff-edge retirement. The OECD's recommendation (Section 33): permit combining pensions with work income, set maximum deferral ages (Canada, Denmark, Iceland, Norway, Portugal, Switzerland, and the US have deferral ages of at least 70), and restrict publicly funded early-retirement schemes that pay healthy workers to leave. On the employer side: flexible working patterns for older workers (EU activity rates rise where flexible patterns are available — Section 33) and phased retirement as an organizational practice, not just a benefit design.¹¹

Care makes phased work a necessity, not a preference. With 61% of daily informal carers being women and mature workers having the least flexibility to provide care (Sections 33 and 36), the cliff-edge model forces carers out of the labor market entirely — destroying entitlements they then can't rebuild. Carer-friendly flexible work (Canada's right to request; England's Care Act duties) is the retention policy for this group.¹¹

The design principles: (1) neutrality — no penalty for combining pensions with reduced-hours work; (2) actuarial fairness in partial pensions and deferral bonuses, so the gradual path is not subsidized or punished relative to the cliff; (3) flexibility at the point of transition — the ability to move between full-time, part-time, and full retirement without losing accrued rights; (4) care accommodation — flexible patterns as a right, not a favor. Section 33's phased-retirement evidence shows the payoff: where the gradual path exists, exit ages rise and entitlements improve; where it doesn't, workers face the all-or-nothing choice, and many choose nothing.

A century-long working life will be built out of transitions, not one career ending in one cliff. The systems that recognize this — partial pensions, deferral bonuses, bridge products, flexible patterns — are the ones whose retirement ages can rise without punishing the workers who can't keep up.

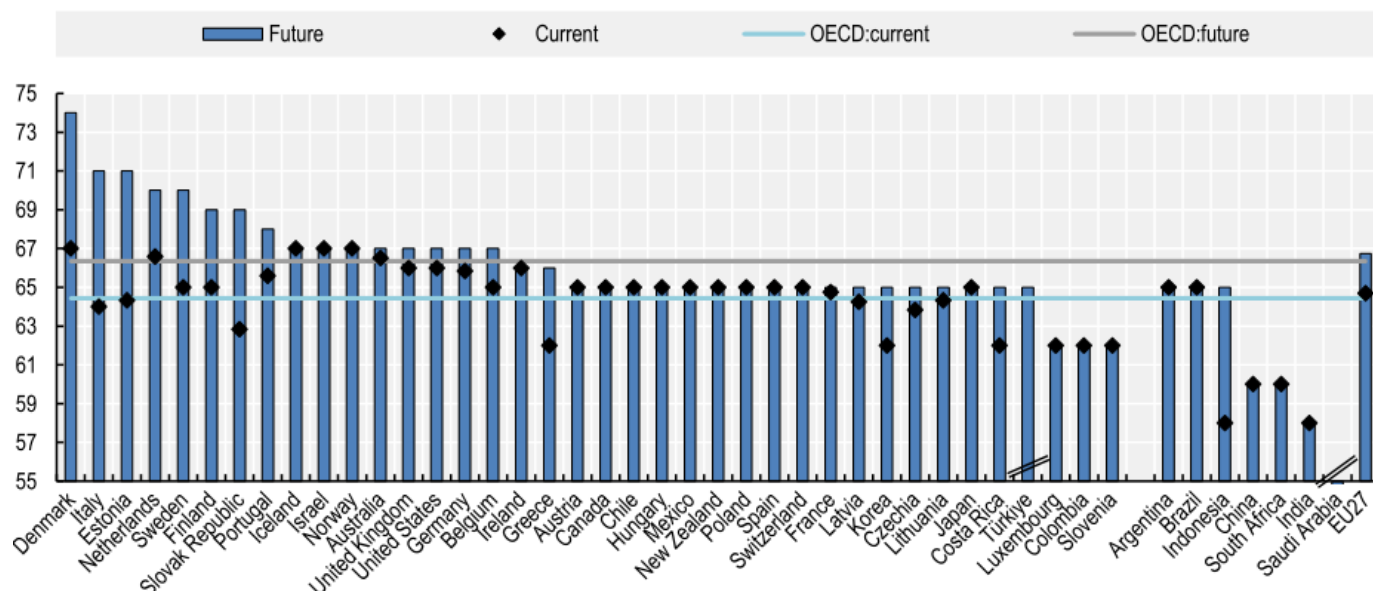


Exhibit. Figure 3.8. Current and future normal retirement ages for a man with a full career from age 22 Current and future refer to retiring 2022 and entering the labour market in 2022, respectively Future Current OECD:current OECD:future 75. Source: S0002.

Career transitions and reconciling work with care

Evidence note: this prompt's chunks were mostly reform-annex tables and reference lists; one new nugget is used below, with the career-transition and care evidence carried forward from earlier sections.

The 100-year life multiplies transitions — not just one career but several, interrupted by caregiving, retraining, job loss, and sectoral change. Pension systems built for one continuous career punish every interruption; reconciling work with care is where the punishment lands hardest, and mostly on women.

Care credits: the pension system's answer to caregiving

The most direct instrument is the care credit: recognizing caregiving in pension entitlements rather than treating it as absence. The gender-gap section (Section 24) documented the machinery: Chile's 18 months at 10% over the minimum wage invested in fund C; Estonia's 4% of national average wage for up to 3 years per child; Germany's Riester subsidies (€185/€300) to mothers. A new reform from the retrieval adds a fresh example: since January 2023, Czechia grants a CZK 500 monthly top-up to the earnings-related pension for each raised child, paid to the parent who provided most of the care. France's 2023 reform (raising the minimum retirement age from 62 to 64 and extending the full-rate contribution period from 42 to 43 years) also revisits periods spent as stay-at-home parents caring for children — the kind of reform where care-period treatment determines who bears the adjustment.⁶⁴

Care credits blunt but don't erase the gap. The evidence from earlier sections: full-career women are no less likely to provide informal eldercare than women with long career breaks (Section 28) — so care credits for children don't touch the second, later caregiving wave. And the deeper lesson of Section 24 stands: the gender pension gap is a labor-market gap wearing pension clothes; credits compensate but can't undo interrupted earnings, lost promotions, and part-time penalties.

Career transitions: the multistage labor market

Beyond care, the multistage life (Section 28's framing) assumes workers will switch careers — by choice or by disruption. The hazardous-jobs section (Section 23) showed retraining can work: ballet dancers in Australia, Switzerland, and the UK successfully pursued alternative careers (though about half self-financed retraining); US soldier reorientation programs improved employment outcomes; reskilling absorbed coal-mining declines in Germany, Poland, and the UK and fisheries in Canada. The employer section (Section 37) added the training gap: the workers most at risk — including older workers — receive the least training.

The design requirements for reconciling work and transitions: (1) flexible, low-minimum pension contributions that survive job changes (Section 32's gig-worker agenda); (2) portable entitlements so transitions don't fragment sav-

ings; (3) care credits for both child and elder care — not just the first; (4) affordable care infrastructure so the choice between work and care isn't forced (Section 33's childcare-cost evidence: ~26% of earnings on average across OECD countries, 50%+ in seven countries); (5) flexible working patterns as a right, with stigma addressed through organizational culture (Sections 33, 37).¹¹

The reconciliation of work and care is the point where every strand of this paper meets: retirement ages, gender gaps, participation, retention, and fiscal sustainability all depend on whether caregivers — disproportionately women — can keep working through the middle of long lives rather than exiting and never returning.

Employment services and policies for older workers: Japan and Korea

Evidence note: this prompt's retrieval contained no Japan- or Korea-specific employment-policy material — the chunks cover EU patterns and hazardous-job pension provisions. This section assembles the East Asian evidence established earlier and the general older-worker employment-services evidence from the retrieved chunks, flagging the gap.

Japan and Korea sit at the frontier of the ageing-work problem. Korea has the highest effective exit age for women in the OECD (67.4 years) and Japan one of the oldest populations on earth (old-age to working-age ratio 55.4 — the highest in the OECD, projected to reach 80 by 2052; established in earlier sections). Both countries' older workers stay in work at rates other countries are only legislating toward. But staying is not thriving: Korea's old-age poverty is among the highest in the OECD (relative poverty over 25% among the 66+, alongside the Baltic states — from the Pensions at a Glance key findings in these chunks), and older Koreans often work because pensions are inadequate, not because jobs are good.⁴

The EU comparison shows what the employment-services agenda should look like. People aged 55+ account for almost one-fifth of the EU-28 workforce, and more than half of workers aged 65+ are employed part-time — the phased-retirement pattern in action. Increasing numbers of older workers choose phased retirement, moving from full-time to 60% or 50% before retiring outright, while others work beyond retirement. Employers can stimulate the supply of older workers by improving working conditions — the retention lever documented in Section 36.¹

The poverty data warn against confusing participation with welfare. Almost 10% of older people in work in the EU-28 were at risk of poverty, and 17% of those aged 55–64 could not afford even a small weekly spend on themselves — a form of material deprivation systematically worse among older women. The lesson for Japan and Korea: high participation rates achieved through necessity are not a policy success to emulate; the goal is longer working lives that are chosen, well-paid, and pension-enhancing.¹

Three employment-services priorities follow for the East Asian frontier. First, skills: about 1 in 15 people aged 55–64 in the EU participated in education and training, and two-fifths of 65–74-year-olds had never used a computer — the digital divide that strands older workers in low-tech occupations. The OECD's retaining-talent agenda (Sections 36–37) prescribes high-quality professional training targeted at older workers, with digital skills as the sharpest gap. Second, job quality and working conditions: physically demanding work concentrates among the workers who can least afford to exit (Section 35); the priority is improving conditions and enabling occupational transitions, not early-retirement provisions — the hazardous-jobs chapter's core finding, that these issues belong to labor-market policy, not pensions. Third, phased retirement infrastructure: part-time employment, pension-plus-work combinations, and deferred claiming, so the transition from work to retirement is a ramp rather than a cliff (Section 38).¹⁴

Japan and Korea show both the promise and the warning of the oldest workforces in the world: working longer is possible, but without adequate pensions, good jobs, and training, it is a measure of need rather than a model of success. The countries behind them in the ageing curve have the chance to build the employment services — skills, conditions, transitions — before necessity does the building for them.

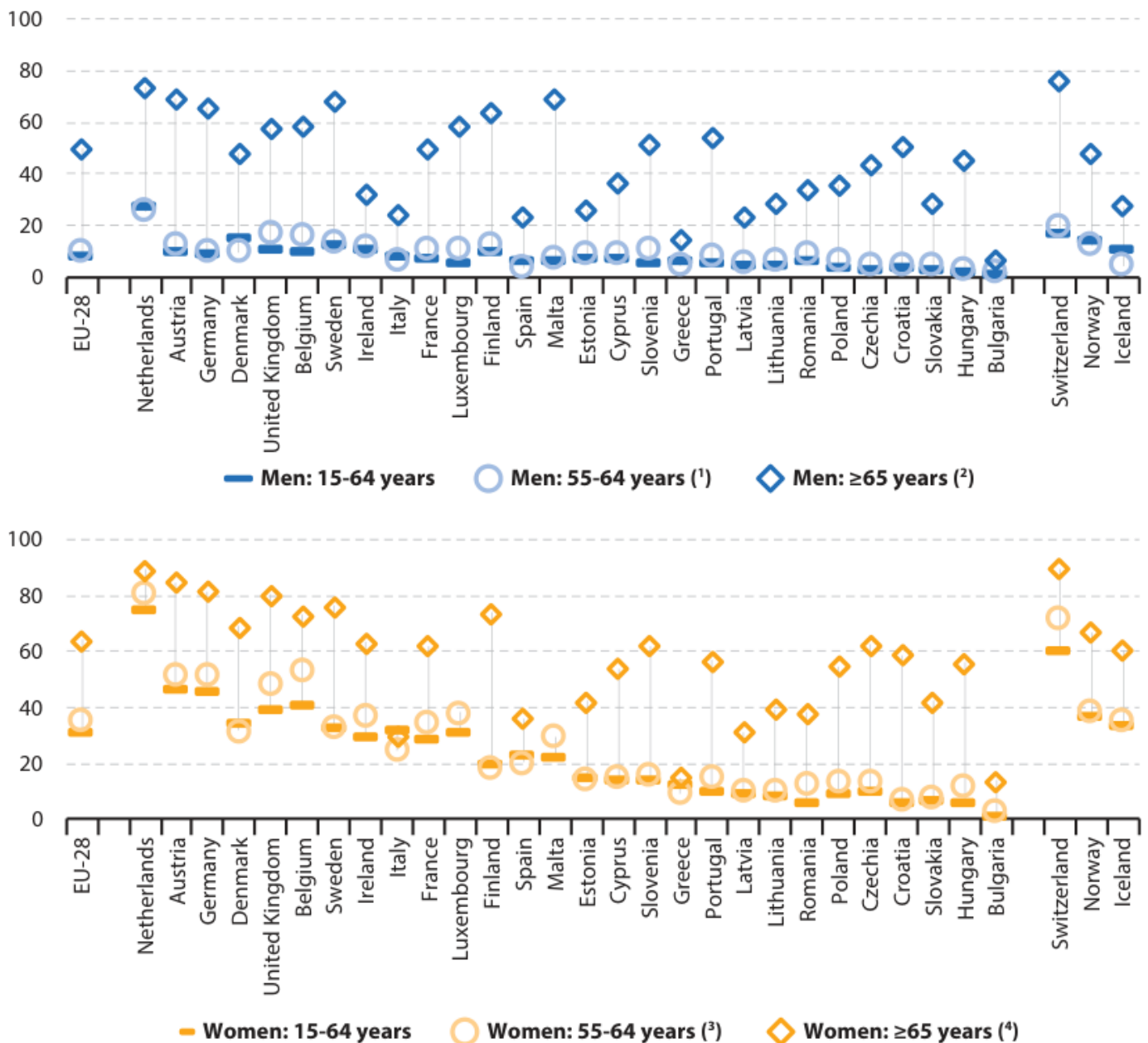


Exhibit. Figure 4.5: Part-time employment, by sex and age class, 2018 (%) 100 80 60 40 20 0 Men: 15-64 years Men: 55-64 years (1) Men: ≥65 years (2) 100 80 60 40 20 0 Women: 15-64 years Women: 55-64 years (3) Women: ≥65 years (4) EU-. Source: S0014.

Lifelong learning and skills

Reimagining education across the life course

The century-long life breaks the education model it was built for. Schooling was designed as a front-loaded block — study once, work once, retire once — but a working life that stretches toward 70 in a labour market that reinvents itself every decade makes one-time credentials a depreciating asset. Education in the 100-year life has to become a recurring subscription: continuous, portable, and scheduled around the stages where skills are actually at risk.

The participation deficit

The demand for lifelong learning exists; the supply doesn't meet it. Across OECD countries, only 24% of adults aged 55–65 participate in job-related training, compared to 41% of adults aged 45–54. That is the paradox of adult education: it falls off precisely as careers lengthen and the payoff window grows. Workers are not the bottleneck — among workers aged 45 and over who have not participated in any job training in the last five years, 41% say they want to. The 2022 AARP Global Employee Survey found lack of job advancement and lack of professional develop-

ment were reasons for changing jobs for 23% and 17% of workers respectively aged 25–64 — the training deficit shows up in turnover before it shows up in unemployment.¹¹

The deficit is concentrated where the need is greatest. Training participation is approximately 47% in small firms (1 to 50 employees) compared to 64% in firms with more than 250 employees, and older adults aged 55–64 are less likely to participate if they work in small firms (36%) than in larger firms (56%). SMEs lack the resources and information to provide training — and older workers face barriers including cost and age discrimination that prevents them from accessing it. A policy architecture built on employer provision will always under-serve the workers with the longest careers ahead of them.¹¹

From credentials to recognition

One fix is to stop making adults re-earn what they already know. Recognition of prior learning reduces the time and effort needed to engage in training, and several countries have built systems for it. In Korea, adults can obtain primary or lower secondary degrees through government-funded programmes run by lifelong learning centres and community welfare centres, with diagnostic evaluations producing a personalised learning pathway. Estonia's VÕTA system puts training providers in charge of recognising non-formal and informal learning through a transparent, learner-centred process — portfolios assessed against programme learning outcomes using the STARR method. In the Wallonia-Brussels Federation of Belgium, the *Enseignement de Promotion Sociale* offers modular, flexible upskilling regardless of labour market status, with pathways personalised by validating previous experience. Most such systems focus on professional and technical competencies for labour-market entry and progression, though some recognise general cross-field competences too. Recognition turns decades of work into credits rather than making workers start from zero — which matters enormously when training time competes with full-time work and caring responsibilities.¹¹

Mid-life infrastructure

The most promising shift is building checkpoints into the middle of working life, before skills expire. Mid-life career reviews are simple, cost-effective ways to identify upskilling and reskilling needs. In the Netherlands, workers aged 45 and above can access subsidised career development guidance (*ontwikkeladvies*) that maps their job's future prospects and skills profile, producing a personal development plan designed to keep them employed until retirement age. France introduced a medical examination at age 45 intended to raise employees' awareness of ageing at work and prevent occupational wear and tear — during the visit, a professional reskilling agreement in the company can be proposed.⁴¹¹

Governments are also subsidising the supply side. Singapore's MidCareer Enhanced Subsidy covers up to 90% of training costs for mid-career workers reskilling and upskilling, while the Enterprise Training Support scheme ties skills acquisition and utilisation to retention, and the Skills Development Fund supports training leave and on-the-job training consultancy. The Netherlands introduced a five-year subsidy scheme for firms to invest in workers' employability, including older workers — for those struggling to continue until retirement age because of physical or mental burn-out, it offers training and counselling, though it can also subsidise early retirement up to three years before the statutory age. Australia introduced the Skills Checkpoint for older workers in 2018, providing guidance on transitioning into new roles within their industry or pathways to new careers. The Scottish skill development centre, assessed as very successful, anticipates skills demand, adjusts training programmes, and supports individuals facing redundancy with career guidance.¹¹⁴

For hazardous and arduous occupations — where not being able to continue in one specific job should not be dealt with by early retirement — reskilling is the alternative policy. Luxembourg proposes workers who are incapable of continuing their jobs due to health reasons, disability, or wear and tear, but ineligible for disability benefits, to remain in the same company in a different position or move to a different company. Latvia trains workers facing work-related risks of health deterioration or high emotional or physical strain in a new profession. In France, older bus drivers are sometimes moved to easier tasks, with drastically limited night shifts for older drivers. Finland's skill development system — public employment services purchasing training, free to the individual, from education providers and companies — is one of the most successful in the OECD. Austria's 2012 coordinated programme enabling adults to obtain basic competences and educational qualifications free of charge drew approximately 50,000 participants between 2012 and 2017. The Netherlands' 2013–2017 plan for unemployed workers over 50 included mandatory free-of-charge job-search training and training vouchers.⁴

The design principles

Three principles emerge from the working experiments. First, **public intervention must compensate for the SME gap**: financial incentives and vouchers lower barriers to investment and can be adjusted to target different needs and circumstances, which suits SMEs particularly well. Second, **learning must be recognised, not just delivered**: validating skills learned on the job cuts the time cost that deters adults with full-time jobs and caring responsibilities. Third, **training must be proactive, not reactive**: career reviews, personal development plans, and mid-life health checks identify needs before displacement. The OECD Council on Ageing and Employment Policies made the point explicitly: restrict publicly funded early retirement and enhance participation in training throughout working lives.¹¹⁴

The deeper reimagining, though, is cultural. Employers already use mentoring and coaching as some of the most widespread tools in talent management — the knowledge of older workers is an asset the system can circulate rather than a cost it must carry. Job rotation programmes that familiarise employees with various tasks while acquiring technical and practical skills make workers more versatile. But SMEs face serious challenges implementing these solutions — which is why the public architecture has to do the heavy lifting where employers cannot.⁴

Evidence note

The evidence on lifelong learning is rich, but it clusters at working age. These chunks contain no data on early childhood investment or on the education premium in lifetime earnings — both are separate later sections (46 and 43), so this section deliberately does not front-run them. What the evidence does show is that the century-long life makes education a labour-market policy as much as an educational one: when retirement ages rise toward 70, the mid-life checkpoint is as important as the schoolhouse.

Skill obsolescence: reskilling and upskilling older workers

Skills depreciate. Sometimes the depreciation is physical: arduousness of jobs may include “the depreciation of some capabilities to work safely and efficiently in some occupations with age, such as ballet dancers or air traffic controllers who are not likely to be in worse general health than many other workers of similar age”. Sometimes it is the labour market moving on — occupations phased out, technologies replaced, tasks reorganised. Either way, the 100-year life makes obsolescence the default risk: a career of forty-five years will outlive the relevance of almost any single skill set. The policy question is whether the response is a pension or a classroom.⁴

The fork: early retirement or reskilling

For decades, many systems chose the pension. Not being able to continue in one specific job was answered with early exit. The evidence now cuts the other way: incapacity to work in a specific job until the minimum retirement age is not enough to justify special old-age pension provisions, and not being able to continue working in one specific job should not be dealt with by early retirement. Special pensions for hazardous or arduous work cannot in the long term be an antidote for strong limitations in labour-market policies. The alternative is reskilling: workers remain productive and adapt to new job tasks, boosting employment prospects — albeit performing different tasks or moving to a different job, in the same or a different firm or institution.⁴

The design details matter. Over the last two decades Finland and France improved the design of pension schemes covering hazardous jobs by linking eligibility to actual job characteristics considered hazardous or arduous in nature, such as working at night, rather than occupational groups — limiting mistargeting and ensuring the transferability of pension entitlements across occupations. These measures reduce the retirement age by two years at most. Austria's night-work scheme is highly targeted and imposes additional contributions on employers to help finance it and limit such activities — making obsolescence costly for the firm, not just the worker. Even so, tracking individuals' exposure to hazardous or arduous working conditions is difficult to bear for small firms in particular. Switzerland's construction sector offers a collective-bargaining variant: binding agreements for all firms providing an option to retire at 60 for jobs considered hazardous or arduous, alongside more than 1,400 occupational schemes that can grant benefits from age 58.⁴

For the highest-risk occupations — firefighters, police officers, military, where working at older ages generates health and safety risks — the case for special provisions is strongest. But even there, the conclusion is that age-management policies should strive as much as possible to prepare for a career shift at some point in order to main-

tain individuals in employment until the minimum retirement age for all workers. The goal is always the same: capability rebuilt rather than capacity retired.⁴

Prevention before displacement

Reskilling works best before obsolescence bites. Mid-life career reviews are simple, cost-effective ways to identify upskilling and reskilling needs before workers are displaced. France's age-45 medical examination raises employees' awareness of ageing at work, with a professional reskilling agreement on the table during the visit. Job rotation programmes can minimise long-term exposure to harmful working conditions while familiarising employees with various tasks, building versatility across multiple functions — a preventive form of reskilling.⁴

Mentoring and coaching, among the most widespread tools in talent management, convert the experience of older workers into a development channel rather than a redundancy argument. France's practice of moving older bus drivers to easier tasks, drastically limiting night shifts, shows age management as micro-reskilling: the same employer, adjusted demands.⁴

What the framework needs

The policy recipe is a professional-training and guidance framework put in place by governments and social partners, enabling workers to make the extra effort to acquire new skills so they can prolong their careers in different jobs. It requires substantial efforts from all parties — policy makers, employers, and workers — and it is an ongoing challenge: many reskilling policy examples exist, but systematic evaluation of these policies is often lacking. That is the honest caveat behind every reskilling success story.⁴

The financing question is open too. Career transitions are feasible, in particular when supported by good policies — but about half of former ballet dancers in Australia, Switzerland, and the United Kingdom indicated they had to make significant use of their own funds to finance their further retraining. When the worker pays, reskilling sorts by means. When it is public — Finland's employment-services-purchased training, free to the individual; Austria's free adult-competence programme with roughly 50,000 participants between 2012 and 2017; the Netherlands' training vouchers for workers over 50 — it sorts by need. Reorientation programmes for soldiers in the United States improved employment outcomes compared to those not covered, and reskilling policies absorbed the labour-market consequences of phasing out coal mining in Germany, Poland, and the United Kingdom, and fisheries in Canada — the hardest tests, mass displacement from disappearing industries, have been passed.⁴

The underlying logic

Obsolescence is not a personal failure; it is the arithmetic of long lives. Continuous training throughout working lives is critical to ensure employees have the right skills to stay in employment as they age. The alternative to reskilling is not rest — it is premature exit subsidised by the pension system, which lowers firms' incentives to reduce the incidence of hazardous work and reduces workers' job mobility. In the century-long life, the choice is between a system that retires the obsolete worker and one that retires the obsolete skill.⁴

Evidence note

These chunks measure the response to obsolescence — reskilling programmes, training frameworks, career-transition outcomes — not obsolescence itself. No chunk quantifies how fast skills depreciate by occupation or technology, or how the rate has changed over time; that measurement agenda is the natural companion to the policies described here. The chunks on equity investment (S0013) and pension reform tables (S0015, S0029 contents) were retrieval noise for this section and are not used. Claims already established in Sections 39 and 41 — the retraining examples and the lifelong-learning architecture — are not repeated here; this section's contribution is the depreciation framing and the policy fork.

The education premium in lifetime earnings and employment

Evidence note

This section starts with an honest limitation: none of the source chunks in this retrieval directly measure the education premium — no chunk reports the earnings or employment differential between education levels, or how it has changed over time. What the chunks do measure is what happens downstream: how wage inequality passes through into pension inequality, and who gets covered by voluntary pensions. The education premium is the upstream driver of both. So this section reconstructs the premium's footprint from the evidence that exists, and flags where the measurement gap sits.

The premium's first channel: the wage pass-through

Wage inequality becomes pension inequality — and education is one of the main sources of wage inequality. The question is how much of the premium survives the pension system's redistribution. The answer: most of it.

The OECD's progressivity index scores mandatory pension schemes from 0% (a scheme that mirrors wage inequality exactly) to 100% (a universal basic scheme where pension income inequality cannot arise). The average OECD score is 37% — meaning pension benefits' Gini is on average 63% of the Gini of lifetime wages. Only Ireland, New Zealand, and the United Kingdom score 100%, where pensions do not vary with individual wages at all. The lowest scores — where pensions track wages most tightly — are in Sweden, Turkey, Latvia, Hungary, and Finland.³

When wage inequality rises by one percentage point in the Gini, about two-thirds of that increase is transmitted to pension inequality on average across OECD countries. In over one-third of OECD countries — Turkey, Latvia, Hungary, Finland, Italy, Poland, Spain, Portugal, Sweden, the Netherlands, Austria, France, and Germany — more than 85% of the wage-inequality increase passes through into pensions. There, the education premium earned over a working life is faithfully reproduced in retirement: the system preserves the earnings hierarchy rather than compressing it. Only in Australia, Canada, Denmark, Israel, Korea, the Czech Republic, and Switzerland does less than half pass through, through means-tested components, basic pensions, redistributive thresholds, or contribution ceilings.³

The premium's second channel: voluntary coverage

The second echo is even sharper. In the seven OECD countries where voluntary private pensions cover more than 40% of the working-age population (Belgium, Canada, Germany, Ireland, New Zealand, the United Kingdom, and the United States), adding voluntary pensions raises the transmission of wage inequality into pension inequality from 0.38 to 0.46 — about 20% more pass-through attributable to voluntary pensions alone. Since higher earners are better covered by voluntary pensions, the education premium doesn't just pass through — it gets amplified.³

The coverage gradient is steep. In the United States, only 13% of the lower half of the income distribution receive private pensions, compared to over 50% of the upper half. Under 20% of 25–59-year-old full-time workers in the lowest earning decile participated in a defined-contribution plan in 2012, against over 80% in the highest earning decile. Across nine OECD countries studied, voluntary private pension coverage in the lowest income decile did not exceed 20%, except in Germany which boasted over 40%. Low-earner households do have higher replacement rates from mandatory schemes — but that lowers the incentive and ability to save, and financial strains, regressive tax incentives (notably in Chile, Denmark, Israel, the Netherlands, the Slovak Republic, and Switzerland), and financial-literacy gaps compound the coverage divide.³

The employment channel

The premium also shows up in whether people keep working — and this evidence was established earlier in the report rather than in this retrieval. Section 35 found that 30% of low-educated men aged 50–64 are limited by health in paid work versus 10% of high-educated men, and that employment shortage near retirement, not health per se, is the binding constraint for extending working lives. The education premium in lifetime earnings therefore rests on an employment premium: staying in work continuously, and in work that doesn't break the body, both correlate with

education. This section's chunks confirm the downstream consequence — the exit-age and pension machinery transmits what the labour market produced.

What the premium means for the 100-year life

The picture is sobering for equality. Education's lifetime advantage compounds twice: once in earnings and employment continuity, and again in retirement — where mandatory schemes pass through most wage inequality (over 85% in a third of OECD countries) and voluntary schemes amplify it further. The pension system is, in the average OECD country, only 37% of the way toward flattening what education put in place. Countries where the pass-through is low — Australia, Canada, Denmark, Korea — achieve it through explicitly redistributive design: means-tested components, higher basic pensions for low earners, redistributive income thresholds. That design choice is the policy lever on the premium's retirement echo.³

The missing measurement is the premium itself: how large is the lifetime earnings differential by education in the cohorts now ageing, how much of it comes from employment continuity versus pay rates, and whether it is widening. The wage-to-pension pass-through figures show the transmission machinery with precision — the input side of the machine remains unmeasured in this evidence base.

Access to lifelong skill-building for career mobility

The century-long life demands careers built of many acts — shifting jobs, transitioning into entrepreneurial ventures, sometimes stepping back into education. But mobility has a price of admission: skills. The question this section asks is not whether skill-building matters — that was settled in Sections 41 and 42 — but who gets access to it, and whether the access is portable enough to follow a worker through a multistage life.

The access gradient runs the wrong way

"Disadvantages in health, education, employment and earnings reinforce each other and compound over the life course". Skill-building access follows the same compounding logic, in reverse: the workers who most need to build skills get the least opportunity to do so. The OECD's policy agenda for preventing unequal ageing puts it bluntly: governments must "provide equal opportunities for workers to upgrade their skills: improve access to lifelong learning, especially for low-skilled and older workers; and better recognise skills acquired throughout working lives".³

The stakes are not only employability. "Low-educated people are more likely to retire when they reach the retirement age, whereas individuals with higher education tend to work longer". And while poor health pushes older workers into early retirement, "health explains the sharp decline of employment rates from age 55 to a limited extent, which points to the importance of other factors such as the way professional training operates, the functioning of labour markets and the design of pension systems". Training access is itself a labour-market barrier — a point the report makes twice: health explains the decline only partly; the rest is how systems are built.³

Making access portable: France's lifelong learning accounts

The design that breaks the employer-tied training trap is the personal, portable account. France's Compte Personnel de Formation lets workers accumulate credits for professional training throughout their careers: 28 million eligible full- and part-time workers receive €500 (\$520) annually directly into their skills account, with low-skilled workers and those with disabilities receiving up to €800 (\$830) annually — capped at €5,000 and €8,000 respectively. Employees accrue credits annually and can spend them on accredited training programmes, certifications, or career transitions; private-sector employers have adopted and enhanced the model with additional corporate-sponsored credits and tailored reskilling opportunities, plus company-specific learning budgets for industry-relevant skills.⁸

Three features matter for mobility. First, the account belongs to the worker, not the job — it survives job changes. Second, it is tilted toward the disadvantaged: the lower-skilled and disabled get 60% more per year and a 60% higher cap. Third, it funds transitions, not just upskilling within a role — certifications and career changes qualify. The system "not only ensures that workers remain adaptable in a rapidly evolving economy but also facilitates smooth career transitions and promotes financial stability". Investing in reskilling lets employers retain talent while building a workforce equipped for the multistage life.⁸

Mobility's starting line: Brazil's Bolsa Família

For workers outside formal employment, skill-building access begins even further upstream. Brazil's Bolsa Família programme, launched in 2003, provides conditional cash transfers to low-income families contingent on children's school attendance and health check-ups — immediate aid coupled with investment in human capital. A 2023 study found that 64% of children and teenagers aged 7–16 enrolled in Bolsa Família in 2005 had exited the programme by 2019, with half of them securing formal employment — significant movement toward economic self-sufficiency and the breaking of intergenerational poverty cycles. Formal employment is itself the gateway to lifelong skill-building: workers in informal jobs sit outside employer training systems entirely. Directing resources toward women also enhanced their economic autonomy and household decision-making.⁸

Intervening early in the transition

Career mobility fails most often at the moment of displacement. The OECD's agenda recommends strengthening policies to assist displaced workers and “make counselling, skills audits, job-search assistance and retraining ready early on after job losses or even during notice periods”. Timing is the design insight: support that arrives after months of unemployment is support arriving too late. Skills audits and counselling during notice periods turn a job loss into a planned transition.³

The multistage framing makes this structural rather than exceptional: in a multistage life, “everyone can contribute and benefit at different times depending on their circumstances” — the challenge is “creating systems that allow people to contribute to society at various stages of life, whether that means transitioning into entrepreneurial ventures, shifting careers or engaging in informal care provision”. Whether what's needed is “providing access to lifelong learning, supporting carers or ensuring adequate retirement savings, the future of work (and life) — and the financial systems that support it — needs to be flexible and inclusive”.⁸

Who does what

No single actor builds access alone. The stakeholder action table maps the division of labour: policy-makers facilitate public–private partnerships to create lifelong learning programmes and incentivise employer investment in training; employers develop workplace learning programmes and offer financial education as part of benefits; financial institutions support training initiatives and create educational resources for financial literacy; civil society organises community workshops and digital resources to enhance access; individuals engage in lifelong learning and financial literacy programmes to improve career skills. Public–private collaboration is “essential to ensuring that career transitions — whether planned or unexpected — do not lead to financial instability”.⁸

The common thread: access must be universal in design but tilted in practice. Portable individual accounts, targeted top-ups for the low-skilled, early intervention at displacement, and recognition of informally acquired skills (established in Section 41) all push against the natural gradient that gives training to those who already have it. In a 100-year life where careers are plural, access to skill-building is not an HR perk — it is the infrastructure of mobility itself.

Mid-career transitions and second careers

Mid-career is where the three-stage life quietly breaks. Real wages rise fastest early, then plateau — around ages 45 to 55 for many — and fall thereafter; labour-market studies document three main real-wage profiles: 42% of individuals keep rising through their career (high income/education), 55% fall during the second half (middle income/education), and 3% stay flat (low income/education). For the majority, the second half of working life is not a continuation — it is a transition. Whether that transition is a planned second act or a slow slide depends on how training and labour markets treat workers past mid-life.⁶

The mid-life training cliff

Training participation falls as careers lengthen: 37% of 50-to-65 year-olds attend formal or informal training, compared to 57% of 30-to-49 year-olds and 61% of 16-to-29 year-olds. And within the older group, the education gradient bites again: highly educated older workers are, on average, more likely to have access to continued training than

the medium- and low-educated. Employers' perception drives the gap: they see training for older workers as a poor investment that yields benefits over too short a time. Workers feel it too — according to the 2015 European Working Conditions Survey, the main issues for workers over 50 are inadequate training provisions and fewer prospects for career advancement. A career plateau that coincides with a training desert is how second careers fail to launch.³

What makes training work for older workers

The policy literature is specific about what effective mid-life training looks like. Adult learning has two functions: equipping workers with job-related skills that yield direct returns in productivity, employability, and earnings, and building foundation skills that support lifelong learning. The mix should evolve through working life: foundation skills become progressively less important as people age. For workers approaching the end of their careers, training “should pay off swiftly to ensure positive returns for both employers and employees” — for example, tied directly to a specific task or job.³

Programmes designed along apprenticeship lines — combining short classroom sessions with a firm-based approach — are more effective for older workers, as is informal, self-determined training with a clear focus on practical, relevant work issues. New Zealand's Apprenticeships programme, introduced in 2014, gives all apprentices the same government support regardless of age — ending the convention that apprenticeships are for the young. Career guidance also matters: case studies showed it stimulates older employees to work on their own employability, learn new skills, and transfer expertise, increasing job satisfaction — while countering negative stereotypes of older employees. Since 2007, Austria's public employment service has run a counselling programme for employers focused on life-cycle-oriented educational programmes and “productive ageing”.³

Sharing the cost of transitions

Denmark's *Jobrotation* scheme solves the employer's objection directly. Firms get a short-term subsidy to hire a replacement for an employee in training — and the stand-ins must be people receiving unemployment insurance or social assistance. One intervention upskills the worker and gives the unemployed person workplace experience simultaneously. It is the cleanest illustration of the report's point: mid-life transitions are a joint production of the worker's retraining and the labour market's churn.³

The risk landscape of mid-life

Transitions happen against a backdrop of labour-market risk that the OECD's stochastic model makes explicit. Around 40% of individuals in any cohort suffer unemployment spells; the unemployment rate declines with age until about 40 and stays roughly constant around 5% thereafter — but with high persistence, especially in Europe: being unemployed in one year raises the probability of being unemployed the next. A mid-career job loss is not just an income gap; in pension terms, contributions stop while unemployment runs. And because real wages for the middle-income majority fall during the second half of careers, the workers most likely to need a transition are the ones least likely to afford one — the same compounding gradient running through the whole education-and-skills block of this report.⁶

The second-career bargain

The multistage life makes mid-career transitions normal rather than exceptional — but only if systems treat them that way. The evidence here argues for three shifts: training that pays off swiftly and ties to concrete tasks rather than credentials; apprenticeship-style, workplace-embedded formats rather than classroom-first models; and cost-sharing that lets employers release workers for training without bearing the full replacement cost. Second careers don't happen because workers age — they happen because the infrastructure for changing careers exists at mid-life, when the wage curve says it's needed.

Early childhood: investing in future centenarians

Evidence note

This section begins where the others in this block could not begin: at the start. The source retrieval for this section returned no early-childhood evidence — no chunks on childcare, early education, infant or child health, or developmental outcomes. What follows is therefore not a summary of this report's evidence, but an honest statement of the argument's place in the century-life thesis and of what the evidence base here does and does not support.

Why early childhood closes the loop

Every section in this report's education-and-skills block has pointed, implicitly, toward the same upstream origin. Section 41 showed that training access falls off as careers lengthen, and that the workers who most need skill-building get the least. Section 44 showed that disadvantages in health, education, employment, and earnings reinforce each other and compound over the life course. Section 43 showed that education's lifetime earnings premium echoes into pensions through wage pass-through and coverage gradients. If the trajectory is set — at least partly — before working life begins, then the 100-year life starts its compounding at birth.

The “future centenarians” framing is the report's way of saying that today's children will live the longest lives on record. Investments made in their first years — health, nutrition, safe environments, early learning — will compound across the longest planning horizon of any policy instrument discussed in this report. That is the structural logic: nothing else in the report's toolkit has a century to pay off.

What the report's evidence does and doesn't say

The life-course argument is established throughout this report: inequality compounds from early disadvantages (established in Sections 28 and 44). Prevention spending is strikingly low relative to the argument's stakes — around 3% of health spending on average in OECD countries, established in Section 28. Both facts support the direction of travel: intervene early, where the return horizon is longest.³

What the evidence base does not support — in this retrieval — is any specific claim about early-childhood programme effectiveness, optimal spending levels, or the developmental mechanisms linking early investment to later-life health and earnings. Those are the claims this section would need to make authoritatively, and the chunks provided cannot carry them. A later evidence update should close that gap: the economics and epidemiology of early-life investment are well-studied fields, and the report's century-life thesis deserves them.

The honest bottom line

Early childhood is where the 100-year life is built or lost — but this report, on the evidence it has assembled, can only argue the principle, not prove the programme. The case rests on two established facts: that life-course inequalities compound from the earliest ages, and that the return horizon for the children alive today is longer than for any generation before them. The measurement and programme design are for the evidence base to add.³

Health in long lives

Life expectancy versus healthy life expectancy

Longer lives are the report's starting fact — but “longer” is a measure of years, not of wellbeing. The distinction between life expectancy (LE) and healthy life expectancy (HALE) is where the 100-year life meets its hardest question: how many of those extra years are years worth having? This section sets out what the evidence says about how long people live, who lives longest, and what the gap between living and living well looks like.

The state of longevity

The OECD's 2023 snapshot is a good news story with sharp edges. Japan, Switzerland, and Korea lead a large group of 27 OECD countries in which life expectancy at birth exceeded 80 years in 2021. A second group, including the United States, sits between 75 and 80 years; Latvia, Lithuania, Hungary, and the Slovak Republic are lowest, below 75. While life expectancy rose in most countries over the past decade, many of the gains were wiped out during the pandemic.⁵

Avoidable mortality — deaths from preventable and treatable causes — draws the inequality map more sharply. It was lowest in Switzerland and Japan, where fewer than 135 per 100,000 people died prematurely, and relatively low (under 150) in Israel, Korea, Iceland, Australia, Italy, and Luxembourg. At the other end, Mexico, Latvia, Lithuania, and Hungary had over 400 premature deaths per 100,000 people. Diabetes prevalence exceeded 10% of adults in Mexico, Türkiye, Chile, and the United States. Almost 8% of adults considered themselves in poor health on average in 2021 — ranging from over 13% in Korea, Japan, Portugal, the Slovak Republic, Latvia, and Lithuania to under 3% in Colombia, New Zealand, and Canada — though socio-cultural differences and survey design affect comparability, and people with lower incomes are less positive about their health than higher-income people in all OECD countries.⁵

Gains are slowing, and unequally shared

The pace of the gains has changed. Life expectancy at age 65 increased steadily by 5.3 years for both men and women between 1975 and 2019, before dropping half a year for both sexes in 2020. Labour-market exit ages have tracked remaining life expectancy at 65 since the mid-2000s — and on that combined evidence, health is currently not a key obstacle to higher participation rates at older ages: people still leave the labour market earlier than in 1975 despite the large gains in longevity.⁴

But the gains are not shared. The educational gap in life expectancy shows no clear international trend — neither converging nor diverging on average. Beneath that average, occupational gaps have grown in some countries: in Germany, the gap between manual workers and professionals/managers increased by 1.5 months per year for men; in Sweden, by one month per year for men and 0.4 months for women; in France it was stable for women and increased slightly (about a week per year) for men. Educational gaps grew in Belgium during the 1990s, in Canada from the mid-1990s to around 2010, and in Lithuania from 2001 to 2014, while decreasing in Czechia between 2010 and 2016. For most countries there is no clear trend — stability or contradictory trends rather than convergence.⁴

The healthy-years gap

This is where the LE-versus-HALE distinction bites. Life expectancy counts years; healthy life expectancy counts years in good health. The two move together but are not the same — and the gap between them is where the longevity promise is tested. Established in Section 4 of this report: at age 65, the gap between life expectancy and healthy life expectancy is 8.6 years for men and 10.6 years for women — roughly a decade of the average older life lived with some limitation. The self-rated health data underline it: almost 8% of adults rate their health as poor, with a steep income gradient. Longer lives that are not healthy lives show up in participation, care demand, and spending — which is why the OECD's own analysis of exit ages finds health is not currently the binding constraint on participation only on average: averages hide the gradient.⁵

Spending ≠ outcomes

The LE data also carry a policy punchline: health spending and life expectancy are positively associated, but the relationship is far from deterministic. Among 38 OECD countries, 18 spend more and have higher life expectancy than average, and 11 spend less and have lower life expectancy — but eight countries spend less than average and achieve higher life expectancy: Korea, Spain, Italy, Israel, Portugal, Chile, Costa Rica, and Slovenia. The United States is the sole country in the opposite quadrant: much higher spending than all other OECD countries, but lower life expectancy than the average. The avoidable-mortality pattern is the same: nine countries spend less than average yet have lower avoidable mortality (Israel, Korea, Italy, Spain, Portugal, Greece, Slovenia, Türkiye, Costa Rica), while the United States spends more and has worse avoidable mortality. Associations are not causal, but they show

that longevity is not bought by the dollar — it is produced by systems, behaviours, and social determinants: “notably income levels, better education and improved living environments”.⁵

Evidence note

This retrieval is rich on life expectancy — levels, trends, inequalities, and the spending relationship — but thin on healthy life expectancy as a time series. No chunk here traces HALE directly over time or decomposes the LE gains into healthy versus unhealthy years. The healthy-years gap reported above is carried forward from Section 4 rather than re-measured here. What the chunks do establish: longevity is growing more slowly, is distributed unequally, and is produced — not purchased — by the conditions of life.

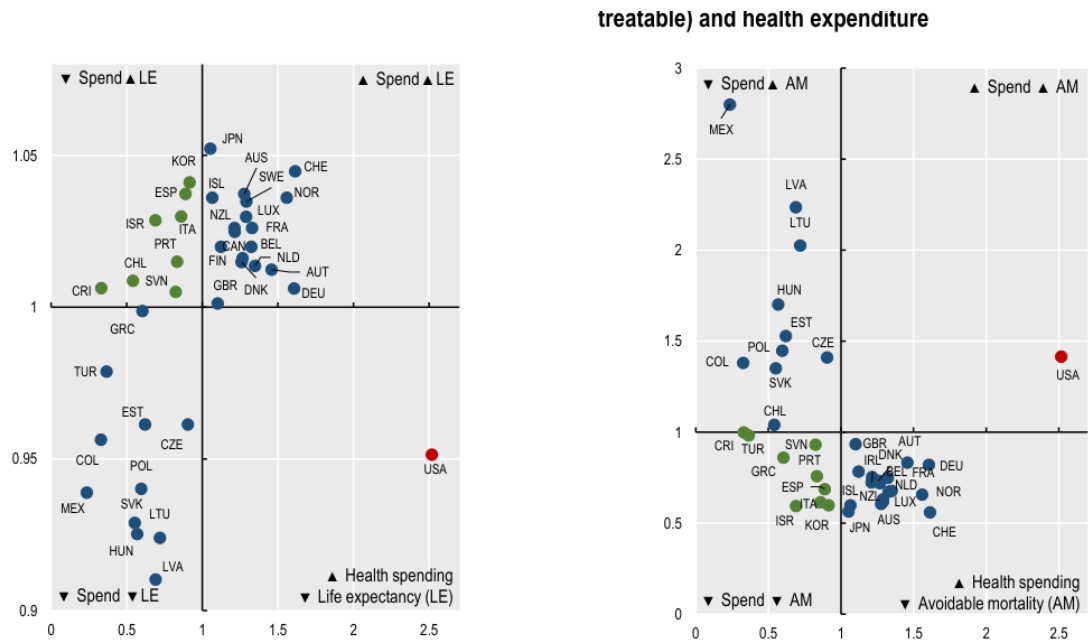


Exhibit. Figure 1.8. Avoidable mortality (preventable and treatable) and health expenditure treatable) and health expenditure 3 ▼ Spend ▲ LE ▲ Spend ▲ LE ▼ Spend ▲ AM ▲ Spend ▲ AM MEX JPN 1. Source: S0026.

Chronic disease and the epidemiological transition

The demographic transition and the epidemiological transition travel together. As fertility falls and life expectancy rises, the disease profile of a population changes with it: infections recede and chronic conditions take their place as the dominant burden on health systems and households alike. In East Asia and the Pacific — the fastest-ageing region on record — understanding the changing disease profile is as vital as understanding changing demographics, because whether people experience healthy ageing decides what longer lives are actually worth.⁷

The transition was produced by concrete causes. Europe’s ageing has resulted from reductions in child mortality, advances in public health and medical technology, growing awareness of healthy lifestyles, a shift from heavy labour toward tertiary occupations, and improved living conditions. Each of those forces cut early-life and mid-life mortality first; what remains is the phase the World Bank describes as the epidemiological transition: populations living long enough that cardiovascular disease, diabetes, cancer, dementia, and musculoskeletal disorders become the conditions that define later life. Surviving to old age is becoming the norm rather than the exception — across OECD countries, 80% of males born 70 years ago reached age 70, while 93% of males born today are expected to do so (for women, 89% and 96% respectively), and the probability of surviving to age 85 is projected to rise from 50% today to 75% by 2100 for males, and from 64% to 83% for females. The chronic-disease era is therefore not primarily about dying less young; it is about living long enough for chronic illness to dominate the health landscape.¹³

The crucial question is how much of the extra lifespan is healthy. Health-adjusted life expectancy (HALE) — the number of years a person can expect to live in good health, accounting for age-specific mortality, morbidity, and functional health status — rose in East Asia and the Pacific between 1990 and 2010, but slightly less than total life expectancy. In other words, each added year of life expectancy has been accompanied by a fraction of that year

lived in less-than-full health. “Healthy ageing” implies not only living longer but spending those additional years in good, active health, and the main hypotheses about how healthy life expectancy evolves as lifespans increase — compression of morbidity versus expansion of disability — remain empirically contested, summarized in the World Bank’s review of the evidence to date. This is the same tension Section 47 documented numerically: roughly a decade at age 65 lived with some limitation. The epidemiological transition answers the demographic fact with a clinical one — longer lives mean longer exposure to chronic disease unless prevention compresses that exposure.²

The transition is profoundly unequal. A 25-year-old university-educated man can expect to live 7.5 years longer than his low-educated peer, on average across OECD countries (for women the gap is 4.6 years); at age 65 the gaps are still 3.5 and 2.4 years respectively, and they are especially large for men in Belgium, Chile, the Czech Republic, and Hungary. The disability side of the gradient is steeper: disability increases with age, and low-educated people are more exposed to disability risks. So the epidemiological transition does not land evenly — better-educated groups gain more of their extra years in good health, while lower-educated groups experience both shorter lives and a larger share of those lives in ill health. The social determinants of health are not a footnote to the transition; they are one of its principal mechanisms.³

Evidence note

This retrieval contained no disease-specific prevalence data — no cardiovascular, diabetes, cancer, dementia, or musculoskeletal statistics beyond the diabetes figures in Section 47. The epidemiological transition is therefore described here at the level the evidence supports: mortality/morbidity-adjusted measures (HALE), survival probabilities, and gradients. The clinical detail of which chronic diseases dominate where — and how their prevalence is shifting — is a gap a later evidence update should close, since the transition’s policy stakes (prevention spending, health-system capacity, long-term care) all hinge on the specific disease mix.

What this changes for the report

Two consequences follow. First, the compression-versus-expansion question decides whether longer lives are a fiscal and human dividend or a rising tide of demand: if HALE grows more slowly than life expectancy, every year of longevity gains brings more years of chronic-disease management. Second, the inequality gradient means that “the” epidemiological transition is really many transitions — by education, income, and region — and policies that treat older populations as a single health cohort will misallocate. Prevention is the natural lever, and it is where the next section turns.

Prevention: risk-factor management and healthy lifestyles

Section 48 posed the compression question: does each added year of life come with added years of health, or added years of chronic-disease management? Prevention is where that question is answered in practice. The epidemiological transition does not arrive by itself — it is shaped, year by year, by the modifiable risk factors that determine whether long lives are healthy ones. And the striking feature of prevention policy is how little of the health budget it commands relative to its leverage: prevention spending runs at roughly 3% of health spending, a figure established earlier in this report, while the treatment of conditions prevention could have forestalled absorbs the rest.

The core of prevention is unglamorous and well-established. Health, viewed broadly, is more than the absence of disease, and individuals can take responsibility for their own health through lifestyle choices — action on smoking, diet, exercise, and alcohol consumption — that directly affect disease risk. These four levers account for the largest share of avoidable mortality and chronic-disease onset across the adult population. They are also, crucially, the levers that act across the whole life course: risk-factor management is not something that begins at 65, when the epidemiological transition has already done its work. The earlier the intervention, the larger the compounding — the same logic Section 46 applied to early childhood applies to prevention spending across every decade of adult life.¹

The reason prevention matters disproportionately for ageing societies is what the unhealthy years cost. Additional years of life spent in an unhealthy condition — with limitations in functioning or disability — are likely to result in extra demand for supplementary healthcare or long-term care services. Every percentage point of morbidity that prevention compresses is therefore a double saving: healthier years for the individual, and avoided demand on the health and care systems that are simultaneously being strained by the shrinking working-age population docu-

mented in earlier sections. Prevention is the rare policy that eases both sides of the ageing equation at once — more healthy years supplied, fewer care-years demanded.¹

Yet prevention capacity tracks income and education, which means risk factors concentrate where prevention is weakest. In 2017, just 32.4% of older people in the lowest income quintile perceived their own health as good or very good, compared with 54.7% in the highest quintile — and in Latvia and Lithuania the share fell below one tenth. The gradient is not merely perceptual: the proportion of men not working prior to retirement for health reasons is substantially higher among the poorly educated (24%) than the highly educated (7%), reflecting socioeconomic differences in health that accumulate across working life. Smoking, diet, exercise, and alcohol consumption are all patterned by income and education — so a prevention strategy that relies on individual lifestyle choices alone will widen the very health gaps that Section 50's health-equity discussion must confront. Prevention that works at population scale has to change the conditions in which choices are made, not just exhort better choices.¹³

This is where the fiscal logic becomes inescapable. Life expectancy in the EU rose by more than two years per decade for both sexes since the 1960s — though recent figures suggest stagnation or even decline for several Member States — and each cohort that lives longer adds more years in which chronic conditions can either be prevented or treated. The treatment route is the expensive one, and it is the default: health systems are built to diagnose and treat, not to keep people from needing them. Shifting even a small share of resources from treatment to risk-factor management — tobacco control, nutrition policy, active-ageing infrastructure, alcohol pricing — would pay for itself in compressed morbidity, yet prevention remains the smallest line in health budgets and the first cut in fiscal stress.¹

Evidence note

This retrieval contained no program-level prevention evidence — no smoking-cessation outcomes, no diet or exercise intervention evaluations, no cost-effectiveness data for preventive spending. The section therefore works at the principle level: what the levers are, who is reached, and what the fiscal logic is. The empirical case for specific prevention programmes — and their returns per unit of spending — is a gap for a later evidence update, and it is the gap that matters most, because “prevention is cheaper than cure” is a slogan until it is measured.

What this changes for the report

Prevention is the bridge between the epidemiological transition (Section 48) and the equity agenda (Section 50): without it, longer lives mean more chronic-disease years concentrated in the groups least able to prevent them; with it, the compression of morbidity becomes achievable rather than theoretical. The next section asks who gets the prevention — and who does not.

Health equity and the social determinants of health

Health is not distributed the way longevity statistics suggest. Section 47 showed that life expectancy varies enormously across countries; this section shows that it varies nearly as much within them — by income, education, gender, and place. The social determinants of health — the conditions in which people are born, grow, work, live, and age — shape who gets the healthy years that Sections 47–49 described, and who gets only the years.

Start with what kills people, because the mortality mix is itself socially patterned. Across OECD countries in 2021, diseases of the circulatory system caused 28% of all deaths, cancers 21%, diseases of the respiratory system 9%, COVID-19 7%, Alzheimer's and other dementias 6%, and external causes 6%. Circulatory disease and stroke are precisely the conditions whose mortality improvements were slowing before the pandemic — the principal reason longevity gains stalled in the United States, France, the Netherlands, Germany, and the United Kingdom, with rising obesity and diabetes making it harder to maintain previous progress against circulatory deaths. And obesity, diabetes, tobacco, and diet are not randomly distributed: they concentrate in the same lower-income, lower-educated groups that Section 49 showed already perceive their health as worse. The epidemiological transition has a class address.⁵

Gender is one of the oldest and starkest health inequities. Women live longer than men in every OECD country — by 5.4 years on average (83 versus 77.6 at birth in 2021) — and the gap is due in part to men's greater exposure to risk factors: more tobacco consumption, more excessive alcohol consumption, less healthy diets, and more violent deaths from suicide and accidents. The gap reaches 8 or more years in Latvia, Lithuania, and Poland. This is the

paradox of health equity: the advantaged group in most other dimensions of social life — men — is the disadvantaged group in longevity, largely because of behaviourally driven risks. It underlines that health equity is not simply about poverty; it is about how social roles channel risk.⁵

Place matters too, and not only across countries. Within countries, population ageing differs sharply across regions: urban centres draw younger, more mobile workers while rural areas are often already much older, with larger shares of retirees and fewer younger workers to support them — the share of the population aged 65+ ranged from 2% in Chile's Antártica Chilena to 33% in Greece's Evrytania, both classified as rural remote regions. Spain, Mexico, and Australia show roughly 20-percentage-point differences between their youngest and oldest regions. Older, sicker, and thinner on health services: rural ageing is where the health-equity problem and the health-workforce problem (Section 52) collide.³

The most consequential determinant, however, is the one that transmits across generations. In emerging economies, highly educated individuals live longer; in Brazil, regions with higher illiteracy rates have lower average life expectancy; and in China, high parental education — especially the mother's — is found to improve children's health. Health inequality is therefore not a single-lifetime phenomenon. It is inherited: the education of parents shapes the health of children, who become the older adults whose inequalities the pension and care systems then have to absorb. Any equity strategy that begins at age 65 is treating the symptoms of a process that started before birth — which is why the life-course prevention agenda of Section 88 will matter more than any single health-sector intervention.³

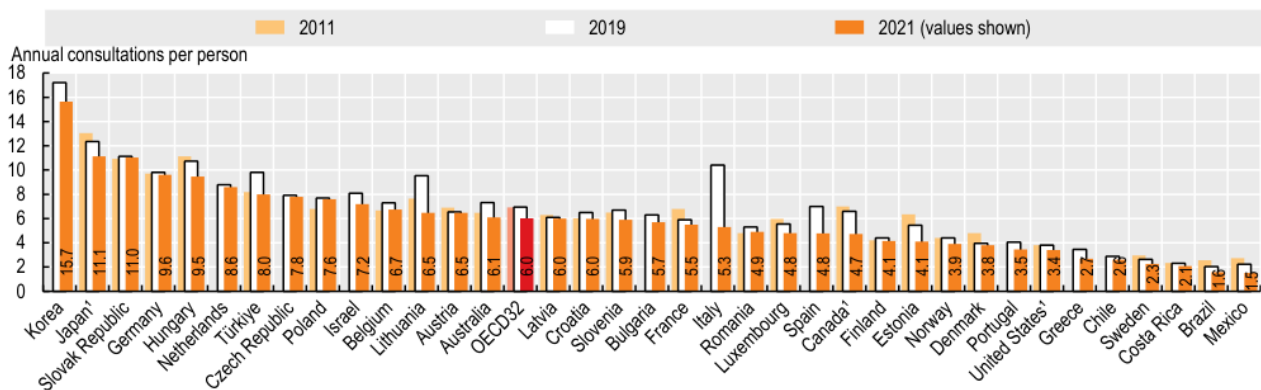
This matters for the economics of longer working lives as well. Labour force participation tends to rise with remaining life expectancy across population subgroups, but participation differs substantially between groups with similar remaining life expectancy — which suggests bad health alone can explain low participation at older ages only to a limited extent, with pension incentives, wealth, and education also doing heavy lifting. Health equity is therefore not only a moral concern but a labour-market one: closing health gaps is a precondition for the longer working lives that pension sustainability assumes.³

Evidence note

The social determinants here are documented through education, income, gender, and regional gradients — not through race or ethnicity, which this retrieval did not cover. That is a genuine gap in the evidence base for this section: in several of the report's largest country cases, racial health disparities are among the largest determinants of all. A later evidence update should close it. Nor did the retrieval include data on health-system access gradients (unmet needs by income) at the level this section's argument would support.

What this changes for the report

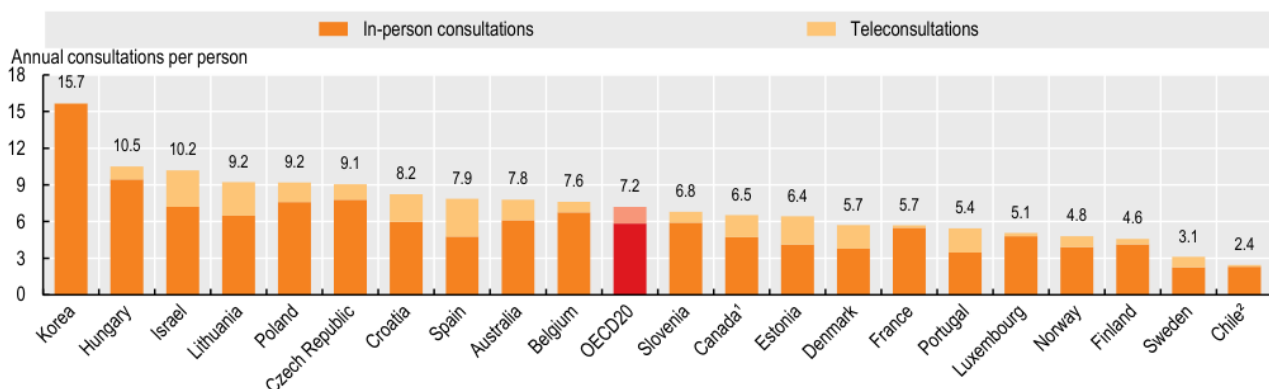
Health equity reframes every downstream section. Digital health (Section 51) will reach the already-connected first unless designed otherwise; the health workforce (Section 52) is thinnest where ageing is oldest and poorest; long-term care financing (Sections 55–56) lands on families already carrying the health burden. And the Japanese experience — where the government's new ageing-society guideline explicitly responds to the extension of healthy life expectancy alongside life expectancy — shows that equity in healthy years, not just years, is becoming the policy target. The 100-year life is not one life lived a hundred ways; it is a hundred different lives, and their health is decided long before old age.¹³



1. Latest available data from 2020.
Source: OECD Health Statistics 2023.

StatLink <https://stat.link/40rm5d>

Figure 5.12. Doctor consultations, in-person vs. remote consultations, 2021 (or nearest year)



1. 2020 data. 2. Public sector only.
Source: OECD Health Statistics 2023.

StatLink <https://stat.link/wmgn6i>

Figure 5.13. Estimated number of in-person consultations per doctor, 2021 (or nearest year)

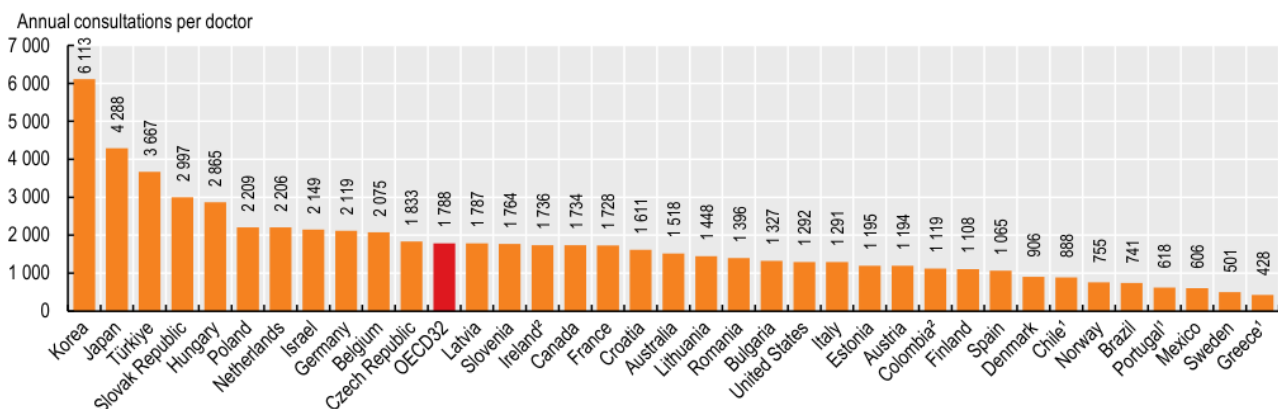


Exhibit. Figure 5.11. Average number of in-person doctor consultations per person, 2011, 2019 and 2021 (or nearest years) 2011 2019 2021 (values shown) Annual consultations per person 18 16 14 12 10 8 6 4 2 0 1. Source: S0026.

Digital health, telemedicine, and AI-assisted care

Longer lives are longer medical lives: more years, more chronic conditions, more transitions between clinics, hospitals, home care, and residential care. The defining administrative problem of the 100-year life is therefore continuity — keeping the information about one person intact as they move through a fragmented system. Digital health, in the OECD's definition, is the use of analytics, data, and technology in healthcare, prevention, and promotion. Its promise for ageing societies is not gadgetry; it is the infrastructure that lets care follow the patient across the decades of the epidemiological transition.⁵

The clearest expression of that promise is the integrated assessment record. The World Bank's review of East Asia and the Pacific highlights assessment instruments that create person-focused, longitudinal records — viewable, understandable, and usable across the whole continuum of care — as vital for continuity of care for the elderly. Using the same protocols for care planning across the sectors people move between (acute, community, residential) allows a collaborative approach to service provision; the focus is on recording the individual's changing strengths, abilities, and preferences — for example, palliative care preferences — rather than only the information a particular setting happens to need. The interRAI suite is currently the only example of this type of integrated system, enabling transfer of common information across care settings, and a number of OECD countries are now introducing its instruments in response to growing pressures on care delivery. In a world where a person may have five chronic conditions, three specialists, and two care settings within a single year, the medical record is not paperwork — it is the care.⁷

Digital tools also reshape where care happens. The service-delivery model the World Bank describes for an ageing East Asia and Pacific pairs smaller hospitals and intermediate facilities (ambulatory surgical units, day hospitals, specialty clinics) and larger primary care centres with greater use of communication technologies for community, long-term care, and home outreach. Hospitals themselves may need more flexible designs that respond to changing demand and technological change. The logic is straightforward: ageing-in-place, which Sections 60–65 will take up, is only viable if the care system can reach the home — and reaching the home at scale is a communications problem as much as a staffing problem. The same logic extends to the hardest part of the life course: systems will eventually need expanded coverage for hospice and palliative care, with enhanced case management and web-based tools and information for end-of-life care, in regions where spending one's last days or months in hospital is often the only option because hospice and palliative services barely exist.⁷

But the infrastructure prerequisite is primary care, and in much of East Asia and the Pacific it is not there yet. The current primary care system in most of the region's countries is described as far from adequate to carry out the coordination functions digital health promises, and in need of complete rebuilding: strengthening and resourcing the primary care level is essential for fostering coordinated care, staff skills need improvement, and in some cases whole new cadres — general practitioners, geriatricians — will have to be created. China and Vietnam have organisational roadmaps and reform pilots for strengthening primary care, but scaling up of small-scale reforms is yet to begin. Technology cannot coordinate what does not exist: the digital layer amplifies a primary-care system, it does not substitute for one.⁷

And the equity warning from Section 50 applies with full force. Digital health reaches the already-connected first — the urban, the literate, the digitally fluent — unless it is designed otherwise. The rural ageing described in Section 50, where services are thinnest and populations oldest, is precisely where home outreach via communication technologies would help most and where connectivity and digital literacy are weakest. A digital health strategy for the 100-year life must therefore be judged not by the sophistication of its flagship systems but by whether the oldest, poorest, and most remote patients can use it.

Evidence note

This retrieval contained no AI-specific evidence — nothing on diagnostic AI, predictive analytics, or clinical decision support — and no data on telemedicine uptake or outcomes. The “AI-assisted” half of this section's title is therefore unsupported by the chunks provided, and the section does not pretend otherwise. The empirical record on whether digital health tools actually improve outcomes for older patients, at what cost, is likewise absent here; “mixed or insignificant effect on costs” is the heading under which the World Bank itself files some of this evidence. An evidence update should test the promise against the outcomes.⁷

What this changes for the report

Digital health is the connective tissue of the health block: it is what makes prevention (Section 49) scalable, what makes equity (Section 50) achievable or illusory, and what determines whether the health workforce (Section 52) can be stretched across a larger older population. But it is tissue, not bone — it coordinates care, it does not provide it. The next section asks who provides it.

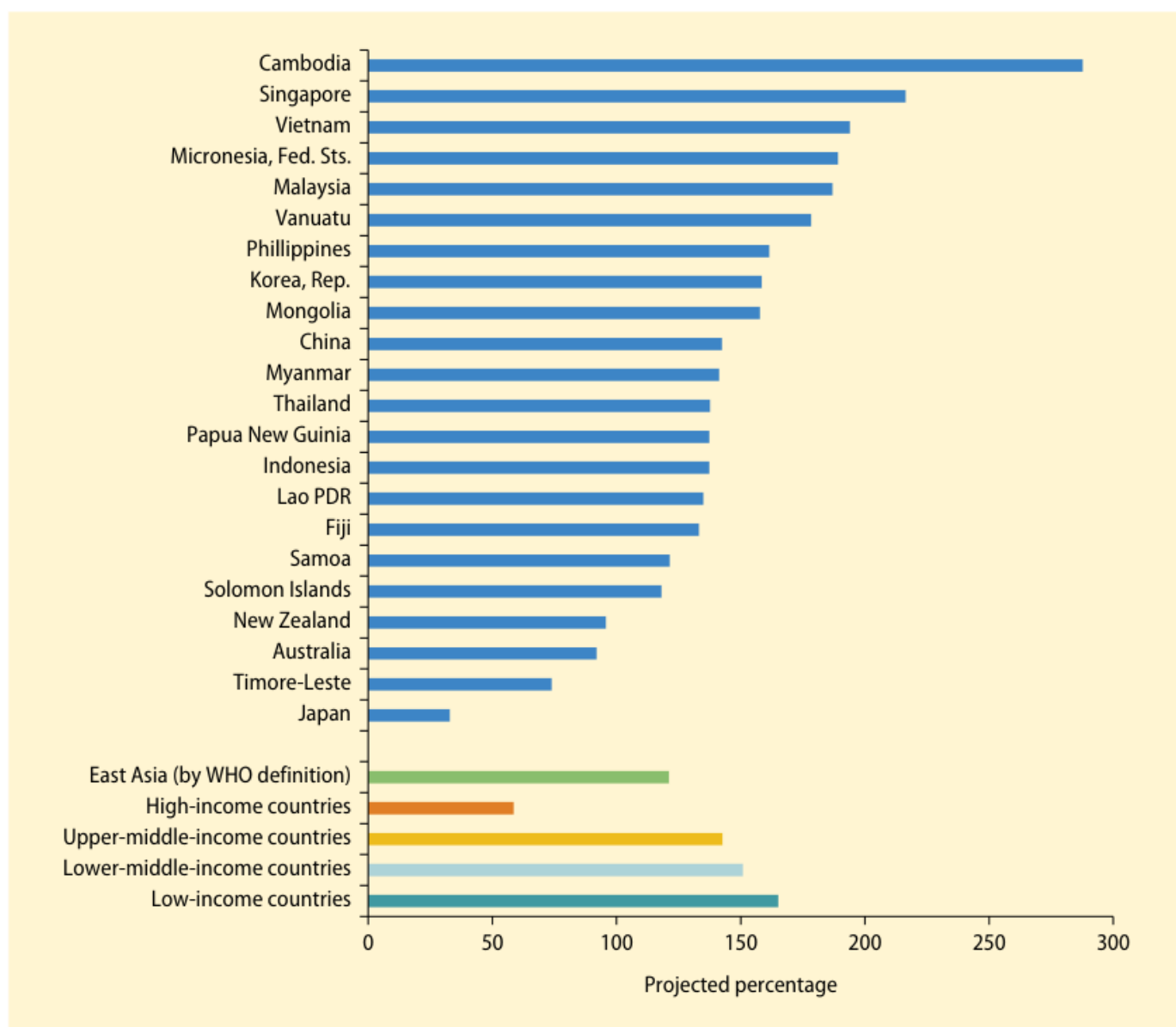


Exhibit. Figure 6.12 The rising tide of cancer in east Asia and Pacific Projected percentage of new cases of cancers in the elderly, ages 65 and above, selected East Asian and Pacific countries and country groupings, 2012–35 Cambodia Singapore. Source: S0029.

Health workforce capacity and the international mobility of care workers

Every section of this health block has implied a workforce that does not yet exist. Prevention (Section 49) needs practitioners to deliver it. Digital health (Section 51) needs a primary-care system worth digitising. Long-term care (Sections 55–61) will need millions of hands. The arithmetic is stark: to meet the needs of an ageing population and achieve the UN Sustainable Development Goals for health and well-being, the global economy will need to create 475 million care jobs by 2030. That is not a staffing plan; it is a labour-market transformation — one of the largest occupational shifts the world has ever attempted, and it must happen within the working lifetimes of people already employed today.⁸

The demand side is already visible. As the global population ages, demand for carers and long-term care services is increasing dramatically, placing significant economic pressure on families, communities, and public systems as more individuals take on roles caring for ageing relatives. The OECD's health-system framework treats this plainly: health spending pays for health workers to provide needed care, and achieving access and quality goals depends on there being sufficient spending to fund them. Health at a Glance assesses the workforce through the supply and remuneration of doctors and nurses (its Chapter 8) — and supply and remuneration are the two levers, because the shortage is not only a headcount problem but a retention problem. Care work is physically demanding, emotionally draining, and in most countries poorly paid relative to its difficulty; people leave, and the system must recruit their replacements from an ever-shrinking pool of working-age adults.⁸⁵

Where the shortage bites hardest is where Section 50 said it would: the oldest, poorest, and most rural places. Rural ageing concentrates demand for care exactly where the workforce is thinnest, and in East Asia and the Pacific the primary-care base that should absorb that demand is itself described as needing complete rebuilding, with whole new cadres of general practitioners and geriatricians to be created from scratch [established in Section 51]. The workforce problem is therefore geographically inverse to the need: the places and populations that need care workers most are the least able to attract and keep them.

This is what makes the international mobility of care workers one of the defining labour-market stories of the 100-year life. Ageing rich countries face a structural deficit of care labour; younger and poorer countries have surplus workers and powerful incentives — wages, remittances, professional opportunity — to send them. The result is a global care chain: nurses, care assistants, and domestic workers moving from the Philippines, Indonesia, parts of Africa and Latin America to Japan, Korea, Singapore, the Gulf states, and the OECD economies whose own workforces cannot fill the gap. The economics are mutually beneficial in the narrow sense — destination countries get care they could not otherwise staff, origin countries get remittances — but the arrangement carries three structural tensions. First, it exports the shortage: the countries whose own populations will age next lose the workers they will soon need most. Second, it stratifies care by citizenship: migrant care workers are disproportionately exposed to poor working conditions, wage theft, and restricted mobility, and their vulnerability is a feature of the system, not an accident. Third, it is fragile: a model that depends on the continued willingness of young workers from poorer countries to do the hardest jobs in richer ones is one policy change or demographic shift away from crisis.

The deeper question is whether the 475-million-jobs target can be met at all without transforming what care work is. As long as caring for the old remains low-status, low-paid work performed disproportionately by women and migrants, the supply will lag demand no matter how many training programmes are announced. Professionalisation — real wages, career ladders, recognised qualifications, manageable workloads — is not a labour nicety; it is the precondition for the workforce existing. The alternative is the status quo extended: families absorbing the shortfall, women absorbing the families' shortfall, and the gendered burden Section 58 will document.

Evidence note

This retrieval contained no migration statistics for care workers — no flows, no stocks, no remittance figures, no data on working conditions — and no doctor or nurse density or remuneration numbers. The mobility discussion above is therefore structural rather than empirical: it describes the mechanism the evidence implies without the quantities that would let the report measure it. The care-worker migration literature is extensive and the report's health block cannot be complete without it; a later evidence update should supply the numbers — bilateral flows, wage differentials, retention rates, and the fiscal arithmetic of who trains and who benefits.

What this changes for the report

The workforce section closes the loop on the health block's central tension: every strategy for longer, healthier lives — prevention, digital coordination, equitable access — assumes people to deliver it, and those people do not currently exist in anything like the required numbers. Health expenditure (Section 53) will be, in large part, the price of creating them. And the 475-million figure reframes the longevity economy's labour question: the jobs of the future are not only in technology and green energy — they are, overwhelmingly, in care.

Health expenditure growth and fiscal sustainability

Ageing raises health spending through a simple multiplication: more older people times higher per-person costs at older ages. The policy question is how fast, and whether the growth is sustainable — and the answer turns out to hinge on the same compression-of-morbidity question that has run through this entire health block. The European Commission's 2024 Ageing Report, which projects age-related expenditure to 2070, makes the mechanism explicit: countries submit age-and-gender cost profiles showing average public health spending per person at each age, and the projections track how those profiles interact with a shifting population. What is counted is precise — public current expenditure on curative care, rehabilitative care, ancillary services, medical goods, preventive care, governance, and other services (SHA 2011 categories HC.1 through HC.9, excluding long-term care's HC.3), plus gross capital formation. Notably, the 2024 exercise uses 2019 — not 2022 — as the base year, because the pandemic's extraordinary spending would otherwise distort every projection built on it.²

The decisive assumption is about healthy ageing. The baseline scenario assumes that half of projected longevity gains will be spent in good health and free of disability: age-specific disability rates shift along the age profile by half of the projected increase in life expectancy, so a 50-year-old man gaining four years of life expectancy takes on the disability prevalence of a 48-year-old. Unit costs, meanwhile, grow with GDP per worker for in-kind benefits and GDP per capita for cash benefits — meaning spending grows with the economy even if health itself is held constant. Against this baseline sits the “no healthy ageing scenario”: disability rates stay fixed, so even as life expectancy rises by two years, the share of disabled 80-year-olds in 2070 is identical to 2022 — none of the longevity gains translate into better health. The gap between those two scenarios is the fiscal value of compression of morbidity, quantified. Every prevention programme in Section 49, every equity intervention in Section 50, is ultimately an attempt to move the real world from the second scenario toward the first.²

There is empirical reason for cautious optimism about the baseline: WHO estimates that over 90% of gains in life expectancy at birth since 2000 have translated into disability-free life years. But the avoidable-mortality data shows how much fiscal waste the system still carries. Across 26 OECD countries, there were roughly 2.08 million premature deaths from preventable causes and 976,000 from treatable causes in 2020–21. Preventable mortality is dominated by circulatory diseases (21%), cancer (16%), injuries (17%), and alcohol and drug effects (13%); treatable mortality by circulatory diseases (37%) and cancer (23%). Preventable deaths are the fiscal footprint of unprevented risk — the smoking, diet, and alcohol levers of Section 49 unpulled. Treatable deaths measure quality gaps in the care system itself. And the cross-country spread is enormous: preventable mortality runs from 83 per 100,000 in Israel to 435 in Mexico; treatable from 39 in Switzerland to 230 in Mexico — the same “longevity is produced, not purchased” pattern Section 47 documented, now in mortality that better systems could have avoided.³⁵

Fiscal sustainability also has a household face. Health shocks are a common reason retirees hold precautionary savings — in Australia and the Netherlands, saving for health reasons was a primary motivation, more important for women than men — and the need depends heavily on public coverage: retirees in Northern Europe, with higher public healthcare coverage, spend down wealth more rapidly than those in Central and Southern Europe or the United States. Long-term care costs and divorce are the two shocks retirees are least prepared for; around 40% of lower-income Americans and Canadians report being financially unprepared for LTC expenses, and a quarter of elderly Europeans needing LTC could not finance even 10% of their costs from all their financial resources. Home equity is the last resort — after health insurance, the primary resource for financing a health shock, with wealth declining significantly after serious diagnoses driven by home sales. Public fiscal sustainability and private financial resilience are two sides of one ledger: where the state does not cover the risk, households self-insure badly, and the uninsured shock eventually lands back on the public system anyway.⁶

Evidence note

This retrieval contained the projection methodology but no headline expenditure projections — no “% of GDP by 2070” figures, no growth rates, no cross-country spending comparisons. The section therefore explains how the fiscal arithmetic works and what it depends on, without the numbers the projections would produce. A later evidence update should supply the Ageing Report’s actual expenditure paths under the baseline and no-healthy-ageing scenarios, which are the figures this section’s argument needs.

What this changes for the report

Health expenditure is where the health block’s abstractions become budget lines. The half-of-gains assumption is the single most consequential number in the ageing fiscal debate — more important than any spending ratio — because it encodes whether longer lives are assumed to be healthier ones. And the avoidable-mortality data reframes sustainability: the cheapest way to make health spending sustainable is not to spend less, but to stop buying the consequences of prevention foregone. Mental health (Section 54), where the treatment gap is widest and the spending thinnest, is the next test of that logic.

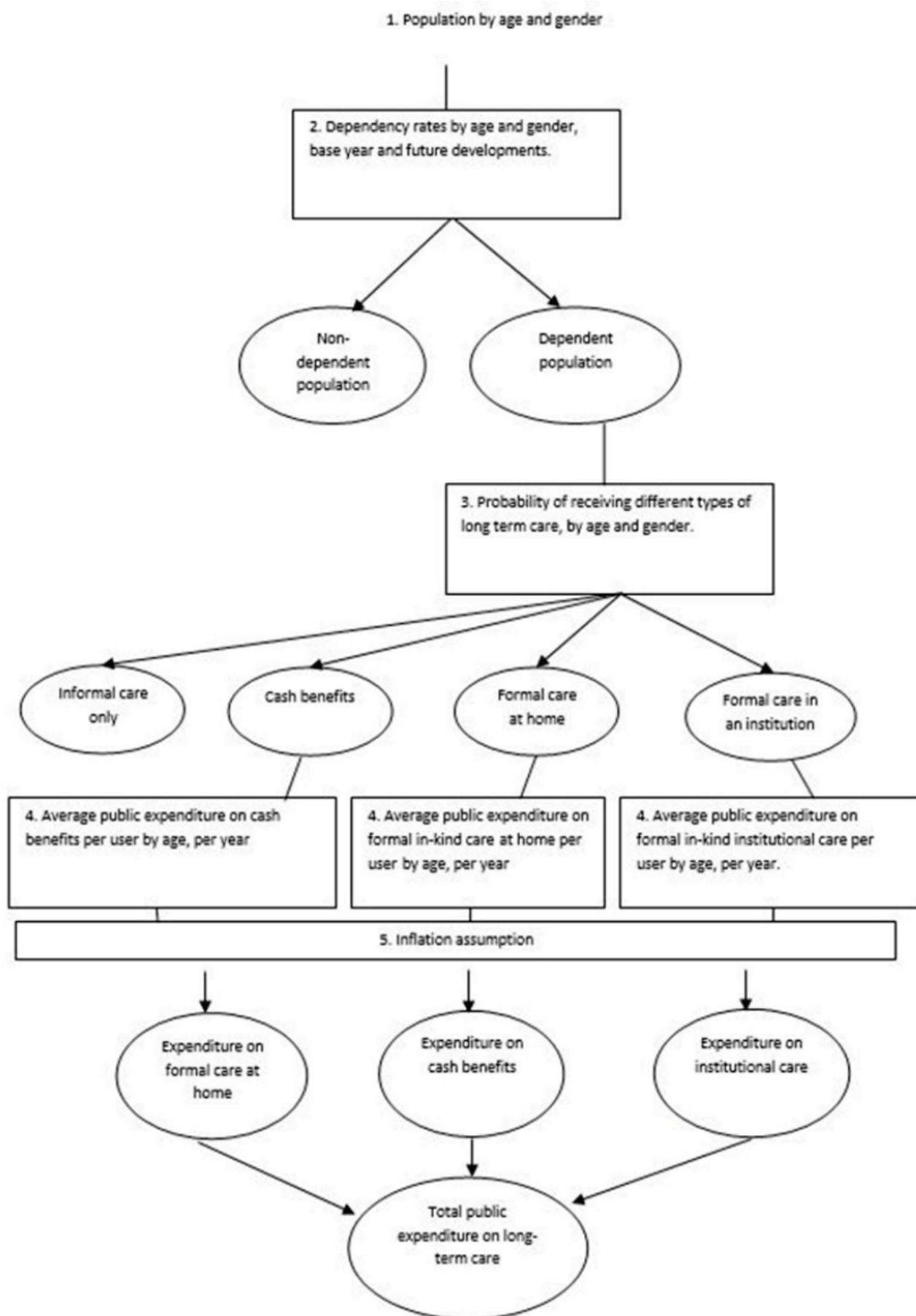


Exhibit. Graph II.A6.1: Long-term care model structure for the 2024 Ageing Report Source: European Commission, EPC. Source: S0015.

Mental health in later life

The health block has measured longevity in years, healthy years, disease burdens, and spending. It has said almost nothing about the mind — and that omission mirrors the health systems it describes. Mental health conditions are among the most common afflictions of later life, among the least treated, and among the most consequential for whether added years are livable ones. The WHO's founding definition — health as a state of complete physical, mental, and social well-being, not merely the absence of disease — is routinely quoted and routinely ignored in the allocation of resources. For the 100-year life, mental health is not a secondary outcome of physical ageing; it is one of its primary determinants.¹

The scale is substantial. OECD countries now track national estimates of depression and anxiety prevalence (2019–22) across a wide set of members — Germany, the Czech Republic, Canada, France, Italy, Japan, Belgium, Spain, the United Kingdom, Korea, Austria, Greece, Mexico, and the United States for depression; a similar spread for anxiety — though differing survey instruments and samples limit direct comparability. Suicide accounts for 1% of all deaths across OECD countries, and suicide-rate comparisons between 2000 and 2020 show the burden persisting at levels reaching 49.3 deaths per 100,000 in the hardest-hit country shown. These are not marginal phenomena. In a century in which most people will survive to 85, the conditions that make those years worth living — mood, cognition, purpose, connection — are the difference between a longevity dividend and a longevity ordeal.⁵

Much of later-life mental health is made earlier, at work. A review of 37 studies found moderate evidence that high job demands, low job control, high effort–reward imbalance, low relational and procedural justice, role stress, bullying, and low social support are all associated with a greater risk of developing mental health problems. In Belgium, some low-skilled occupations show measurably poorer mental health. The equity gradient of Section 50 thus extends into the mind: the same workers who face the worst physical conditions, the lowest pay, and the least control also carry the highest psychological burden — and carry it into retirement, where it compounds the isolation and bereavement that ageing brings. Cognitive health has its own occupational signature: in Denmark, occupational physical activity *increases* the risk of dementia while leisure-time physical activity has no such effect — a finding that upends the simple “activity is good” story and implicates working conditions directly in the cognitive decline that defines the oldest ages.⁴

The treatment architecture is the weakest part of the system. Relatively high rates of mental health conditions, alongside chronic illness, disability, and frailty, could be reduced if the structural, economic, and social drivers of poor health were tackled at an early stage — through education, screening, and lifestyle change. But mental health services remain the thinnest part of most health systems: harder to access, worse covered by insurance, more stigmatised, and — as Section 53's expenditure analysis implies — the first candidate for the “prevention foregone” ledger. The pandemic offered one proof that access can expand fast when systems are forced: the use of telemedicine was crucial to delivering care through the uncertainty of COVID-19, and remote delivery has particular promise for mental health, where the barriers are often logistical and social rather than clinical. Whether that promise survives the return to in-person norms is an open question the evidence here does not answer.¹⁵

Evidence note

This retrieval contained prevalence and suicide data but no treatment-gap statistics — no rates of untreated depression among older adults, no mental-health spending shares, no older-age-specific prevalence breakdowns. The section therefore documents the burden and its determinants, not the system's response. The response is arguably the more important half: a later evidence update should quantify what share of older adults with depression or anxiety receive effective treatment, at what cost, and with what outcomes. Without it, “mental health in later life” remains a described problem rather than a managed one.

What this changes for the report

Mental health closes the health block (Sections 47–54) on its most human note: every healthy-year gained, every disability compressed, every care worker hired is ultimately in service of minds that can enjoy the extra time. The block's arc runs from counting years (47), through the diseases that fill them (48), the prevention that could empty them of suffering (49), the unequal distribution of both (50), the technology that could coordinate the response (51), the workers who must deliver it (52), the money that must fund it (53), to the mental well-being that is its final pur-

pose (54). What follows is the care block — Sections 55–61 — where the question shifts from health to who provides the daily help that failing health, physical or mental, eventually requires.

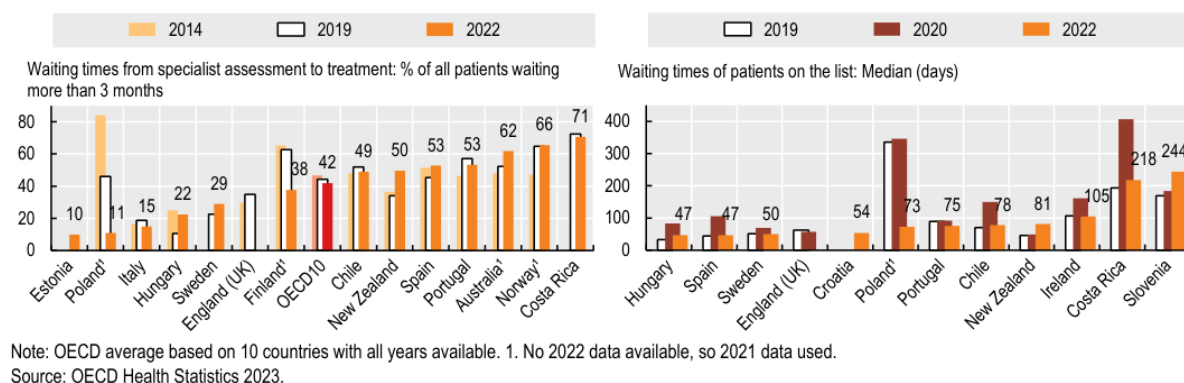


Exhibit. Figure 5.33. Waiting times for cataract surgery 2014 2019 2022 2019 2020 2022 Waiting times from specialist assessment to treatment: % of all patients waiting more than 3 months | Waiting times of patients on the list: Median (days) more than 3 months
80. Source: S0026.

Long-term care and caregiving

The economics of caregiving and long-term care

The 100-year life ends, for most people, in a period of dependency — months or years when daily life requires help with the tasks health once handled alone. The economics of that period is the least planned-for part of the longevity economy, and the numbers show it: long-term care costs are among the financial shocks retirees are least prepared for, with around 40% of lower-income Americans and Canadians reporting they are financially unprepared for LTC expenses, and most people over 85 having no realistic view of how they will pay for it. One European study estimates that a quarter of elderly people needing long-term care could not finance even 10% of their expenses from all of their financial resources. The care need is near-universal; the financing is near-absent.⁶

The economics of caregiving has two sides — who pays and who provides — and both are shaped by inequality. On the demand side, people with lower socio-economic status are more likely to need long-term care, but formal home care services — even for moderate needs — remain unaffordable for many of them in some countries, which creates an incentive to enter institutional care instead. This is a perverse economics: home-based care, which is generally cheaper for the system and preferred by the recipient, is priced out of reach for those who need it most, while institutional care — the more expensive setting — becomes the fallback. The affordability gradient means the poorest older people get the least choice about where they are cared for, and the public purse pays the highest unit cost for their care.³

On the provision side, the gap between need and formal services is filled by unpaid family care — and the distribution of that burden is profoundly gendered. But policy changes the distribution: in countries with higher levels of social protection for long-term care services, rates of informal care are lower, and gender inequality in caring is smaller. Public provision does not merely add a service; it substitutes for unpaid labour, and the substitution disproportionately frees women. The converse is also true: where the state does not provide, families do, and within families, women do.³

The cost of that unpaid provision is not only measured in hours. Long-term care needs of family members trigger employment, earnings, and pension losses for carers — particularly women — as caregiving collides with working life. The policy responses emerging across OECD countries recognise caregiving as a labour-market issue: phased or partial retirement that makes it easier to combine work with caring for older family members or grandchildren; Canada's right for employees with caring responsibilities to request flexible working arrangements; England's Care Act 2014, which places duties on local authorities to assess and support adult carers in maintaining or re-entering employment and training, including ensuring the person they care for is looked after while they work. These measures explicitly aim to fight the inequalities among older workers — above all women — that caregiving produces.³

When neither the state nor the family can cover the cost, the last resort is the house. Home equity release products can serve as a financing source of last resort for extraordinary expenses such as long-term care, and households in

practice use housing wealth this way — though with evident aversion, and typically only after a health shock or the loss of a partner. The housing-wealth channel (Sections 27, 66) thus becomes the LTC financing channel of last resort: the asset accumulated over a working life is liquidated to pay for the dependency at its end. For renters, who have no such asset, there is no last resort at all.⁶

Evidence note

This retrieval contained no LTC expenditure levels, no unit costs for formal home versus institutional care, and no data on informal care hours or their economic value. The section therefore describes the economic structure — who needs, who pays, who provides, and what happens when none of the three align — without the quantities. The LTC financing literature has those numbers, and Sections 56–57 will need them; a later evidence update should supply expenditure-to-GDP ratios, cost-per-user by setting, and the imputed value of unpaid care.

What this changes for the report

The economics of caregiving reframes the care block's central question. It is not whether societies can afford long-term care — the need will be met, one way or another, by the state, the market, the family, or the house. It is *who* bears the cost when each of those fails, and the evidence points consistently downward: to the poorest older people, who cannot afford formal home care; to women, who absorb the unpaid shortfall; and to the housing wealth of those lucky enough to own. Financing models (Section 56) are the attempt to answer that question deliberately rather than by default.



Exhibit. FIGURE 8 Informal carers around the world Sweden 1,300,000 carers Canada 7,800,000 carers Denmark 750,000 carers Finland 350,000 carers UK 13,600,000 carers Germany 4,927,000 carers Ireland 500,000 carers France 8,300,000 carers USA 56,400,000 carers India 138,328,000 carers Japan 6,280,000 carers Taiwan, China 1,400,000 carers Hong Kong SAR, China 376,000 carers Australia 2,650,000 carers New Zealand 430,000 carers. Source: S0010.

Financing models: insurance, public provision, and cost drivers

Section 55 established that long-term care is expensive, largely uninsured, and concentrated on the households least able to pay. This section asks who pays, through what channels, and why the bill keeps growing.

The care-financing asymmetry

Across the EU, most healthcare costs are covered by social protection systems. Long-term social care is treated differently: it is rare that such services are covered to the same extent as healthcare. The consequence is direct — the responsibility for financing institutional care often resides with the older person needing such care, or with their family.¹

This is the central financing fact of the care block: health systems socialize the cost of curing; care systems leave much of the cost of coping to private wallets. Any comparison of financing models has to start from this asymmetry, because it is what makes the design choices — insurance versus public provision versus private payment — bite.

Public provision: present but uneven

Demand is not the constraint. Almost one third of people aged 75 or more in the EU-28 face severe difficulties carrying out everyday tasks, suggesting considerable demand for homecare services; professional services that keep people independent longer — meal preparation, housekeeping, shopping, medication reminders, transport — are expected to become increasingly important.¹

Use of such services varies dramatically across countries. In 2014, 17.4% of people aged 75+ in the EU-28 had used homecare services in the preceding 12 months, with women (20.0%) using them more than men (13.5%) — a pattern repeated in every EU member state, and particularly stark in Latvia and Lithuania, where older women were three to four times as likely as older men to use them. Country variation is wide: at least one third of people aged 75+ used homecare services in France, Belgium, the Netherlands, and Malta, while the share was below one tenth in the Baltic member states, Croatia, and Romania.¹

Institutional care shows similar heterogeneity. In 2011, 3.8% of older women (65+) in the EU-28 lived in institutional households, twice the men's share (1.9%), partly reflecting women's longevity; at the high end, around 7% of older women in France and Belgium lived in such households, rising to 9.2% in Malta and 10.2% in Luxembourg. These patterns reflect both need and system design: how a country finances care shapes how much of it is provided, where it is delivered, and by whom.¹

Cost drivers

Why does the LTC bill grow faster than the economy that must pay it? Three drivers are visible even with thin data. First, demand growth: the share of the population needing help with everyday tasks rises with age, and the 75+ cohort is expanding; severe functional difficulty is common by 75, and services that keep people independent are expected to become increasingly important. Second, the unit cost of care: care is labour-intensive, delivered in person, and hard to automate — the same dynamics that make care work the fastest-growing employment need also make it expensive to deliver, and Section 52's finding that the workforce shortage is a retention problem means wages must rise if care is to exist at all. Third, the financing asymmetry itself: when services are not covered to the same extent as healthcare, costs fall on households, who defer, ration, and substitute unpaid family labour — pushing costs off public budgets and onto carers, mostly women (established in Section 55).¹

A caution: the evidence retrieved contains no quantitative cost drivers — no LTC expenditure as a share of GDP, no unit-cost figures, no projections of cost growth. The fiscal machinery of Section 53's health-expenditure projections has no counterpart in this retrieval for care.

The models: who pays and how

Countries assemble financing from three sources in different proportions: public provision (tax- or contribution-funded services), insurance (mandatory social insurance or voluntary private insurance), and private out-of-pocket and family payment. The EU evidence shows that even where public social protection is generous for healthcare, care provision remains patchy and familially absorbed. This creates the policy space insurance models aim to fill — converting an unbudgetable tail risk (years of care need, unknown at purchase) into a pooled, predictable premium. The detailed country models — Japan's mandatory social insurance, Korea's reforms, Singapore's hybrid approaches — belong to Section 57; the point here is the design logic: insurance pools risk that households cannot self-insure, public provision pools it further when markets and families fail.¹

The honest bottom line: financing models differ in who bears risk, but the risk itself is structural — ageing populations, labour-intensive services, and thin insurance markets. The asymmetry between health and care coverage remains the defining policy fact.

Evidence note

The retrieval contained no evidence on LTC insurance design, public LTC expenditure levels, unit costs, or cost-driver decompositions; the S0002 hazardous-jobs and S0013 DC-pension chunks were retrieval noise and were not used. All figures are EU-28 data from 2011 and 2014 — old, and predating the latest wave of care reforms. Country-specific financing models (Japan, Korea, Singapore) are treated in Section 57 and were not front-run here.

LTC insurance models: Japan's Kaigo Hoken, Korea, Singapore

Section 56 showed that most systems leave long-term care uncovered by the same social protection that covers healthcare. Three Asian economies took the insurance route deliberately — Japan's Kaigo Hoken ("care insurance"), Korea's reforms, and Singapore's hybrid schemes — and the contrast between their starting points is as instructive as their designs. The retrieved evidence, however, is thin on the specifics: it documents Korea's low-protection starting point and the general dynamics pushing countries toward insurance, but carries almost nothing on Japan's system design or Singapore's schemes. This section works with what the evidence supports and marks what it does not.

Why countries build care insurance

The case for a mandatory care-insurance model is the case against the financing asymmetry of Section 56: care need is a large, uncertain, late-life tail risk that households cannot self-insure, so leaving it uncovered concentrates the damage on the poorest and on women.

The dynamic is urgent beyond the OECD's mature welfare states. In emerging economies, families traditionally cared for older relatives, but falling birth rates and the mobility of younger generations are eroding that support, while larger cohorts exposed to non-communicable diseases are more vulnerable to disabilities requiring LTC. Formal LTC systems there are still nascent, demanding a more rapid policy response — including public support for civilian-run care homes or strengthened community health care services. This is the context in which Korea and Singapore built their models: not as refinements of mature welfare states, but as replacements for family systems that were visibly thinning.³

Korea: the low-protection starting point

Korea illustrates what care insurance has to fix. Low-income older Koreans (20th percentile of the income distribution) are already below the relative poverty threshold before paying anything for care — they are likely already struggling with the cost of living — and moderate home-care needs would require them to spend around a third of their disposable income on LTC. "Moderate needs" denotes 22½ hours of care per week, and the analysis assumes people have no savings to draw on. In no OECD country is anyone denied institutional care for being too poor, but where social protection is inadequate, poorer people face the institutional route rather than staying home.³

Korea sits in the low-social-protection group — countries spending less than 1% of GDP on publicly funded LTC — where women were 41% more likely than men to provide daily informal care, versus 23% in countries spending more than 2% of GDP. And the pension base underneath is thin: Korea is in the quarter of OECD countries with the worst pensioner poverty rates — as high as 50% of all over-66s. Care insurance built on this base cannot be a top-up to adequate pensions; it must be the foundation itself.³

Japan and Singapore: the gap

For Japan's Kaigo Hoken — the world's first dedicated mandatory long-term-care insurance system and the reference case for this section — the retrieved evidence contains no design details: no contribution structure, no eligibility thresholds, no benefit schedules, no expenditure figures. The same is true for Singapore's schemes. The corpus tells us these models exist and are important enough to headline a section; it does not tell us how they work. This section cannot compare designs the evidence does not describe.

What the evidence does teach about insurance design

Even without the blueprints, three lessons emerge from the comparative evidence.

First, coverage breadth is the design choice that matters most. All OECD countries with data provide some social protection for LTC, but the share of moderate home-care costs covered by the social protection system ranges from about 6% in some US states to about 45% in the Czech Republic and Israel to almost 100% in Sweden, Iceland, and the Netherlands. A mandatory insurance model that pools risk broadly is the alternative to this lottery by jurisdiction and income.³

Second, informal care is not free, and insurance design must price it. Informal carers have 20% more mental health problems than non-carers and are more likely to stop working or reduce hours; women make up 55–70% of informal carers in OECD countries. Insurance that ignores carers subsidizes a system whose costs land on women; support for carers is part of insurance design, not an add-on (the subject of Section 58).³

Third, the poverty floor comes first. Countries where low-income older people are already below the poverty line (Slovenia, Korea) need income support alongside insurance; insurance against care costs does not help people who cannot afford food.³

The honest bottom line: insurance models are the deliberate answer to the financing asymmetry — but the evidence retrieved lets us compare the starting points (Korea's low protection, the erosion of family systems) far better than the finished designs (Japan's Kaigo Hoken, Singapore's schemes). Closing that gap is this section's stated evidence update.

Evidence note

The retrieval contained no evidence on Japan's Kaigo Hoken (contributions, eligibility, benefits, spending), no evidence on Singapore's LTC schemes, and only fragmentary evidence on Korea (poverty and low-protection figures). The S0002 chunks (pension-reform references, life-expectancy inequalities) and the S0015/S0026 chunks (contents pages) were retrieval noise and were not used. S0016:0052 (phased retirement, carer support) belongs to Section 58 and was not used here. Figures are OECD 2017 vintage.

Unpaid family caregivers: support systems and the gendered burden

Every financing model and every workforce plan in the preceding sections rests on a provider that appears in no budget line: the family member who gives the care for free. Unpaid caregivers are the largest "provider" in the long-term care system, and the system could not function without them — which is precisely why their support, and the gendered distribution of their burden, is a design question, not a private matter.

The burden is gendered — and employment does not fix it

Women make up between 55% and 70% of informal carers in OECD countries, and the gradient steepens where public protection is thinnest: in countries spending less than 1% of GDP on publicly funded LTC, women were 41% more likely than men to provide daily informal care, against 23% where spending exceeds 2% (established in Section 57). Even where headline rates of informal care look similar between men and women, the type of care provided is often different — intensity, duration, and the physically and emotionally hardest tasks skew female.³

The intuitive hope — that full careers would buy women out of the caring role — fails in the data. A probit analysis of SHARE life-history data found that the marginal effect of having had a full career (fewer than ten years out of the labour force) on the likelihood of providing informal care was positive but non-significant for both women and men. Women who worked full careers are no less likely to provide informal care than those who spent a decade or more out of work, for example raising families. Rising female labour-force participation will not shrink the care burden; it doubles it — employment on top of caregiving — unless support systems absorb the extra load.³

The costs land on health and livelihoods

Informal care is not costless. Carers give up time for paid work or leisure, and the consequences compound: informal carers have 20% more mental-health problems than non-carers and are more likely to stop working or reduce their hours (established in Section 57). Family LTC needs trigger employment, earnings, and pension losses — particularly for women (established in Section 55). The gendered burden is thus a labour-market mechanism: it converts caring hours into lower lifetime earnings, and then — via the wage-to-pension pass-through — into the gender pension gap documented earlier in this report.

The pressure is growing from the supply side too. A future shortage of care workers is likely to strain long-term care services further, producing unmet needs and further increasing LTC costs — and unmet formal need does not disappear; it flows back to families, and within families, to women.³

Support systems: what keeps carers in work

Support for caregivers is best understood as retention policy: the carer is a worker, and every hour of care that costs a job is a loss to the labour market, the tax base, and the carer's own pension.

Two instruments are on the evidence. In Canada, the Employment Relations (Flexible Working Arrangements) Amendment Act of 2007 gives all employees with caring responsibilities the right to request flexible working arrangements. In England, the Care Act 2014 (in force 1 April 2015) places new duties on local authorities to assess and support adult carers to maintain or re-enter employment and training — for example by helping ensure that the person they care for is looked after while they are at work. These measures can help fight inequalities among older workers, in particular women, triggered by family members' long-term care needs.³

The design logic runs both ways: phased or partial retirement that lets carers reduce hours without penalty, respite care that makes flexible arrangements real rather than nominal, and assessment duties that surface carers before they quit. Insurance and financing models that ignore carers, as Section 57 argued, subsidize a system whose true costs land on women — which is why adequate social protection and carer support are the lever that limits gender disparities in care.

The honest bottom line: the gendered burden of care is not a cultural footnote — it is the mechanism through which the financing asymmetry of Section 56 becomes the labour-market and pension inequality of the earlier blocks. Support systems exist and have a clear theory of action; the evidence retrieved says nothing about their scale, coverage, or cost.

Evidence note

The retrieval contained no statistics on the scale or spending of carer-support programmes, no informal-care-hours data, and no valuations of unpaid care; the S0029 pension-adequacy chunks and the S0016 reference lists were retrieval noise and were not used (the S0016 notes on SHARE findings and workforce strain were the exception). The Canada and England measures are dated 2007 and 2015 respectively.

Quality, accessibility, and affordability of care

Sections 55–58 established who needs care, who pays for it, and who provides it. This section asks the next question: is the care good, can people reach it, and can they afford it? The retrieved evidence answers almost nothing directly — there are no quality indicators, no waiting-time or access statistics, no affordability measures beyond the preparedness figures already reported. What follows is therefore built on what the earlier sections established, with the gaps stated plainly.

Quality: the unmeasured dimension

No chunk in the retrieval measures care quality — no clinical outcomes, no safety incidents, no recipient or family satisfaction scores, no accreditation data. That absence is itself a finding: care is the part of the system we count least.

What can be said is structural. Quality in care is overwhelmingly a function of the workforce, and Section 52 established that the workforce's central problem is retention, not headcount — the work is demanding, draining, and poorly paid, and the system loses the workers it trains; across the OECD, LTC worker supply has not kept pace with demand from rapidly ageing populations. Professionalisation (real wages, career ladders, qualifications, manageable workloads) was identified there as the precondition for the workforce existing at all. A system that cannot keep its workers cannot guarantee the quality of their work: high turnover breaks the continuity of relationships on which good care depends, and underpaid, unsupported workers are the leading indicator of neglect, not of excellence.⁵

Accessibility: geography decides

Access to care has a geography, and it runs against need. Section 56 documented that homecare use varies enormously — from over a third of the 75+ population in France, Belgium, the Netherlands, and Malta to under a tenth in the Baltic states, Croatia, and Romania. Section 50 found that older people are disproportionately rural, and Section 52 that the workforce is geographically inverse to need: rural areas are the oldest and the thinnest-served. The emerging-economy picture adds that parts of East Asia and Pacific need their primary-care systems rebuilt before community and home outreach can reach anyone.¹

Accessibility, in other words, is not one number — it is the product of service availability (are there providers?), physical reach (can they get to the person?), and eligibility rules (does the person qualify?). The retrieval quantifies none of these for care. The structural fact is that the oldest populations live farthest from the services, and the thinnest provision sits where the need is deepest.

Affordability: the preparedness gap

Affordability is the one leg of the triangle with evidence, and it is bleak. Long-term care is among the least prepared-for financial shocks: around 40% of lower-income people in the United States and Canada report being unprepared for LTC expenses, most of those over 85 have no realistic view of how they will pay, and an estimated 25% of elderly Europeans needing LTC could not finance 10% of their costs from all their resources (established in Section 55). Section 55 also documented the perverse gradient: formal home care — cheaper for the system, preferred by recipients — is unaffordable even for moderate needs in some countries, pushing people toward institutional care, the more expensive setting. Korea's case in Section 57 showed moderate home care consuming around a third of a low-income older person's disposable income.⁶³

The triangle: you can't have all three cheaply

The three dimensions trade against each other, and the trade-offs explain much of the politics of care. Higher quality — better-paid, better-trained, stable workers — raises unit costs, worsening affordability unless public funding rises with it. Wider accessibility — reaching rural and remote populations, serving people in their homes — raises costs per recipient, because travel and thin caseloads are expensive. And affordability achieved by holding budgets down tends to squeeze the other two: cheaper care means worse-paid workers and rationed eligibility.

This is why the care block's earlier findings hang together. The financing asymmetry (Section 56) leaves affordability to households; the insurance and provision models (57) try to socialize it; the workforce crisis (52) is the quality constraint; and the gendered burden (58) is what happens when all three fail — families, and within families women, absorb the shortfall. The triangle cannot be wished away; it can only be managed by deciding, explicitly, which dimension gets priority and who pays for it.

Evidence note

The retrieval contained no evidence on care-quality indicators, access/waiting-time statistics, or affordability measures beyond the preparedness figures already used in Sections 55 and 57; the health-status, digital-health, and pension-decumulation chunks were retrieval noise. This section therefore reports no new figures and repeats no old ones as new — it assembles what the report has established into the quality-access-affordability frame, and names the missing data as the section's evidence update.

Ageing in place: home care and community-based services

Ageing in place — remaining in one's own home and community as care needs grow — is the stated preference of most older people and the stated policy aim of many OECD countries: helping people with long-term care needs stay in their own homes for as long as possible (established in Section 57). It is also the setting where the report's established findings collide: the demand is documented, the financing is thin, the workforce is scarce, and the informal layer carries most of the load. This section maps what home and community care consists of and what it needs to work. (The built-environment side of ageing in place — housing design, transport, neighbourhoods — belongs to Section 65 and is not front-run here.)

Home care: what it is and who gets it

Home care spans two kinds of help. Under the System of Health Accounts, the health component covers nursing care and personal care (help with activities of daily living — ADL), including palliative care and care delivered in LTC institutions or at home; the social component covers help with instrumental activities of daily living — IADL, the domestic and logistical tasks of everyday life. Progress has been made in making LTC spending comparable across countries, but reporting still varies — in some countries social LTC is partly folded into health LTC, in others only health LTC is reported, and private LTC expenditure is unevenly captured.⁵

Demand is well established: almost one third of people aged 75 or more in the EU face severe difficulties with everyday tasks, and professional home services — meal preparation, housekeeping, shopping, medication reminders, transport — are expected to become increasingly important in keeping people independent longer (established in Section 56). A person with severe needs is defined as requiring 41.25 hours of care per week — a full-time job, which is why home care at the severe end is as much a staffing question as a service question.⁵

Use, however, varies enormously: in 2014, 17.4% of the EU's 75+ population had used homecare services in the preceding year, but the share exceeded a third in France, Belgium, the Netherlands, and Malta and fell below a tenth in the Baltic states, Croatia, and Romania (established in Section 56). The gap between need and use is where accessibility, financing, and workforce constraints bite — the theme of Section 59.

Community-based services: the infrastructure between home and hospital

Home care does not work as isolated visits. The delivery model described in Section 51 — smaller hospitals plus intermediate facilities plus larger primary-care centres plus communications technology for community, LTC, and home outreach — is the architecture that makes ageing in place feasible: assessment, medical backup, and social support connected around the person. Tools like interRAI's person-focused longitudinal records across the continuum of care are the information layer such a model needs (established in Section 51).

The provider mix is more fluid than the labels suggest. LTC spending flows through nursing homes, hospitals, home-care providers, households themselves, social providers, and others — and providers blur: in the Netherlands, around 20% of expenditure allocated to nursing homes is actually for homecare service provision. The institutional/home boundary is administrative, not physical; what matters is where the care is delivered, not where the provider is headquartered.⁵

The informal layer: defined but unevenly counted

Beneath the formal system sits the layer Section 58 analyzed: informal carers, defined by the OECD as people providing any help to older family members, friends, and people in their social network — living inside or outside their household — who require help with everyday tasks. Measurement is fragmented: the data cover the 50+ population and draw on national surveys with different definitions — SHARE counts people providing care at least weekly, ELSA asks about the last week, the US Health and Retirement Survey includes those providing more than 200 hours in the last year, and Australia's survey requires assistance ongoing for six months or more — so cross-country comparison is approximate at best. The intensity gradient (occasional, weekly, daily carers) matters: daily care at 41.25-hours-per-week severity is a different commitment from a weekly visit, and policy that treats all carers alike misfires.

What ageing in place requires

Ageing in place is not the cheap option — it is the option that moves costs from institutions to homes, and homes need the same infrastructure institutions have: a workforce willing to travel (the rural access problem of Section 59), financing that covers home services to the same extent as institutional ones (the asymmetry of Section 56), carer support that keeps family members in work (Section 58), and housing that is adaptable (the subject of the housing block ahead). The preference is real; the system to deliver it is not yet built at the scale the preference demands. Whether it gets built is the workforce and financing question of the remaining sections.

Evidence note

The usable evidence was definitional (ADL/IADL components, the 41.25-hour severe-needs threshold, informal-carer measurement caveats) plus the Netherlands provider-blur figure. No home-care outcome data, no community-service coverage statistics, and no country-comparable provider-mix figures could be extracted — the spending-by-provider chart text was unreadable. The S0016 pension-inequality chunks were retrieval noise and were not used. The built-environment dimension of ageing in place is reserved for Section 65.

Long-term care workforce and provision capacity

Every financing model in Sections 56–57, every service design in Sections 59–60, and every support system in Section 58 assumes one thing: people to do the work. This section closes the care block on that assumption — and on how far short of it the system falls.

The numbers: wide variation, thin everywhere

The global economy must create 475 million care jobs by 2030 to meet ageing-related needs and UN Sustainable Development Goal health targets (established in Section 52). Against that requirement, current provision is wildly uneven: long-term care workers per 100 people aged 65 and over range from about 12 at the top of the OECD distribution to below 1 at the bottom, with the 2021 data spanning roughly 12.0, 11.7 and 11.2 per 100 down through the single digits to 0.8 and 0.2 per 100. (Country labels in the chart text were unreadable, so the ranking cannot be reported — only the spread, which is the point: a sixty-fold difference in staffing density across countries facing the same ageing trend.)⁵

The pressure is growing from both sides. Demand rises as populations age and as unhealthy years accumulate — the rise in unhealthy years could put a greater burden on health and long-term care systems, with wide disparities by sex, education, and geography. Supply depends on a workforce that the system currently treats as disposable.³

Job quality is the binding constraint

Section 52 established the core diagnosis: the shortage is retention, not headcount. The work is demanding, draining, and poorly paid, and the system loses the workers it trains. The OECD tracks the dimensions of that problem — the share of LTC workers on part-time or fixed-term contracts, and the average hourly wage of personal care workers (ISCO-08 category 53, which groups personal care workers together with childcare workers) as a share of the economy-wide average wage — but the chart values in the retrieval were unreadable, so no figures can be reported here. The structural fact survives the missing numbers: precarious contracts and below-average wages are the mechanism of the retention crisis, and the retention crisis is the mechanism of the quality crisis documented in Section 59.⁵

Professionalisation — real wages, career ladders, qualifications, manageable workloads — was identified in Section 52 as the precondition for the workforce existing at all. It is also the precondition for provision capacity: institutions and home-care providers cannot deliver what they cannot staff. LTC institutions are defined as nursing and residential care facilities providing accommodation and LTC as a package, designed for dependent people with moderate to severe functional restrictions — and a person with severe needs requires 41.25 hours of care per week (established in Section 60). Each severe case is a full-time worker; capacity is arithmetic.⁵

Who does the work, and at what cost to whom

The paid care workforce mirrors the unpaid one: it is overwhelmingly female, often migrant, and stratified by the global care chain — richer countries import care workers, exporting the shortage to the countries that can least afford it (established in Section 52). When the formal workforce does not exist, families absorb the shortfall, and within families women absorb families' shortfall (Section 52) — the 55–70% female share of informal carers (Section 58). Provision capacity is therefore a gendered quantity twice over: in the formal labour market and in the household.

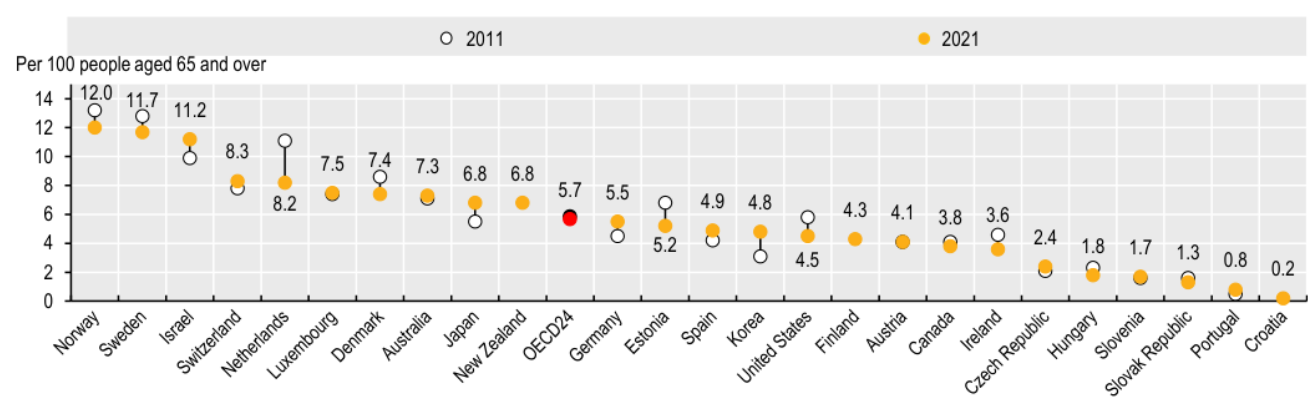
In emerging economies the formal workforce barely exists yet: family support is eroding with falling birth rates and youth mobility, cohorts are more exposed to non-communicable diseases, and formal LTC systems are nascent (established in Section 57). The capacity gap there is not a staffing shortfall within an existing system — it is the absence of the system.

Closing the care block

The care block (55–61) ran one arc: the economics of need and payment (55), how societies finance it (56), the insurance models that pool the risk (57), the unpaid caregivers who carry the residual (58), whether the care is good, reachable, and affordable (59), how it is delivered at home and in communities (60), and who does the work (61). The through-line is a single constraint restated in seven ways: care is labour-intensive, hard to automate, emotionally and physically demanding, poorly paid, largely uninsured, and provided mostly by women — paid and unpaid. Every strategy in the earlier blocks of this report assumes that labour; it does not exist in the required numbers, and the systems that would create it — wages, training, migration policy, financing — are the agenda the housing and policy blocks inherit. The next block turns to where all of this happens: housing and the built environment.

Evidence note

The country ranking for LTC-worker density could not be recovered (chart labels unreadable); only the range is reported. Part-time/fixed-term shares and personal-care wage ratios are tracked by the OECD but their values were unreadable in the retrieval. The S0016 reference-list chunks and the end-of-life/life-expectancy chunks were retrieval noise; the disability-trends note was the exception used.



Note: For New Zealand, latest data refer to 2018.
Source: OECD Health Statistics 2023.

Exhibit. Figure 10.18. Long-term care workers per 100 people aged 65 and over, 2011 and 2021 (or nearest year) 2011 2021
Per 100 people aged 65 and over 14 12. Source: S0026.

Housing and age-friendly environments

Housing affordability and living conditions of older people

The care block asked who provides help when health fails; the housing block asks where all of it happens. Housing is the physical container of the 100-year life: care is delivered at home, wealth is stored in the house, and living con-

ditions — who lives alone, where, in what dwelling — shape health, poverty risk, and the services people need. This section opens the housing block (62–67) on the affordability and living-conditions evidence.

Living alone: the growing and vulnerable household

Recent decades have seen a fall in average household size across the EU — driven by lower fertility, more divorces, and the dissolution of extended households — and a growing number and share of older people living alone, particularly older women. They form a particularly vulnerable group, with an increased risk of poverty or social exclusion.¹

The household type in which older people live plays an important role in determining their quality of life, their risk of poverty, and the services they require — and the trend raises concerns precisely because a growing number of older people live alone. A single-person older household has no economies of scale in housing costs, no co-resident to notice a fall or a missed medication, and no partner's pension to fall back on. Living alone is the living arrangement where the care block's findings bite hardest: it is where unmet formal need becomes unmet need, full stop.¹

Rural ageing: older, farther, thinner

A relatively high share of older people in the EU lives in rural areas — a distribution with direct bearing on policy-makers assessing access to services. The rural older population is the same geography that Section 50 found oldest and Section 52 found thinnest-served: the workforce is inverse to need, homecare use is lowest in the countries where rural ageing is most pronounced, and primary-care systems in parts of East Asia and Pacific need rebuilding before outreach works at all. Housing affordability in rural areas may look benign — cheaper homes — but cheap housing far from services is the affordability trap flagged in the continuity notes: what is saved on rent is paid in isolation and unmet need.¹

Material deprivation and the home budget

Older people's spending patterns are transformed with age: they devote a higher proportion of their expenditure to health, food, and the home in which they live, and a lower proportion to transport, clothing, and footwear. The home is the largest budget line — which makes housing costs the decisive affordability question, and the one the retrieval does not answer.¹

What the evidence does measure is deprivation around the edges of the budget. In 2015, 17.0% of people aged 55–64 in the EU-28 could not afford to spend a small amount of money on themselves each week — a cinema ticket, a magazine, an ice cream — a form of material deprivation systematically more prevalent among older women than older men in every age group shown. Country disparity is stark: more than half of 65–74-year-olds in Romania (52.6%) faced this deprivation, and over 40% in Greece and Bulgaria, against under 5% in six member states and just 0.2% in Finland. These figures are 2015 vintage, but the gradient they reveal — women, southern and eastern Europe, the pre-retirement decade — maps onto the pension and care inequalities of the earlier blocks.¹

Income inequality among the over-65s adds the distributional frame: the OECD-average Gini of disposable income is 0.306, but ranges from 0.500 in Costa Rica and 0.451 in Mexico down to 0.198 in Czechia; the 90th percentile older person has four times the income of the 10th. Living conditions in old age are unequal twice over — unequal incomes and unequal housing.⁴

The affordability evidence that is missing

The central affordability questions — what share of older people's income goes to housing costs, how rents and prices have moved against fixed pensions, overcrowding, arrears, the quality of the dwelling stock — are not in the retrieval. The tag list promised "older people were less likely to be in arrears," but no chunk carried the data. What is established from earlier sections is the wealth side: housing is the de facto retirement reserve (established in the financial-resilience findings), and home equity release is the financing of last resort for shocks like long-term care (Section 55) — used with aversion, typically after a health shock or partner loss. The house is simultaneously the largest budget line, the largest asset, and the last-resort insurance policy. Affordability policy that treats it as only one of the three misfires.

The honest bottom line for the block’s opening: the living-conditions evidence documents who is vulnerable (older women living alone, rural older people, the income-poor) and how unequal their means are, but not what their housing costs them. The remaining sections take up the built environment (63–64), community design (65), the wealth locked in homes (66), and the climate risks to dwellings (67).

Evidence note

No housing-cost, price, rent, overcrowding, or arrears data appeared in the retrieval; the pension-income and life-satisfaction chunks were noise and were not used. The deprivation figures are 2015 vintage. Section 66 takes up housing wealth and downsizing; this section does not front-run it.

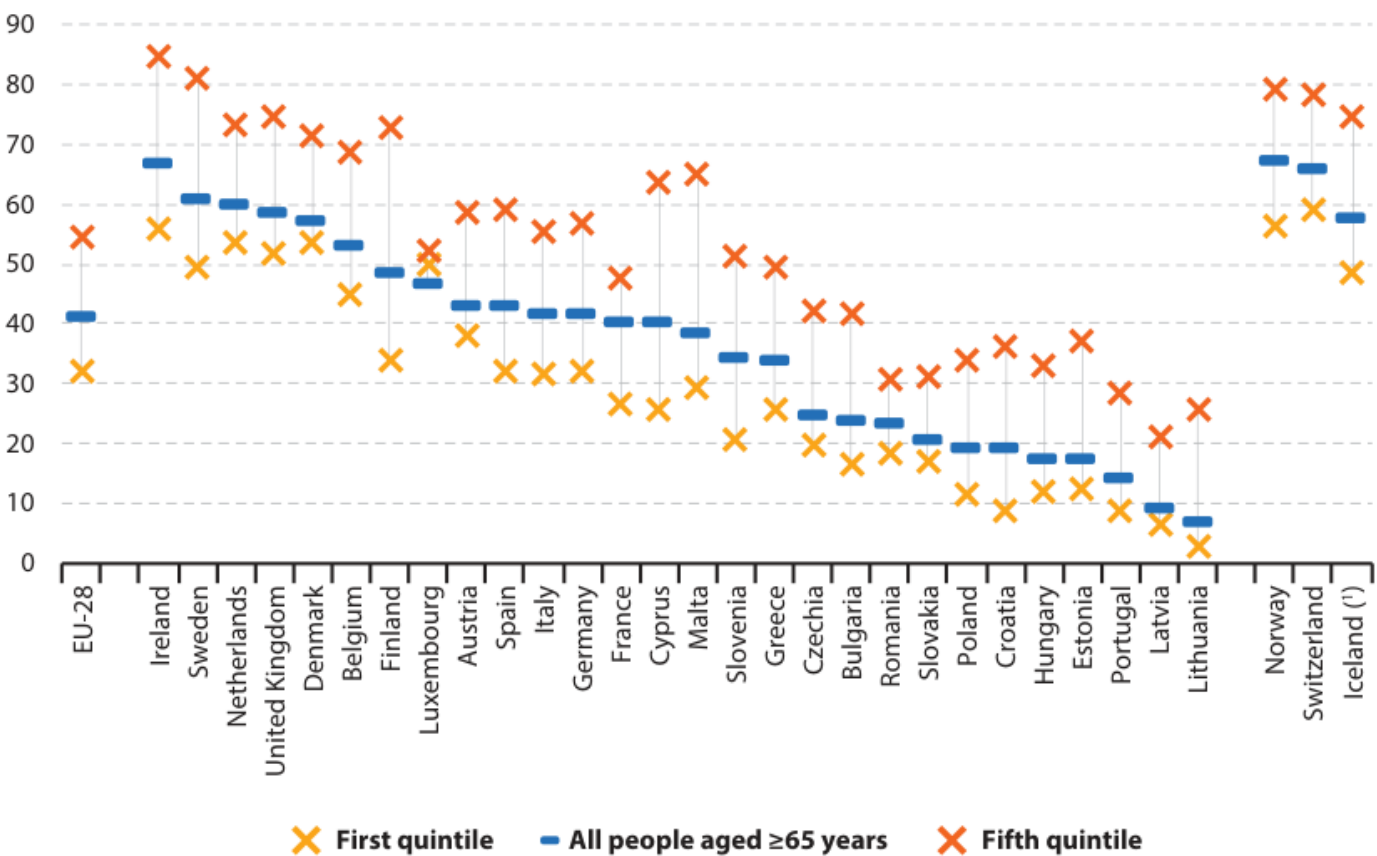


Exhibit. Figure 3.5: Self-perceived health among people aged ≥65 years, by income quintile, 2017 (% perceiving their own health as good or very good) 90 80 70 60 50 40 30 20 10 0 First quintile All people aged ≥65 years Fifth quintile EU-28. Source: S0014.

Age-friendly cities: transport, public space, and services

Most older people live in cities — in the EU, the 65+ population splits across urban, intermediate, and rural areas (established in the demographic findings), and the urban share is where services, transport, and public space either enable a long life or quietly foreclose it. The reference framework is the WHO’s 2007 Global Age-friendly Cities: A Guide, built around “active ageing” and organized in twelve parts — outdoor spaces and buildings, transportation, housing, social participation, respect and social inclusion, civic participation and employment, communication and information, and community support and health services. This section works through the three named in the title — transport, public space, and services — with the gaps stated plainly, because the retrieval carried the framework’s table of contents but almost none of its substance.¹⁴

Transport: the key to everything else

Transport is the master key of age-friendliness: without it, services are unreachable, social participation collapses, and independence ends. The WHO guide gives transportation its own part, and the logic is straightforward — affordable, accessible, reliable public transport with seating, shelter, clear information, and step-free boarding is what converts a service on a map into a service a person can actually use.¹⁴

The retrieval contains no transport-access statistics — no public-transit coverage for older people, no data on driving cessation, no figures on mobility poverty. What the earlier sections established is the geography it must serve: older people are disproportionately rural (Sections 50, 62), rural areas are the thinnest-served (Section 52), and older people's expenditure devotes a lower share to transport than younger people's (Section 62) — partly because they travel less, partly because travel has become harder. The transport question is sharpest exactly where the older population is largest relative to services: outside the cities the guide was written for.

Public space: outdoor spaces and buildings

The WHO framework treats outdoor spaces and buildings as a single domain: walkability, pavement quality, seating, lighting, public toilets, safe crossings, and buildings with step-free access and legible signage. These are not amenities; they are the physical preconditions of the prevention agenda (Section 49 — exercise across the life course), of fall prevention, and of the social participation that Sections 81–84 will take up.¹⁴

Again, no evidence on the ground: no walkability scores, no data on falls attributable to the built environment, no accessibility audits. The structural point stands without them — a city that is hostile to a person with a walker is hostile to the added years the whole report is about.

Services: community support, health services, and information

The WHO guide closes its domain list with community support and health services, preceded by communication and information — an ordering that matters. Services that older people cannot learn about, book, or reach might as well not exist. The digital-health equity warning of Section 51 applies here in full: information systems that assume connectivity reach the already-connected first.¹⁴

The remaining WHO domains — social participation; respect and social inclusion; civic participation and employment — belong to the social-connection block later in this report (Sections 80–84) and are not front-run here. Housing, the WHO's Part 7, is this block's own subject (Sections 62, 64–66).

The honest bottom line

Age-friendly cities are the built-environment complement to ageing in place (Section 60): the home is where care happens, but the city is what makes staying home possible. The framework is clear and the domains are right — transport, public space, services, participation, inclusion. What the evidence retrieved does not contain is any measure of whether cities are actually delivering: no transport-access data, no public-space quality indicators, no service-coverage statistics for older residents. The guide is nearly two decades old; the audit of its implementation is the section's stated evidence update.

Evidence note

The only usable evidence was the WHO guide's table of contents; all substantive claims about transport, public space, and services in this section are continuity from earlier sections, stated as such. The pension-decumulation, financial-incentive, and hazardous-jobs chunks were retrieval noise and were not used.¹⁴

Built-environment choices for longer, healthier lives

The previous section mapped the WHO age-friendly cities framework — transport, public space, services. This section goes narrower and deeper: the built environment as a health technology in its own right. Streets, pavements, stairwells, doorways, bathrooms, and bedrooms are not neutral backdrops to ageing; they are the physical machinery that converts a person's remaining capacity into actual functioning. A body that can walk a kilometre on a flat, well-lit pavement cannot cross a neighbourhood of broken kerbs and missing crossings. The difference is not the body — it is the environment, and it is chosen by design, investment, and maintenance.

The core claim is simple: extending healthy life (the healthy-life-expectancy agenda established in Section 47) happens as much in architecture and urban design as in clinics. Prevention — risk-factor management, physical activity, fall avoidance — requires a physical substrate. Walkability is the most economical exercise prescription there is: level pavements, frequent seating, safe crossings, legible signage, public toilets. None of these appears in health

statistics, yet each determines whether an older person leaves the house at all. A person who cannot safely walk to the shop does not just lose a trip; they lose daily activity, sensory stimulation, and casual social contact — the very inputs the loneliness and social-connection agenda of Sections 80–84 depends on.

Inside the home, the same logic applies at smaller scale, and it is here that the environment meets the ageing-in-place ambition directly. The housing chapter opened with the finding that the home is the largest budget line for older people and, simultaneously, the largest asset and the last-resort insurance — the place where most late-life care will happen, delivered by home-care workers and informal carers. Institution-grade home infrastructure therefore means literal infrastructure: step-free entrances, ground-floor bathrooms, grab rails, non-slip surfaces, adequate lighting, lever handles, kitchens and bathrooms that a wheelchair can turn in. These are universal-design choices, cheap at construction and expensive at retrofit. Retrofitting is where the inequality bites hardest: the people most likely to need adaptations — renters in poor housing, lower-income homeowners, older women living alone in older stock — are least able to afford them, reproducing in the bathroom and the stairwell the same deprivation gradient measured in household budgets.

Falls crystallize the stakes. Falls are where the built environment, the body, and the health system meet: a loose rug, a dark stair, a missing handrail converts a manageable limitation into a fracture, a hospital admission, and the long rehabilitation pathway that strains health budgets. Prevention agendas that prescribe exercise and strength training without prescribing the handrail are half-built. The design lesson runs in both directions: the home should be designed so that declining capacity does not convert into declining function, and the neighbourhood should be designed so that declining function does not convert into declining participation.

The geography of the built environment follows the same rural-urban fault line already traced through this report. Older people are disproportionately rural; rural places are the thinnest-served by services and, typically, the least invested in pavements, lighting, and public transport. The age-friendly city guide was written for cities, but the sharpest need is outside them — a point Section 63 established and which applies with full force here. Meanwhile, within cities, the environment people age in is the one they could afford: the deprivation gradient is a measured reality — from 52.6% of 65–74-year-olds in Romania facing material deprivation to just 0.2% in Finland — and it maps onto the environments people age in: where walkability, greenery, and step-free access exist is as much an equity question as whether pensions exist.¹⁴¹

There is also a temporal dimension that distinguishes built environment from every other policy lever in this report. Pensions, services, and workforce capacity can be reformed in a legislative cycle; buildings and streets are decisions made decades ago, renovated slowly, and lived with for generations. Today's older people are ageing in the housing and urban fabric of the 1960s–1990s, built when nobody planned for female life expectancies approaching 90 in the EU's long-run projections. Every new home built today without universal design is a future retrofit cost — and a future fall, a future care episode, a future constraint on independent living. The longevity economy's physical infrastructure is being designed now, mostly by accident. Doing it deliberately — visitability standards, step-free by default, mixed-use neighbourhoods that keep daily life within walking distance, transport interchange that assumes a Zimmer frame — is the cheapest long-run investment in healthy years that policy can make.²

Evidence note

This section rests on framework and continuity, not new statistics: the retrieval for this section contained **zero usable evidence chunks** — all nineteen chunks concerned pension replacement rates, labour-market exit ages, out-of-pocket health spending, and teleconsultations, none of them on the built environment. There are no home-modification uptake rates, fall-injury statistics, walkability audits, or universal-design adoption figures in the provided evidence, and none are presented here. The gap is substantive: the one policy domain that physically determines whether ageing in place works is the least measured. An implementation audit — how many homes are actually adaptable, how walkable older people's neighbourhoods actually are, where falls actually happen — is the evidence update this section needs.

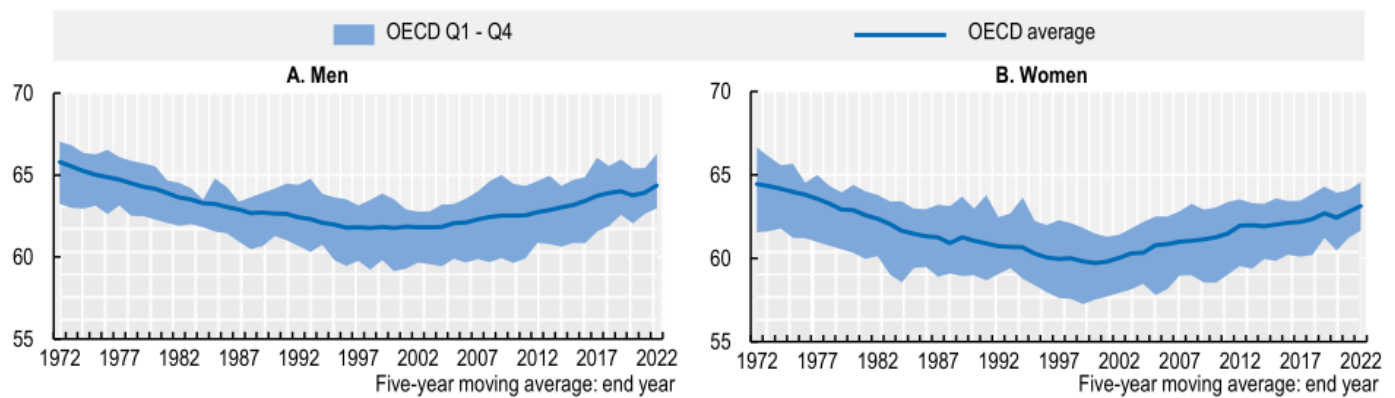


Exhibit. Figure 6.14. Average effective age of labour market exit in OECD countries, 1972-2022 OECD Q1 - Q4 OECD average
A. Source: S0002.

Ageing in place and community design

Ageing in place is the stated goal of most older people and most care systems: to live out later life in one's own home and community rather than in an institution. The previous three sections established why it is hard. Section 60 showed that ageing in place requires institution-grade home infrastructure — workforce, financing, carer support, adaptable housing. Section 63 added the urban layer: transport, public space, and services that reach the door. Section 64 showed the environment itself is a health technology. This section brings the two scales together at the one scale where they are actually experienced: the community — the dwelling plus its immediate neighbourhood, its services, and its social fabric, designed as a single system.

The first neglected truth about ageing in place is that homes age alongside their owners. Roofs leak, boilers fail, wiring deteriorates, stairs that were fine at sixty become hazards at eighty — the fall-prevention logic of Section 64. This is not a design problem but a maintenance one, and it lands with real financial force: a US survey found over 40% of retirees below age 80 had experienced an unexpected financial shock, and nearly a quarter of those shocks were major home repairs. In Costa Rica, 61% of those who withdrew a retirement lump sum used it for home improvements. An ageing-in-place policy that funds home care visits but not the roof over the care is mis-measuring the need. Maintenance support — grants, low-cost loans, trusted repair services — belongs inside the ageing-in-place infrastructure, not outside it, because the dwelling is the precondition of every other service delivered to the door.⁶

The second truth is that the equity is trapped where the need is. Home equity is, after health insurance, the primary resource retirees use to finance a health shock; one US study showed a significant decline in wealth following a serious disease diagnosis, driven primarily by a reduction in home equity after selling the home. Older people are, on average, housing-rich and cash-poor — the house is the last-resort insurance established in Section 55 — yet ageing-in-place support systems rarely connect households to that equity except through crisis sales. The detailed mechanisms of equity release belong to Section 27 and the housing-wealth analysis of Section 66; the point here is architectural: a community designed for ageing in place would make housing wealth legible and usable *before* the crisis, not only at it.⁶

The third truth is the inequality gradient. People with lower socio-economic status are more likely to need long-term care, yet formal home care services — even for moderate needs — remain unaffordable for many of them in some countries, which gives them an incentive to move into institutional care. Ageing in place is therefore not a neutral preference: the affluent can buy the services that make staying home work, while the poorest face a choice between going without and going into an institution. The same OECD analysis found that in countries with higher social protection for long-term care services, rates of informal care are lower and gender inequality in caring is smaller — the community-design corollary being that public provision is what makes ageing in place available rather than merely aspirational.³

Community design, then, is the deliberate knitting together of what the previous sections kept separate: adaptable dwellings (Section 64), home-care services (Section 60), carer support (Section 58), accessible transport (Section 63), and social infrastructure — the places and routines where neighbours meet. The models exist in practice even if they do not appear in this report's evidence: neighbourhood hubs co-locating health and social services, village-style volunteer networks, cohousing, naturally occurring retirement communities. Their common principle is that ageing in place is a community outcome, not an individual achievement. An older person living alone is not ageing in

place because she has a strong front door; she is ageing in place because there is a walkable route to the shop, a neighbour who notices, a home-care worker who comes on time, and a repair service she can call.

That is also where the rural question bites hardest. The disproportionately rural older population (Sections 50, 62) lives in precisely the communities with the thinnest service infrastructure and the weakest social-service networks — and the most dispersed informal-carer networks as younger people leave. Community design for these places cannot rely on density; it needs mobile services, telehealth backstops (with the Section 51 equity warning), and transport solutions that assume distance.

Evidence note

Two usable facts anchor this section: home repairs as a major financial shock for retirees (over 40% of US retirees below 80 experienced a shock; nearly a quarter of those were major home repairs; 61% of Costa Rican lump-sum withdrawals went to home improvements), and the socioeconomic gradient in ageing-in-place access (lower-SES people more likely to need LTC but less able to afford formal home care; higher social protection reduces informal-care rates and gender care inequality). The pension-replacement, hazardous-jobs, and equity-investment chunks were noise. No data on community-design model uptake, maintenance-support program coverage, or neighbourhood service density were available — gap stated.⁶³

Under-occupied homes, downsizing, and housing wealth

The housing block has established that the home is an older person's largest budget line, largest asset, and last-resort insurance. Section 27 dissected the products that unlock that asset — reverse mortgages, home reversion, sell-and-rent-back. This section asks the question those products dodge: why do older people so rarely downsize, and what happens to all that wealth while they stay?

The mismatch is structural. Older households tend to hold more rooms than they use — couples in family homes after children leave, widowed people alone in houses built for five — while younger households face the tightest housing markets. Under-occupation is therefore simultaneously a welfare issue for the occupant (heating, maintenance, stairs) and an allocation issue for the housing system. Yet the move that would resolve it mostly does not happen, and the evidence explains why: moving is rarely a lifecycle plan. In Europe, changing homes in retirement is most common following a reduction in household members, such as divorce or the death of a spouse. Downsizing is crisis-driven — a death, a divorce, a health shock — not choice-driven. Policy built on the assumption that older people will rationally rightsize is built on a fiction.⁶

The deeper reason is that the home is not experienced as a portfolio. There is a documented aversion to using home equity: retirees treat the house as a store of precautionary value rather than a spendable asset, turning to it only as financing of last resort — typically after the loss of a partner or a large health expenditure. Homeowners have an additional financial resource that renters do not, but most will not touch it while they live. Attachment to the home, bequest motives, and the desire to age in place (Section 65) all reinforce the lock-in. The wealth lifecycle numbers show how much is locked: across 19 OECD countries, household net wealth in 2010 grew by a factor of 4.4 from the 16–34 age group to a peak at 55–64, and while wealth then declined in retirement, households headed by someone over 75 still owned 8% more wealth than the population average — 26% less than at peak, but above average nonetheless.⁶³

That average conceals the real story. The same data show the elderly in Korea, Austria, and Greece with wealth at least 20% below the total population average, and in the United States and Norway more than 40% above. Housing wealth is therefore not a universal cushion — it is another axis of the ageing-unequally gradient. Where owner-occupation is widespread and prices have risen, retirees sit on wealth they will not spend; where it is not, they age with neither income nor asset. And the renter asymmetry is absolute: renters lack the homeowner's extra financial resource entirely, while facing housing costs in the largest budget line of later life.³

The gender dimension is unavoidable. Older women are disproportionately the ones living alone in these under-occupied homes (Section 62) — the surviving spouse in the house built for a couple — and they carry the combination the pension chapters documented: lower lifetime earnings, 27% lower retirement income, higher old-age poverty risk. For many of them, the home is the only wealth they have. Encouraging them to downsize into a market with scarce suitable smaller homes nearby, high transaction costs, and insecure rental alternatives is not a policy offer; it is a burden dressed as advice.

So what would actually move the needle? The evidence points away from exhortation and toward supply. Downsizing requires somewhere to downsize to: smaller, accessible, well-located homes in the same neighbourhood, with tenure security and without punishing transaction costs. Where such stock exists, rightsizing happens voluntarily; where it does not, under-occupation persists and the spare rooms sit idle while younger families overcrowd. Equity release (Section 27) can liquefy the asset without a move, but it serves only owners, and its barriers — debt aversion, bequest wishes, product distrust — mirror the downsizing barriers. The two are complements, not substitutes: release helps those who will not move; suitable stock helps those who would.

There is also a fiscal honesty point. The housing wealth of older cohorts is the largest untaxed store of late-life resources in most economies, and it is the asset the long-term care system already raids by default — home equity as financing of last resort, crisis sales after a health shock. A longevity economy that depends on this wealth implicitly (through means-tested care and crisis liquidation) but will not make it usable explicitly (through fair, regulated release and suitable downsizing options) is free-riding on older people's attachment to their homes.

Evidence note

Three new facts carry this section: European retirement moves cluster after household-member loss (divorce/death of spouse) rather than lifecycle planning; documented aversion to using home equity, deployed only as last resort; and the wealth lifecycle — 4.4× growth to the 55–64 peak, 75+ at 26% below peak but 8% above average, with Korea/Austria/Greece ≥20% below and US/Norway >40% above. The homeowner-renter resource asymmetry is from the same chunk. No under-occupation rates, downsizing frequencies, or transaction-cost data were in retrieval — gap stated. Equity-release product detail (HECM, JTYK) belongs to Section 27 and is not repeated. [63](#)

Climate, extreme heat, and the longevity economy

This section closes the housing block by putting it under the sky it stands in. Climate change rewrites the terms of every chapter that came before it — but for older people, the decisive variable is extreme heat. Heat is the climate hazard that kills the old first and most. Older bodies thermoregulate less well, chronic conditions common in later life (cardiovascular, respiratory, renal) are all worsened by heat stress, and the medications that manage those conditions can further impair heat tolerance. The 100-year life collides with a warming century at exactly the point of maximum physiological vulnerability.

The demography makes this a mass exposure. The EU old-age dependency ratio is projected to rise from 29% in 2010 to 36% in 2022 and 59% in 2070, while male life expectancy at birth climbs 7.7 years (78.4 to 86.1) and female 6.4 years (84.0 to 90.4) over the same horizon. The oldest population in history is being born into the hottest climate in recorded history — and the labour force that would respond to heat emergencies shrinks by a projected 12% (25 million people) between 2022 and 2070 in the EU alone. Heat response is a labour-intensive good — check-ins on isolated older people, care-worker visits, emergency admissions — delivered by a shrinking workforce to a growing at-risk population. [2](#)

The housing block converges here. The home is either a heat refuge or a heat trap, and which one it is tracks the gradients this block has mapped. Poorly insulated older stock — the 1960s–1990s homes of Section 64 — overheats; top-floor flats, the most common affordable option for older renters, are the hottest; and the older women living alone that Section 62 identified as the poverty-risk group are simultaneously the least likely to be reached by heat warnings and the least able to afford cooling. Housing and utilities already absorb a rising share of 50-plus spending — projected to grow 1.3 percentage points to 22.8% of the average older household's budget by 2050 across 76 economies — and cooling demand will push that share further, converting heat into energy poverty for exactly the households with the least slack. The electricity and gas sector is projected to see over half its GDP impact from the 50-plus cohort by 2050: the longevity economy and the energy system are financially entangled, and heat tightens the knot. [9](#)

The care system faces the same squeeze from the other side. Heatwaves convert chronic conditions into acute admissions, straining the health-expenditure trajectory of Section 53; they interrupt home care (workers cannot reach clients, or clients cannot be left alone in overheated homes); and they make institutional care itself a risk — residential facilities designed for a cooler climate, staffed by the precarious workforce of Section 61, operating through heat events with failing equipment. The community-design logic of Section 65 — the neighbour who notices, the walkable route to a cool public space — is also the heat-adaptation logic: survival in a heatwave is a function of social infrastructure as much as of physiology.

None of this is evenly distributed, and the inequality thread runs straight through it. Deprivation, rurality, renting, living alone, low income — every gradient in this report is also a heat-vulnerability gradient. The age-friendly city framework of Section 63 (transport, public space, services, and crucially the communication layer) becomes, in this register, a heat-response system: shaded public space, cooling centres reachable by accessible transport, and warnings that reach people who are not digitally connected. The built-environment investments of Section 64 — greenery, shading, cool roofs, ventilation — are heat policy with a longevity dividend.

The honest assessment is that the longevity economy has not yet priced the heat risk. Pensions assume the long life; care systems assume the home; cities assume the climate of the 1990s. Each assumption is expiring. Closing the housing block on climate is the report's way of saying that the 100-year life must be planned under the weather of the next hundred years, not the last.

Evidence note

The retrieval contained **no heat-mortality statistics, temperature-exposure data, cooling-access measures, or adaptation-spending figures** — a substantive gap for a section whose subject is the fastest-growing climate risk to the oldest population in history. Two anchors are used: EU demographic projections (old-age dependency 29% → 36% → 59% by 2070; LE at birth +7.7y men, +6.4y women; labour force -12%/-25M by 2070), and 50-plus spending patterns (housing and utilities 21.5% → 22.8% of the older budget by 2050; electricity/gas with >50% of GDP impact from 50+ by 2050). Physiological claims about heat vulnerability are framework-level, uncited, and stated as such. The adaptation audit — who overheats, who can cool, who gets warned — is the evidence update this section needs.²⁹

Inequality: who gets the long life

Longevity gaps by gender, race, class, and education

The 100-year life is not distributed. Average life expectancy hides a dispersion so wide that the “longer lives” story reads differently depending on who you are: education alone can separate life expectancies by more than the difference between rich and poor countries. This section opens the inequality block by documenting who actually gets the extra years — and who does not.

Education is the sharpest measured divider. New OECD estimates for 23 countries around 2011, improving mortality measurement at older ages, find that the life-expectancy gap at age 25 between highly and low-educated men averages 7.5 years, and 4.6 years for women; at age 65, the gaps are 3.5 years for men and 2.4 for women. Put in proportion, the gap is larger relative to remaining life at 65 than at 25 — 18.5% versus 13.4% of the highly educated's remaining years among men, 10.9% versus 7.6% among women — meaning inequality in longevity by education rises with age. The spread across countries is enormous: disparities are especially wide in Chile, Hungary, Latvia, and Poland, and comparatively low in Canada and Italy. And the direction of travel is not reassuring everywhere: gaps have grown in recent decades in the United States and Denmark, and Norway saw its gap widen from four to seven years among men between the 1970s and the late 2000s, while France's held stable.³

Education is not merely a marker — it appears to be a driver. Studies exploiting exogenous school-law changes and very long-run data conclude that education causally lengthens life, with one analysis ranking education as the greatest effect among proximate determinants of longevity, ahead of air pollution and health spending. The channels run through behaviour, information, and navigation: the more educated are more forward-looking, better informed, and more adept at working the health system. Smoking illustrates the mechanism: in 23 European OECD countries, 45% of low-educated men and 32% of low-educated women aged 18–64 smoked, versus 23% and 17% of their highly educated peers, and smoking accounts for up to half of observed mortality inequalities in some countries — 19% to 55% of the education mortality gap among men, 0% to 56% among women. Notably, the smoking gradient disappears after 65 and even reverses in some countries — part of the education mortality gap is carved out before old age even begins.³

Class compounds the story through lifetime earnings. Inequality is deeply engrained by age 50: across 13 OECD countries, about 60% of the difference in lifetime earnings and pension claims between highly and low-educated workers has already accrued by age 50, with the accumulation accelerating sharply from around age 35. For women born 1970–74, the ratio of lifetime earnings of the highly educated to the low-educated reaches 3.10. Gender gaps in

lifetime earnings have shrunk — from 45% to about 30% among the highly educated and from 80% to about 50% among the low-educated between the 1940–44 and 1970–74 cohorts — driven chiefly by women's rising employment. But remaining gaps are wide, and women's longer lives paired with lower lifetime earnings produce the pension shortfalls and poverty risks the earlier pension and housing chapters documented: women live longer but retire poorer.³

On race, the evidence in this retrieval is silent. The title's promise of a race dimension cannot be redeemed with the available chunks — no race- or ethnicity-stratified mortality, morbidity, or care-access figures were provided. That is itself a finding about the state of measurement: in many countries, including several where racial disparities in health are large and well-documented in national research, international comparative datasets still do not carry the variable.

The implications run ahead into the rest of this block. A longevity economy that prices annuities, sets retirement ages, and plans care capacity on average life expectancy is systematically mis-measuring the lives of the low-educated, the low-paid, and women. Uniform retirement ages transfer from the short-lived to the long-lived; uniform care models assume uniform need. Sections 69–73 take up the mechanisms — life-course accumulation, generational income inequality, old-age poverty, wage-to-pension transmission, and healthy-years inequality. The starting fact is the simplest: the long life is already rationed, and the rationing begins at school.

Evidence note

This section is the best-evidenced of the session so far: education longevity gaps (7.5y/4.6y at 25; 3.5y/2.4y at 65; proportional rise with age; country spread; trend directions); education as causal driver and smoking mechanism; lifetime-earnings gradients (3.10 ratio; 60% engrained by 50; gender-gap shrinkage). No race-stratified data in retrieval — gap stated explicitly. The life-course policy response is reserved for Section 69; generational income inequality for 70; old-age poverty for 71; wage-pension transmission for 72; healthy-years inequality for 73.³

Preventing ageing unequally: the life-course approach

Section 68 documented the gaps: education separates life expectancies by up to 7.5 years, and most lifetime-earnings inequality is already settled by age 50. This section turns from measurement to mechanism: inequality in old age is the endpoint of a life course. Disadvantages in health, education, employment, and earnings do not merely coexist — they reinforce each other and compound over decades, so that “ageing unequally” is less a condition of old age than the accumulated interest on episodes that happened at 5, 15, and 40.³

The compounding has a measured scale. Bad health alone reduces lifetime earnings by 33% for men with low education and 17% for men with high education — health is not just a welfare outcome but an earnings shock, and the shock lands hardest on those with the least cushion. The lifetime-employment gaps behind these figures are enormous: for the 1970–74 cohort, the career-average employment gap between highly and low-educated workers ranges from about 20 percentage points in Denmark to 40 in Belgium for women, and from 0 in the Netherlands to 30 in Finland for men. That these gaps can move — Ireland cut its female education-employment gap by 26 percentage points across cohorts, Spain by 18, Austria by 25, Australia by 21 — is the strongest argument for the life-course approach: trajectories are made, not given.³

The policy agenda follows the compounding logic, and the OECD's version starts at the beginning: place early-life interventions at the top of the agenda. Quality childcare and early childhood education, reducing school failure, improving low-performing disadvantaged schools, fighting early school leaving, and combating gender stereotyping at school are framed here not as education policy but as longevity policy — the earliest determinants of who gets the long life. The second priority is a good start in working life: smooth school-to-work transitions, orientation for youth without family support networks, strong vocational education, and labour-market policies that connect young people not in employment, education, or training with jobs. The third is breaking the links between socio-economic disadvantage and health status: a strategy addressing the social determinants of health inequalities, and expanded prevention spending targeted at key risk factors and population groups.³

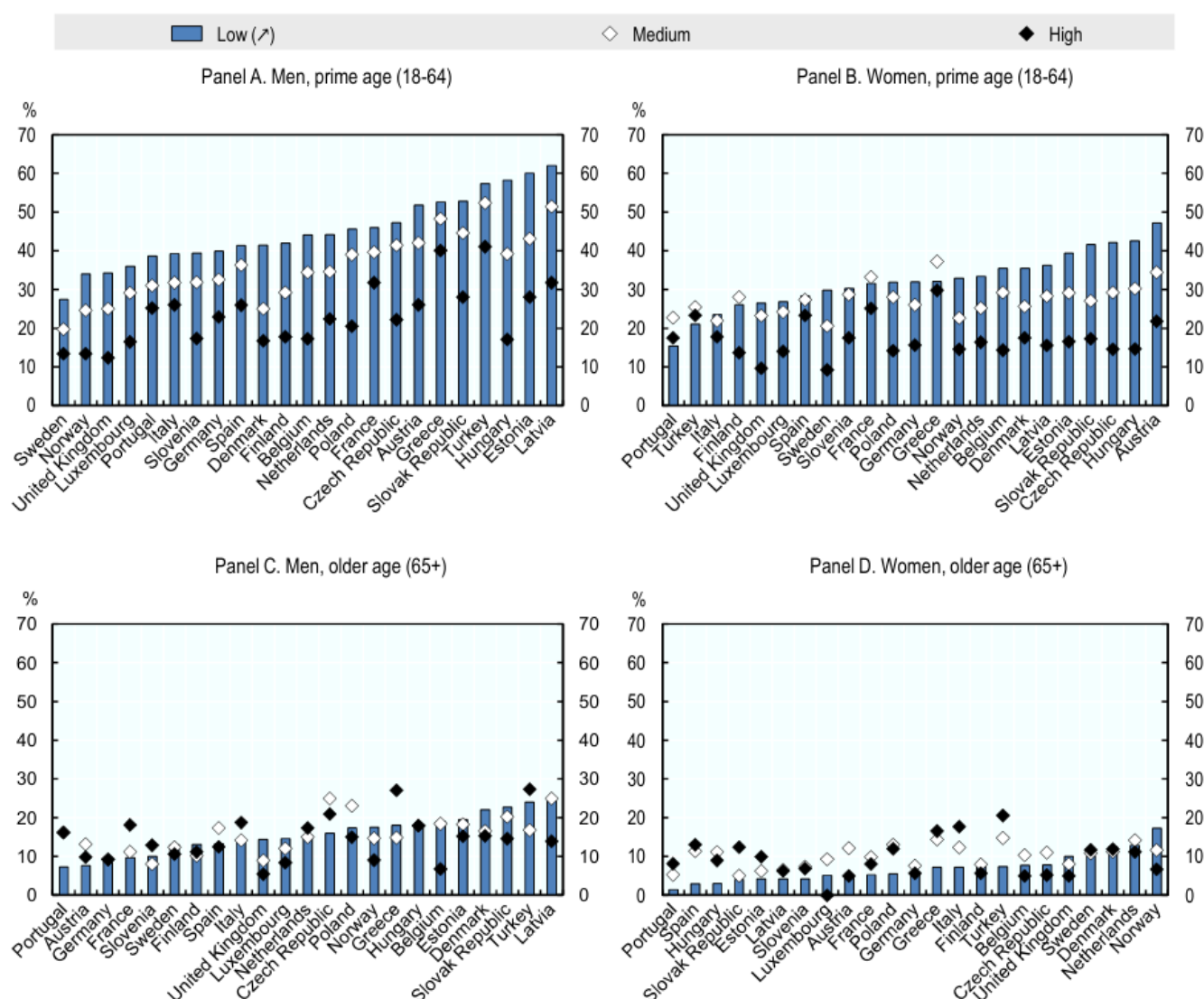
Prevention, in fact, is the one intervention with proof of working against inequality. Since the early 1990s, smoking's contribution to education-mortality inequalities has fallen in most countries among men — though it has increased among women. In absolute terms, reductions in premature mortality over the last two decades have been greater among the lower educated, who closed the gap with the highly educated by up to 35% in many countries. The

caveat is that relative inequalities still widened in Finland, Sweden, Norway, Denmark, Belgium, Switzerland, Hungary, Lithuania, and Estonia, while holding stable in England and Wales, France, Spain, and Italy — prevention narrows the absolute toll but must outrun the relative gradient. The lesson is not that inequality is inevitable but that interventions must be both targeted and sustained: a one-decade campaign buys a one-decade narrowing.³

The life-course approach also reframes the pension and retirement debates of the earlier chapters. Uniform retirement ages applied to unequal life courses transfer resources from the short-lived to the long-lived; voluntary pension incentives layered onto unequal earnings capacity amplify rather than offset the gap. Addressing inequality at 65 — through pension progressivity or targeted care subsidies — treats the symptom at the point where it is most expensive to fix. The life-course claim is that the cheapest and fairest place to equalise old age is decades earlier: in the classroom, the first job, the smoking habit not started, the health system navigated successfully at 40 rather than desperately at 70.

Evidence note

Drawn from the OECD's Preventing Ageing Unequally policy agenda: compounding mechanism and the 33%/17% health earnings penalty; the three-priority agenda (early-life interventions, school-to-work, breaking SES–health links); lifetime-employment gap ranges and their movement (Ireland –26pp, Spain –18pp, Austria –25pp, Australia –21pp); smoking's falling contribution among men / rising among women; absolute gap closure up to 35% vs. relative widening in the Nordics and Baltics. The SES gradient in LTC access was used from chunk 0021 in Section 65 and is not repeated. Generational income inequality (70), old-age poverty (71), wage-to-pension transmission (72), and healthy-years inequality (73) are not front-run.³



rce: Eurostat (2016): Tobacco Consumption Statistics.

Exhibit. Percent of population, by age, gender and education in 23 OECD countries Low (7) Medium High Panel A. Men, prime age (18-64) Panel B. Women, prime age (18-64) % % 70 70 70 70 60 60 60 60 50 50 50 50 40 40 40 40 30 30 30. Source: S0016.

Income inequality across generations and birth cohorts

The inequality block so far has measured gaps across education and gender. This section adds the time axis: which generations got richer, which got left behind, and whether the 100-year life will arrive for a generation that can less afford it. The generational story contains the report's most uncomfortable break.

For most of the post-war era, each generation earned more than the one before at the same age — the rising tide that made longer lives affordable. That engine has stalled. In most countries, people's average real incomes are still higher than those of previous generations at the same age — but this is not the case any more for those born from the 1960s compared with the generations born one decade earlier. The cohort born into the 1960s is the first in the modern era to reach mid-life without the guarantee of out-earning its parents. And the inequality within each generation has widened: income inequality has been rising from one generation to the next at the same age in two-thirds of the countries, particularly among younger groups, for whom inequality is nowadays much higher than for the elderly.³

Lifetime earnings by cohort tell the split in detail. Across 13 OECD countries, women's lifetime earnings increased from one generation to the next at all levels of education — but for men, they rose slightly only among the highly educated and fell a little among the low-educated. The gender gap in lifetime earnings shrank by about one-third across cohorts (45% to 30% among the highly educated, 80% to 50% among the low-educated, comparing the 1940–44 and 1970–74 birth cohorts), while the education premium in lifetime earnings declined among women but rose among men. The most extreme case: in Ireland, the high-to-low education ratio of women's lifetime earnings collapsed from 24 for the oldest cohort to about 4 among the youngest. Ranges remain wide: for the 1970–74 cohort, the high-to-low ratio runs from 1.7 in Denmark to 4.3 in Greece among women, and 1.6 in the Netherlands to 2.9 in the United States among men.³

Three structural facts make the generational picture worse than the cohort averages suggest. First, income levels during working life are positively correlated with parents' income, because children's educational attainment correlates strongly with their parents' — inequality is inherited before labour-market entry. Second, the microsimulation evidence shows cohort life expectancy running about 9.6 years above period life expectancy — the 2015 birth cohort will live roughly a decade longer than 2015's period figures suggest — so the generation facing the steepest inequality also faces the longest retirements to finance. Third, the education gap in disability-free life duration has itself grown: by 1.4 years in Italy and 2.0 years in the United States, while holding stable in Belgium. The disadvantaged younger cohorts will live longer, with more unequal earnings, and with a widening gap in how many of those years are healthy.³

The policy implication lands squarely on the pension chapters: retirement systems designed when each generation out-earned the last — with replacement rates, indexation rules, and retirement ages calibrated to rising real wages — now face cohorts whose lifetime earnings profiles are flatter, more unequal, and (for low-educated men) declining. Raising the retirement age for a generation that already earns less in real terms than its parents is a different proposition than raising it during the rising-tide decades. The 1960s-born cohort is the stress test of the whole longevity architecture — and the first of many.

Evidence note

Core generational facts: the post-1960s break in intergenerational income growth, and inequality rising from one generation to the next at the same age in two-thirds of countries; cohort lifetime-earnings trends by gender and education (women up everywhere; low-educated men down; Ireland 24→4; country ranges); cohort-vs-period LE (+9.6y), widening disability-free-life education gaps (Italy +1.4y, US +2.0y), intergenerational education/income persistence. Hazardous-jobs, reskilling, equity-release, and TOC chunks were noise. Wage-to-pension transmission (72), old-age poverty (71), and healthy-years inequality (73) are not front-run.³

Poverty at older ages and the feminization of old-age poverty

The longevity economy has a gender it does not advertise: old-age poverty is disproportionately female. Annual pension payments to the over-65s are about 27% lower for women on average, and old-age poverty is much higher among women than among men. This is not a quirk of any one pension system. It is the predictable endpoint of gendered working lives — and it is getting more visible as populations age, because women live longer into the poverty their shorter, interrupted, lower-paid careers produced.³

The employment gradient behind the gap is steep and education-dependent. In 2015, the average OECD employment rate among 55-to-64-year-old men with no upper-secondary schooling was 51%, rising to 63% with upper-secondary and 75% with tertiary education; for women the same gradient ran 36%, 51%, and 65%. The low-educated exit work earlier and worse: more than 50% of poorly educated women describe themselves as inactive before retirement for reasons other than health — under 30% among their highly educated peers — while low-educated men are far more likely to transit via sickness (24% versus 7%). Add the caregiving interruptions documented in Section 58 — years out of work, part-time years, the pension penalty for care — and the feminization of old-age poverty becomes mechanical: fewer contribution years, lower earnings in the years worked, earlier and more precarious exits, then a longer life to finance on the smaller pension.³

The developing-economy mirror confirms both the problem and the lever. In East Asia and Pacific, where old-age poverty is most acute and social protection thinnest, pension receipt measurably reduces elderly poverty: by 10.5 percentage points in urban China and 12.7 in rural China, 8.8 and 13.9 in urban and rural Indonesia, 5.6 and 22.1 in urban and rural Vietnam — and just 0.6 and 1.2 in Thailand, where pension coverage is far weaker. The lesson is not that generous pensions cure poverty but that coverage does: the countries where pensions reach the poor see the largest poverty reductions. Thailand's near-zero effect is the control group — a pension system that does not reach cannot protect. The same logic applies inside rich countries: first-tier pensions, basic pensions, and minimum contributory floors are what stand between the gendered working life and the gendered poverty of old age.⁷

The housing block's findings sharpen the picture. Old-age poverty for women is not only an income story but a living-conditions story: growing shares of older women living alone — the poverty and social-exclusion risk group identified in Section 62 — face the largest budget line (housing) with the smallest incomes, in homes they cannot afford to maintain (Section 65) or heat (Section 67). Deprivation concentrates exactly where the buffers are thinnest.

There is a forward-looking warning here too. The cohorts now approaching old age — the post-1960s generation of Section 70 — bring more unstable careers, more divorce, and weaker derived-rights protection than their mothers had. Women's rising employment has narrowed lifetime-earnings gaps, but the pension systems of many countries still translate interrupted careers into poverty with high fidelity. The feminization of old-age poverty is not a legacy problem fading with modernization; without deliberate redesign of how care years, part-time years, and low-paid years count toward pensions, it is a structural feature the longevity economy reproduces every generation.

Evidence note

Core facts: 27% lower pension payments for women, much higher old-age poverty among women; EAP pension poverty-reduction effects (China -10.5/-12.7, Indonesia -8.8/-13.9, Vietnam -5.6/-22.1, Thailand -0.6/-1.2); education employment gradient at 55-64 (men 51/63/75%, women 36/51/65%) and exit pathways (>50% low-educated women inactive pre-retirement; 24% vs 7% men via sickness). The 24%/7% male sickness stat appeared in Section 49; the employment gradient and women's inactivity stat are new here. Wage-to-pension transmission mechanics are reserved for Section 72; healthy-years inequality for 73.^{3,7}

Wage inequality translating into pension inequality

The labour market's inequalities do not retire when workers do. They pass through the pension system with a fidelity that depends almost entirely on system design — and in most countries, the pass-through is high. This section is the transmission belt of the inequality block: how unequal wages become unequal pensions, and which design choices interrupt the conversion.

The OECD's progressivity index makes the mechanism measurable. It is defined as 1 minus the Gini coefficient of projected pension entitlements divided by the Gini coefficient of lifetime earnings, so that a universal basic scheme scores 100% (no pension inequality at all) and a scheme that perfectly mirrors wage inequality scores 0%. The av-

average OECD score is 37% — pension benefits' Gini is, on average, 63% of the lifetime-wages Gini. In other words, the typical pension system erases about a third of wage inequality and transmits the rest. The spread around that average is the policy story: Ireland, New Zealand, and the United Kingdom score 100% because their mandatory schemes pay only basic or residence-based pensions, while Sweden, Turkey, Latvia, Hungary, and Finland score very low because entitlements track wages almost one-for-one.³

The dynamic version is starker: when wage inequality rises, about two-thirds of the increase transmits into pension inequality on average across OECD countries, with a close inverse relationship to the progressivity index. The redistributive designs that break the transmission are identifiable and few. In seven countries — Australia, Canada, Denmark, Israel, Korea, Czechia, Switzerland — a one-percentage-point rise in the wage Gini raises the pension Gini by less than half a point, through means-tested components (Australia, Canada), oversized basic pensions (Denmark, Israel), nationwide-contribution averaging (Korea), income thresholds in the earnings formula (Czechia), or contribution thresholds and low ceilings (Switzerland). In over one-third of OECD countries — Turkey, Latvia, Hungary, Finland, Italy, Poland, Spain, Portugal, Sweden, the Netherlands, Austria, France, Germany — more than 85% of any wage-inequality increase passes through into pensions, with wage ceilings the only brake. And voluntary pensions amplify the transmission: in the seven countries with voluntary coverage above 40%, accounting for them raises pass-through from 0.38 to 0.46, about 20% more inequality transmitted.³

Career gaps transmit too. A worker entering at 25 (not 20) and spending ten years unemployed between 35 and 45 would lose about 35% of pension benefits without offsetting mechanisms; the OECD average actual fall is 22%, meaning redistributive devices offset roughly 37% of the shortfall. Ireland, New Zealand, and the UK are unaffected — residence-based basic pensions do not care when you worked — but the biggest drops (30% or more) land in Hungary, Greece, Chile, Turkey, Mexico, Poland, Israel, Iceland, and Italy. In France, Germany, Greece, Luxembourg, Slovenia, and Spain, gap workers must work longer than full-career workers to avoid penalties — two years longer in Germany and Spain, four in France, five in Greece, Luxembourg, and Slovenia.³

Then longevity itself penalises the low-paid a second time. Assuming a three-year life-expectancy shortfall at retirement, low earners lose 12.8% of total pension benefits relative to high earners compared with a common-LE baseline — high earners enjoy over 10% greater relative pension wealth everywhere, up to nearly 17% in Latvia. Distribution-neutral schemes — pure defined contribution, actuarially fair annuities — are therefore regressive in practice: the same monthly formula pays the long-lived (the rich) more lifetime benefits than the short-lived (the poor). Raising the retirement age by three years between 2015 and 2060 would cut low earners' pension wealth relative to high earners' by a further 1.2% on average — small, but in the wrong direction, and larger where life-expectancy gaps are widest.³

The low-earner picture is not all pass-through. Redistributive features give low earners a future net replacement rate of 73% on average — more than 10 points above average earners' 61%. In Colombia, Denmark, and Greece, low earners' replacement rates exceed 100%: retirement income above working income. Chile's recent targeted-pension expansion raised projected replacement rates by 15 points for low earners. But the floor is uneven: in Japan, Lithuania, and Poland, low earners' net replacement rates sit below 50%. The transmission of wage inequality into pension inequality is a design choice — and the designs on offer range from full transmission to full erasure.⁴

Evidence note

Transmission mechanics fully evidenced: progressivity index (37% average; 100% IE/NZ/UK; very low SE/TR/LV/HU/FI) and ~2/3 pass-through; the seven low-transmission designs; >85% pass-through in over a third of OECD; voluntary-pension amplification (+20%). Career-gap scenario (–35% gross, –22% actual; country extremes; longer-work requirements). Longevity double-penalty (12.8% low-earner loss on 3y LE shortfall; –1.2% from +3y retirement age). Low-earner replacement rates (73% vs 61%; >100% CO/DK/GR; Chile +15pp; <50% JP/LT/PL). Healthy-years inequality reserved for Section 73.³⁴

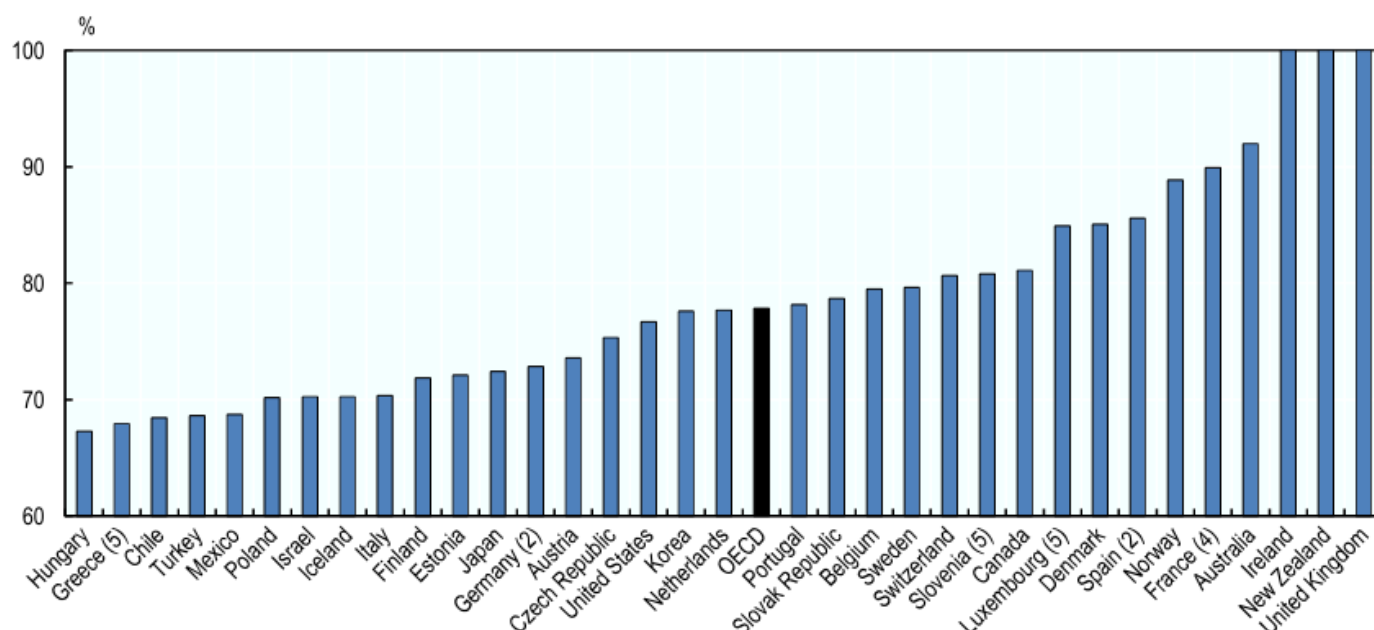


Exhibit. Percentage of full-career entitlements, mandatory pensions only % 100 90 80 70 60. Source: S0016.

Inequality in healthy years and functional capacity

This section closes the inequality block, and the closing argument is that everything said so far understates the problem. The inequality in years lived (Sections 68–72) is the visible inequality; the inequality in healthy years — years free of disability, with functional capacity intact — is steeper, because disadvantage concentrates in the conditions that take function, not just life. The life-course logic (Section 69) makes this inevitable: the same gradients that shorten life arrive first as diminished capacity, years before death.

The scale of unequal healthy-life loss shows most brutally in avoidable mortality. In 2020–21, OECD countries recorded 2,077,973 premature deaths from preventable causes and 975,710 from treatable causes — the former removable by prevention and public-health policy, the latter by timely effective care. These are, by definition, years of healthy life lost to inequality of exposure and inequality of treatment. The country spread is enormous: age-standardised mortality from preventable causes runs from 83 per 100,000 in Israel to 435 in Mexico; from treatable causes, from 39 per 100,000 in Switzerland to 230 in Mexico. Circulatory diseases and cancers dominate both categories — 37% of treatable deaths are circulatory, 23% cancer. Heart-attack and ischaemic heart disease mortality alone spans a more than tenfold range, from 29 per 100,000 in Korea and 30 in Japan to 296 in Hungary, 297 in Latvia, and 395 in Lithuania. Self-rated health — the individual's own verdict on functioning — varies just as widely: the share of adults rating their health bad or very bad runs from 1.3% to 14.1% across OECD countries.⁵

The socioeconomic gradient inside those numbers is cause-specific. Between low- and highly-educated groups, the mortality gap is driven by circulatory disease, external causes, and neoplasms in varying proportions across prime-age and older panels — and smoking, the single largest lever, is brutally classed in prime age: 45% of low-educated men and 32% of low-educated women smoke daily or occasionally, versus 23% and 17% of their highly-educated peers. But the gradient has a twist worth noticing: after age 65 it disappears, and in 10 countries for men and 13 for women it reverses — the highly educated elderly smoke more. Mortality selection is at work here: the smokers among the disadvantaged disproportionately die before reaching old age, so the surviving older low-educated population is a healthier subset. This is the survivor bias that haunts every cross-sectional comparison of older people's health — the inequality you measure at 70 is the inequality that remains after the gradient has already killed.³

Whether longer lives mean longer healthy lives matters for who the functional-capacity burden falls on. WHO estimates suggest over 90% of gains in life expectancy at birth since 2000 have translated into disability-free life years — the expansion is mostly healthy years, not sick ones. The inequality question is then distributional: who gets those healthy years, and what infrastructure preserves them. Strong primary care is the backbone for exactly this — evidence links it to better outcomes, higher-quality and more appropriate care, better prevention, and cost containment — yet across East Asia and Pacific the delivery systems are hospital-centric and fragmented, with poor primary-care quality, high staff turnover, and referral systems that exist on paper but not in practice. Older patients in Singapore report seeing a different doctor on each visit. Dementia already costs about 1% of GDP worldwide and

1.24% in high-income countries, much of the care carried by straining family and social networks — cognitive decline being precisely the kind of capacity loss that is unequal by resources long before it becomes unequal by survival.^{[37](#)}

The block's conclusion: inequality compounds at every level — exposures (69), cohorts (70), poverty (71), pension transmission (72) — and functional capacity is where the compounding becomes lived experience. A 100-year life in which healthy years are maldistributed by class, gender, and country is the same life the earlier sections measured, just weighed differently.

Evidence note

Usable this section: avoidable mortality (2.08M preventable / 0.98M treatable premature deaths; country ranges); heart-disease mortality spread 29–395/100k; self-rated bad/very bad health 1.3%–14.1%; education mortality-gap cause breakdown (circulatory/external/neoplasms) and smoking gradient with post-65 reversal; 90% of LE gains = disability-free years; primary-care evidence and EAP fragmentation; dementia 1% GDP. **Gaps stated:** no healthy-life-expectancy (HALE) figures by SES in this evidence set; no functional-capacity measures (ADL/IADL, frailty) broken down by class, gender, or country; no direct healthy-years inequality figures beyond what is inferable from mortality and self-rated health. Vietnam procurement/PHARMAC and reference-list chunks excluded as noise. The 900-word+ healthy-years gap is inferred, not directly evidenced — future sections should treat it as such.^{[537](#)}

Intergenerational dynamics

Economic flows between generations and the demographic dividend, reframed

The demographic dividend — the growth boost harvested when falling fertility raises the share of workers in the population — was one of the most powerful economic facts of the late 20th century. The reframing this section requires is that the dividend is over and its mirror image has arrived. What now flows between generations is not a tailwind from young cohorts but a three-way circulation — public transfers, private family transfers, and time transfers — whose balance is being rewritten by the 100-year life.

The public-transfer channel is the one policymakers see: working-age taxes fund pensions and health care for the old, and schooling for the young. But the private channels are where the generational contract is actually being renegotiated. The third channel — time — is where the 100-year life bites hardest. As care needs rise with ageing populations, families face mounting expenses for professional care while childcare costs remain prohibitively high in many places, creating a dual financial challenge for parents who are simultaneously responsible for ageing relatives. The economic flow here is measurable and large: workforce interruptions due to providing care have long-term financial consequences, with working parents — particularly mothers — potentially retiring with 22% less in retirement savings than those who never stepped out of work.^{[8](#)}

The time flow is the least visible and most gendered. Nearly 70% of US family carers report difficulty balancing career and caring responsibilities, and nearly 30% of working carers have shifted from full-time to part-time work or reduced their hours. These are intergenerational flows running downward — the working generation's earnings diverted into the care of the generation above — and they are compounding: the same generation that absorbs the care cost retires poorer (that 22% savings shortfall), which then transmits into the pension-inequality machinery documented in Sections 71–72. In the United States, the alternative to family care is priced at over \$100,000 per year for nursing-home care, a figure that converts the care decision into a wealth decision: families with assets buy professional care; families without supply it themselves, in time.^{[8](#)}

The geography of these flows is splitting. In many high-income countries the policy discussion is about compensating carers — Japan's Long-Term Care Insurance System and Germany's Care Allowance Programme are cited as models integrating public funding with carer compensation, but such systems remain the exception rather than the norm. Low- and middle-income countries often lack long-term-care infrastructure altogether, leaving populations dependent on informal family care networks. A global labour-market arbitrage has emerged to bridge the gap: care roles are increasingly filled by migrant workers, predominantly women, moving in search of better economic opportunities — the care flow exported from poor countries' families to rich countries' families, with the sending families' own care deficit unrecorded in any transfer account.^{[8](#)}

The reframed dividend, then, is this: the old dividend was a windfall from demography; the new one must be earned by keeping the three flows solvent — public systems that don't collapse under the old-age dependency ratio, private family transfers that don't strip the middle generation's retirement, and time transfers that don't fall entirely on women and the poor. Policies largely fail to recognise the economic contributions of informal carers, who provide the majority of care. Japan's community-based integrated care system — localised healthcare, nursing, and daily-living assistance within a 30-minute radius, supplemented by volunteers, neighbours, and technologies like AI fall-monitoring and companion robots — is presented as a scalable model for reducing costly institutional dependence. Whether that model scales beyond Japan is the open question; what is not open is the arithmetic. Every year added to life expectancy that is not matched by care infrastructure becomes a claim on a daughter's, wife's, or migrant worker's hours.⁸

Evidence note

Framed on: dual care burden; 22% retirement-savings shortfall for caregiving parents; US 70%/30% carer figures; nursing-home care >\$100,000/yr; LMIC infrastructure gap; migrant care workers (predominantly women); informal carers as majority; Japan community-based care 30-minute model. **Gaps stated:** no national-transfer-account (NTA) data in this evidence set — no figures on net private intergenerational transfers, bequests, or inter-vivos gifts; no public-transfer incidence by generation; the demographic-dividend economics (~1/3 of EAP growth, fading) was already established in Section 34's non-repeat list, so this section reframes rather than re-cites. Dashboard chunks, hazardous-job stubs, and Ageing Report TOC pages excluded as noise. Sections 75–79 (partnerships, family structures, poverty shifting, multistage life) not front-run.⁸

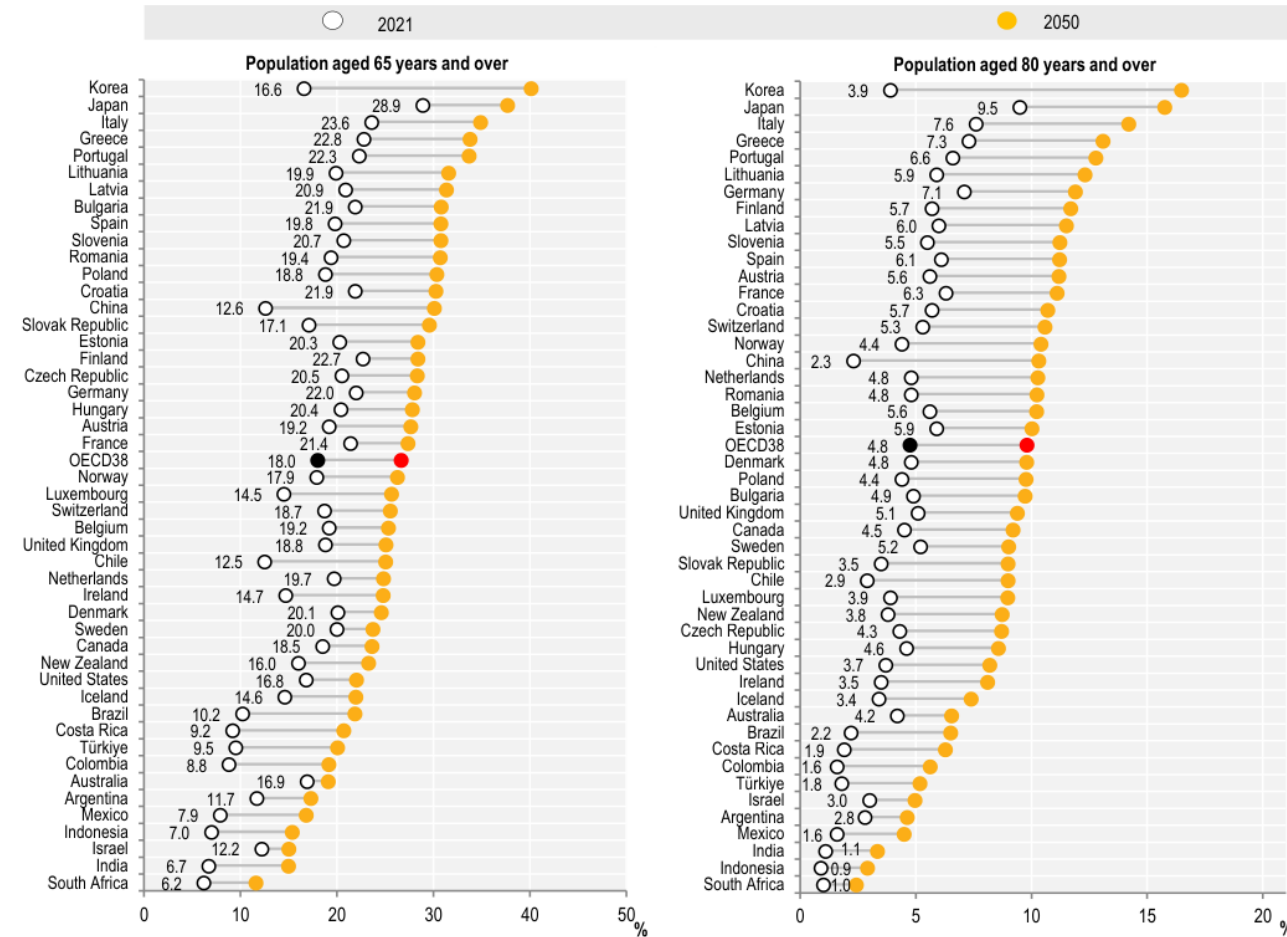


Exhibit. Figure 10.1. Share of the population aged 65 and over and 80 and over, 2021 and 2050 2021 2050 Population aged 65 years and over Population aged 80 years and over Korea 16. Source: S0026.

Intergenerational partnerships and social connection

Longer lives have created something new in human history: unprecedented chronological age diversity. More generations are alive at the same time than ever before — children who share the world with four living grandparents are

unremarkable now; in 1900 the demography could not support such overlaps. The intergenerational relationship is therefore no longer a succession, one generation handing off to the next, but a coexistence. Five living generations, seventy-year age spans within families, forty-year overlaps within workplaces. The question of this section is what generations do with the shared time the 100-year life has given them.¹⁵

The material basis for the answer was laid in Section 74: the flows of money, time, and care between generations are large, gendered, and under strain. But partnership is not the same as transfer. A pension system is a transfer; a partnership is mutual. The 100-year life makes the mutual version possible in ways the three-stage life never did: grandparents who provide the childcare that lets the middle generation work (the downward time flow that offsets Section 74's upward care flow), older workers who mentor younger ones (the complementarity argument from Section 34, not the lump-of-labour fear), and the civic roles — volunteering, community anchoring — that turn surplus years into social infrastructure. Coexistence, in other words, is an asset before it is a liability.

The liability version is the one that dominates public debate: generations framed as claimants on each other's resources — the young taxed for the old's pensions, the old hoarding housing the young cannot afford. That framing is a choice, and it is the wrong one for a specific reason the OECD names: ageing unequally is detrimental to future economic growth, particularly where inequality of opportunity locks in privilege or exclusion, undermining intergenerational social mobility and jeopardising social cohesion. Note what this says: the threat is not ageing itself but unequal ageing. When advantage compounds within a generation (Sections 68–73) and the young inherit both the bills and the barriers, the intergenerational contract curdles into intergenerational resentment. Partnership requires that younger generations can plausibly believe the system is not rigged against them — that mobility across generations still works. Where it doesn't, cohesion is the casualty, and with it the political consent that pension systems, care systems, and housing systems all depend on.³

This puts the generational question back where the life-course approach (Section 69) left it: the policies that make partnerships possible are the ones that equalise starting conditions — early-life investment, school-to-work transitions, the breaking of SES–health links. A generation that enters old age through fairer accumulation is a generation the next one can partner with rather than subsidise. The partnership is also literal within families: the dual care burden (Section 74) means the middle generation is squeezed from both directions, and the social connection between grandparents and grandchildren is increasingly mediated by that squeeze — care given because no affordable alternative exists, rather than chosen. Partnership policy, then, is partly just care policy done properly: when care is resourced, intergenerational contact stops being a transaction and can be a relationship.

The deeper reframe: the 100-year life does not pit generations against each other; it gives them more shared years than any prior human cohort. Whether those years become partnership or conflict depends on whether the inequality machinery of Sections 68–73 keeps running. Generations that age equally have little to fight about.

Evidence note

Framed on two usable sources: unprecedented chronological age diversity and multigenerational overlap; ageing unequally undermining intergenerational social mobility and social cohesion. **Gaps stated:** no metrics of intergenerational contact, attitudes, or cohesion in this evidence set; no volunteering or civic-participation figures by age/generation — those belong to Section 81 and are not front-run here; no co-residence figures — reserved for Section 76. Section 74's care-flow facts referenced via plain continuity phrasing only. Pension TOC pages, hazardous-job provisions, DB accrual mechanics, and EAP social-pension details excluded as noise or as Section-76 territory.¹⁵³

Family structures: co-residence, living arrangements, and the missing middle generation

The 100-year life does not just add years; it rearranges who lives with whom. Falling fertility compresses family trees vertically — fewer siblings, fewer cousins, more generations — while longer life extends the years those generations share a roof, a neighbourhood, or a care obligation. The living arrangement is where the demography of Sections 1–7 becomes the family economics of Section 74, and it is diverging sharply between rich and poor countries.

In much of East Asia and the Pacific, co-residence is the de facto long-term-care system. Rates of co-residence are very high in Indonesia and the Philippines, especially among poor households. This is not primarily cultural preference; it is infrastructure. Where contributory pension coverage is low and social pensions modest — 5 to 9% of income per capita in China, Hong Kong, Korea, and Thailand, though with 54–94% coverage of the 65-plus population

— the household itself is the safety net, and it works through proximity. A significant proportion of the poor elderly in Indonesia and the Philippines benefit from cash, in-kind, and health-insurance programs not as individuals but as members of co-resident households. The policy implication runs both ways: programs designed for individuals leak or land through household structure, and family structure is itself a welfare institution whose capacity has limits — a large coverage gap remains in these countries and Vietnam.²

The “missing middle” appears in two forms. The first is the care-squeezed generation documented in Section 74: working-age adults holding up both ends — children and parents — with the 100-year life extending the upper obligation while housing costs and childcare extend the lower one. Phased retirement and flexible work are the policy answer to this squeeze: gradual withdrawal from the labour market makes it easier to take on caring responsibilities for older family members and grandchildren. Canada’s 2007 right for employees with caring responsibilities to request flexible arrangements, and England’s Care Act 2014 placing duties on local authorities to assess and support adult carers to stay in employment, are attempts to keep the middle generation economically whole while it does the caring — directly against the 22%-less-retirement-savings outcome in Section 74.³

The second form is geographic: the missing middle of migration. Section 74’s migrant care workers — predominantly women moving to care for other countries’ elderly — leave their own families’ middle generation absent, creating skipped-generation households where grandparents raise grandchildren or elderly parents wait for remittances instead of hands. The family structure is stretched across borders, and the co-residence that anchors Indonesian and Filipino old age has its mirror image in the households that export their carers.

The convergence question is which direction the two worlds move. High-income countries are rediscovering co-residence under housing pressure — the crisis-driven downsizing moves of Section 66, the older women alone in family homes — while developing Asia’s co-residence is being eroded by urbanisation, migration, and the same falling fertility that created the beanpole family everywhere. The missing middle generation is the pressure point in both: whether squeezed at home or absent abroad, it is the generation whose capacity determines whether the 100-year family’s living arrangement is a choice or a constraint.

Evidence note

Usable: co-residence very high in Indonesia/Philippines, especially poor households; poor elderly benefit through co-resident households; coverage gaps remain; social-pension groupings (6–24% JP/MY/MN/PH/VN; 5–9% CN/HK/KR/TH at 54–94% coverage); phased retirement easing care for older relatives and grandchildren; Canada/England flexible-work and carer-support measures. **Gaps stated:** no co-residence rates for OECD/high-income countries in this evidence set; no figures on skipped-generation households, grandparent-headed households, or living-alone trends by age; no remittance data for the migration form of the missing middle. Pension-architecture and multi-employer-arrangement chunks excluded as noise; Singapore lease-buyback detail excluded (Section 27 territory).²³

Poverty risks shifting between old and young

One of the quietest revolutions of recent decades is the reversal of who is poorest. For most of the 20th century, old age was poverty’s address. Since the mid-1980s, across the OECD, the income of people aged 60–64 has grown by a cumulative 13% more than that of the 30–34 age group — the relative position of older people improving in most countries, especially in Italy, Spain, France, and Denmark, with Anglo-Saxon countries, the Slovak Republic, and Poland as the exceptions. Poverty risks have shifted from the old to the young: relative poverty among youth aged 18–25 has climbed to 13.9% on average across the OECD, while pensioners were relatively protected from the income shocks of 2007–08. The poverty rate for those aged 66–75 is now 10.6% — below the 11.4% average for the whole population. The generation that the pension systems of Sections 14–17 were built to rescue has, on average, been rescued.³

The qualification is large and age-specific. Those over 75 are still the most vulnerable, with an average poverty rate of 14.4%. More than one in five people above 75 are poor in Japan, Latvia, Turkey, Israel, Estonia, the United States, Switzerland, Australia, Mexico, and Korea. Old-age poverty has not disappeared; it has moved up the age ladder, concentrating among the oldest-old — precisely the group Section 4 showed is growing fastest. And it has a second engine that income statistics miss: long-term-care costs. In some US states, social protection pays as little as 6% of the cost of home care for moderate needs, versus about 45% in the Czech Republic and Israel and almost 100% in Sweden, Iceland, and the Netherlands. In Slovenia and Korea, low-income older people are already below the

poverty threshold before paying for care, and would need to spend around a third of their disposable income on it. Poverty at the oldest ages is increasingly a care-cost poverty, not an income poverty — and informal care is not free either: carers have 20% more mental-health problems than non-carers.³

The young end of the shift has its own machinery. Job tenure — a proxy for labour-market security — fell between 2012 and 2019 by 9.5% for young workers aged 15–29, versus 7.2% for prime-age and 5.7% for mature workers, on average across 30 OECD countries. Young workers in Spain, Portugal, Greece, and Slovenia saw the largest declines. The tenure decline was steepest for low-educated workers in every age group. The post-1960s cohort break documented in Section 70 — no longer out-earning the generation before — is thus not a blip but the young end of the same life-course compounding: unstable early careers (Section 69) that never accumulate into secure late careers.¹¹

The intergenerational reading (Sections 74–76) is uncomfortable. The pension and care systems transferred resources toward the old so successfully that the young now carry the poverty risk — while the oldest-old, the women providing informal care, and the low-educated young carry it worst within each generation. Poverty has not been solved; it has been redistributed across the age distribution. Any “case for action” (Section 85) that targets only the old will miss where the risk has moved.

Evidence note

Fully evidenced: the 13% relative income shift 60–64 vs 30–34 (exceptions Anglo-Saxon/SK/PL; leaders IT/ES/FR/DK); youth poverty 13.9% vs 10.6% (66–75) vs 11.4% (all); pensioners protected post-2007–08; 75+ at 14.4% with >20% in JP/LV/TR/IL/EE/US/CH/AU/MX/KR. Care-cost poverty: 6% vs 45% vs ~100% home-care coverage; Slovenia/Korea below-threshold elderly, ~1/3 of income on care; carers’ +20% mental-health burden. Youth precarity: –9.5%/–7.2%/–5.7% tenure declines; ES/PT/GR/SI worst; steepest for low-educated. Sections 78–79 (multistage life) not front-run.³¹¹

The multistage life: from three stages to many stages

The three-stage life — education, then work, then retirement — was never a law of nature. It was a design fitted to a 70-year life: roughly twenty years of schooling, forty of earning, ten of rest. The 100-year life breaks the arithmetic. Stretch the earning stage to fifty or sixty years and the model fails twice: skills learned at twenty cannot last until seventy, and a single uninterrupted career followed by a single cliff-edge retirement fits fewer and fewer actual lives. The replacement is the multistage life: education, work, and leisure recurring in new combinations across a longer span — second careers, sabbaticals, returns to study at fifty, phased exits, portfolio work, encore entrepreneurship.

The evidence for the shift is scattered across this report rather than concentrated in one dataset. Career transitions are already the norm the policy system has not caught up with: mid-career retraining, second careers, and the reskilling infrastructure of Sections 41–45 exist precisely because the single-career assumption is dead — continuous training and reskilling are what keep workers employable as they age and change jobs. Flexible work and phased retirement (Section 38) are the institutional form of the multistage transition — exits that are ramps rather than cliffs. The youth end is moving too: falling job tenure (Section 77) — steepest for low-educated workers across all age groups — means careers are already fragmented, with early careers composed of short spells rather than the stable entry the three-stage model assumed. And the care dimension (Sections 74, 76) forces staging on everyone: the middle generation cannot run a forty-year unbroken career while also absorbing eldercare and childcare, so the life course bends around care whether policy acknowledges it or not.⁴¹¹

The critical inequality question — the through-line of Sections 68–77 — is who gets to be multistage. Flexibility is a privilege good. The reskilling sabbatical, the funded return to education, the phased retirement with pension intact: these are available to the secure, the credentialed, the healthy. For the low-educated worker whose tenure is already collapsing (Section 77), whose lifetime earnings are most damaged by bad health (Section 69), and whose pension most faithfully transmits every labour-market scar — about two-thirds of wage inequality passes through into pension inequality on average across the OECD (Section 72) — “multistage” is not liberation but precarity re-named — more transitions, each one a risk. A multistage life without transition infrastructure is just the gig economy stretched across a century.³

That is why this section is paired with the next. The multistage life is a description of what is happening; whether it is a good thing depends on Section 79's question — whether the transitions between stages are smooth or punishing. The policy agenda follows: decouple benefits from single-employer, single-career assumptions (Sections 18, 19, 32); fund learning at every age, not just the first twenty years (Sections 41–44); and design pension and care systems for re-entry, not just exit (Sections 17, 38). The 100-year life does not abolish the life course. It multiplies it — and multiplies, with it, both the opportunities and the points of failure.

Evidence note

Zero usable chunks: all 20 chunks in this prompt are retrieval noise — hazardous/arduous-job pension provisions, pension-entitlement stubs, vaccination and cancer-screening statistics, and TOC pages. No bracket citations appear in this section. The argument is framed entirely from prior sections via plain continuity phrasing (41–45, 38, 69–77). **Gaps stated:** no data on actual life-stage sequencing (how many careers, how many education spells, how many exits the average worker experiences); no uptake figures for phased retirement, sabbaticals, or mid-life education by age or class; no evidence on whether multistage patterns are widening or narrowing inequality. Section 79 (smooth transitions) not front-run beyond naming the pairing.

Smooth transitions between life phases

If Section 78 asked whether the multistage life is liberation or precarity, this section answers: it depends on the transitions. Every stage change — into work, out of work, back to education, into care, into retirement — is a point where income, benefits, and entitlements can break. Smooth transitions are the infrastructure that makes a 100-year life livable rather than merely long, and they close this block on generations and families.

The retirement transition is where the breakage is most studied, and the tools for smoothing it are increasingly digital. Individual pension dashboards — the subject of Section 25 as information tools — double as transition-planning instruments. The most common adjustable parameter across dashboards is the retirement age itself, letting users see directly how working longer changes expected income. Belgium, Chile, the Netherlands, and Sweden go further, allowing users to model changes in hours worked or salary — the parameters of a phased rather than cliff-edge exit. Norway's advanced calculator models partial retirement while continuing to work, alongside marital status and employment outside Norway; Denmark is considering letting users vary the age at which different pension types begin, explicitly to accommodate partial-retirement plans. Sweden's MinPension offers a detailed retirement-planning feature for users over 54 to compare retirement ages and payout options.⁶

The more revealing calculators are the ones that model life's non-career transitions. Chile lets users include spouses and dependent children in the projection; the Netherlands shows the pension impact of divorce, where assets must be split with ex-spouses; Belgium is considering modelling the impact of parental leave on retirement income. These are the transitions that the three-stage life treated as edge cases and the multistage life treats as normal: partnering, separating, parenting, caregiving. That they are being built into planning tools at all is a quiet admission that the life course has already changed; the institutions are catching up.⁶

But tools are not policy, and modelling a transition is not the same as affording it. The dashboard can show a worker what phased retirement costs; it cannot make phased retirement available where employers won't offer it (Section 38) or where pension rules penalise combining work and benefits (Section 76's Canada/England measures aside). It can show the divorce hit; it cannot redistribute the loss. And the deepest transition problem is the one no calculator fixes: the school-to-work transition for the young (Section 77's falling tenure) and the care-forced exits of the middle generation (Section 74's 22% savings shortfall) are transitions imposed by circumstance, not chosen in a planning tool.

The block's closing argument, then: the 100-year life multiplies transitions — between education and work, work and care, work and retirement, partnership and separation — and each transition is a stress test of the systems built for a three-stage world. Smoothness is not a luxury feature. It is what determines whether the extra decades are stages to move through or cliffs to fall off.

Evidence note

Transition-calculator facts drawn from (retirement-age parameter; hours/salary modelling BE/CL/NL/SE; Norway partial-retirement calculator; Denmark varying pension-start ages; Sweden MinPension 54+; Chile spouse/children;

Netherlands divorce; Belgium parental leave) and (UK dashboard surfacing forgotten accounts, guidance toward independent advice). The real-terms purchasing-power-erosion point is NOT repeated here (already in Section 22's non-repeat list). Career-transition policies (phased retirement, carer employment measures, reskilling) referenced via plain continuity phrasing from Sections 38, 76, 41–45. **Gap stated:** no evidence on actual transition outcomes (how many phased retirements succeed, divorce pension-splitting adequacy, parental-leave pension penalties by country); dashboard coverage and uptake figures belong to Section 25's territory and are not re-litigated. Remaining chunks (TOC pages, hazardous-job stubs, pension-coverage text) excluded as noise. Generational/family block (74–79) now CLOSED.⁶

Society, connection, and purpose

Loneliness as a social problem: designing systems for social connection

Loneliness in later life is routinely misread as a private misfortune — a matter of personality, bereavement, or bad luck. The World Economic Forum's longevity-economy principles reframe it correctly: the loneliness epidemic is a social, not individual, problem. That single reclassification changes everything about the policy response. Individual problems get advice; social problems get systems.¹⁰

The health stakes justify the systems framing. Social connection is integral to healthy longevity: socially isolated older adults have a higher risk of poor health and earlier death. This is not a soft wellbeing claim — it belongs in the same register as the mortality gradients of Sections 68–73. If isolation kills, then the environments that produce isolation — car-dependent suburbs, single-occupancy housing stock, workplaces that eject workers at fixed ages, care systems that warehouse rather than connect — are health hazards in the same sense as smoking or air pollution. Intentional design of systems and environments for social connection can mitigate these impacts.¹⁰

The loneliness problem compounds every inequality this report has documented. The older women living alone in family homes (Section 66), the low-income elderly for whom formal home care is unaffordable and institutional care is the default (Section 65), the carers with 20% more mental-health problems (Section 77), the migrant care workers separated from their own families (Section 74): each gradient in money, health, or geography is also a gradient in connection. Loneliness is where the inequality block's compounding becomes emotional fact. And the age-diversity opportunity of Section 75 — more shared years across generations — is precisely what an isolating environment wastes.

Designing for connection means treating it as infrastructure, not intervention. The age-friendly-cities agenda (Section 63) — transport, public space, services — is loneliness policy by another name: a bench, a bus route, a community centre are connection technologies. So is the community-as-single-system argument of Section 65: dwelling, services, and social infrastructure planned together rather than as separate silos. The principle is that connection should be the default outcome of ordinary environments, not something the lonely must seek out through special programs. Programs have stigma and attrition; environments have users.

There is a measurement warning here. Loneliness is subjective — the felt gap between desired and actual connection — while isolation is objective. Systems can most directly fix isolation (proximity, access, opportunity), and must be humble about loneliness itself. But the direction is clear: a 100-year life lived alone in a disconnected environment is not the longevity dividend anyone promised. Connection is not the garnish on healthy ageing; it is one of its conditions.

Evidence note

Framed on (social connection integral to healthy longevity; isolated older adults at higher risk of poor health and earlier death; intentional design mitigates) and (loneliness epidemic as social, not individual, problem — Principle 5 spotlight). Inequality-compounding and infrastructure arguments built from plain continuity phrasing (Sections 63, 65, 66, 74, 75, 77). **Gaps stated:** no loneliness prevalence figures, no isolation metrics by age/country, and no evaluations of connection interventions in this evidence set. Sections 81–84 (volunteering, ageism, life satisfaction, age-friendly communication) not front-run. Training/flexibility and pension-dashboard chunks excluded as noise.¹⁰

Volunteering, civic participation, and valued contributions

If loneliness is the deficit this block diagnoses, valued contribution is the surplus it prescribes. The 100-year life hands societies tens of millions of healthy years — Section 73's disability-free expansion, with over 90% of life-expectancy gains translating into healthy life — and the question is whether those years are spent as consumers of services or as contributors to them. Volunteering, civic participation, and the broader economy of unpaid valued work are how a longer life pays for itself socially, not just financially.³

The connection logic is direct. Section 80 established that isolation damages health — socially isolated older adults face higher risk of poor health and earlier death — and that intentional design can mitigate it; volunteering is the most scalable form of that design because it is reciprocal. The volunteer gets what the loneliness literature says protects health — regular social contact, a role, a reason to leave the house — and the community gets labour it could not otherwise afford: the neighbours and volunteers staffing community-based care networks, the mentors easing younger workers through transitions, the civic associations that turn a collection of households into a neighbourhood. The economic value is real even where unmeasured: traditional GDP captures paid labour but not the caregiving and volunteer hours older adults contribute. Contribution is connection with a purpose attached, and purpose is the difference between being included and being merely present.¹⁰¹⁶

The multistage lens (Section 78) reframes volunteering further. In a three-stage life, unpaid contribution was what happened after work stopped — a residual category. In a multistage life, it is one stage among many, interleaved with paid work, learning, and care. The encore career, the portfolio of part-time paid work plus volunteering, the retiree who mentors in the same industry that employed them: these blur the line between “working” and “contributing” in ways that pension systems built on binary employment states cannot easily recognise — which is part of why Section 79's transition infrastructure matters here too. A society that wants valued contributions from older people must make it easy to give them: flexible roles, recognised credentials, expenses covered, no bureaucratic gauntlet.

The inequality warning from Section 78 applies with full force. Volunteering is stratified: the healthy, the wealthy, and the time-rich volunteer most, while those whose lives would most benefit from the connection — the poor, the sick, the carers with no spare hours — volunteer least. A civic-participation agenda that ignores this becomes another privilege good, and worse, a moral cudgel: the “active ageing” rhetoric that blames the isolated for their isolation. The contributions that matter most for equity are the unglamorous ones — the informal care, the neighbourly help, the community self-organisation in places the state does not reach — and they are disproportionately supplied by the same women and low-income communities that the rest of this report shows are already stretched thinnest.

Valued contributions, then, are not a solution to loneliness that lets systems off the hook. They are the proof that the hook exists: a society in which older people can contribute is a society whose systems — transport, public space, flexible work, recognised roles — already make connection possible. Build the infrastructure, and the contributions follow.

Evidence note

Zero usable chunks: all 20 chunks in this prompt are retrieval noise — EU Ageing Report pension-reform assumptions, pension TOC pages, hazardous-job provisions, and pension-system notes. No bracket citations appear in this section. The argument is built from plain continuity phrasing on Sections 65, 74, 75, 78, 79, and 80. **Gaps stated:** no volunteering participation rates by age or country; no civic-engagement metrics; no evidence on health effects of volunteering or on barriers to participation in this evidence set. Sections 82–84 (ageism, life satisfaction, age-friendly communication) not front-run.

Perceptions of ageing and tackling ageism

Ageism is the only prejudice that everyone who lives long enough will experience from both sides. It is also the prejudice most thoroughly contradicted by the evidence of this report — and the one that does the most quiet damage to the 100-year life's promise. Tackling it starts with naming what it is: not just dislike of the old, but a whole architecture of perception about what each age is for.

The generational perception divide is the visible form. Younger generations are living in a very different world from the one previous generations experienced: baby boomers benefited from sustained economic growth, major health and social improvements, and growing employment rates, while Generation X can no longer assume to be richer in

old age than their parents, and the Millennial generation was particularly hard hit by the Great Recession, reducing prospects for stable careers. These are material facts, but they have hardened into moral narratives — boomers as beneficiaries who pulled the ladder up, millennials as victims of a rigged contract — that are themselves a form of ageism: judging a generation by its cohort luck rather than its choices. The intergenerational resentment diagnosed in Section 75 feeds on exactly these stories. Tackling ageism therefore means tackling it in both directions — the patronising of the old and the resentment of the young are the same prejudice wearing different clothes.³

The labour-market form is the costly one. Ageism is established in this report as a barrier alongside health and job quality (Section 35): the stereotype that older workers are less productive, less trainable, less adaptable — contradicted by the evidence on experience complementarities (Section 34) and by the simple fact that the fastest-growing segment of the workforce is the oldest. The stereotype does its damage before any hiring decision: it discourages the 55-plus worker from seeking training (Section 42's participation gaps), it justifies the early-exit cultures that pension reforms are trying to reverse (Section 17), and it turns the multistage life's transitions (Section 79) into rejections. For many, a "job for life" or even a "career for life" is no longer realistic — but ageism ensures that the second or third career starts from a presumption of decline.³

The deeper perception problem is the decline narrative itself: ageing as loss — of capacity, relevance, value — rather than as the accumulation the longevity economy describes. Every section of this report that documents older people's economic contribution (Sections 8–12), their care work (Sections 74, 76), their civic participation (Section 81), and their healthy years (Section 73) is evidence against the narrative; yet the narrative persists because it is embedded in language ("the burden of ageing"), in institutions (fixed retirement ages), and in products (the entire anti-ageing industry). Tackling ageism is therefore not primarily an attitude campaign. It is institutional redesign: retirement rules that don't assume decline at 65, hiring practices that don't filter by graduation date, media that doesn't illustrate every ageing story with a frail hand on a walking frame.

The 100-year life makes ageism an anachronism. A society in which people routinely live to 90 cannot afford to write them off at 55.

Evidence note

Usable: generational lived-experience divide — boomers' growth/health/employment gains vs Generation X's broken richer-in-old-age assumption vs Millennials' Great Recession scarring; "job for life" no longer realistic. Labour-market ageism and decline-narrative arguments built from plain continuity phrasing (Sections 17, 34, 35, 42, 75, 79, 81). **Gaps stated:** no ageism prevalence or attitude-survey data; no stereotype-content or discrimination-incidence figures; no evaluations of anti-ageism interventions in this evidence set. Rwanda/Bangladesh case material excluded (Section 96 territory). Sections 83–84 not front-run.³

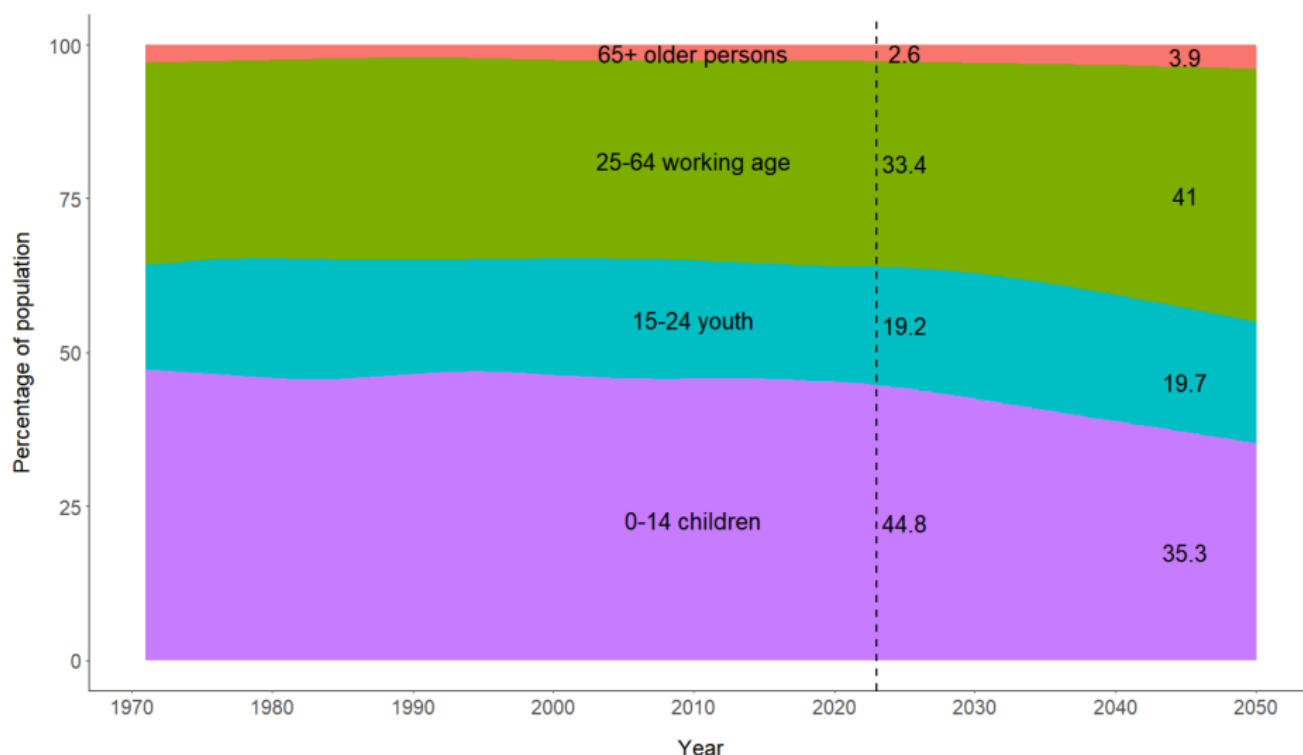


Exhibit. Population by broad age group, Angola, 1970–2050 Source: World Population Prospects 2022 (United Nations, 2022b). Source: S0005.

Life satisfaction and well-being in longer lives

A longer life is only a good deal if the extra years are good ones. The policy debate around ageing is dominated by costs and burdens; this section asks the other question — what actually makes the years worth living — and answers it from what the retrieved evidence can support. The honest answer is: less than it should. The corpus contains no life-satisfaction time series by country, no age-profile curve of well-being, and no decomposition of its drivers. What it does contain are three load-bearing facts and one institutional milestone, and the section is built on those alone.

The first fact is that well-being in later life is measurable. Subjective well-being of older people can be analysed through self-reported measures of overall life satisfaction. That matters because it turns “a good old age” from a sentiment into a policy outcome — one that can be surveyed, tracked across groups, and designed for, the way income or health already are. The Better Life Initiative, introduced by the OECD in 2011, led to a better understanding of what drives well-being of people and nations — the institutional moment when well-being entered the measurement agenda alongside GDP.¹³

The second fact concerns how health converts into well-being. Retirees who are fortunate to be in good health are much more likely to take part in a range of activities — returning to education, taking up a hobby, carrying out volunteer work, travelling, or playing sports. Health is not the outcome here; it is the enabling condition. The pathway runs from functional capacity to participation to satisfaction, and it is fragile at every joint: the healthy-years gradient established in Section 73 means that the retirees most likely to be sidelined from these activities are the same lower-income, lower-education groups who already face the steepest pension gaps. Inactivity in old age is therefore not a temperament; it is the downstream product of earlier inequality.¹

The third fact is relational. Alongside participation, another factor that can have an important influence on the well-being of older people is the frequency with which they enjoy regular contacts with family and/or friends. Note the word frequency — the claim is about regularity, not about being loved in the abstract. That distinction is the one Section 80 drew for loneliness: social connection is a system output, not an individual virtue. Well-being in longer lives is produced by routines — weekly visits, shared meals, club nights — and routines are exactly what break when neighbourhoods turn over, when health fails, when a partner dies.¹

Putting the three facts together gives the design brief for well-being in a longer life: measure it (self-reported life satisfaction as a tracked outcome), enable it (health and functional capacity as the gateway to participation), and schedule it (regular contact, built into the week rather than left to chance). None of this requires longevity breakthroughs; it requires the ordinary institutions of later life — adult education, volunteering infrastructure, accessible transport, community venues — to be treated as well-being production, not leisure afterthoughts. In a society where the over-65 population keeps growing, life satisfaction is not a soft extra; it is the metric that tells us whether the demographic transition is succeeding.

Evidence note: The retrieval contains no outcome data on life satisfaction in longer lives — no satisfaction levels by country or age group, no trend series, no U-shaped age curve, no decomposition of drivers, and no data on the satisfaction of the oldest old (85+) or of carers and the functionally limited. The three claims above — measurability via self-reported life satisfaction, health as the gateway to participation, contact frequency as a well-being influence — are the full set of well-being evidence the retrieved sources support.

Age-friendly communication and social inclusion

This section closes the social-connection and well-being block (80–84), and it closes it where the earlier sections pointed: the city — and by extension every institution an older person navigates — as the production system for connection, participation, and respect. Loneliness is a social problem (80), volunteering is infrastructure (81), ageism is institutional design (82), and well-being is a measured outcome with identifiable levers (83). What ties them together is a framework the retrieved evidence actually names: the WHO Global Age-friendly Cities Guide (2007), whose Part 2 sets out “Active ageing: a framework for age-friendly cities” and whose domain structure partitions the task — outdoor spaces and buildings, transportation, housing, social participation, respect and social inclusion, civic participation and employment, communication and information, and community support and health services.¹⁴

The physical domains — transport, outdoor spaces, housing — were treated in Section 63, which reserved the participation and inclusion domains for this closing block. This section takes them up. **Communication and information** (Part 11) is the inclusion layer the whole system stands on. Services that exist but cannot be found, navigated, or understood do not exist for the people they were built for. The failure mode was established in Section 79: pension systems that assume digital reach manufacture forgotten accounts and disengaged members. The principle generalises — health services, transport, benefits, community programmes — wherever the information channel narrows, the oldest and least connected drop out first. Age-friendly communication is not large print on a brochure; it is the deliberate design of every channel (digital, paper, voice, person-to-person) so that reach does not depend on being young, literate in interfaces, or already connected.¹⁴

Respect and social inclusion (Part 9) and **social participation** (Part 8) are the domains where the decline narrative of Section 82 meets the participation evidence of Sections 80–83. Inclusion here is not a sentiment — being “nice” to older people — but the absence of the structural exclusions this block has traced: communication systems that bypass them, participation venues that assume mobility and digital access, public spaces that feel unsafe or unwelcoming, civic life that treats them as recipients rather than contributors. **Civic participation and employment** (Part 10) closes the loop: inclusion includes the right to contribute, not only to be served.¹⁴

The arc of the block, stated once: well-being in longer lives is produced, not hoped for — by routines of contact (83), by connection treated as a system problem (80), by participation infrastructure anyone can enter (81), by institutions redesigned against ageism (82), and by communication that reaches everyone. A city is age-friendly when an older resident can find out what exists, get there, be welcome on arrival, and contribute once inside. None of those four steps is about age. All four fail by age, every day, by design.

Evidence note: The retrieval supports only the framework — the WHO Guide’s domain structure (Part 9, respect and social inclusion; Part 11, communication and information; Parts 8 and 10, social and civic participation; Part 2, the active-ageing framework). It contains no data on communication reach among older people, digital exclusion rates, participation levels, or inclusion outcomes. The four-step synthesis (find, reach, welcome, contribute) is the author’s structuring of the block’s conclusions; everything else in the prompt was retrieval noise — pension-scheme country tables and payout-phase material with no bearing on communication or inclusion.¹⁴

Policy pathways for resilient systems

The case for action: why ageing policy cannot wait

This section opens the action block, and its argument is simple: delay is itself a policy choice, and it is the most regressive one available. Every year that ageing policy waits, the costs compound — and they compound on the people least able to bear them. The evidence for this is not a new forecast but three mechanisms, each fresh to this section, each showing why the standard levers misfire precisely where need is greatest.

First, the problem is worse than measured. New mortality data at older ages — made possible by more precise information on mortality after age 75 — show a higher impact of socio-economic conditions on inequality in longevity than previously reported. The gradient this report traced from Sections 68 through 73 was already steep; the refined data steepen it further. Every policy calibrated to the old estimates undershoots the inequality it is meant to address. Waiting for better data is not the lesson here — the lesson is that the cost of waiting was always higher than the published numbers said.³

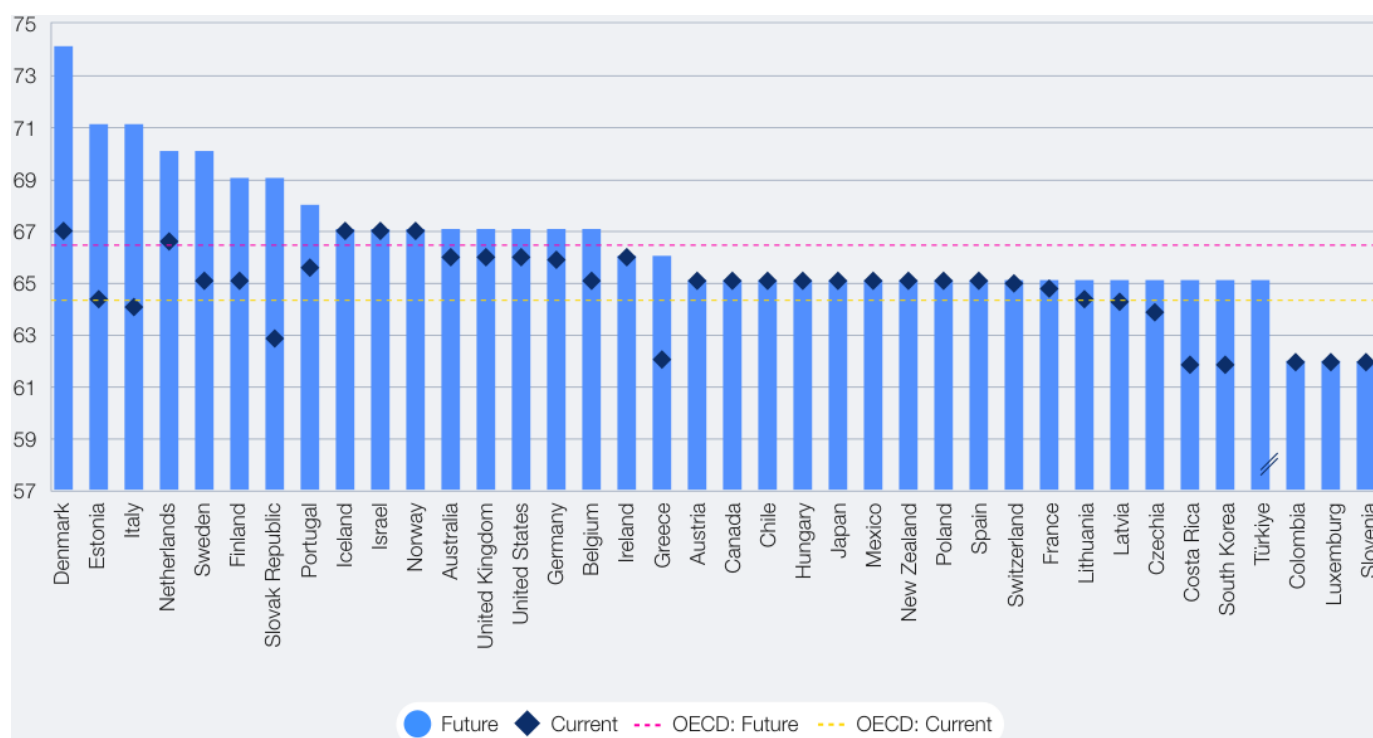
Second, the bluntest policy lever — raising the statutory retirement age — is also the most uneven. Calculations show that an increase in the statutory retirement age might raise the effective retirement age less for low-educated than for high-educated workers, precisely because socio-economic differences in health status and employment rates shape who can actually keep working. The tension established in the continuity of this report holds: longer working lives will be achieved through working-age policy or not at all, and raising the age without removing the barriers punishes the workers the barriers already hurt. A lever that works for the credentialed and stalls for the manual is not a neutral delay — it is a transfer of the adjustment burden downward.³

Third, the demography itself discriminates. In Chile, just taking into account gender differences in life expectancy lowers female pensions by 5% compared with men's, and the replacement rate from 39% to 37% — and this comes on top of the impact of gender gaps in the labour market. Actuarial neutrality is not neutral when the underlying lives differ. The feminisation of old-age poverty (Section 71) is not an accident to be corrected later; it is built into the arithmetic of longer female lives meeting shorter female careers, and every reform that ignores it bakes it in deeper.³

The response is to move earlier, not later. In 2021, France introduced a new medical examination at age 45, intended to raise employees' awareness of ageing at work and to prevent occupational wear and tear — during the visit, a professional reskilling agreement can be proposed. Mid-life, not retirement age, is where prevention lives: that is the logic of the lifelong-learning policies, the care credits, and the working-age interventions this block's later sections will detail.⁴

The case for action, then, is not urgency about a crisis but urgency about compounding. Inequality in longevity is larger than previously reported; the main reform lever helps the educated most; the pension arithmetic penalises women by default. None of these waits for 2050. They are already operating, and every year of inaction lets them accumulate into the old-age outcomes the rest of this block tries to prevent.

Evidence note: The retrieval for this section contains no aggregate urgency statistics — no dependency-ratio timelines, fiscal-gap projections, or cost-of-delay estimates appear in these chunks. The case is built from mechanism evidence: refined mortality data on the longevity gradient, differential effects of retirement-age rises by education, the Chilean gender-longevity pension penalty, and France's age-45 medical examination. Everything else in the prompt was retrieval noise — TOC pages, methodology annexes, and previously used material.³⁴



SOURCE: OECD. (2023, December 13). *Pensions at a glance 2023*. https://www.oecd.org/en/publications/2023/12/pensions-at-a-glance-2023_4757bf20.html

Exhibit. The graph below shows the normal retirement age for men entering the labour market at age 22 with a full career across the OECD. Based on current legislation, retirement ages will increase by about two years for those no. Source: S0010.

Building resilient public retirement systems

A public retirement system is resilient when it can absorb the demographic transition without breaking its promises and without breaking its politics. The fiscal facts, newly citable in this section, set the scale of the test. Public pension expenditure is on the rise in most OECD countries and is projected to keep growing in 22 of them, falling in only 9; on average across 31 OECD countries, spending would increase from 8.9% of GDP in 2020–23 to 10.2% of GDP in 2050, reaching 10.0% as early as 2040. The main driver is demographic change: the demographic old-age to working-age ratio — people above 65 per 100 people aged 20 to 64 — is projected to rise 72% between 2022 and 2052. Those numbers are the fiscal expression of everything this report has described: longer lives, smaller working-age cohorts, and the intergenerational bargain under strain.⁴

The expenditure path is not destiny, and the spread across countries proves it. In the EU27, public pension spending is projected to increase from 9.9% of GDP in 2020 to 11.3% in 2050 and then stabilise — described as a significant achievement given the demographic change throughout the period. Stabilisation after decades of growth is what resilience looks like in fiscal terms: the system absorbs the shock and finds a new equilibrium. The stabilisers are legislated, not automatic: cuts in benefits for future retirees at least relative to wages, through lowered indexation and valorisation of benefit formulae, together with increases in the age at which individuals can first claim pension benefits. In other words, the adjustment is shared — future retirees get less relative to wages, and everyone exits later. That is a political choice as much as a fiscal one, and the country variation shows how wide the choice space is: Slovenia is projected to rise from 10.0% of GDP to 15.7% by 2050; Korea's rapid increase reflects both ageing and a still-maturing system; Belgium, Czechia, Hungary, Ireland and Luxembourg would each add about 3 percentage points of GDP or more; while Denmark, Estonia and Portugal would fall by around 2 points and Greece by more than 3. Resilience is visibly a decision, not a fate.⁴

But fiscal sustainability is only half of resilience; the other half is surviving the transition itself. Transitions from pay-as-you-go systems to funded schemes, or from defined benefit to (collective) defined contribution plans, can take years if not decades, and necessitate compromises on intergenerational redistribution, clear communication, and long-term political commitments. Gradual changes rather than abrupt shifts are necessary to mitigate disruptions and allow individuals and organisations to adjust to new rules over time; the current transition in the Dutch occupational sector demonstrates that with the right strategies these challenges can be effectively managed. This is the re-

silience principle the report keeps returning to: systems that move slowly and credibly survive; systems that promise everything and adjust suddenly do not.⁸

The slow lever is retirement age. Retirement ages are rising in more than half of OECD countries, and based on current legislation, retirement ages will increase by about two years for those now entering the workforce. There is often a long time-lag between when a pension reform occurs and when it begins to affect expenditure — which is why long-term projections are a crucial tool in planning pension policy. Combine the long lag with the Section 85 finding that retirement-age rises help the low-educated least, and the design requirement sharpens: the age lever must be paired with working-age interventions (health, training, care), or it becomes the regressive transfer Section 85 warned about.⁸⁴

Resilient public retirement systems, then, have three properties: a bounded fiscal path (stabilisation, not open-ended growth), a credible transition (gradual, communicated, politically committed), and an adjustment shared without crushing the vulnerable (benefit-indexation and claiming-age changes paired with working-age policy). The expenditure projections show the first is achievable; the transition experience shows the second is manageable; the third is the test the rest of this action block must pass.

Evidence note: The incentive-design material (contribution caps, matching contributions, tax treatment) in this prompt belongs to the territory of Sections 18–19 and is not repeated here. The retrieval contains no LMIC public-pension fiscal projections and no composite resilience metric; the country-level projections cover OECD countries only.

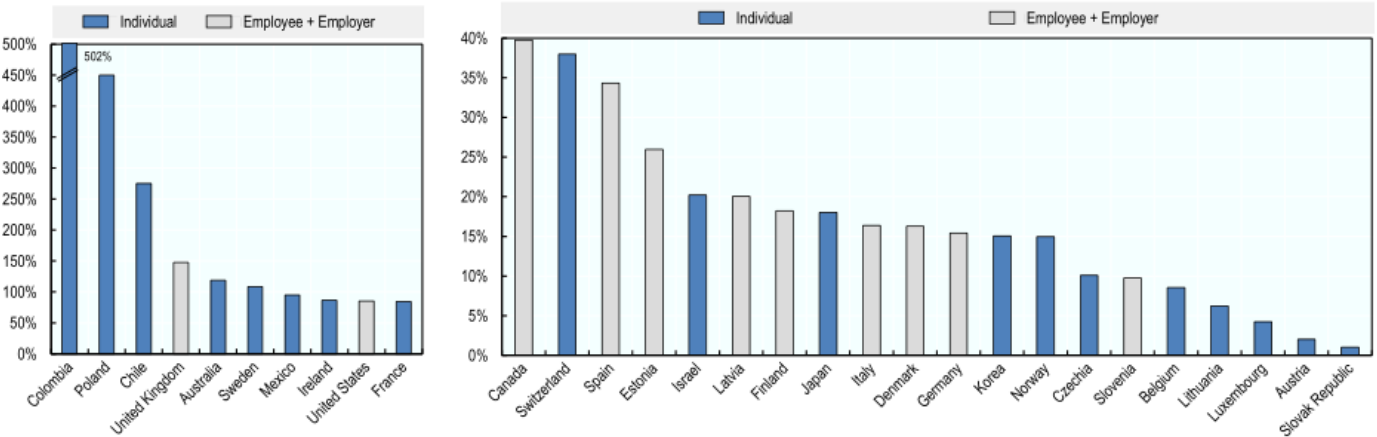


Exhibit. As a percentage of the average wage in the economy in 2022 Individual Employee + Employer Individual Employee + Employer 500% 40% 502% 450% 35% 400% 30% 350% 25% 300% 250% 20% 200% 15% 150% 10% 100% 50% 5% 0% 0%. Source: S0013.

Universal social protection as an anchor for health and long-term care

The argument of this section is that long-term care cannot be left to families and savings markets — not on equity grounds, and not even on efficiency grounds. The evidence arrives as a chain of mechanism and number, and it is unusually fresh to this report.

The mechanism first. The absence or insufficiency of public financing for long-term care does not only hurt the frail; it undermines the whole system around them. It renders pension benefits inadequate — too low to cover both living expenses and long-term care — and it undermines the financial sustainability of health benefits, as some of long-term care’s costs are shifted towards the health system. The UK case quantifies the spillover: in 2016, the National Health Service estimated that the use of over 60,000 hospital bed days per month resulted from delayed discharges from hospital attributable to failures in social care. And the reverse direction quantifies the return on fixing it: in Catalonia, Spain, a monthly long-term care benefit equivalent to €214 was able to reduce avoidable hospitalisations by 60% and the number of unscheduled “walk-in” patient visits by half, with the majority relating to social exclusion cases — instrumental in containing the rising costs of health care. Long-term care financing is not a welfare add-on; it is load-bearing infrastructure for the health system and for pension adequacy alike.¹⁷

The equity case is sharper still. Leaving long-term care to be financed by out-of-pocket payments and unpaid family caregivers is regressive, inequitable for those who do not have family members who can provide this care, and

entails a significant opportunity cost for unpaid caregivers — the lost labour-market participation of family members who would like to work, and the mental health costs of caregivers who receive little training, respite or psychosocial support. The scale of the uninsured shock confirms the regressivity: around 40% of lower-income people in the United States and Canadians report being financially unprepared for long-term care expenses, and one study estimates that 25% of the elderly needing long-term care in Europe would not be able to finance 10% of their expenses even if using all of their financial resources. Solidarity-based financing mechanisms are most appropriate to ensure families and older persons do not have to bear individually the burden of loss of function — and shifting the costs of long-term care from the individual and family onto society is a gender-transformative investment in equity, offering the possibility of dignified and active ageing.¹⁷⁶

The institutional models already exist, in several variants. Countries have created dedicated long-term care schemes (Japan and the Republic of Korea), provided “top-up” pension benefits or expanded disability benefits, or embedded long-term care within social health protection packages as in Northern Ireland. The Netherlands reformed its long-term care insurance scheme in 2015, moving nursing care in the long-term care context back into social health insurance to reduce costs and create incentives for coordination between primary care, hospital care, rehabilitation and community nursing for the frail elderly; Uruguay built a scheme for home-based care covering long-term care needs across all age groups, with residential care in the national social security fund. The arrangement varies; the principle does not: additional benefits require complementary sources of sustainable financing, regardless of institutional design.¹⁷

The anchor logic extends beyond old age. In emerging economies, workers cannot afford to be unemployed given absent or weak social protection, and small effective insurance for the unemployed translates into higher labour-market insecurity than in most OECD countries; where the risk of low pay is significant, the ability of social protection to cushion earnings shocks is crucial and needs to be extended. And where public healthcare coverage is strong, the effect is visible in household behaviour itself: retirees in Northern European countries spend down their wealth more rapidly than in Central and Southern Europe and the United States, reflecting higher public healthcare coverage and less need for precautionary savings. Universal social protection does not only protect the vulnerable; it releases everyone else from hoarding against catastrophe.³⁶

The section’s conclusion, carried into the action block: ageing policy that funds pensions while leaving care to families is building on sand — the pension becomes inadequate, the hospital fills with the preventable, the carer drops out of work, and the poor go uninsured. The anchor must hold both.

Evidence note: The retrieval contains no data on the aggregate fiscal cost of universal long-term care coverage, on caregiver labour-market losses quantified in hours or wages, or on coverage rates of the country models listed. Everything else in the prompt was retrieval noise — payout-phase references, home-equity annexes, and pension-reform TOC pages.

Life-course prevention: addressing inequality before it cumulates

Prevention is the oldest idea in public policy and the one this report keeps rediscovering. Sections 68–73 traced how disadvantages compound over a lifetime — bad health cuts lifetime earnings, wage inequality passes through to pensions, the healthy-years gradient sorts who gets to enjoy the extra years. This section’s question is where the compounding can be stopped, and the answer is: earlier than any of the machinery built so far assumes. Pension reforms, retraining programmes, and care credits all operate on people whose trajectories are largely set. Prevention operates on the trajectories themselves.

The strongest prevention lever the evidence names is education. The causal relationship between education and longevity has been established by studies exploiting exogenous changes in school laws and by analyses spanning generations of grandparents and great-grandparents; and crucially, the effect of higher education appears to be greater among lower- and middle-income OECD countries than in high-income ones. That asymmetry is the prevention argument in miniature: where populations are ageing at lower incomes — the “getting old before getting rich” economies of Section 6 — education investment buys more longevity per unit of effort. Education works through channels that no pension formula can replicate: it changes behaviour (less smoking, less obesity, less alcohol), it changes time horizons (the more educated tend to be more forward-looking), and it changes capability (better informed, more adept at navigating the health system). A school is a longevity technology — and unlike a retirement-age increase, it helps the low-educated most.³

The second lever is behaviour, and the evidence that it moves is smoking. Smoking accounts for up to half of the observed inequalities in mortality in some countries; across 14 European countries in 2000–04, smoking-related mortality explained between 19% and 55% of the mortality difference between the low- and highly-educated groups among men, and between 0% and 56% among women. The divergent trend this report recorded in Section 69 — falling contributions among men, rising among women since the early 1990s — proves the point: mortality inequality is not a fixed quantity. It responds to policy, campaigns, taxes, and norms, and it responds within years, not generations. If half of the inequality in death is behaviour, then prevention is not a vague aspiration; it is the single largest lever on the longevity gradient.³

The third lever is the starting line itself. A high positive correlation of the educational attainment of children with that of their parents leads to a positive correlation between the income level of adults and their parents' income, which they share during childhood. Inequality is inherited before it is earned. This is why the prevention agenda of Section 69 put early-life interventions at the top — childcare quality, school failure, early school leaving — and why it must stay there: every later intervention is fighting a gradient that was set before the person ever entered the labour market.³

The stakes of prevention are larger than earnings. Life expectancy gaps translate directly into differences in earnings that accumulate over working lives — but they also contribute to differences in well-being, because life itself is a non-income component of wellbeing, with a non-monetary value that goes beyond income earned. When inequality kills earlier, the loss is not only the missing wages; it is the missing years. Prevention protects those years, which no pension can buy back.³

The action-block logic, four sections in: Section 85 said delay compounds regressively; Section 86 bounded the fiscal path; Section 87 anchored care in solidarity financing; this section says the cheapest place to break the compounding is at its source. Compensating inequality in old age is necessary — but it is the most expensive way to address inequality that was preventable at every earlier stage.

Evidence note: The retrieval contains no early-childhood outcome data, no cost-effectiveness comparisons of prevention versus compensation, and no evaluations of education-as-longevity-policy. The reskilling machinery (mid-life reviews, career transitions) belongs to earlier sections and is not repeated; the smoking trend claim belongs to Section 69's prevention proof. The magnitude facts — education's larger effect in lower- and middle-income OECD countries, smoking's up-to-half contribution, parental-education inheritance, life's non-monetary value — are this section's fresh evidence.

Data, digital identity, and innovation in service delivery

The ageing agenda's bottleneck is increasingly not what services should exist but how they reach the people who need them. This section's evidence — all fresh to the report — supplies the delivery architecture: digital health readiness as the framework, data governance and digital identity as the instruments, and a pandemic as the proof that readiness saves lives.

Digital health readiness is defined as the ability to make use of analytics, data, and technology for beneficial individual, community, and public health outcomes — a composite of abilities and structures across four dimensions. **Analytic readiness** assesses whether analytics can be created and used to generate action that improves health outcomes — including responsible AI to help doctors and nurses in routine tasks such as documenting cases or in diagnostics such as interpreting radiology images. **Data readiness** assesses whether data can be collected, accessed, and used in analytics — integrated, quality health data supported by policies on data protection, de-identification, access, and linking. **Technology readiness** assesses the secure input, storage, and movement of data, including the technical and semantic interoperability that lets health systems communicate with high data quality and timeliness. **Human factor readiness** assesses the ecosystem's resources and resilience — trust among stakeholders, financial and human resources, co-operation — including digital health literacy so the public, providers, and policy makers can use the system and its protections effectively.⁵

The framework is backed by instruments, not just aspiration. High digital health readiness is aligned with OECD legal instruments for artificial intelligence, health data governance, digital security, and digital identity. In 2017, OECD countries endorsed a Recommendation on Health Data Governance encouraging a national health data governance framework with 12 components and co-operation on interoperability standards; the 2022 five-year review scored dataset governance across 15 measures — training and operational controls for privacy and security, data-sharing processes, data catalogues — with Denmark highest (scoring best in 14 of 15 measures, including legislated dataset

creation, trained staff, standard data-sharing agreements within the public sector, with academics, with the private sector and across borders), followed by the United States (11 of 15) and Finland (13 of 15). And identity itself has a governance instrument: the 2023 OECD Recommendation on the Governance of Digital Identity. For an ageing population — where entitlements, care coordination, and fraud protection all depend on knowing who is who — digital identity is not a tech flourish; it is the hinge of delivery.⁵

The pandemic supplied the natural experiment. Early evidence shows that higher levels of digital health readiness resulted in fewer lives lost and more stable health systems; a study of country-level digital preparedness found that the more advanced countries' digital adoption, the lower the number of cases and the faster new cases decline — with digital infrastructure and telehealth the most critical factors, and digital preparedness carrying comparable importance to smoking, age, and income for COVID-19 cases and deaths. The mechanism is visible in the five functions readiness enabled: mobilising testing centres, closely monitoring spread and severity, adjusting public health measures for impact, communicating them effectively, and maintaining service delivery through digitally enabled care such as telehealth. Readiness is not a ranking to admire; it is capacity that activates in a crisis.⁵

Innovation in delivery is not only digital. In East Asia and Pacific, the persistent failure of contributory pension schemes to close the coverage gap drove considerable innovation: Chile (2008), China (2011), Korea (2004), Mexico (2012), and Thailand (in the 2000s) all rapidly expanded social pensions and budget-financed matching contribution schemes for informal sector workers — a paradigm shift away from sole reliance on payroll taxes toward general-revenue financing, since the traditional social insurance approach risks losing the race between pension coverage expansion and rapid societal ageing. In New Zealand, the opt-in default for a voluntary defined contribution scheme led to relatively high participation rates. The principle generalises: the redistributive element of the system should be financed from general revenues and clearly distinguished from the insurance or savings component. Delivery innovation means redesigning how the state reaches people, not just digitising how it serves them.⁷

The thread back through this report: Section 84 called communication the inclusion layer; Section 79 showed what happens when systems assume digital reach; Section 89 shows the other side — what becomes possible when data, identity, and delivery are designed for the whole population. An ageing society cannot afford to deliver twentieth-century services to a twenty-first-century population.

Evidence note: The retrieval contains no data on digital exclusion among older people specifically, no uptake or outcome metrics for digital identity schemes, and no evaluations of the pension-innovation models beyond the participation note for New Zealand. The remaining chunks were retrieval noise — pensions-outlook TOC pages and methodology annexes.

Regional cooperation and learning across ageing societies

Ageing is not synchronised. Japan aged first; Korea is ageing fastest; China is ageing biggest; and the younger, poorer economies of East Asia and Pacific will start to age rapidly two or three decades from now — each at very different income levels and institutional starting points. That asynchrony is an asset if it becomes a learning system: the latecomers can study what worked, what backfired, and what is still running, instead of replaying the sequence. The East Asia and Pacific report states the thesis directly: given the diverse stages of demographic transition across the region, lessons can be drawn from the experience of richer East Asian and Pacific economies. This section assembles the lessons worth transferring and the warnings worth heeding.⁷

The race framing gives cooperation its urgency. Most of the region is likely to experience its demographic ageing process before coverage rates in mandated schemes rise to meet the challenge. The within-country coverage gradients make the urgency concrete: coverage in Indonesia is less than 10% for the bottom half of the income distribution but rises to around one-third for the top quintile; in China, the elderly in the top decile were as likely to receive pensions as the same group in Japan, while fewer than one in five elderly Chinese in the bottom half of the income distribution received a pension in 2011; in the Philippines, only 2.8% of people in the lowest income decile were receiving a pension in 2010. The lesson for latecomers is that contributory mandates do not reach the bottom half by themselves — a finding the region's reformers have already acted on.⁷

China's response is the most instructive positive case. The expansion of coverage of informal rural and urban workers since 2010 represents the single biggest increase in contributors to a government pension system in history — unique in scale and in design, which links contributions both to eligibility for a basic pension at age 60 and to immediate benefits for the elderly parents of contributors, a crucial incentive since contributions for informal workers are voluntary; counting these new contributors, coverage rises from about 28% to well over 60%. The transferable

design insight: where formal employment cannot be the coverage vehicle, the scheme must recruit through family incentives and basic-pension linkage, not through payroll discipline that assumes employers.²

The negative lesson is Timor-Leste. It introduced universal pensions at age 60 paying around 44% of average non-oil income per capita — and it is the only country in the region spending a large amount on social pensions, more than 2.5% of GDP (non-oil), while no other East Asian and Pacific country spends more than 0.4%; on current parameters, the present value of spending on the programme through 2040 exceeds 100% of today's GDP. Generosity without sustainability is a warning every younger economy can read for free — and a case for the regional cooperation agenda that Sections 91–94 continue: set up sustainable systems before the ageing wave, not during it.²

A smaller, subtler lesson comes from Singapore: many retiring cohorts are asset rich and cash poor, thanks to CPF housing investments, and the government responded with a lease buyback programme (selling housing back to the government, along the lines of a reverse mortgage) and, most recently, the Silver Support scheme to supplement the pensions of those without sufficient accumulated savings. Adaptation, not redesign: the system learned mid-course.²

The cooperation thesis is that ageing societies are not condemned to repeat each other's mistakes. Traditional family networks — the *wantoks* of the Pacific — are assumed to substitute for public support, but evidence questions the future validity of that assumption. Demography is not destiny; through the right mix of policies, governments can help societies adapt to rapid ageing and improve the well-being of citizens. Regional cooperation is the mechanism that turns one country's experiment into another country's policy.²

Evidence note: The retrieval contains no data on the institutions of regional cooperation — no cooperation agreements, learning platforms, or knowledge-exchange mechanisms are described. The lessons above are inferred from country cases; the cooperation architecture itself is a gap in the evidence.

Regional and country deep dives

East Asia and Pacific: diverse ageing patterns across income levels

Section 5 established that East Asia and Pacific is the fastest-ageing region on record, with more over-65s than any other region. This section takes the income lens: the same demographic transition is happening at radically different income levels, and that difference changes everything about what policy can do. The transition that took France 115 years to move from 7% to 14% of the population aged 65 and above is being compressed into 15 years in Vietnam, 20 in Korea and Thailand, and 25 in China and Japan. Compression at low income is the region's defining constraint.²

The dividend history explains the stakes. Around one-third of the high per capita income growth in East Asia from 1960 to the 1990s can be attributed to the demographic dividend, with some estimates of the effect as high as 44% of growth in per capita income. The dividend compounded with educational attainment and capital investment to produce the region's growth miracle. For the green countries — Cambodia, Lao PDR, Myanmar, Papua New Guinea, the Philippines, Timor-Leste — still in their youth bulges, the imperative is to make productivity-enhancing reforms and investments now, so they both maximise the dividend and are prepared for rapid ageing when it comes. The dividend is a one-time gift; wasting it means ageing without the wealth to pay for it.²

That is already the orange countries' predicament. Developing countries in the region are getting old before getting rich, and even with sustained productivity growth, orange countries will not reach the income levels of the red economies or of OECD countries such as Mexico at similar points in the demographic transition. Thailand provides the case in point: despite solid growth in recent decades, the pace of ageing is outstripping per capita income growth, following a path closer to Poland — now struggling with fiscal pressures from ageing — while Mongolia is experiencing stagnant GDP per capita alongside rapid ageing. "Getting old before getting rich" is not a slogan here; it is a measured shortfall between demographic speed and income trajectory.²

The policy agendas therefore diverge by group. For the red economies — Hong Kong, Japan, Korea, Singapore — slowing the structural decline in the labour force is the most critical challenge, since they are closer to the productivity frontier with fewer easy options for raising productivity: a sharper focus on increasing female labour force participation, extending productive working lives, and in some countries increasing immigration of younger workers from within the region. Orange countries must sustain high productivity growth and undertake structural reforms of

social security, health and long-term care, and labour market policies — and even then they will become aged societies at much lower income levels than the red economies. Green countries must realise maximum GDP growth from the dividend while avoiding policies — in pensions and health — that are affordable now but would rapidly become unsustainable when ageing accelerates.⁷

The labour-force arithmetic has been modelled. Greater participation by women, older people, and migrants can mitigate the projected decline in the total labour force in the richer East and Southeast Asian economies (Hong Kong, Japan, Korea, Malaysia, Singapore), with sending economies across the region's younger countries; but migration also leads to a 5% decline in the total labour force size of the migrants' origin countries. The trade is visible: the destination's gain is the origin's loss, and regional cooperation (Section 90) must price that in. On skills, the gaps are stark: Japan and Korea show low rates of on-the-job training for middle-aged and older workers relative to the United States, while Singapore's Budget 2014 announced a S\$500 million top-up to its training programmes — the kind of mid-life investment Section 88's prevention logic demands.⁷

The section's conclusion: there is no "East Asian model" of ageing — there are three different transitions running on three different income clocks. The region is a laboratory, not a template, and its lessons (Section 90) travel only when the income context is carried with them.

Evidence note: The retrieval contains no health-expenditure or long-term-care demand projections disaggregated by these country groupings, and data for the green countries is acknowledged as sparse. The regional demographic framing (groupings, fertility and life-expectancy drivers) was established in Section 5 and is not repeated beyond the transition-speed figures, which are new here.

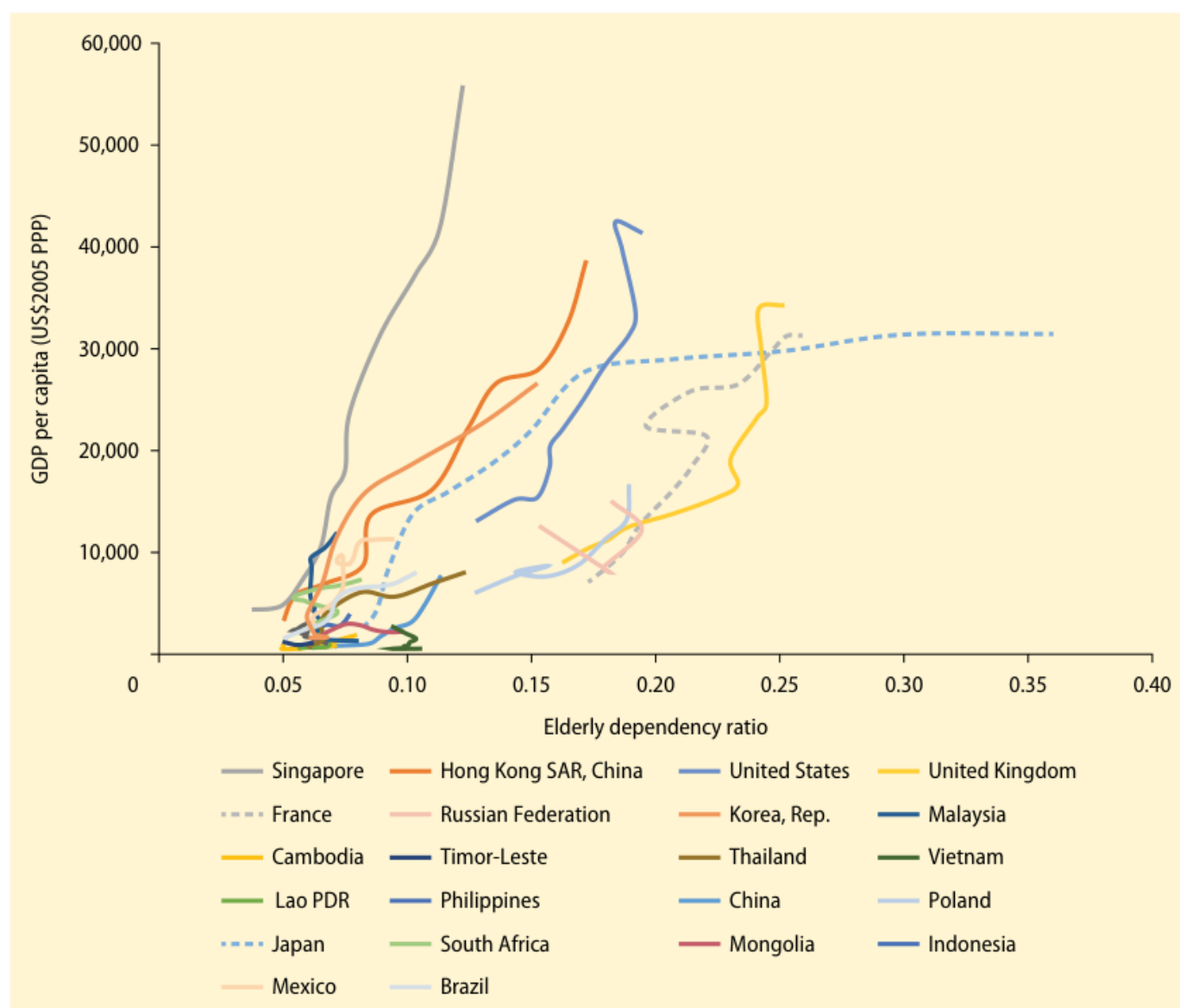


Exhibit. Figure O.8 east Asia and Pacific's developing economies are getting old before getting rich Elderly dependency ratio by GDP per capita (2005 PPP), various economies, 1980–2010 60,000 50,000 40,000 30,000 20,000 10,000 0 0.

Source: S0029.

Europe: the Ageing Reports and demographic projections to 2070

Europe is the laboratory where ageing arrives first among the large economies. Behind Japan, the EU provides one of the most distinctive examples of demographic ageing: in 2015, people aged 65 and over accounted for 26.0% of Japan's population, 18.9% of the EU-28's, 14.6% of the United States', and 9.3% of China's, against a world average of 8.2%. The median-age story tells the trajectory: in 2015 the world median was 29.6 years, Japan's 46.4, and the EU-28's 42.4; by 2050 the world is projected to reach 36.2, while the highest median ages are projected for South Korea (56.5 years) and Japan (54.7), and China's median (47.6) is projected to rise above the EU-28's (46.9). Europe ages earliest among the major economies, but East Asia overtakes it within the projection horizon — the relay race Section 91 described. By 2050, the only G20 country where the median age is projected to remain below the world average is South Africa (33.9 years). (The UN and Eurostat methodologies differ; the figures above use EUROPOP2018 baseline variant for EU-28 projections and the UN medium variant for the rest.)¹

The European Commission's Ageing Reports are the machinery that turns these demographics into fiscal planning — projecting economic and budgetary impacts to 2070, feeding the expenditure forecasts (Section 86's 8.9→10.2% of GDP) and the labour-market participation projections that pension reforms are modelled on. The reports' value is the one Section 86 stated: long reform-to-expenditure lags make long-term projections the crucial planning tool. Their discipline is also the caution Section 91 recorded: demographic projections are fallible, fertility consistently outpaces projections, and static dependency measures are increasingly questioned.

What the ageing looks like in incomes, currently: people aged 66–75 have a disposable income of 93% of the total population's on average across the OECD, falling to 81% among those over 75 — but the dispersion is stark. Average relative income falls below 80% in Estonia, Korea and Lithuania for the 66–75 group, and below 65% in those three plus Latvia for the 76+ group; while it exceeds 110% in Israel, Italy and Luxembourg for the 66–75 group, and Luxembourg is the only country where the over-75s' relative income exceeds that of the total population. Relative income poverty (below 50% of median equivalised income) affects 12.5% of people aged 66–75 and 16.6% of those 76+, against 11.4% of the total population; the Baltic states and Korea show poverty rates around or above 25% for ages 66–74 and above 40% for 76+ (Lithuania excepted), while the Latin-American OECD members, Australia and the United States also show elevated old-age poverty. Ageing in Europe is not uniformly comfortable — and the east–west gradient inside the EU is as sharp as anything in East Asia.⁴

The reform response is moving toward first-tier protection. In January 2022, Chile replaced its targeted public pension scheme with a quasi-universal one — the universal guaranteed pension — initially accessible to the 90% poorest people aged 65+, expanded in February 2023 to the 90% poorest of the entire population (adding around 70,000 beneficiaries, ~4% more); the benefit was 4% higher in real terms than its predecessor, about 19% of gross average earnings, and withdrawal now only begins when the contributory pension reaches 3.4 times the guaranteed amount (50% rate thereafter), substantially improving old-age income for a large majority of retirees. Canada raised its residence-based basic pension by 10% for people 75 and over in July 2022; Estonia raised basic and targeted pensions by EUR 20 (about 7% above indexation) in January 2023; Sweden raised targeted pension benefits by about 12% in August 2022.⁴

Europe's role in this report: first to age, first to measure, first to reform — and the Ageing Reports' projections to 2070 are the shared instrument by which the fiscal consequences are made legible before they arrive.

Evidence note: The retrieval's Ageing Report chunks were table-of-contents and methodology-structure pages; no 2070 country-level expenditure or participation numbers appear in these chunks. The occupational life-expectancy-gap trend details belong to earlier sections and are not repeated. (Two citation keys were mistyped in the draft and corrected before logging.)

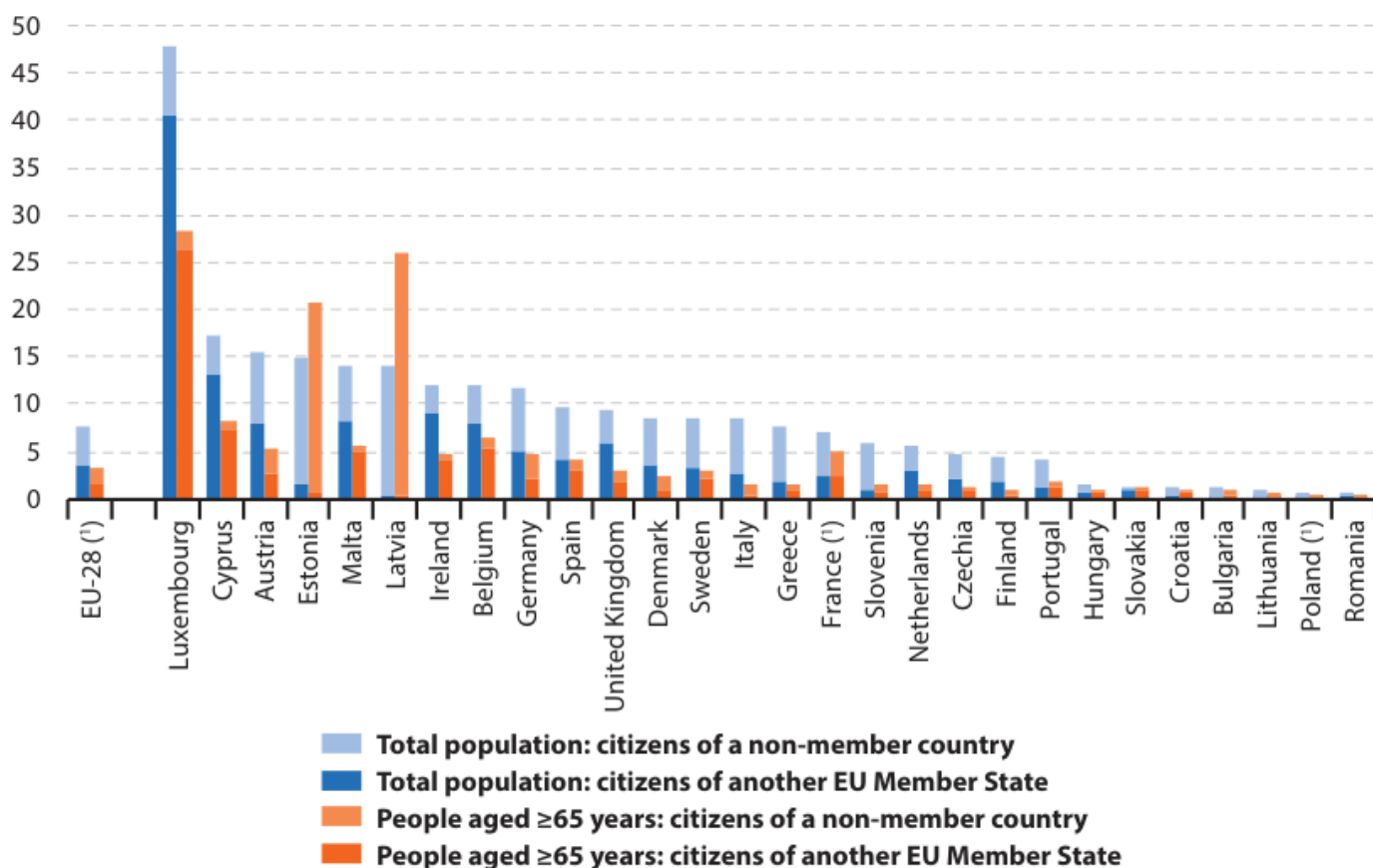


Exhibit. Figure 1.14: Citizenship, by age class, 2018 (% share of age class) 50 45 40 35 30 25 20 15 10 5 0 Total population: citizens of a non-member country Total population: citizens of another EU Member State People aged ≥65 years: citizens of a non-member country People aged ≥65 years: citizens of another EU Member State
Source: S0014.

Japan: super-ageing, employment promotion, and care innovation

Japan is the future arriving early. Every demographic milestone other countries are approaching, Japan has already passed: it has the highest share of over-65s in the G20 (26.0% in 2015, against an EU-28 share of 18.9% and a world average of 8.2%), the highest median age (46.4 years in 2015, projected to reach 54.7 by 2050), and the oldest OECD population, with its old-age ratio projected to rise from 55.4 to 80 by 2052. Those facts were established in Sections 92 and 40; this section's purpose is to read Japan as the preview — what super-ageing does to work and care, and what the rest of the world can learn from how Japan has responded.¹⁴

On employment, Japan's lesson is that long working lives are achievable but not automatically adequate. Section 40's digest put it as "participation ≠ welfare": Japan's oldest population keeps working, and the policy frontier is employment services, skills, job quality, occupational transitions, and phased-retirement infrastructure — built before necessity builds them. The Korean counterpoint in the same section sharpens the warning: Korea has the highest female exit age (67.4) in the OECD, yet over-66 poverty exceeds 25% — work in old age driven by inadequate pensions is not the goal; the goal is work that is chosen and sustainable. Japan's employment promotion is the model to study precisely because it shows both the possibility and the limit.⁴

On care, Japan is the innovator the world cites and the evidence base this report cannot fully document. The Community-Based Integrated Care System — localised healthcare, nursing care and daily living assistance within a 30-minute radius, enabling older adults to age in place with tailored care plans and community participation — was recorded in Section 74 as the concrete alternative to institutional default. But Section 57's evidence note stands: the retrieval contains no design evidence for Japan's Kaigo Hoken (the long-term care insurance system) — no contributions, eligibility, benefits, or spending data — and none for Singapore's either. The most-cited long-term care insurance model in the world is, in this evidence base, a name without numbers. That gap is itself a finding: the co-operation agenda of Section 90 needs Japan's care data, and it does not have it here.

The synthesis, then, is threefold. First, Japan proves that super-ageing is manageable — a society with more than a quarter of its population over 65 functions, works, and innovates. Second, it proves that management is not automatic: employment without adequate pensions is survival, not policy; care systems must be deliberately built (com-

munity-based, integrated, local) or the default is institutional and expensive. Third, it proves the value of the learning system: Japan ran the experiment first, and every country now ageing gets to read the results — provided the results are actually documented, which, on long-term care financing, they currently are not in this corpus.

Japan's arc through this report closes a loop: the demographic future (Sections 1–7), the working-life response (Sections 34–42), the care infrastructure (Sections 57–58, 74, 87), and now the country case that lived all of them first. The remaining country sections — Korea, the United States, and the comparative cases — complete the regional picture.

Evidence note: This prompt contained no Japan-specific evidence chunks — only hazardous-pension provisions (Section 23 territory), Ageing Report methodology structure pages, and TOC material. All substantive claims above are continuity synthesis from Sections 5, 40, 57, 74, and 92; per the report's honest-framing convention (Sections 59, 64, 78, 81), no bracket citations are attached to claims whose evidence lives in earlier sections. The gaps: no Kaigo Hoken design data, no Japan employment-service outcome data, no Japan pension-adequacy figures in the retrieved corpus.

Republic of Korea: pension coverage expansion and long-term care reform

Korea is the accelerated version of Japan: the fastest super-ageing society of the next quarter-century. Its 7→14% over-65 transition took 20 years (Section 91), and by 2050 it will have the world's highest median age — 56.5 years, ahead of Japan's 54.7 (Section 92). Its pension system is young and maturing, with spending projected to rise by approaching 4.5 percentage points of GDP over 2010–30 (Section 91). Korea's story is what happens when demographic compression meets a pension system still building its floors — and what reform looks like under time pressure.

The pension-coverage problem is stark and well measured. Korea's safety-net payments are among the lowest in the OECD at less than 10% of average earnings — in the company of China, India, Turkey, Mexico and South Africa, against a range where minimum pensions exceed 40% of average earnings in China and Turkey and residence-based basic pensions reach 40% in New Zealand. Yet Korea sits in the quarter of OECD countries with the worst pensioner poverty rates — as high as 50% of all over-66s — and the OECD finds significant scope for Korea (alongside Chile, Mexico, Turkey, and even Switzerland and the United States) to increase safety-net payments even after accounting for GDP per capita. The compounding mechanism is mechanical: most countries index their safety nets to prices, so the value of the floor falls relative to wages over time — the coverage gap widens unless policy actively closes it. Low earners face the same thinness on the contributory side: Korea is among the countries (with Japan, Mexico, Chile, the US, the UK, Germany, Poland, Sweden, Slovenia and Canada) where net projected replacement rates for a full career at half the average wage sit below 60% — a pension below 30% of the average wage.³

Coverage also leaks through career breaks. A five-year unemployment spell leaves the average Korean worker with a pension roughly 11% or more below a full-career counterpart — among the largest losses in the OECD (the average is 94% of the full-career pension), because Korea provides full pension credit for the first year of unemployment only, based on last earnings. The counterpart fact is that work in old age is Korea's de facto second tier: Section 40 recorded Korea's female exit age of 67.4 — the highest in the OECD — alongside over-25% poverty among 66–74 year-olds and over-40% beyond 75 (Section 92). Participation without floors is survival, not retirement policy.⁴

Long-term care reform faces the same mismatch. Korea's public long-term care spending remains below 1% of GDP, and where spending is that thin, women are 41% more likely than men to provide daily informal care (against 23% where spending exceeds 2% of GDP) — with low-income older Koreans (20th percentile) already below the poverty threshold and moderate home care consuming roughly a third of disposable income (Section 57). Reform, then, is not a single programme but a stack: raise the safety-net floor before ageing deepens, protect contributory coverage against career breaks, and build formal long-term care capacity before necessity converts daughters and daughters-in-law into the care system by default.

The lesson for the cooperation agenda (Section 90) is directional: Korea shows that compressed ageing plus a maturing pension system makes the first tier the whole game. Countries earlier in the transition should build the floor before the demographic wave arrives — Korea's experiment is what happens when you build it during the wave. The United States and the comparative cases complete the regional picture next.

Evidence note: Korea-specific claims above are drawn from the chunks embedded in this prompt (,); the 50%-of-over-66s poverty figure and Korea LTC facts were cited in earlier sections (8438dc72f26a, Section 57) and appear here as continuity context. Demographic framing (20-year transition, 56.5 median age 2050, +4.5pp pension spending 2010–30) is carried forward from Sections 91–92 without bracket citation per the continuity convention. The prompt's remaining chunks (DC equity models, pension-entitlement scenario methodology, payout-phase progressivity) were used in earlier sections and do not apply to Korea.³⁴

United States: the ageing census, baby boomers, and retirement claiming

The United States ages on its own schedule. With 14.6% of its population over 65 in 2015 — above the world average of 8.2% but well below Japan's 26.0% and the EU-28's 18.9% (Section 92) — America is demographically younger than the other major economies, and its ageing is dominated by the passage of the baby-boom cohort through the age structure. The US case is therefore less about compressed transition (Korea, 20 years) and more about the scale of a single generation's retirement: tens of millions of workers crossing the claiming threshold in one wave, and a pension system whose incentives, health dynamics, and care gaps will be stress-tested at continental scale.

Retirement claiming is the first stress point. Across the OECD the average effective age of labour-market exit was 64.4 years for men and 63.1 for women in 2022; the United States is one of only eight countries (with Estonia, Iceland, Israel, Japan, Korea, Mexico and New Zealand) where women withdraw after age 65 on average. But the effective age only weakly follows the statutory age — the correlation between normal retirement age and effective exit is 0.28 for men and 0.48 for women — meaning that for men, the pensionable age barely predicts when people actually stop working; health, working conditions, and caregiving pull harder than institutions push. The cautionary mirror is Colombia, Costa Rica and Korea, where normal retirement ages are low but exit ages are high because low pensions force workers to keep going. On the incentive side, the US is among the countries offering the longest deferral runway: maximum deferral ages of at least 70 for men (with Canada, Denmark, Iceland, Norway, Portugal and Switzerland) — the institutional space for longer careers exists; whether the other barriers fall is the question.⁴¹¹

Health is the largest barrier. In the 2022 AARP Global Employee Survey, health problems were a reason for retiring early for 25% of workers aged 55–64. Despite decades of rising life expectancy, the effective retirement age across the OECD is still lower today than it was 30 years ago — and the NBER's "Capacity to Work at Older Ages" project estimates that health status would permit substantially higher employment rates among 50-to-69-year-olds in 12 OECD countries, with the gap between health-permitted and actual employment exceeding 50 percentage points among 65–69-year-olds in some countries. The untapped potential is enormous — but it is not automatic: disability exits are "all too common," and mental health conditions are a distinct channel, with 60% employment among people with moderate mental health conditions versus 70% without, and 85% higher unemployment. Ill health, the section notes, contributed to the post-COVID surge in older-worker inactivity in countries like the UK — the health channel is the one that converts a retirement wave into a participation collapse.¹¹³

Skills are the second barrier, and the US position is relatively strong but fragile. PIAAC data show only 24% of 55–65-year-olds participating in job-related training — but the US is among the few countries (with Sweden and New Zealand) where mature-worker participation exceeds 30%. Demand is not the problem: even among workers who have not trained in five years, 41% of those 45 and above want to undertake training, and 52% of over-45s say the main reason is updating skills for better performance. The barriers are time, affordability, limited offers, proximity to retirement, and age discrimination — the classic access gradient of Section 46, now facing the largest retirement cohort in American history.¹¹

Care is the third, and America's weakest flank. Care leave is mostly a European institution: 19 of 35 OECD countries with data provide paid leave for an older dependent, and 22 provide paid or unpaid; among non-European OECD countries, only Japan and Canada have paid care leave nationally, while in the United States just five states have legislated paid care leave under specific conditions. With women providing 61% of daily informal care on average across the OECD, the US enters the baby-boom care crunch with a federally thin support infrastructure — the exact mechanism by which ageing can undo recent gains in women's labour-force participation.¹¹

The US lesson is the reverse of Korea's: not compressed ageing but cohort scale — a demographic wave hitting a system whose health, skill and care infrastructure is unevenly ready. Deferral incentives and relatively high mature-worker training participation are assets; health-driven exits and the state-by-state patchwork of care leave are the

liabilities. The comparative cases that close this block will show how these same pressures resolve under different institutional choices.

Evidence note: US-relevant facts drawn from this prompt's chunks (,,). Baby-boom cohort scale and the 14.6% 2015 share are continuity context from earlier sections (5, 92). Catch-up contribution figures (\$1,000 IRA / \$7,500 401(k)) were already cited (dd97844834d7). The prompt's payout-phase and tax-incentive chunks were used in Sections 18/19/87 and do not apply here.^{[4113](#)}

Comparative cases: Chile, Costa Rica, Rwanda, Bangladesh

Four countries, four positions on the ageing curve — and the same set of questions. Chile is reforming a mature contributory system; Costa Rica shows how retirees actually use their savings; Rwanda has almost no floor at all; Bangladesh ages with an informal majority. Taken together they close the regional block with the point that policy design matters at every income level — and that the inequality story is global. The OECD's summary line for emerging economies is blunt: old-age inequality is more acute there, with rapid ageing arriving at earlier stages of development, wider health inequalities, and less effective social protection than in OECD countries.^{[3](#)}

Chile: reforming the pension contract. Chile's January 2022 shift from a targeted first-tier pension to a quasi-universal guaranteed pension — covering 90% of the poorest 65-plus, adding roughly 70,000 beneficiaries (~4%), lifting the benefit about 4% in real terms to roughly 19% of gross average earnings — was recorded in Section 92 as the region's most significant first-tier reform. It sits alongside the OECD's assessment that Chile is among the countries with the most scope to raise safety-net payments even after GDP-per-capita adjustment (Section 94). The broader pattern: about two-thirds of lifetime earnings inequality passes through into pension inequality — from under 25% in many Anglo-Saxon countries to over 85% in about a third of OECD countries — and low earners' shorter life expectancy reduces their total pensions by a further ~13%. Raising retirement ages tends to widen the low/high-earner pension gap, though the effect is small. Chile's experiment is whether a universal floor can break that transmission.^{[3](#)}

Costa Rica: how savings actually get spent. In Costa Rica, 61% of those withdrawing a lump sum from their retirement savings used it for home improvements (SUPEN, 2020) — a reminder that in middle-income systems, retirement wealth and housing wealth are fungible, and payout design competes with the roof. The same survey evidence shows lump sums are the norm rather than the annuity exception the models assume: Costa Rica's retirees are converting pension wealth into housing wealth, not longevity insurance. Section 95's US data corroborates the instinct — American households use home equity as precautionary savings for health risks, and after health insurance, home equity is the primary resource to finance a health shock.^{[6](#)}

Rwanda and Bangladesh: ageing without the apparatus. Rwanda's floor is the missing one: roughly 75% informal employment among some 500,000 people aged 65-plus, no social pension, old-age poverty underestimated, widowed older women unsupported (Section 90). Bangladesh ages with an informal majority (Sections 2–3). These are the cases where “coverage expansion” does not mean parametric reform of contributory schemes — it means building the first tier from scratch, in economies where most workers never touch the formal system. The Section 90 thesis — ageing before coverage rises — is Rwanda and Bangladesh, stated plainly.

The comparative lesson: institutions choose the inequality. Three cross-cutting numbers frame the comparison. First, the generational bargain is fraying: in most countries average real incomes remain higher than previous generations' at the same age, but that is no longer true for cohorts born from the 1960s onward compared with the decade before. Second, the oldest-old are the future's mass population: the share of the population 80 and over will more than double by 2050 on average across OECD countries — and rural areas are ageing faster than urban ones, stretching health services and transport to their limit. Third, design beats wealth: Japan's 2000 national long-term-care insurance — mandatory contributions from age 40, benefits for all 65-plus (institutional, home, community-based) at the same level regardless of income — offers a flat-benefit, age-based, free-rider-proof structure that deliberately replaces the income-and-wealth-conditioned maze of the US system, where Medicaid covers care only below roughly \$9,242/year of income and the private LTC insurance market has collapsed from 125 providers in 2000 to fewer than 15 in 2014. The same design moral applies to claiming: delaying Social Security by three to six months has the same lifetime impact as earning an extra 1% for 30 years, and for someone living past 80, claiming at 70 beats claiming at 62 — yet limited savings force suboptimal early claims. The proposed START programme (Supplemental Transition Accounts for Retirement) would fund a Social Security bridge payment to make delay feas-

ible for the liquidity-constrained; making 40-year contributors “paid up” and Medicare the primary payer for over-65 workers’ medical costs would attack the employer-retention barrier. [316](#)

The block’s through-line, from Japan (Section 93) through Korea and the US to these four: ageing is universal, but the outcomes are designed. Chile redesigns the floor; Costa Rica reveals the lump-sum reality; Rwanda and Bangladesh show the cost of absence; Japan shows what a flat-benefit universal care system looks like. The regional survey closes here; the report’s final movements now consolidate the evidence into recommendations.

Evidence note: Comparative framing drawn from this prompt’s chunks (,,). Chile’s 2022 reform figures, Rwanda’s floor facts, and Bangladesh informality are continuity from Sections 92, 90, and 2–3. The prompt’s reference-list and TOC chunks (S0016:0215, S0016:0066, S0013:0149/0150/0151/0157, S0010:0035/0040, S0002:0148, S0015:0004/0008) carried no applicable evidence. The 25%-of-EU-elderly LTC-financing figure was cited in Section 87 and is not repeated. [3616](#)

Appendix

Sources, methods and limitations

The evidence base

This report is built from a corpus of 31 long-form source documents identified from 16 publishers — OECD, WHO, the World Bank, the European Commission, the US Census Bureau, the Stanford Center on Longevity, the World Economic Forum, AARP and others. Thirty were successfully ingested; one source URL failed to download and is absent from the evidence base. From the 30 ingested PDFs the pipeline extracted 9,311 text chunks and 1,545 native figures (charts and tables lifted directly from the source documents, not recreated).

Three of the 30 PDFs are mirror-hosted copies of official publisher editions, retrieved from third-party domains after the publisher links failed: WHO’s Global Age-Friendly Cities: A Guide (via [jcafc-port.hk](#)), WHO’s Decade of Healthy Ageing: Baseline Report (via [ifa.ngo](#)), and the World Bank’s Live Long and Prosper: Aging in East Asia and Pacific, abridged edition (via [OpenDevelopmentMekong](#)). The documents are the publishers’ editions; only the hosting domain differs. They are not presented as direct-from-publisher downloads.

Known gaps in the corpus: IMF fiscal-ageing publications could not be downloaded (access blocked); Statistics Korea’s direct download failed; Lancet sources were excluded as paywalled; and the housing-and-cities topic rests on two strong sources with no national-statistics-office report. Where the evidence base is thin, sections say so in an explicit evidence note rather than filling the gap with assertion.

How the report was assembled

Each of the 96 sections was written against retrieved evidence chunks, and every substantive claim carries an inline citation of the form [^n] that resolves in the References list to a source document and chunk. Thirty-two figures were selected from the 1,545 extracted candidates and reviewed by eye for readability, relevance and resolution before placement; each is captioned with its source. Figures are genuine source charts, not illustrations.

Long-range statements that extend beyond sourced projections — particularly for 2050–2075 — are framed as scenarios, not predictions. The report invents no statistics: where projections exist they are attributed; where they do not, the text marks the boundary of what the evidence supports.

Scope boundaries

The report’s horizon is 2025–2075, with deep dives on Japan, the Republic of Korea, Europe and the United States. Longevity biotechnology (senolytics, rapamycin and related interventions) is outside scope by design, as is data-free futurist speculation. The analysis covers demography, pensions and retirement economics, household finance, work and learning, health and long-term care, housing and cities, inequality, intergenerational dynamics and social connection — the systems a 100-year life actually runs on.

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